Mechanics of materials

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MECHANICS
OF
MATERIALS

By

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CONTAINING 27 CHAPTERS

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Preface

For an efficient, critical and competitive design of a machine or a structure, a thorough knowledge of Strength or Mechanics of Materials is essential. Consequently, the modern engineering curriculum lays greater emphasis on the comprehensive study of the fundamentals as well as advanced concepts of Mechanics of Materials. The book, through its twenty seven chapters, covers almost all the standard topics of strength or mechanics of materials. Each chapter includes topics of both the elementary as well as advanced and specialised nature. Hence it can be used both as a standard text as well as the reference for professional/field designs. Through their long teaching, design and professional experience, the Authors have made an attempt to present the subject matter which lays emphasis on fundamentals, keeping in view the difficulties confronted by the students. Permissible stresses have been adopted as per the latest recommendations contained in various Standards published by the Bureau of Indian Standards (formerly, Indian Standards Institution), New Delhi.

Starting with an introductory chapter on *mechanical properties of materials*, chapters 2 to 5 are devoted on *stress and strain,* with chapter 5 dealing with more advanced topics on *analysis of strain.* Chapter 6 is on *strain energy*, while chapter 7 includes advanced material on *theories of failure*. Most of chapters use the *properties of area* of a plane lamina; hence these properties have been discussed thoroughly in chapter 8. Chapters 9 to 17 deal with *bending aspects of beams*, incorporating bending moment, shear force, bending stresses, shearing stresses, slope and deflection of various types of beams. The effects of *axial loads,* with or without eccentricity, as well as *lateral loads,* have been dealt with in chapters 18 to 20, along with the *design aspects*. Chapters 21 and 22 deals with *torsion of shafts and springs. Pressure vessels*, both thin and thick, have been included in chapters 23 and 24, incorporating both the analysis as well as the design aspects. Chapter 25 includes the analysis of *perfect frames*. Lastly, chapters 26 and 27 deal with the analysis and design of *riveted and welded joints.*

Each chapter, starting with assumptions and theory, is complete in itself and is built up logically to cover all aspects of the particular theory, and in some cases, designs also. Both worked examples and unsolved problems are given in the text, and all these are treated in SI units. Some examples are based on questions set up by various examining bodies, and thankful acknowledgement is recorded here. Though utmost care has been taken to eliminate errors,

some may still remain, and we would be thankful if these are pointed out. The Authors are thankful to Shri M.S. Gahlot for Laser Type Setting and to Shri Kanhya Lal for tracing the diagrams. Finally, the Authors are thankful to Shri R.K. Gupta of Laxmi Publications (P) Ltd. for good printing and excellent get up to the book and that too in a record short period.

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1

Mechanical Properties of Materials

1.1. IMPORTANT MECHANICAL PROPERTIES

The following are the most important *mechanical properties* of *engineering materials:*

(i)	Elasticity	(ii)	Plasticity
(iii)	Ductility	(iv)	Brittleness
(v)	Malleability	(vi)	Toughness
(vii)	Hardness, and	(viii)	Strength

Some of the above properties can not be mutually reconciled; hence no material can possess them all simultaneously. The criteria of suitability (or otherwise) of an engineering material, forming part of either a machine or a structure, is dependent upon the possession of one or more of the above properties. The above properties are assessed, with a fair degree of accuracy, by resorting to *mechanical tests*.

1.2. ELASTICITY

When external forces are applied on a body, made of engineering materials, the external forces tend to deform the body while the molecular forces acting between the molecules offer resistance against *deformation*. The deformation or displacement of the particles continues till full resistance to the external forces is setup. If the forces are now gradually diminished, the body will return, wholly or partly to its original shape. *Elasticity is the property by virtue of which a material deformed under the load is enabled to return to its original dimension when the load is removed.* If a body regains completely its original shape, it is said to be *perfectly elastic.* For any particular material, a critical value of the load, known as the *elastic limit* marks the partial break down of elasticity beyond which removal of load results in a degree of *permanent deformation* or *permanent set* (Fig. 1.1). Steel, aluminium, copper, stone, concrete etc. may be considered to be perfectly elastic, within certain limits.

Stress-Strain relationship : The load per unit area, normal to the applied load is known as *stress* (p). Similarly, the deformation per unit length in the direction of deformation is known as *strain* (ε). The elastic properties of materials used in engineering are determined by tests performed on small specimens of material. The tests are conducted in materials-testing-laboratories equipped with testing machines capable of loading the specimens in gradually applied increments, and the resulting stresses and strains are measured at all such load increments, till the specimen fails. Fig. 1.1 shows one such stress-strain diagram (schematic). In Fig. 1.1(a), the specimen is loaded only upto point A, well within the elastic limit E. When the load, corresponding to point A, is gradually removed the curve follows the same path AO and the strain completely disappears. Such a behaviour is known as the *elastic behaviour.* In Fig. 1.1(b), the specimen

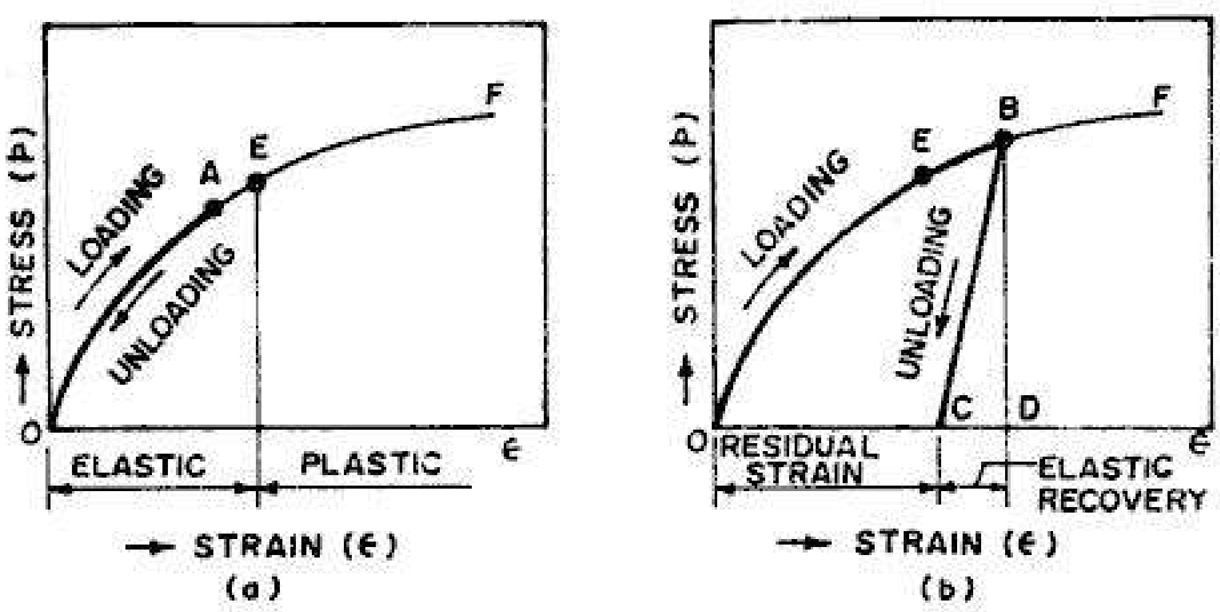


FIG. 1.1. ELASTICITY AND PLASTICITY

is loaded upto point *B*, beyond the elastic limit *E*. When the specimen is gradually *unloaded*, the curve follows path *BC*, resulting in a *residual strain* (*OC*) or permanent strain. Such a behaviour of the material, loaded beyond the elastic limit, is known as partially elastic behaviour. A more detailed discussion of stress-strain curve is given in § 2.4.

Homogeneity and Isotropy : A material is homogeneous if it has *same composition* throughout the body. For such a material, the *elastic properties* are the same at each and every point in the body. It is interesting to note that for a homogeneous material, the elastic properties need not be the same in all the directions. If a material is equally elastic in all the directions, it is said to be *isotropic*. If, however, it is not equally elastic in all directions, *i.e.* it possesses different elastic properties in different directions, it is called *anisotropic*. A *theoretically ideal material* could be equally elastic in all directions, *i.e.* isotropic. Many structural materials meet the requirements of homogeneity and isotropy. We shall be dealing with only the homogeneous and isotropic materials in this book.

1.3. PLASTICITY

Plasticity is the converse of elasticity. A material in *plastic state* is permanently deformed by the application of load, and it has no tendency to recover. Every elastic material possesses the property of plasticity. Under the action of large forces, most engineering materials become *plastic* and behave in a manner similar to a *viscous liquid*. The characteristic of the material by which it undergoes *inelastic strains* beyond those at the *elastic limit* is known as *plasticity*. When large deformations occur in a ductile material loaded in the *plastic region*, the material is said to undergo *plastic flow*. The property is particularly useful in the operations of *pressing* and *forging*. 'Plasticity' is also useful in the design of structural members, utilising its *ultimate strength*.

1.4. DUCTILITY

Ductility is the characteristic which permits a material to be drawn out longitudinally to a reduced section, under the action of a tensile force. In a ductile material, therefore, large deformation is possible before *absolute failure* or *rupture* takes place. A ductile material must, of necessity, possess a high degree of *plasticity* and *strength*. During ductile extension, a material shows a certain degree of elasticity, together with a considerable degree of plasticity. Ductility is measured in the tensile test of specimen of the material, either in terms of percentage elongation or in terms of percentage reduction in the cross-sectional area of the test specimen. The property of ductility is utilised in wire drawing.

1.5. BRITTLENESS

Brittleness implies lack of ductility. A material is said to be brittle when it can not be drawn out by tension to smaller section. In a brittle material, failure takes place under load without significant deformation. Brittle fractures take place without warning and the property is generally highly undesirable. Examples of brittle materials are (i) cast iron (ii) high carbon steel, (iii) concrete (iv) stone, (v) glass, (vi) ceramic materials, and (vii) many common metallic alloys. Fig. 2.2 shows a typical stress-strain curve for a typical brittle material which fail with only little elongation after proportional limit (point A) is exceeded, and the fracture stress (point F) is the same as ultimate stress. Ordinary glass is a nearly ideal brittle material in which the stress-strain curve in tension is essentially a straight line, with failure occurring before any yielding takes place. Thus, glass exhibits almost no ductility whatsoever.

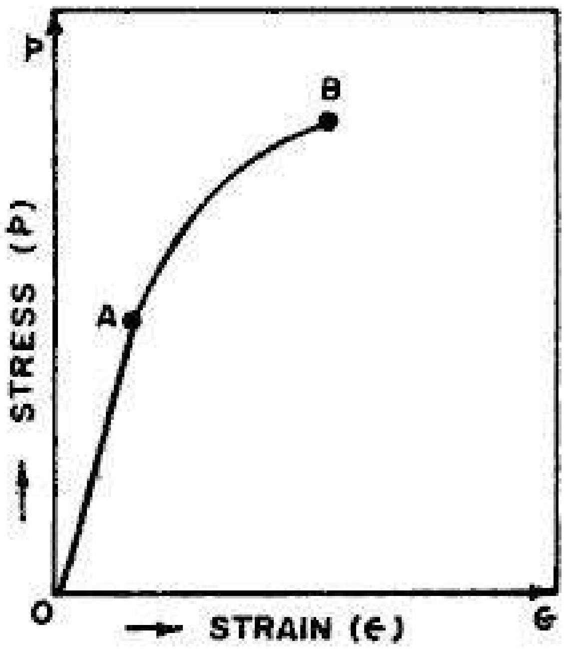


FIG. 1.2. STRESS STRAIN CURVE FOR A BRITTLE MATERIAL

1.6. MALLEABILITY

Malleability is a property of a material which permits the materials to be extended in all directions without rupture. *A malleable material possesses a high degree of plasticity, but not necessarily great strength.* This property is utilised in many operations such as forging, hot rolling, drop-stamping etc.

1.7. TOUGHNESS

Toughness is the property of a material which enables it to absorb energy without fracture. This property is very desirable in components subject to cyclic or shock loading. Toughness is measured in terms of energy required per unit volume of the material, to cause rupture under the action of gradually increasing tensile load. This energy includes the work done upto the elastic limit which is small in comparison with the energy subsequently expanded. Fig. 1.3 shows the stress-strain curves, both for mild steel as well as high carbon steel. The toughness is represented by the area under the stress-strain curve for the material. A common comparative test for toughness is the bend test in which a material is expected to sustain angular bending without failure.

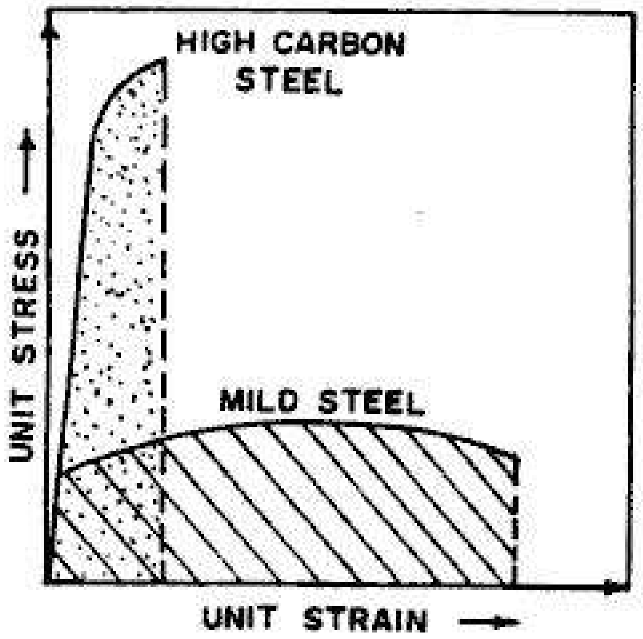


FIG. 1.3. MEASURE OF TOUGHNESS.

1.8. HARDNESS

Hardness is the ability of a material to resist indentation or surface abrasion. Since these resistances are not necessarily synonymous, it is usual to base the estimation of the hardness of a material on resistance to indentation only. Tests on hardness may be classified into (*i*) scratch test, and (*ii*) indentation test. The scratch test consists of pressing a loaded diamond into the surface of the specimen, and then pulling the diamond so as to make a scratch. The *hardness number* is then determined on the basis of (*i*) load required to make a scratch of a given width, or (*ii*) the width of the scratch made with a given load. The *indentation test* consists of pressing a body of standard shape into the surface of the test specimen. In the commonly used Brinnell hardness test a hardened steel ball of a given diameter is squeezed into the surface of test specimen, under a fixed standard load and then surface area of the indent is measured. Brinell's hardness number (B.H.N.) is then given by :

$$\text{B.H.N.} = \frac{P}{\frac{\pi D}{2}\left[D - \sqrt{D^2 - d^2}\right]}$$

where P = Standard load (N) ; D = diameter of steel ball (mm)

d = diameter of the indent (mm)

1.9. STRENGTH

This is the most important property of a material, from design point of view. *The strength of a material enables it to resist fracture under load.* The load required to cause fracture, divided by the area of the test specimen, is termed as the *ultimate strength* of the material, and is expressed in the unit of stress. An important consideration in engineering design is the capacity of the object (such as building structure, machine, air craft, vehicle, ship etc.), usually referred to as *structure*, to support or transmit loads. If structural failure is to be avoided, the loads that a structure actually can support must be greater than the loads it will be required to sustain when in service. Since the ability of a *structure* to resist loads is called *strength*, the governing criterion is that the *actual strength* of a structure must exceed the required strength. The ratio of the actual strength to the required strength is called the *factor of safety*. However, failure may occur under the action of tensile load, compressive load or shear load. Hence it is essential to know the *ultimate strength* of the material in each of these three conditions, and the three ultimate strengths are separately determined experimentally.

1.10. MECHANICS (OR STRENGTH) OF MATERIALS

Three fundamental areas of engineering mechanics (or applied mechanics) are

(*i*) Statics (*ii*) Dynamics

and (*iii*) Mechanics (or strength) of materials.

Statics and dynamics are devoted primarily to the study of the *extended effects* of forces on rigid bodies, *i.e.* the bodies for which the change in shape (or deformations) can be neglected. In contrast to this, *mechanics of materials,* commonly known as *strength of materials* deals with the relation between externally applied loads and their *internal effects* on solid bodies. The *solid bodies* include axially loaded members, shaft in torsion, thin and thick cylinders and shells, beams, and columns, as well as structures that are assemblies of these components. These bodies are no longer assumed to be rigid ; the deformation, however small, are of major interest. In actual design, the engineer must consider both dimensions and material properties to satisfy the requirements of *strength* and *rigidity*. A machine part or structure should neither break nor deform excessively. The purpose of studying *strength of materials* is to ensure that the structure used will be safe against maximum internal effects that may be produced by any combination of loading.

2

Simple Stresses and Strains

2.1. SIMPLE STRESSES

When a body (*i.e.* structural element) is acted upon by external force or load, internal resisting force is set up. Such a body is then said to be in a *state of stress*, where *stress is the resistance offered by the body to deformation*. For further understanding of this internal resistance, consider a *prismatic bar AB* subjected to axial forces at the ends as shown in Fig. 2.1 (*a*). A *prismatic bar* is a straight structural member of uniform cross-section (A) throughout its length (L). In order to know the *internal stresses* produced in the prismatic bar, take a section mn normal to the longitudinal axis of the bar ; such a section is known as a *cross-section*. If we consider the equilibrium of either the left part or the right part at section mn, taken as a free body, the *internal resistance* or the stress (p) offered by the molecules against the external force may be assumed to be uniformly distributed over the whole area of cross-section. Then

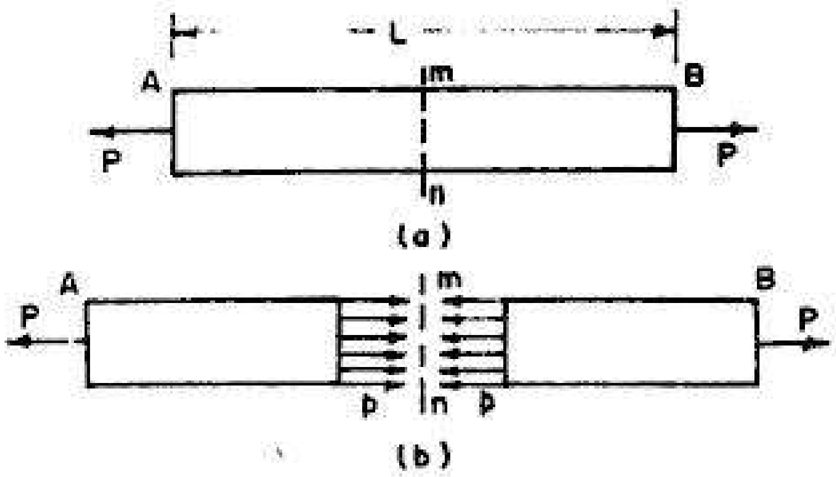


FIG. 2.1. STATE OF STRESS

$$p = \frac{P}{A} \qquad \text{...(2.1)}$$

where p = Internal resistance = stress = intensity of force.

A = Area of cross-section normal to the axis.

As the stress p acts in a direction perpendicular to the cut surface, it is referred to as a *normal stress*. Since the normal stress p is obtained by dividing the axial force by the cross-sectional area, it has the units of force per unit area, such as kN/m^2 or N/mm^2

Saint Venant's principle

We have assumed above that the distribution of stress over the cross-section mn is *uniform*. This assumption is based on Saint Venant's principle. This principle states that except in the region of extreme ends of a bar carrying direct loading, the stress distribution over the cross-section is *uniform*.

Consider a square bar (Fig. 2.2*a*) of section $b \times b$, subjected to axial force P. The stress distribution at section $m_1 n_1$, distant $b/2$ from the end is shown in Fig. 2.2 (*b*), where the maximum normal stress (p_{max}) is found to be equal to 1.387 times the average stress (p_{av}). The stress distribution at section $m_2 n_2$, distant b from the end is shown in Fig. 2.2 (*c*), where p_{max} is found to be $1.027 p_{av}$. Lastly, at section $m_3 n_3$, distant $3b/2$ from the end (Fig. 2.2 *d*), p_{max} is found to be equal to p_{av}. This illustrates Saint Venant's famous principal of *rapid dissipation of localised stresses*. Hence in all practical cases of stress analysis, St. Venant's principle can be safely followed, and the normal stress distribution given by Eq. 2.1 can be assumed.

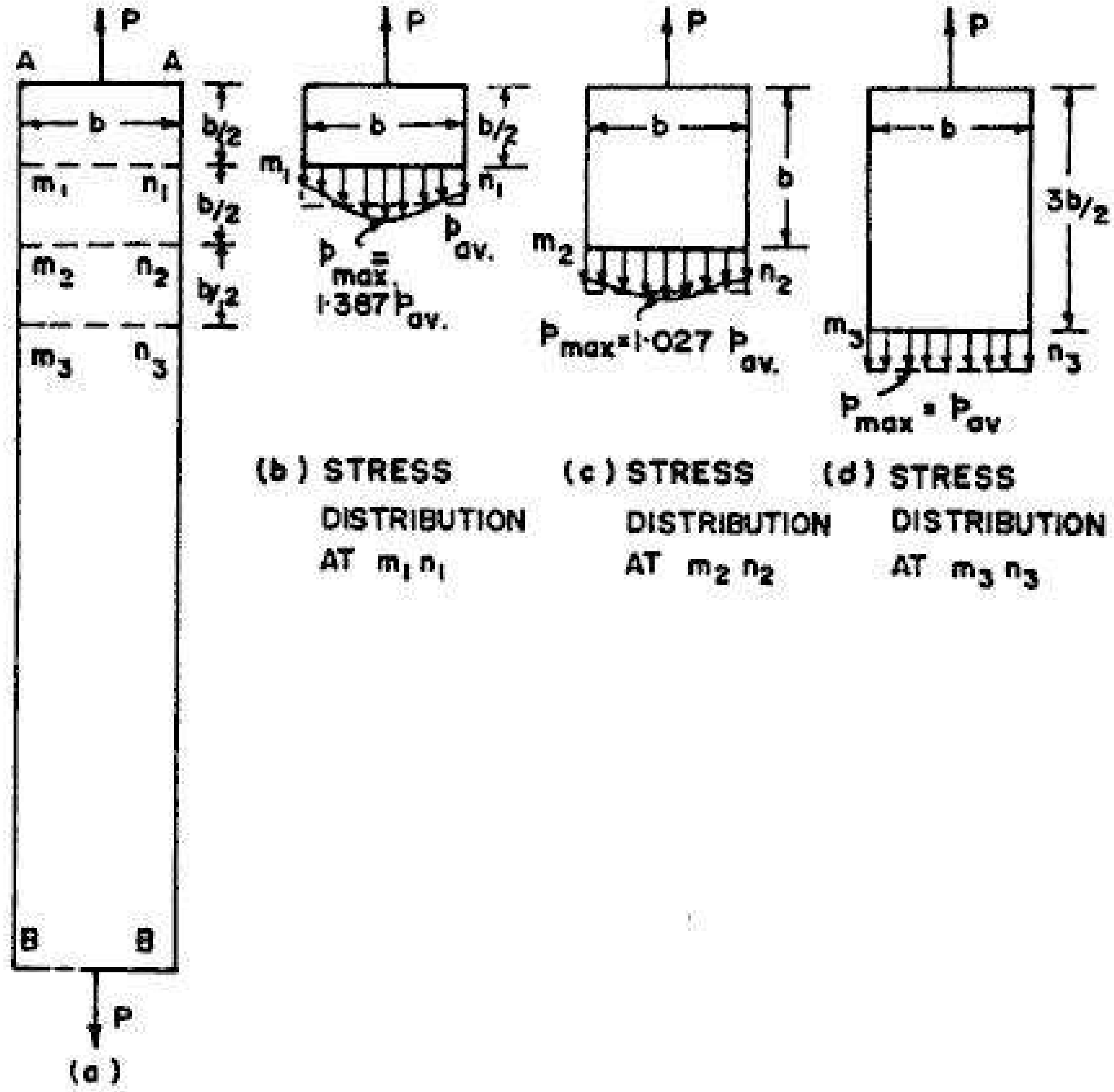


FIG. 2.2. St. VENANT'S PRINCIPLE

2.2. KINDS OF STRESSES

There are the following kinds of stresses

(1) Normal stresses

  (*i*) Tensile stress

  (*ii*) Compressive stress

(2) Shear stress or tangential stress

(3) Bending stress

(4) Twisting or torsional stress

(5) Bearing stress

Normal stresses

When a stress acts in a direction perpendicular to the cut surface, it is known as *normal stress* or *direct stress*. Normal stresses are of two types : (*i*) tensile stress, and (*ii*) compressive stress.

Tensile stress

When a body is stretched by the force P, as shown in Fig. 2.1, the resulting stresses are tensile stresses. Thus tensile stress exists between two parts of a body when each draws the other towards itself. Such a state of stress is shown in Fig. 2.1 where

$$p_{av} = p = \frac{P}{A} \qquad ...(2.1)$$

Compressive stress

If the forces are reversed in direction, causing the body to be compressed, we obtain compressive stresses. Thus, compressive stress exists between two parts of a body when each pushes the other from it. Such a state of stress is shown in Fig. 2.3. where

$$p_{av} = p = \frac{P}{A} \qquad ...(2.2)$$

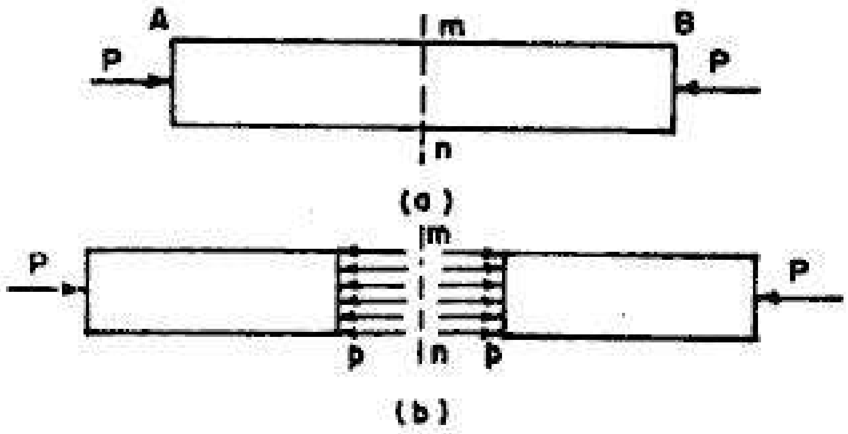


FIG. 2.3. COMPRESSIVE STRESS

Shear stress

Shear stress is the one which acts *parallel* or *tangential* to the surface. Thus, shear stress exists between the parts of a body when the two parts exert equal and opposite forces on each other laterally in a direction tangential to their surface in contact.

Fig. 2.4 (*a*) shows a riveted connection, where the rivet resists the shear across its cross-sectional area (*A*), when subjected to pulls *P* applied to the plates so jointed. Under the action of the pulls *P*, the two plates will press against the rivet in *bearing*, and contact stresses, called *bearing stresses* will be developed against the rivet. A free-body diagram of the rivet (Fig. 2.4 *a ii*) shows these bearing stresses. This free body diagram shows that there is a tendency to shear the rivet along cross section *mn*. From the free body diagram of the section *mn* of the rivet (Fig. 2.4 *a iii*), we see that shear force *V* acts over the cut surface. In this particular case (known as the case of *single shear*), the shear force *V* is equal to *P*. This shear force is, infact, the resultant of the shear stresses distributed over the cross-sectional area of the rivet, shown in Fig. 2.4 (*a iv*).

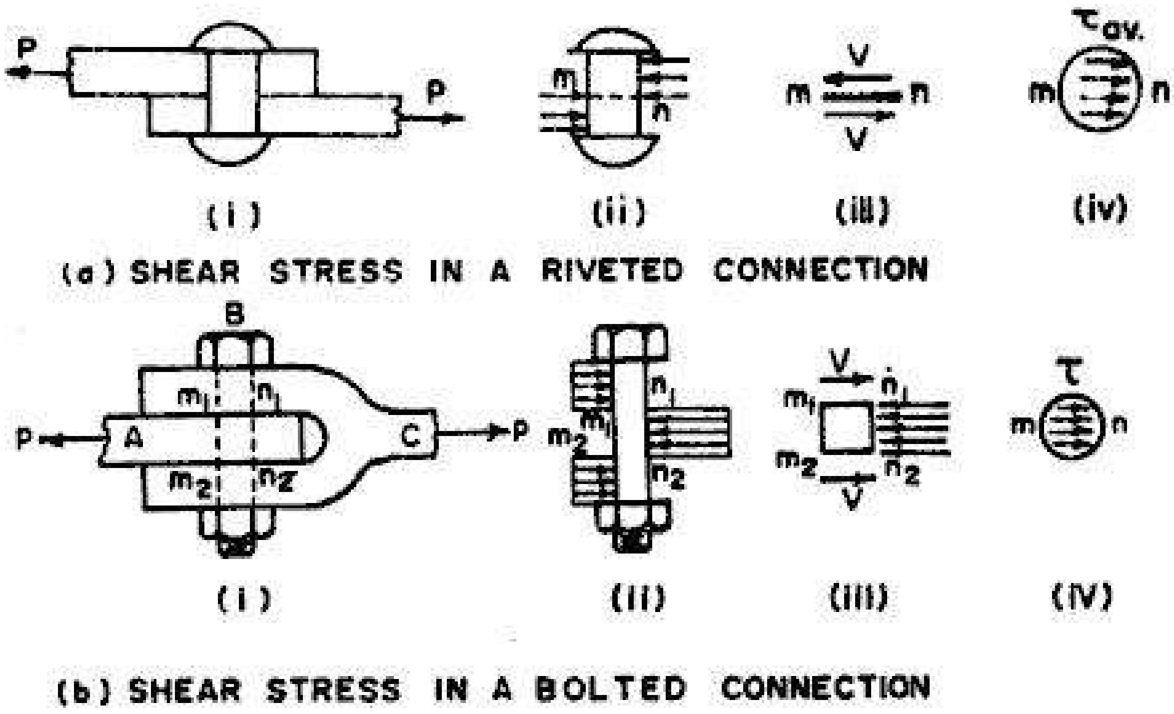


FIG. 2.4. EXAMPLES OF DIRECT SHEAR OR SIMPLE SHEAR

The average shear stress on the cross-section of the rivet is obtained by dividing the shear force *V* by the area of cross-section (*A*) of the rivet :

$$\tau_{av} = \frac{V}{A} = \frac{P}{A} \qquad ...(2.3)$$

Another practical example of shear stress is the bolted connection shown in Fig. 2.4(*b*), consisting of a flat bar *A*, a clevis *C*, and a bolt *B* that passes through holes in the bar and the clevis. Under the action of pulls *P*, the bar and the clevis will press against the bolt in *bearing*, resulting in the development of *bearing stresses* against the bolt, as shown in Fig. 2.4 (*bii*). The bolt will have the tendency to get sheared across sections $m_1 n_1$ and $m_2 n_2$. Fig 2.4(*biii*) shows the free body diagram of the portion $m_1 n_1 - m_2 n_2$ of the bolt, which suggests that shear forces *V* must act over the cut surfaces $m_1 n_1$ and $m_2 n_2$ of the bolt. In this particular case (known as the case of *double shear*), each shear force is equal to $P/2$. These shear forces are in fact the resultants of the shear stresses distributed over the cross-sectional area of the bolt, at sections $m_1 n_1$ and $m_2 n_2$ (Fig. 2.4 *b iv*).

The average shear stress on the cross-section of the bolt is obtained by dividing the shear force *V* by the area of the cross-section (*A*) of the bolt :

$$\tau_{av} = \frac{V}{A} = \frac{P}{2A} \qquad ...(2.3\ a)$$

The examples shown in Fig. 2.4 are the examples of *direct shear* or *simple shear*. Such direct stresses arise in bolted, pinned, riveted, welded or glued joints, wherein the shear stresses are caused by a direct action of the force trying to act through the material. Shear stresses are also developed in an *indirect* manner when members are subjected to bending or torsion.

Bending stresses, torsional stresses and bearing stresses have been discussed in later chapters.

Units of stress

(i) SI system : Since normal stress *p* is obtained by dividing the axial force by the cross-sectional area, it has *units* of force per unit of area. In S.I. units, the unit of force is *newton* expressed by the symbol N, and the area is expressed in square metres (m^2) . Hence the unit of stress is newtons per square metre (N/m^2) or *Pascals* (Pa). *However, newton is such a small unit of stress that it becomes necessary to work with large multiples. Due to this, force is generally expressed in terms of kilo-newton and meganewton, where :*

1 kilo-newton (kN) $= 10^3$ N

1 mega-newton (MN) $= 10^6$ N

1 giga-newton (GN) $= 10^9$ N

Similarly, the stress unit Pascal (*i.e.* N/m^2) is such a small unit of stress that it becomes necessary to work with large multiples. Hence stress is generally expressed in terms of kN/m^2 , MN/m^2 , GN/m^2 and N/mm^2 (MPa). As an example, a typical tensile stress in a steel bar might have a magnitude of 150 N/mm^2 (150 MPa) which is 150×10^6 Pa . A more common form of unit of stress (which is not recommended in SI) is N/mm^2 , which is a unit identical to mega pascal (MPa) Thus, we have

$1\ N/mm^2 = 10^6\ N/m^2 = 10^6\ Pa = 1\ MPa$

(ii) M.K.S. system : In M.K.S. system, the unit of force is in the *gravitational unit*, *i.e.* kilogram force kg(f), commonly expressed by kg only. When force is large, it is expressed by tonnes, where 1 t = 1000 kg. The unit of stress is usually expressed as kg/cm^2.

The following relationships exist :

$$1 \text{ kg(f)} = g \text{ newtons}$$

Taking $$g = 9.807,$$

$$1 \text{ kg(f)} = 9.807 \text{ N} \approx 10 \text{ N}$$

Also, $$1 \text{ tonne} = 9.807 \text{ kN} \approx 10 \text{ kN}$$

Hence $$1 \text{ kg/cm}^2 \approx \frac{10 \text{ N}}{10^{-4} \text{ m}^2} \approx 10^5 \text{ N/m}^2 \approx 10^5 \text{ Pa}$$

or $$1 \text{ kg/cm}^2 \approx \frac{10^5 \text{ N}}{10^6 \text{ mm}^2} \approx 0.1 \text{ N/mm}^2$$

2.3. STRAIN

When a prismatic bar is subjected to axial load, it undergoes a change in length, as indicated in Fig. 2.5. This change in length is usually called *deformation*. If the axial force is tensile, the length of the bar is increased, while if the axial force is compressive, there is shortening of the length of the bar. This *elongation* (or shortening), as the case may be, is the cumulative result of the stretching (or compressing) of the material throughout the length L of the bar. The deformation (*i.e.* elongation or shortening) per unit length of the bar is termed as strain (ε or e). In general, *strain* is the *measure* of the deformation caused due to external loading.

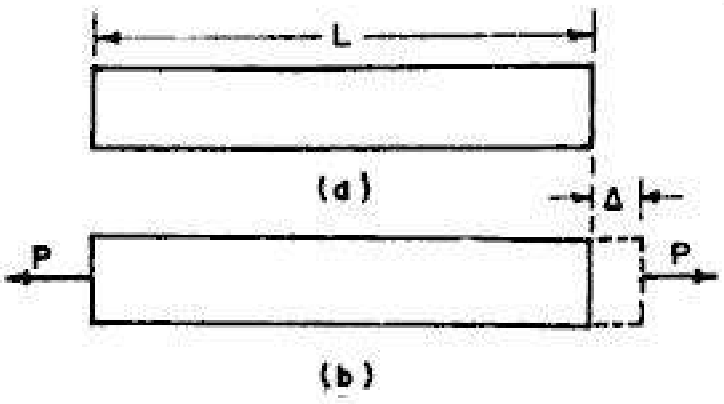


FIG. 2.5. DEFORMATION AND STRAIN

If the bar is in tension, the resulting strain is known as *tensile strain*. Similarly, the strain resulting from a compressive force is known as *compressive strain*. In general, strains associated with *normal stresses* are known as *normal strains*. Similarly, the strain associated with shear stress is known as *shear strain*.

Since strain is the deformation per unit length, it is a *dimensionless quantity*. Thus, it has no units, and therefore, it is expressed as pure number. For example, if the deformation of a bar of 1.6 m length is 1.2 mm, the strain $\varepsilon = \Delta/L$ = 1.2 mm/1.6 × 1000 mm = 0.00075 = 750 × 10^{-6}. Some times, in practice, strain is recorded in forms such as mm/m or μ m/m etc. Thus, strain of the above example is 0.75 mm/m or 750 μ m/m.

Example 2.1. *A prismatic bar has a cross-section of 25 mm × 50 mm and a length of 2 m. Under an axial tensile force of 90 kN, the measured elongation of the bar is 1.5 mm. Compute the tensile stress and strain in the bar.*

Solution

Stress $p = \dfrac{P}{A} = \dfrac{90 \times 1000}{25 \times 50}$

$= 72 \text{ N/mm}^2$ (or MPa)

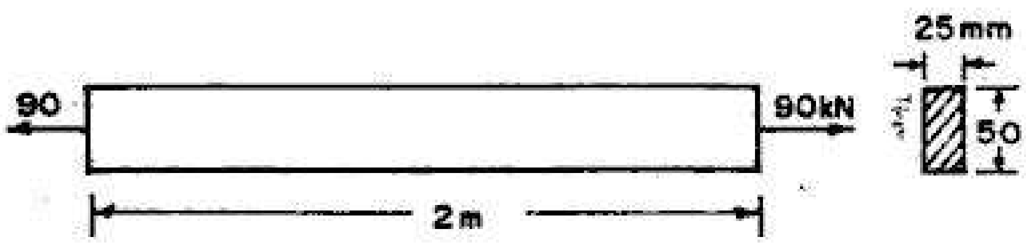


FIG. 2.6

Strain $\qquad \varepsilon = \dfrac{\Delta}{L} = \dfrac{1.5 \text{ mm}}{(2 \times 1000) \text{ mm}} = \mathbf{0.00075 = 750 \times 10^{-6}}$

2.4. STRESS - STRAIN DIAGRAM

The mechanical properties of a material, discussed in chapter 1, are determined in the laboratory by performing tests on small specimens of the material, in the *materials testing laboratory*. The most common materials test is the *tension test* performed on a cylindrical specimen of the material. The loads are measured on the main dial of the machine while the elongations are measured with the help of *extensometers*. The cylindrical specimen has enlarged ends so that they can fit in the grips of the machine. This ensures that failure will occur in the central uniform region, where the stress is easy to be calculated rather than at or near ends where the stress distribution is not uniform. When such a specimen of a ductile material is subjected to a gradually increasing pull in a tension test machine, it is found that the resultant strain is proportional to the corresponding stress upto a limit only and beyond that, the relation is not linear. In investigating the mechanical properties of the material beyond this limit, the relationship between the strain and the corresponding stress is usually represented graphically by a ***tensile test diagram*** or ***stress strain diagram***.

A = Proportional Limit	Oa = Linear Deformation
B = Elastic Limit	Ob = Elastic Deformation
C = Yield Point	bd = Perfect Plastic Yielding
C' = Lower Yield Point	de = Strain Hardening
E = Ultimate Strength	ef = Necking
F = Rupture Strength	

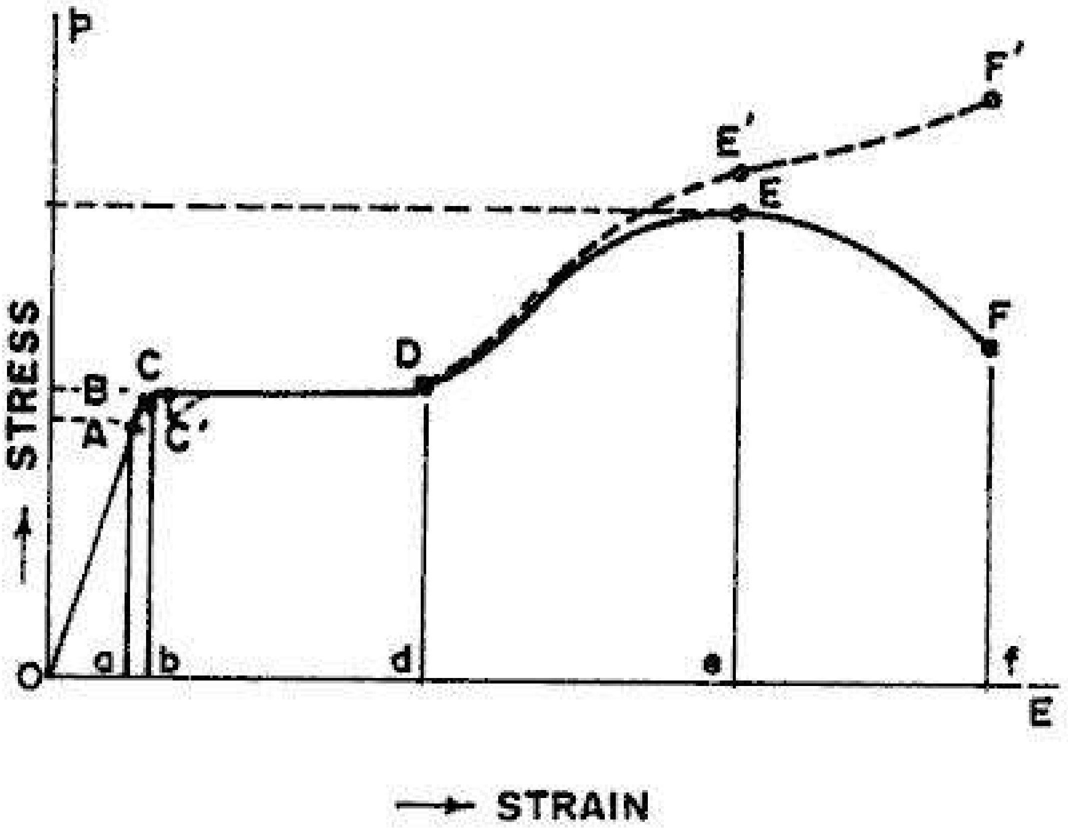


FIG. 2.7. TENSILE TEST DIAGRAM (NOT TO SCALE)

A stress-strain diagram for a typical structural steel in tension is shown in Fig. 2.7 (not to scale), where the strain is plotted along the horizontal axis while stress is plotted on the vertical axis. The diagram begins with a straight line O to A, in which the stress strain relationship is *linear, i.e.* stress and strain are directly proportional. Point A marks the *limit of proportionality* beyond which the curve becomes slightly curved, until point B, the *elastic limit* of the material, is reached. Region AB is the non-linear region in which the stress is not proportional to strain, and the elongation increases more rapidly. However, upto the point B, the removal of load would result in *complete recovery* by the specimen of its original dimensions. If the load is increased further, yielding takes place; point C is the point of sudden large extension, known as the *yield point*. After the yield point stress is reached, the *ductile extensions* take place, the strains increasing at an accelerating rate as represented by C to D.

During the ductile extension, the area of cross-section decreases in practically the same proportion that the length increases, and hence this is the region of *perfect plasticity* or *yielding*. In this region, there is no noticeable increase in the tensile force. The material becomes *perfectly plastic* in this region (C to D), which means that it can deform without an increase in the applied load. For mild steel, the elongation in this region is 10 to 15 times the elongation that occurs between O and A. If the load is further increased, the steel begins to *strain harden*. During strain hardening region, the material appears to regain some of its strength and offers more resistance, thus requiring increased tensile load for further deformation. This is so because *the material undergoes changes in its atomic and crystalline structure* in the strain hardening region. After D, with further increases in loads and extensions, the point E of the *maximum load* or *ultimate stress* (commonly known as the *ultimate strength*) is reached. Up to the maximum load, the bar extends *uniformly* over its parallel length but, if straining is continued, a local reduction in cross-sectional area occurs (*i.e.* formation of waist) and as the load is concentrated at this reduced area, a considerable local extension (known as *concentrated plastic deformation*) also takes place, till the failure or rupture takes place at F. It is customary to base all the stress calculations on the original cross-sectional area of the specimen, and since the latter is not constant, the stresses so calculated are known as *nominal stresses*. The nominal stress is less at rupture load than at the maximum load, as indicated by points F and E respectively. The diagram of *real stresses* (*i.e.* load divided by reduced area of cross-section) would be as shown by the dotted curve in Fig. 2.7. The breaking load divided by the reduced area of section (*i.e. actual stress intensity*) is greater than at the maximum load.

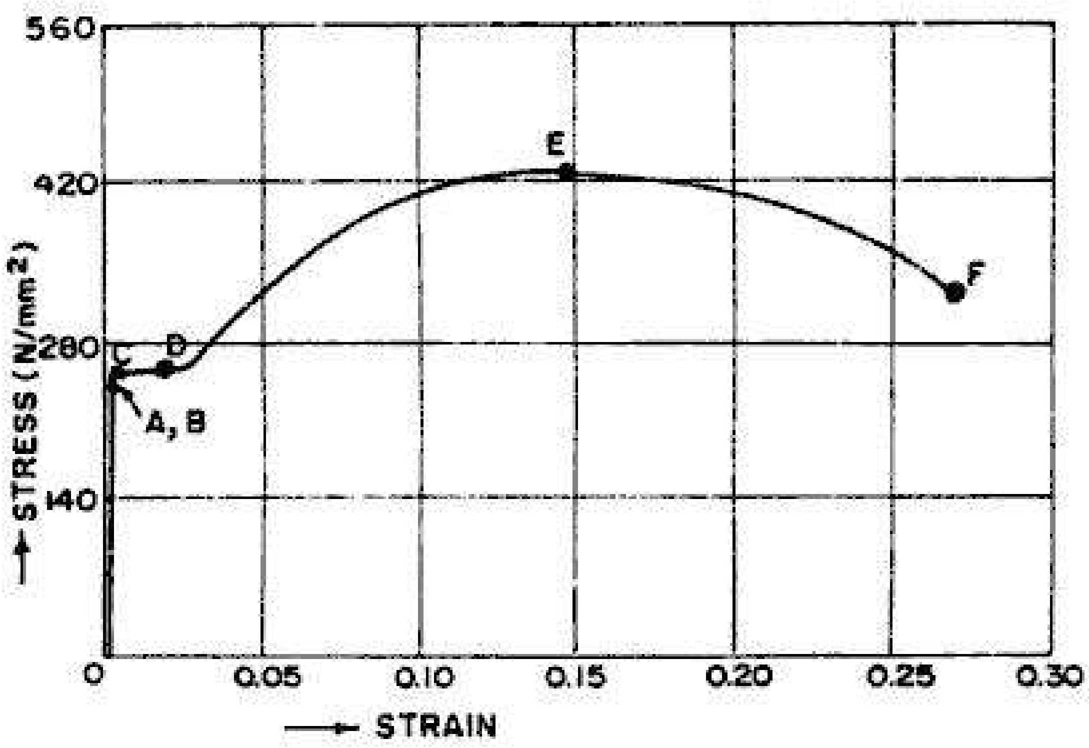


FIG. 2.8. TENSILE TEST DIAGRAM FOR MILD STEEL (DRAWN TO SCALE)

Fig. 2.8 shows a stress-strain curve in tension for mild steel drawn to scale. The strains that occur from C to D are 15 times more than the strains that occur from O to A, and further the strains from D to F are many times greater than those from C to D. Hence, in this diagram, the linear part of the diagram appears to be vertical, with the points A, B and C over lapping.

Stress-strain curves for other materials

Fig. 2.9 shows stress strain curves for steels having carbon contents varying from 0.12 to 1 percent. From these curves, we notice that with increasing carbon content, the curves approach the form characteristic of brittle materials such as cast iron, though the ultimate stress is many times greater.

Fig. 2.10 shows typical stress-strain diagrams for several common materials such as high carbon steel, nickel-chrome steel, mild steel, wrought iron, cast iron, copper and cast aluminium. From these we observe that for steels and wrought irons, proportionality exists almost until

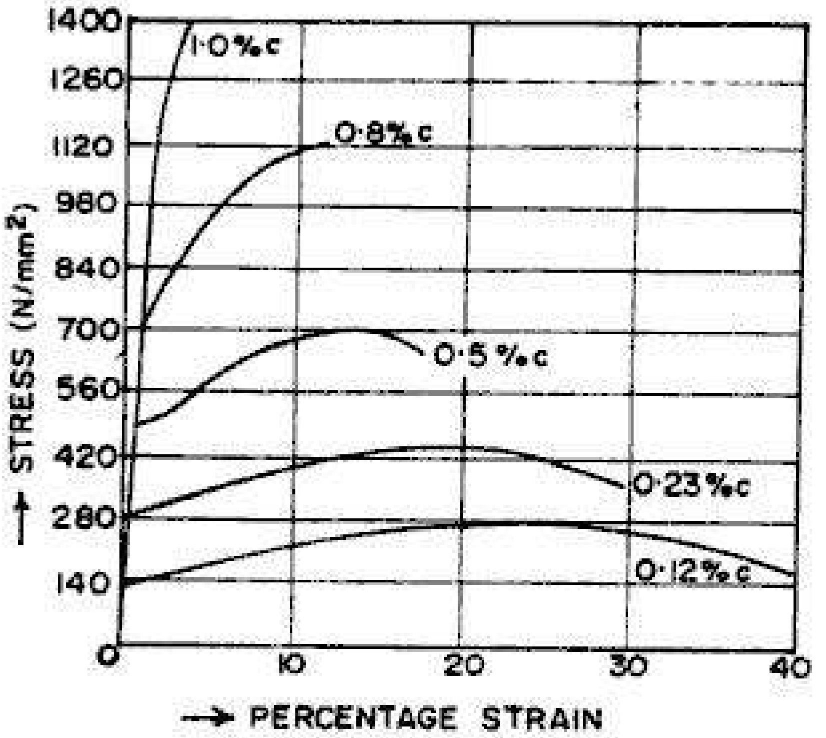


FIG. 2.9. STRESS-STRAIN DIAGRAMS FOR STEELS WITH VARYING PERCENTAGE OF CARBON

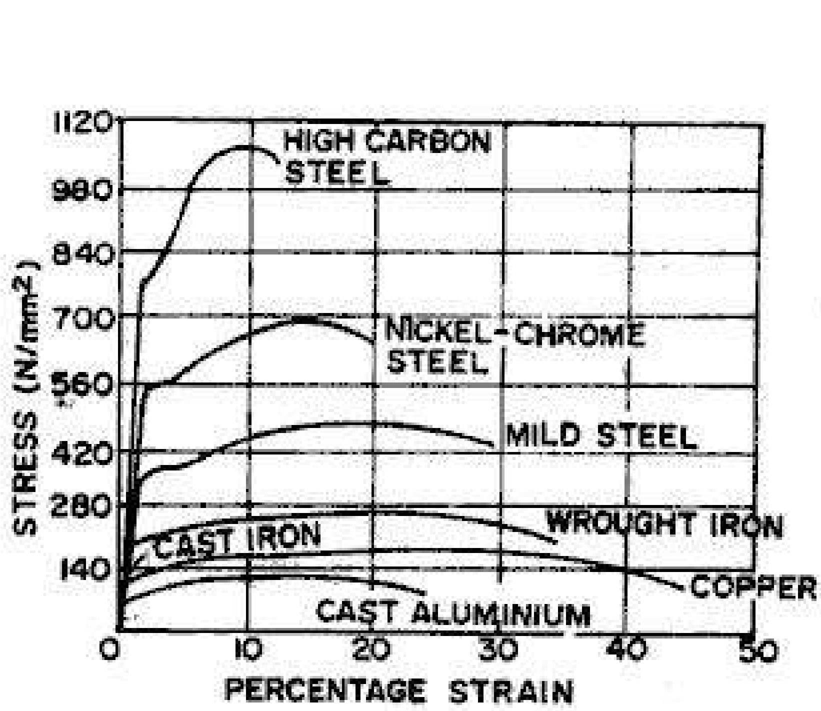


FIG. 2.10. STRESS STRAIN DIAGRAMS FOR SOME COMMON MATERIALS

yielding takes place. However, for copper, cast aluminium and high alloys, no clearly defined limit of proportionality, elastic limit or yield point are exhibited. Cast iron behaves like a brittle material which fails without any visible elongation or reduction in area.

Fig. 2.11 shows typical stress-strain diagram for aluminium alloy, exhibiting considerable ductility, though they do not have clearly definable yield point. Fig. 2.12 shows stress-strain diagrams for hard rubber and soft rubber. The curve for rubber is linear upto very large strains in the vicinity of 0.1 or 0.2. Soft rubber usually continue to stretch enormously without failure, and after that if offers increasing resistance to the load with the result that the curve turns markedly upward prior to failure.

Typical stress-strain curve for brittle material is shown in Fig. 1.2 wherein the material fails in tension at relatively low value of strain. Examples of brittle materials are concrete, stone, cast iron, glass, ceramic materials and many common metallic alloys. Ordinary glass is

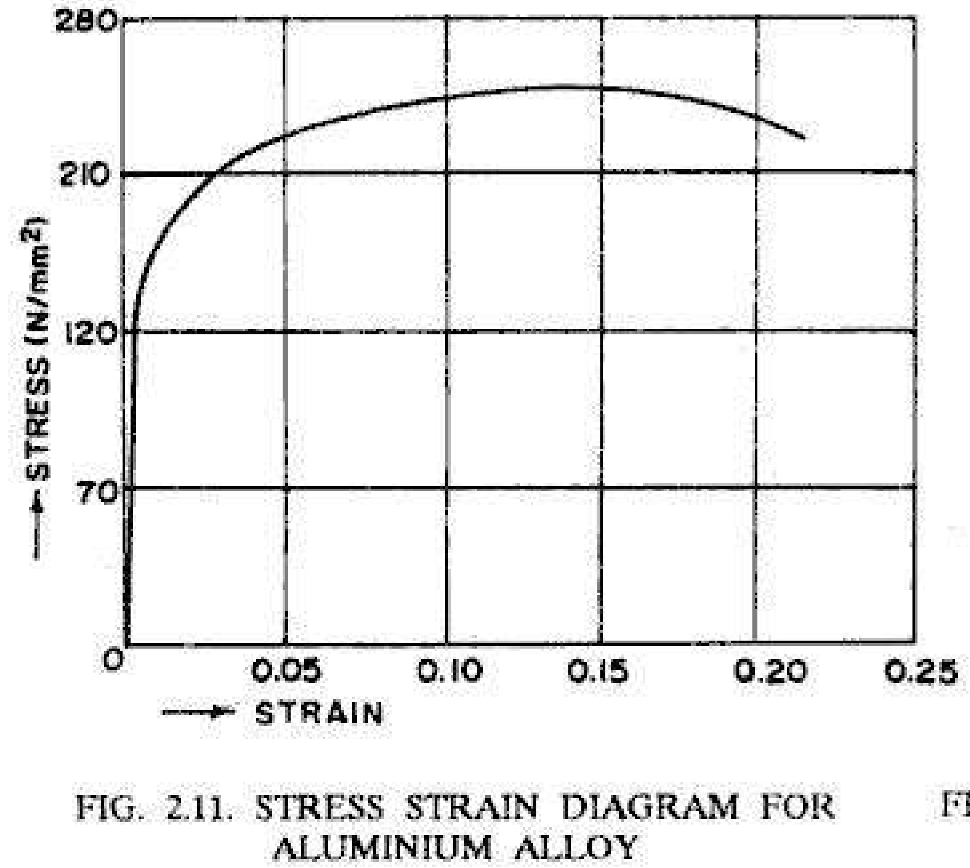


FIG. 2.11. STRESS STRAIN DIAGRAM FOR ALUMINIUM ALLOY

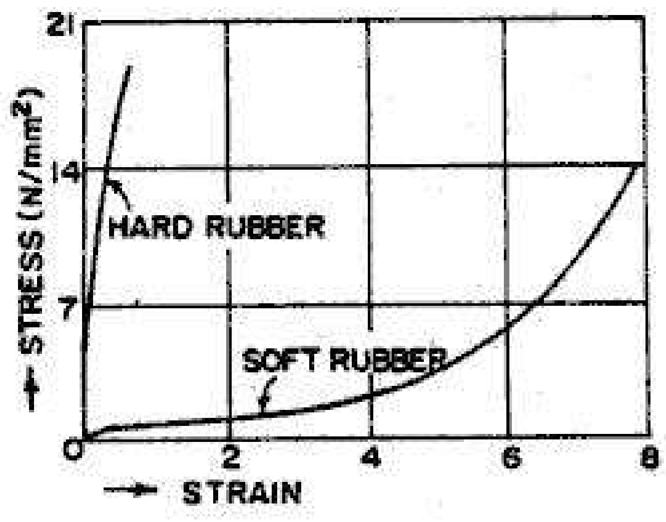


FIG. 2.12. STRESS-STRAIN DIAGRAM FOR RUBBER

a nearly ideal brittle material, exhibiting almost no ductility whatsoever.

Limit of proportionality

Limit of proportionality is the stress at which the stress-strain curve ceases to be a straight line; it is the stress at which extensions cease to be proportional to strain. In Fig. 2.7, point A corresponds to this limit. Robert Hooke's famous law "Ut tensio sic vis", *i.e.* 'As strain, so force' related strain to stress linearly, and did not recognise limit to this proportionality. The proportional limit is important because all subsequent theory involving the behaviour of elastic bodies is based on stress-strain proportionality.

Elastic limit

It is that point in the stress-strain curve upto which the material remains elastic, *i.e.* the material regains its original shape after the removal of the load. Thus, elastic limit represents the *maximum stress* that may be developed during a simple tension test such that there is no permanent or residual deformation after the removal of the load. Its value can be approximately determined by loading and unloading the test specimen till *permanent set* is found on complete removal of the load. This point is represented by point B in stress-strain curve of Fig. 2.7. However, for many materials, elastic limit and proportional limit are almost numerically the same, and the terms are sometimes used synonymously. In cases (such as in Fig. 2.7), where the two are different, elastic limit is always greater than the proportional limit.

Elastic range

This is the region of the stress-strain curve between the origin and the elastic limit. Thus in Fig. 2.7, *ob* is the *elastic range* in which only elastic deformations take place. These deformations disappear on the removal of the load.

Plastic range

This is the region of the stress-strain curve between the elastic limit (B) and point of rupture (F). Thus, in Fig. 2.7, *bf* is the plastic range, in which plastic deformations take place. These deformations are permanent deformations which do not disappear even after the removal of the load. The plastic range consists of three regions : (*i*) region *bd* in which perfect plastic yielding takes place, (*ii*) region *de*, usually called *strain hardening range,* and (*iii*) region *ef* in which non-uniform or concentrated plastic deformation takes place giving rise to *necking* of the specimen. Region *bd* and *de* taken together mark the *uniformly distributed plastic deformation.*

Yield point

Yield point is the point just beyond the elastic limit, at which the specimen undergoes an *appreciable* increase in length without further increase in the load. The phenomenon of yielding is more peculiar to structural steel; other materials do not possess well defined yield point. Careful testing of more ductile materials (like annealed low carbon steel) indicates that there is, in reality, a slight load reduction giving two yield points C and C'(Fig. 2.7), known respectively as *upper* and *lower* yield points. It is possible to obtain yield point in mild steel of the order of 100% greater than the lower yield stress. M.M. Hutchinson obtained upper yield point at 73000 lb/sq in. (486 N/mm^2) and lower yield point at 37000 lb/sq in. (247 N/mm^2) from a tensile test on a 0.038 in. diameter mild steel wire after proper annealing and careful preparation of the test piece. In a tensile test, it is usual to remove the *extensometer* from the specimen at this stage (*i.e.* at yield point) so that the instrument is not damaged by the ensuing large plastic extensions which thereafter are measured by means of dividers and a steel rule or similar other device.

Yield strength

The yield strength of a material is closely associated with the yield point. *Yield strength* is defined as the lowest stress at which extension of the test piece increases without further increase in load. It is indicated by careful testing of the specimen. Many materials do not have well defined yield point. For such cases, yield strength is determined by *off-set method*. As illustrated in Fig. 2.13 (*a*), a line offset of an arbitrary strain of 0.2% (*i.e.* 0.002 m/m), is drawn parallel to the straight line portion of the original stress-strain diagram. Where there is no specific straight line portion (the diagram being continuously curved), the 0.2% offset is drawn parallel to the *initial tangent* of the stress strain curve (Fig. 2.13 *b*). In both the cases, the intersection of the offset line with the curve (*i.e.* point *C*) *defines the yield stress*, some times also known as the *offset yield stress*.

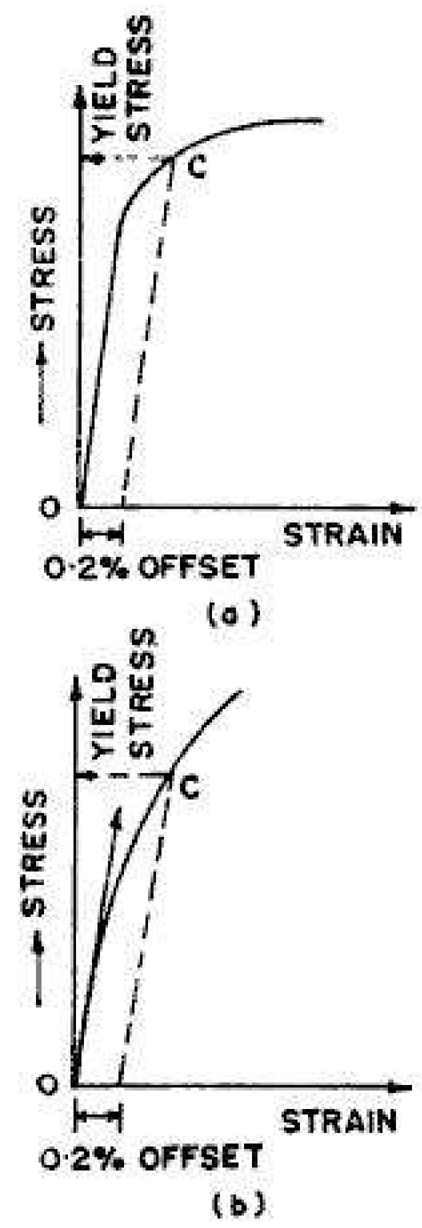


FIG. 2.13. OFFSET METHOD

Ultimate strength

The *ultimate stress* or *ultimate strength*, as it is more commonly called, is the highest point (such as point *E* in Fig. 2.7) on the stress-strain curve. At this highest point, concentrated plastic deformation takes place, resulting in the formation of *neck* or *waist* in the specimen, resulting in decrease in the load.

$$\text{Thus, Ultimate strength} = \frac{\text{Maximum load}}{\text{Original area of cross-section}} \quad ...(2.5)$$

Rupture strength

The *rupture strength* is the stress corresponding to the failure point *F* of the stress-strain curve. For structural steel, it is some what lower than the ultimate strength. This is so because the rupture stress is computed by dividing the rupture load by the original cross-sectional area, while the actual area is very much less because of *necking*. The *actual rupture strength* (point *F'*) obtained by dividing the rupture load by the cross-sectional area at the time of rupture, is very much higher than the *actual ultimate strength* (point *E'*). Although actual rupture strength is considerably higher than the ultimate strength, *the ultimate strength is commonly taken as the maximum stress of the material.*

Proof stress

Proof stress is the stress necessary to cause a non-proportional or permanent extension equal to a *defined percentage* (say 0.1 or 0.2%) of gauge length. Alternatively, proof stress can be expressed as the stress at which the stress-strain diagram departs by a specified percentage of the gauge length from the produced straight line of proportionality. If the specified percentage is 0.1% of the gauge length, the corresponding proof stress is designated as 0.1% *proof stress*. If a certain value for proof stress is specified for a material, and, after loading to that stress and then unloading, the permanent extension is less than the specified percentage of the gauge length, then the material is considered in respect of the minimum proof stress requirement. It should be clearly noted that *proof stress is prefixed by the percentage non-proportional strain it produces*, *i.e.* 0.1% proof stress or 0.2% proof stress. The proof stress for a material is determined (Fig. 2.14) by drawing a line parallel to the linear portion of the stress-strain diagram from a point which is at a distance equal to the specified percentage (say, 0.1 percent) of the strain on the gauge length. The intersection of this line with the stress-strain curve represents the 0.1 percent proof stress for the material.

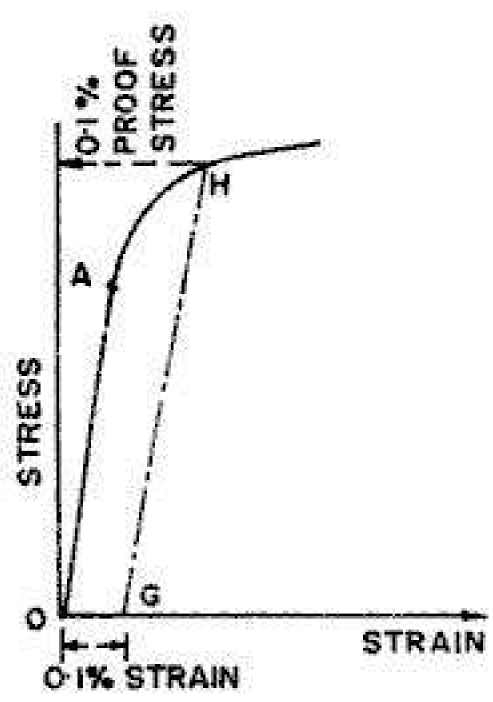


FIG. 2.14. DETERMINATION OF PROOF STRESS

Estimation of ductility of the material

The study of stress-strain diagram indicates that from the yield point to the ultimate strength, the elongation is practically distributed uniformly over the length of the specimen. This region of *uniformly distributed plastic deformation* is typical of ductile material. An advantage of ductility is that visible distortions may occur if the loads become two large, thus providing an opportunity to take remedial measures before an actual fracture takes place. Also, ductile materials are capable of absorbing large amounts of energy prior to fracture. The presence of a pronounced yield point followed by large plastic strains is an important characteristic of mild steel that is some times used in practical design. Other ductile materials include aluminum and some of its alloys, copper, magnesium, lead, molybdenum, nickel, brass, bronze, monel metal, nylon, teflon etc.

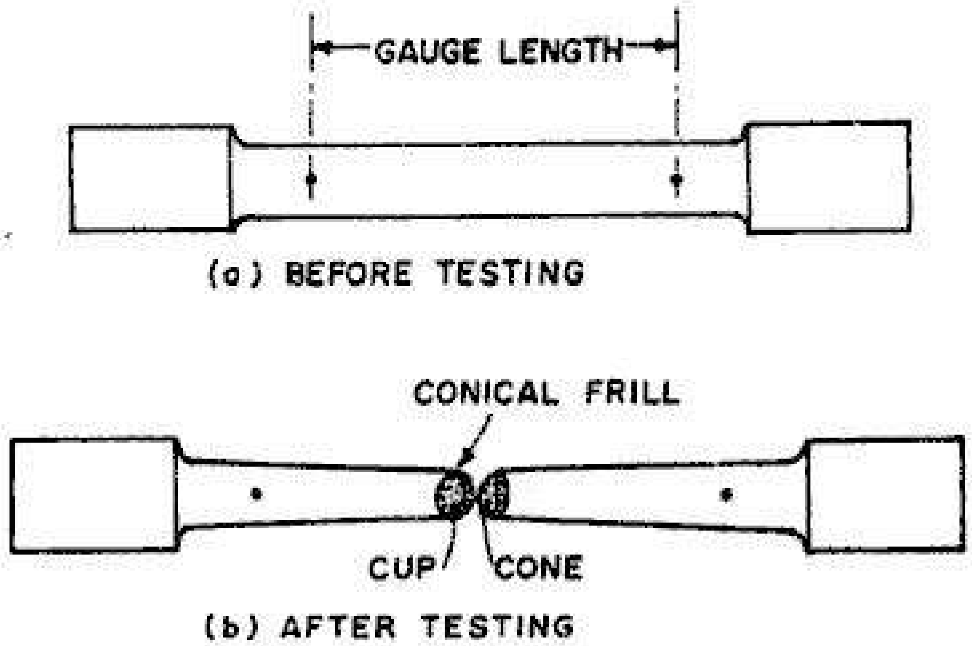


FIG. 2.15. TEST SPECIMEN BEFORE AND AFTER TESTING

Ductility of a material is estimated by two methods.

(*i*) Percentage elongation method

(*ii*) Percentage reduction in area or percentage contraction

The second method is considered to be a better measure of ductility, being independent of gauge length.

Percentage elongation

The percentage elongation is the percentage increase in the length of the gauge length. If L_0 is the original gauge length and L_f the final length between the gauge marks, measured after fracture,

$$\text{Percentage elongation} = \frac{L_f - L_0}{L_0} = \frac{\Delta L_0}{L_0} \times 100 \qquad \text{...(2.6)}$$

Since local yielding occurs before the fracture of the specimen, the percentage elongation depends upon the length of the specimen, (*i.e.*, upon the gauge length). Hence it is always essential to mention the gauge length over which percentage elongation is computed

Percentage reduction in area (or percentage contraction)

Ductility of the material can also be estimated in terms of percentage reduction in the area in the waist at fracture.

$$\text{Thus, percentage reduction in area} = \frac{A_0 - A_f}{A_f} \times 100 \qquad \text{...(2.7)}$$

Where A_0 = original area of test specimen
A_f = area in the waist at fracture

For ductile materials, this reduction is about 50%.

Gauge length : Barba's law

From the study of the tensile test diagram (Fig. 2.7) we observe that the total elongation of a specimen of a given gauge length is equal to the sum of (*i*) *uniform extension* (*i.e.* from point O to point E in Fig. 2.7) and (*ii*) *Local extension* (From point E to point F) due to 'necking' or 'waisting'. It is found that the *uniform extension*, taking place during the elastic and the plastic range, is proportional to the gauge length, while the *local extension* is independent of the gauge length. Due to this, it becomes essential to specify the *gauge length* in a tension test; otherwise, if the gauge length is increased, the effect of local extension would decrease the percentage elongation.

Prof. Unvin verified that the local extension is proportional to the square root of the cross-sectional area. He gave the following expression for the total extension (Δ_T) :

$$\Delta_T = a\,L_0 + b\,\sqrt{A_0} \qquad \text{...(2.9)}$$

where L_0 = gauge length,
A_0 = original area of cross-section of the specimen and a, b are constants.

By means of extensive series of experiments, Barba gave the following law :

Geometrically similar specimens of the same material deform similarly if they are so proportioned that $L_0/\sqrt{A_0}$ is constant.

The following values are given for mild steel :

$$a = 0.2 \text{ and } b = 0.7$$

In order to eliminate any error in comparison of elongation figures, it is recommended in B.S. 18 that the gauge length (L_0) should be equal to $4\sqrt{A}$.

Working stress and factor of safety

The *working stress*, also called the *allowable stress*, is the *maximum safe stress* the material may carry. In design, the working stress should be limited to a value not exceeding the proportional limit of the material so that Hooke's law, on which all subsequent theories are based, is not invalidated. Working stress is based either on the yield point stress or on ultimate strength, by dividing these by suitable factors of safety (n).

Thus, $$\text{Working stress} = \frac{\text{Ultimate strength}}{\text{Factor of safety}} = \frac{p_u}{n} \quad ...(2.8\ a)$$

or $$\text{Working stress} = \frac{\text{Yield stress}}{\text{Factor of safety}} = \frac{p_y}{n'} \quad ...(2.8)$$

For structural steel and other similar ductile materials having definite yield stress, working stress is based on yield strength. For brittle materials, where there is no definite yield point, working stress is based on ultimate strength of the material.

While selecting the factor of safety and hence the working stress for a particular material, the following points are taken into consideration :

(1) ***Nature of loading***

(*i*) Whether it is *dead load* or *live load*
(*ii*) Whether the load is applied *gradually* or *suddenly*
(*iii*) Whether the load is *constant* in type or *alternating*
(*iv*) Whether the load is for *short duration* or for *long duration*

(2) ***Nature of the material***

(*i*) Whether the material is homogeneous and isotropic or not
(*ii*) Whether there are likely to be any weaknesses (such as internal flaws etc.) or not.

(3) ***Environmental factors***

(*i*) Effects of salt water and humidity
(*ii*) Effects of corrosion
(*iii*) Effects of wear

(4) ***Previous case histories*** : Possible results of any failure

(5) ***Workman-ship*** : The possibility of errors occurring in manufacture or fabrication.

As a rule, the factors of safety are not directly specified; working stresses are set for different materials under different conditions of use. Generally, *codes of practice* for different materials are prepared by the Bureau of Standards, specifying the working stresses which are to be used by the designers.

2.5. LINEAR ELASTICITY : HOOKE'S LAW

In Fig. 2.7, we observe that the initial portion (OA) of the stress-strain diagram is straight. The *slope* of this line is the *ratio* of stress to strain, and is constant for a material. In this range, the material also remains elastic. When a material behaves elastically and also exhibits a linear relationship between stress and strain, it is called *linearly elastic. Linear elasticity is a property of many solid materials, including metals, wood, concrete, plastics and ceramics.*

The slope of stress-strain curve is called the *modulus of elasticity* (E):

$$\text{Slope of stress-strain curve} = E = \frac{p}{\varepsilon} \quad ...(2.10)$$

In other words, the linear relationship between stress and strain is expressed by the equation

$$p = E\,.\,\varepsilon \quad ...(2.10\ a)$$

Thus, the *modulus of elasticity* (E) is the constant of proportionality which is defined as the *intensity of stress that causes unit strain*. Thus, modulus of elasticity E has the units same as units of stress.

The equation $p = E \cdot \varepsilon$ is commonly known as *Hooke's law*, named after the famous English scientist Robert Hooke (1635-1703) who was the first person to investigate the elastic properties of various materials such as metals, wood, stone, bones etc. Robert Hooke's famous law *"Ut tensio sic vis"*, *i.e.* *"As strain, so force"* related strain to stress and did not recognise a limit to this proportionality.

Originally, Hooke's law specified merely that stress was proportional to strain. It was Thomas Young in 1807, who introduced a constant of proportionality, which later came to be known as *Young's modulus of elasticity*.

A common variation of Hooke's law is obtained by replacing stress p by its equivalent P/A and replacing ε by Δ/L, in Eq. 2.10. Thus,

$$\frac{p}{\varepsilon} = E$$

or

$$\frac{P/A}{\Delta/L} = E$$

which gives

$$\Delta = \frac{PL}{AE} = \frac{pL}{E} \qquad \text{...(2.11)}$$

Eq. 2.11 is the most commonly used version of Hooke's law applied for direct stress (tensile or compressive).

Most metals have high value of E and hence the strains are always very small. For example, mild steel has a value of E approximately 2.05×10^5 N/mm^2 under normal working conditions. At this value, strain $\varepsilon = p/E = 150/2.05 \times 10^5 = 0.00073$, when $p = 150$ N/mm^2. This means that a bar of 1 m length will change in length by 0.73 mm only. On the other hand, rubber, though it does not obey Hooke's law accurately, has a low value of E and will undergo considerable deformation at moderate stress values, as is evident from Fig. 2.12.

2.6. PRINCIPLE OF SUPERPOSITION

Many times, a structural member is subjected to a number of forces acting, not only at the ends, but also at intermediate points along its length. Such a member can be analysed by the application of principle of superposition. According to the principle of superposition, the resulting strain will be equal to the algebraic sum of the strains caused by the individual forces acting along the length of the member. Thus, if a member of uniform section is subjected to a number of forces, the resulting deformation (Δ) is given by

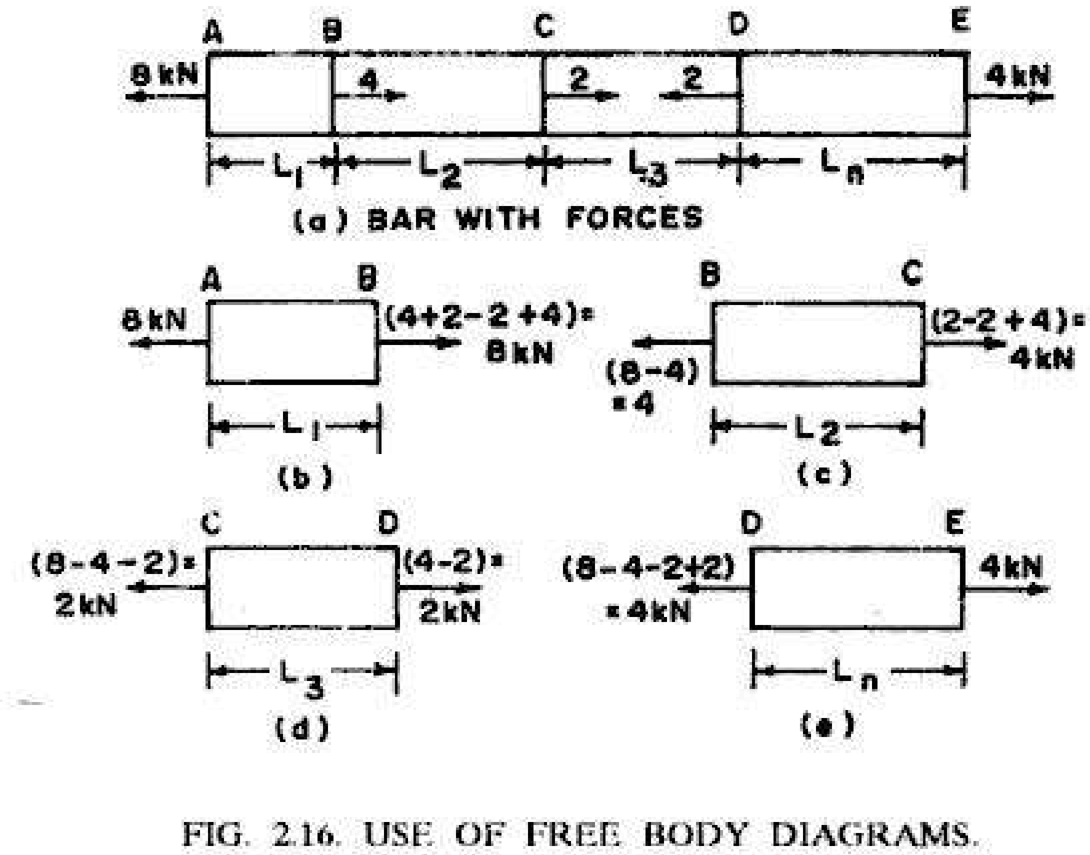


FIG. 2.16. USE OF FREE BODY DIAGRAMS.

$$\Delta = \Sigma \frac{PL}{AE} = \frac{1}{AE}\left[P_1 L_1 + P_2 L_2 + \ldots + P_n L_n\right] \qquad ...(2.12)$$

where P_n = force acting on section n

L_n = length of section n

The computations are conveniently done by the use of free body diagrams. Fig. 2.16 (*a*) shows a bar *AE* with four different sections of lengths L_1, L_2, L_3 and L_n ($= L_4$) and acted upon by several forces. The free body diagrams shown in Figs. 2.16 (*b*), (*c*), (*d*) and (*e*) for the four sections of length.

For the first section *AB* of length L_1 , the net force to the left end = 8 kN (tensile), while the net force acting to the right end $=(4+2-2+4) = 8$ kN . Thus the first section of length L_1, is subjected to a tensile force of 8 kN. Hence its extension will be $\Delta_1 = \frac{8L_1}{AE}$.

For the second section *BC* of length L_2, the net force at end $B = (8 - 4) = 4$ kN while the net force at end $C = (2 - 2 + 4) = 4$ kN. Thus the second section of length L_2 is subjected to a tensile force of 4 kN, and its extension will be $\Delta_2 = \frac{4L_2}{AE}$.

Similarly, for the third section *CD* of length L_3, the net force at end $C = (8 - 4 - 2) = 2$ kN while the net force at end $D = (4 - 2) = 2$ kN. Thus, the third section of length L_3 is subjected to a tensile force of 2 kN and its extension will be $\Delta_3 = \frac{2L_3}{AE}$.

Like wise, the last section *DE* of length L_n is subjected to a tensile force of 4 kN and its extension will be $\Delta_n = \frac{4L_n}{AE}$.

According to the principle of superposition, the total extension of the bar is given by

$$\Delta = \Delta_1 + \Delta_2 + \Delta_3 + \Delta_n = \frac{8L_1}{AE} + \frac{4L_2}{AE} + \frac{2L_3}{AE} + \frac{4L_n}{AE}$$

2.7. BARS OF VARYING SECTIONS

When a structural member having varying areas of cross-section along its length is subjected to an axial force *P*, the total deformation will be equal to sum of deformations of individual sections under the action of axial force *P*.

Thus, with reference to the bar of Fig. 2.17, the total elongation is

$$\Delta = \Delta_1 + \Delta_2 + \Delta_n = \frac{P}{E}\left(\frac{L_1}{A_1} + \frac{L_2}{A_2} + \frac{L_n}{A_n}\right) \qquad ...(2.12)$$

Similarly, if the bar of varying sections is subjected to various forces, both at the ends as well as at intermediate points, principle of superposition can be applied and the total deformation can be computed by drawing the free body diagrams of individual sections. The total deformation for such a bar is given by

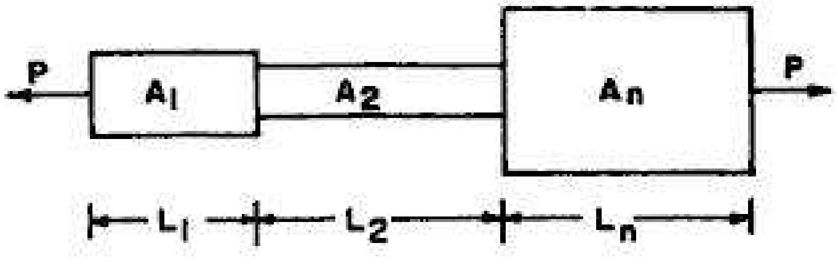


FIG. 2.17. BAR OF VARYING SECTIONS

$$\Delta = \Sigma\Delta = \frac{1}{E}\left(\frac{P_1 L_1}{A_1} + \frac{P_2 L_2}{A_2} + \ldots . \frac{P_n L_n}{A_n}\right) = \frac{1}{E}\Sigma\frac{PL}{A} \qquad ...(2.14)$$

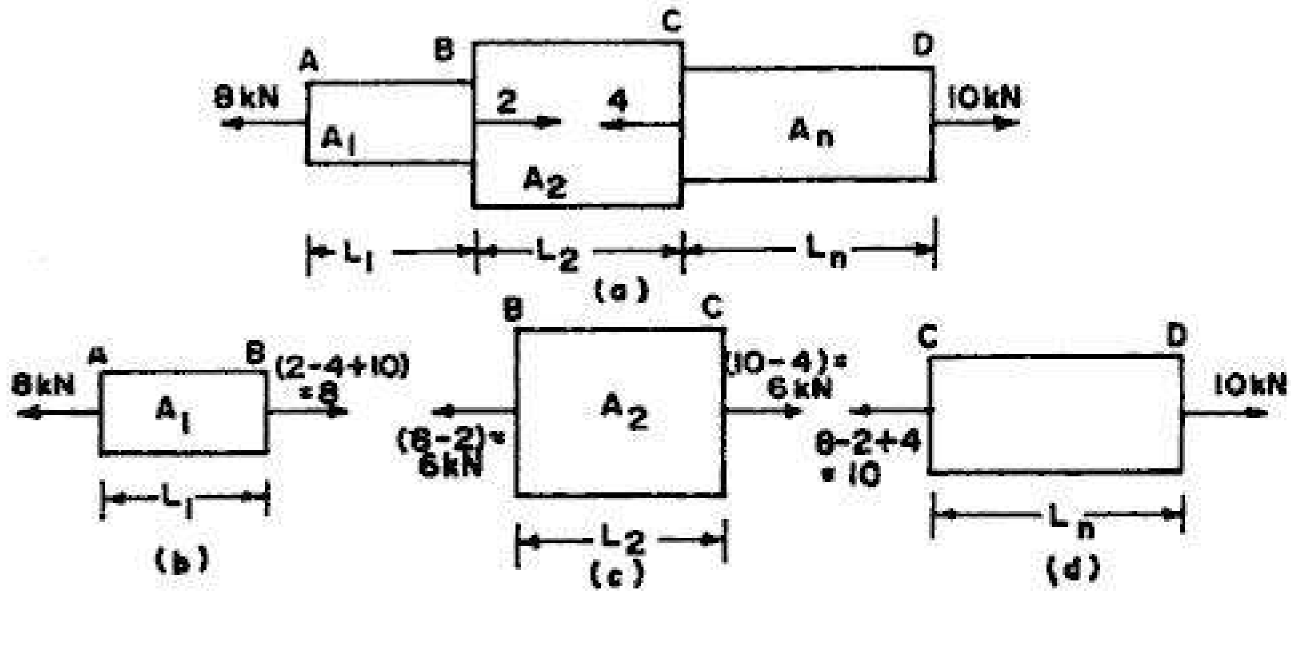


FIG. 2.18. BARS OF VARYING SECTIONS WITH MULTIPLE LOADS

Thus with reference to the free body diagrams of bar of Fig. 2.18(a), we have

$$\Delta = \Sigma\,\Delta = \frac{1}{E}\Sigma\frac{PL}{A} = \frac{1}{E}\left(\frac{8L_1}{A_1} + \frac{6L_2}{A_2} + \frac{10L_3}{A_3}\right)$$

2.8. BARS OF VARYING SECTIONS OF DIFFERENT MATERIALS

If, however, a bar of varying section is composed of different materials for its different sections, the final deformation under the action of various forces acting at different sections can be computed by drawing free body diagrams of individual sections and then applying the principle of superposition.

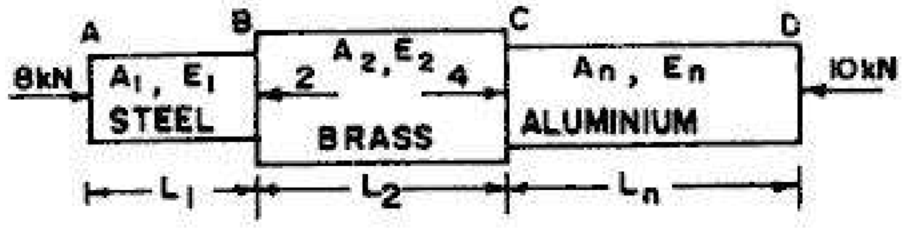


FIG. 2.19. BAR OF VARYING SECTIONS OF DIFFERENT MATERIALS

The total deformation for such a bar is given by

$$\Delta = \Sigma\,\Delta = \Sigma\frac{PL}{AE} = \frac{P_1 L_1}{A_1 E_1} + \frac{P_2 L_2}{A_2 E_2} + \ldots . \frac{P_n L_n}{A_n E_n} \qquad ...(2.15)$$

With reference to the bar of Fig. 2.19 made of different materials, the total contraction is given by

$$\Delta = \frac{8L_1}{A_1 E_1} + \frac{6L_2}{A_2 E_2} + \frac{10L_3}{A_3 E_3}$$

Example 2.2. *A circular rod of 12 mm diameter was tested for tension. The total elongation on a 300 mm length was 0.22 mm under a tensile load of 17 kN. Find the value of E.*

Solution

Stress $$p = \frac{P}{A} = \frac{17 \times 10^3}{\frac{\pi}{4}(12)^2} = 150.31 \text{ N/mm}^2$$

Strain $$\varepsilon = \frac{\Delta L}{L} = \frac{0.22}{300} = 7.333 \times 10^{-4}$$

$$\therefore \quad E = \frac{\text{Stress}}{\text{Strain}} = \frac{150.31}{7.333 \times 10^{-4}} = \mathbf{2.05 \times 10^5 \ N/mm^2 = 210 \ kN/mm^2}$$

Example 2.3. *A rod of variable sections, shown in Fig. 2.20 is subjected to a pull of 1000 kN at the ends. Find the extension of the rod, taking $E=2\times10^5$ N/mm².*

Solution

From Eq. 2.13, $$\Delta = \frac{P}{E}\left[\frac{L_1}{A_1} + \frac{L_2}{A_2} + \frac{L_3}{A_3} + \ldots\right]$$

Here, $$A_1 = \frac{\pi}{4}(100)^2 = 7854 \text{ mm}^2$$

$$A_2 = \frac{\pi}{4}(70)^2 = 3848.5 \text{ mm}^2$$

$$A_3 = \frac{\pi}{4}(90)^2 = 6361.7 \text{ mm}^2$$

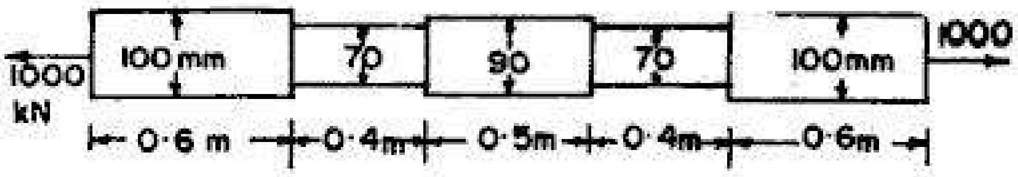


FIG. 2.20

$$\therefore \quad \Delta = \frac{1000\times10^3}{2\times10^5}\left[\frac{600+600}{7854} + \frac{400+400}{3848.5} + \frac{500}{6361.7}\right] = 5\,[0.1528+0.2079+0.0786]$$

$$= \mathbf{2.196 \ mm}$$

Example 2.4. *A steel bar of variable section is subjected to forces as shown in Fig. 2.21. Taking $E = 205$ kN/m², determine the total elongation of the bar.*

Solution

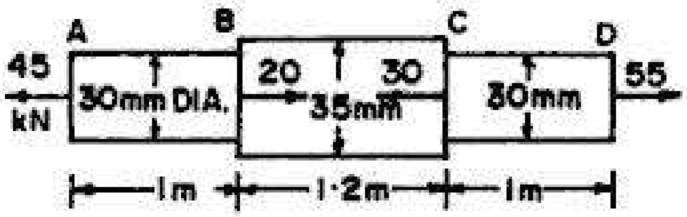


FIG. 2.21

$$A_1 = \frac{\pi}{4}(30)^2 = 706.86 \text{ mm}^2 \ ; \ L_1 = 1000 \text{ mm}$$

$$A_2 = \frac{\pi}{4}(35)^2 = 962.11 \text{ mm}^2 \ ; \ L_2 = 1200 \text{ mm}$$

$$A_3 = \frac{\pi}{10}(30)^2 = 706.86 \text{ mm}^2 \ \ L_3 = 100 \text{ mm}$$

The free body diagrams of the three sections are shown is Fig. 2.22.

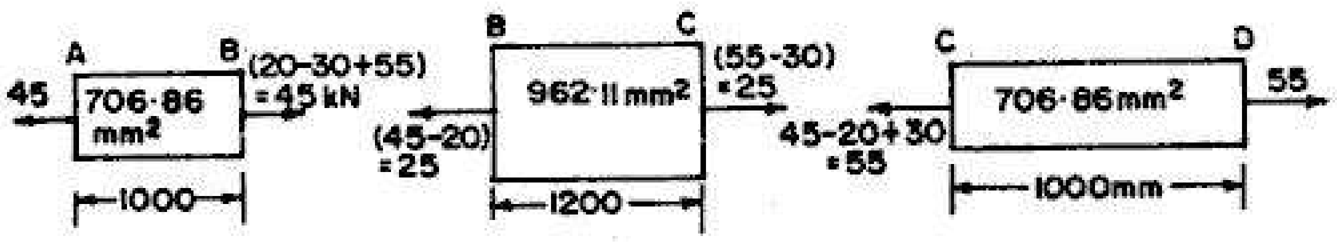


FIG. 2.22. FREE BODY DIAGRAMS

$$\therefore \quad \Delta_1 = \frac{P_1 L_1}{A_1 E} = \frac{45 \times 1000 \times 1000}{706.86 \times 205 \times 1000} = 0.311 \text{ mm}$$

$$\Delta_2 = \frac{P_2 L_2}{A_2 E} = \frac{25 \times 1000 \times 1200}{962.11 \times 205 \times 1000} = 0.152 \text{ mm}$$

$$\Delta_3 = \frac{P_3 L_3}{A_3 E} = \frac{55 \times 1000 \times 1000}{706.86 \times 205 \times 1000} = 0.380$$

Total $\quad \Delta = \Delta_1 + \Delta_2 + \Delta_3 = 0.311 + 0.152 + 0.380 = \mathbf{0.843 \text{ mm}}$

Example 2.5. *A member ABCD is subjected to point loads P_1, P_2, P_3 and P_4 as shown in Fig. 2.23.*

Calculate the force P_2 necessary for equilibrium if $P_1 = 10$ kN, $P_3 = 40$ kN and $P_4 = 16$ kN. Taking modulus of elasticity as 2.05×10^5 N/mm^2, determine the total elongation of the member.

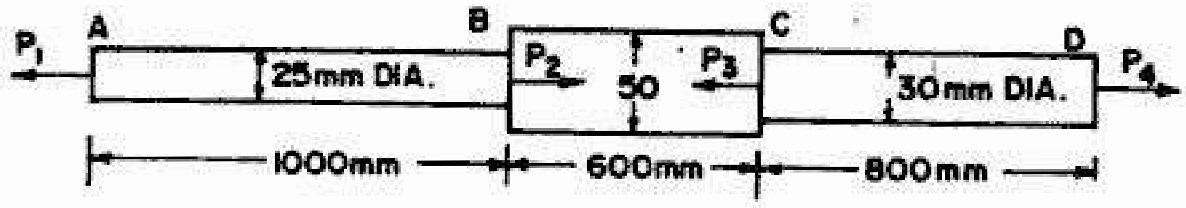


FIG. 2.23

(Based on Jadhavpur University, 1976)

Solution

For the equilibrium of the bar,

$$P_1 + P_3 = P_2 + P_4$$

or $\quad 10 + 40 = P_2 + 16$, From which $P_2 = \mathbf{34 \text{ kN}}\ (\rightarrow)$

Now from Eq. 2.14, $\Delta = \dfrac{1}{E}\Sigma\dfrac{PL}{A}$

The free body diagrams for the three portions of the bar are shown in Fig. 2.24.

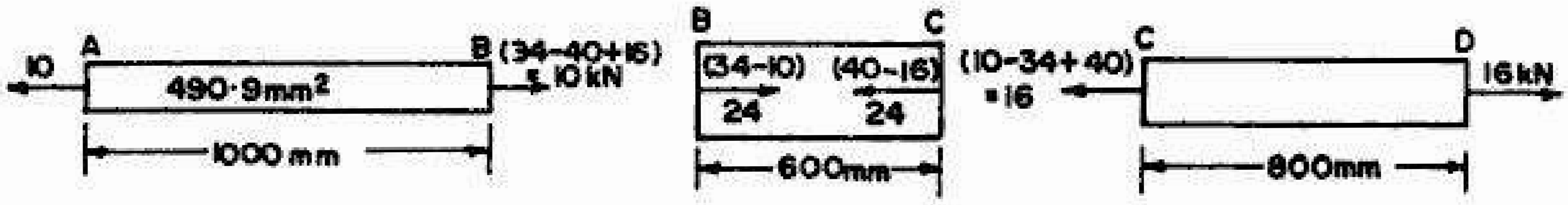


FIG. 2.24. FREE BODY DIAGRAMS

$$A_1 = \frac{\pi}{4}(25)^2 = 490.9 \text{ mm}^2 \;;\; A_2 = \frac{\pi}{4}(50)^2 = 1963.5 \text{ mm}^2 \;;\; A_3 = \frac{\pi}{4}(30)^2 = 706.9 \text{ mm}^2$$

$$\Delta_1 = \frac{10 \times 1000 \times 1000}{490.9 \times 2.05 \times 10^5} = 0.099 \text{ (elongation)}$$

$$\Delta_2 = \frac{24 \times 1000 \times 600}{1963.5 \times 2.05 \times 10^5} = 0.036 \text{ (contraction)}$$

$$\Delta_3 = \frac{16 \times 1000 \times 800}{706.9 \times 2.05 \times 10^5} = 0.088 \text{ (elongation)}$$

Total $\Delta = \Delta_1 - \Delta_2 + \Delta_3 = 0.099 - 0.036 + 0.088 = \mathbf{0.151}$ **mm.**

Example 2.6. *A steel bar of 2 m length and uniform section of 600 sq. mm is suspended vertically and loaded as shown in Fig. 2.25. Taking $E = 2.05 \times 10^5$ N/mm², determine the total elongation of the bar.*

Solution

$$\text{Reaction } R = 20 + 30 + 40 = 90 \text{ kN}$$

Hence force P_1 on portion $AB = 90$ kN (tensile)

Force P_2 on portion $BC = 30 + 40 = 70$ kN (tensile)

(or $= 90 - 20 = 70$ kN)

Force P_3 on portion $CD = 40$ kN (tensile)

From Eq. 2.12, total deformation $\Delta = \dfrac{1}{AE}\left[P_1L_1 + P_2L_2 + P_3L_3\right]$

$$\therefore \Delta = \frac{1}{600 \times 2.05 \times 10^5}\left[90 \times 10^3 \times 750 + 70 \times 10^3 \times 500 + 40 \times 10^3 \times 750\right]$$

$$= \frac{1}{1230 \times 10^5}\left[675 \times 10^5 + 350 \times 10^5 + 300 \times 10^5\right] = \mathbf{1.077 \text{ mm}}$$

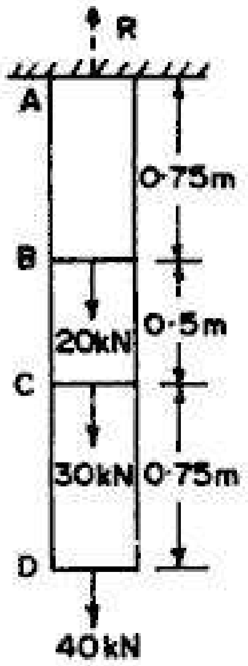


FIG. 2.25

Example 2.7. *A member ABC is formed by connecting a steel bar of 20 mm diameter to an aluminium bar of 30 mm diameter, and is subjected to forces as shown in Fig. 2.26. Determine the total deformation of the bar, taking E for aluminium as 0.7×10^5 N/mm² and that for steel as 2×10^5 N/mm².*

Solution

From Eq. 2.15, $\Delta = \Sigma \dfrac{PL}{AE} = \dfrac{P_1L_1}{A_1E_1} + \dfrac{P_2L_2}{A_2E_2}$

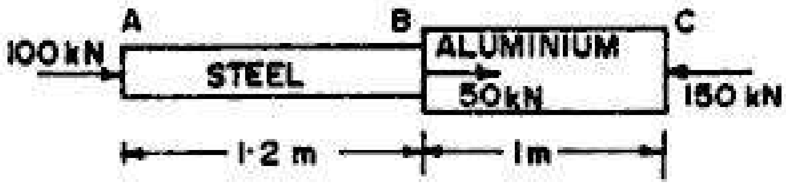


FIG. 2.26

Portion AB

Force $P_1 = 100$ kN (or $150 - 50 = 100$ kN), compressive;

$L_1 = 1200$ mm

$A_1 = \dfrac{\pi}{4}(20)^2 = 314.16 \text{ mm}^2$

$E_1 = 2 \times 10^5 \text{ N/mm}^2$

$\therefore \quad \Delta_1 = \dfrac{100 \times 10^3 \times 1200}{314.16 \times 2 \times 10^5} \approx 1.91$ mm (contraction)

Portion BC

Force $P_2 = 100 + 50 = 150$ kN; $L_2 = 1000$ mm

$A_2 = \dfrac{\pi}{4}(30)^2 = 706.86$; $E_2 = 0.7 \times 10^5 \text{ N/mm}^2$

$\Delta_2 = \dfrac{150 \times 10^3 \times 1000}{706.86 \times 0.7 \times 10^5} = 0.303$ mm (contraction)

$\therefore$ Total $\quad \Delta = \Delta_1 + \Delta_2 = 1.91 + 0.303 = \mathbf{2.213\ mm}$ (contraction)

Example 2.8. *A member formed by connecting a steel bar to an aluminium bar is shown in Fig. 2.27. Assuming that the bars are prevented from buckling sidewise, calculate the magnitude of force P, that will cause the total length of the member to decrease 0.20 mm. The values of elastic modulus for steel and aluminium are 210 kN/mm² and 70 kN/mm² respectively.*

(Based on Oxford University)

Solution

Let us use suffix 1 for steel bar and suffix 2 for aluminium bar.

$\therefore \quad A_1 = 40 \times 40 = 1600\ \text{mm}^2$

and $\quad A_2 = 80 \times 80 = 6400\ \text{mm}^2$

$L_1 = 500$ mm; $L_2 = 600$ mm; $E_1 = 210 \times 10^3\ \text{N/mm}^2$;

$E_2 = 70 \times 10^3\ \text{N/mm}^2$

Total $\quad \Delta = P\left[\frac{L_1}{A_1E_1} + \frac{L_2}{A_2E_2}\right]$, where $\Delta = 0.2$ mm

$$\therefore \quad 0.2 = P\left[\frac{500}{1600 \times 210 \times 10^3} + \frac{600}{6400 \times 70 \times 10^3}\right]$$

$$= P\left[1.488\times10^{-6} + 1.339 + 10^{-6}\right]$$

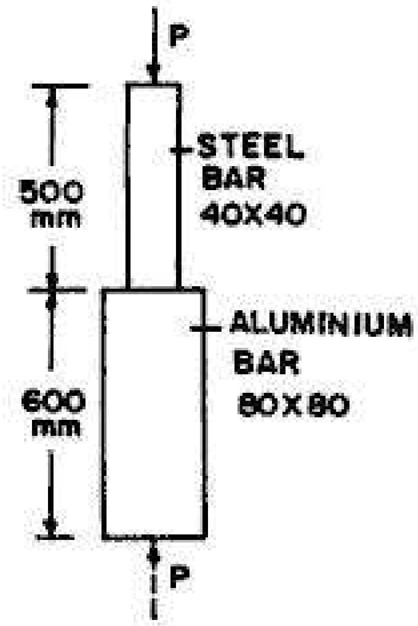


FIG. 2.27

From which $\quad P = 70737$ N $= \mathbf{70.737\ kN}$

Example 2.9. *A bar, shown in Fig. 2.28 is subjected to a tensile force of 200 kN at each end. Find (a) the diameter of middle portion if the stress in the middle portion is limited to 150 N/mm², and (b) the length of the individual portions if the total elongation of the bar is limited to 0.30 mm. Take E = 200 kN/mm².*

Solution

Force on central bar = 200 kN (tensile)

Let the diameter of central portion be d_2.

$$\therefore \quad A_2 = \frac{\pi}{4}d_2^2$$

$$\therefore \quad \text{Stress} = \frac{200 \times 10^3}{\frac{\pi}{4}d_2^2}$$

$$= \frac{254648}{d_2^2}\ \text{N/mm}^2$$

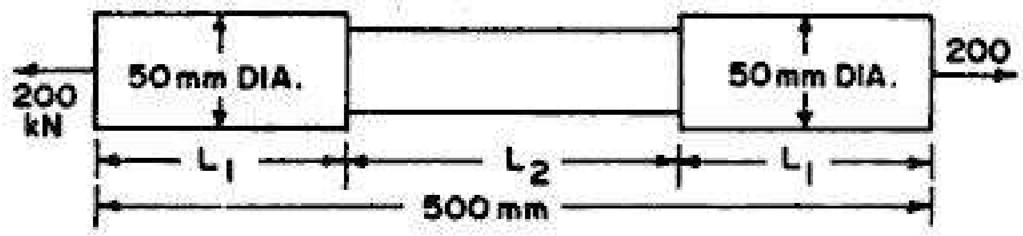


FIG. 2.28

But this should not exceed 150 N/mm².

$$\therefore \quad 150 = \frac{254648}{d_2^2}$$

From which $d_2 = \mathbf{41.2\ mm}$. Hence $A_2 = \frac{\pi}{4}(41.2)^2 = 1333.3\ \text{mm}^2$

Also, $$A_1 = \frac{\pi}{4}(50)^2 = 1963.5 \text{ mm}^2$$

Let the length of middle portion be L_2 mm

Hence total length of the two end portions $= 2L_1 = (500 - L_2)$ mm.

$$\therefore \quad \Delta = \frac{P}{E}\left[\frac{2L_1}{A_1} + \frac{L_2}{A_2}\right] = \frac{200 \times 10^3}{200 \times 10^3}\left[\frac{500 - L_2}{1963.5} + \frac{L_2}{1333.3}\right]$$

But Δ is limited to 0.3 mm.

$$\therefore \quad 0.3 = \frac{500 - L_2}{1963.5} + \frac{L_2}{1333.3} = 0.2546 - 5.093 \times 10^{-4} L_2 + 7.5 \times 10^{-4} L_2$$

or $$2.407 L_2 \times 10^{-4} = 0.0454$$

From which $$L_2 = \mathbf{188.62 \text{ mm}}$$

$$\therefore \quad L_1 = \tfrac{1}{2}(500 - 188.62) = \mathbf{155.69 \text{ mm}}$$

Example 2.10. *A tensile load of 50 kN is acting on rod of 50 mm diameter and length of 5 m. Determine the length of a bore of 25 mm that can be made central in the rod, if the total extension is not to exceed by 25 percent under the same tensile load.*
Take $E = 2.05 \times 10^5 \text{ N/mm}^2$.

Solution

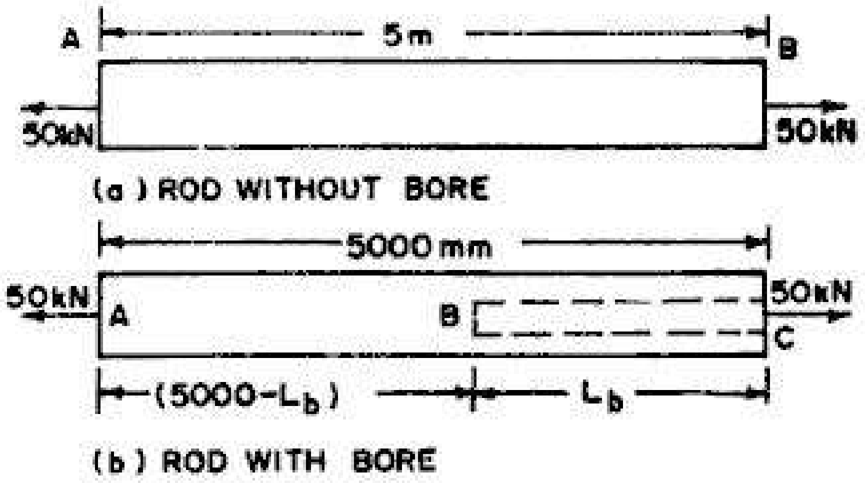


FIG. 2.29

(a) Rod without bore

$$A = \frac{\pi}{4}(50)^2 = 1963.5 \text{ mm}^2$$

$$L = 5 \times 1000 = 5000 \text{ mm}$$

$$\therefore \quad \Delta = \frac{50 \times 10^3 \times 5000}{1963.5 \times 2.05 \times 10^5} = 0.621 \text{ mm}$$

(b) Rod with bore (Fig. 2.29 *b*)

Let the length of the bore be L_b mm.

Hence length of unbored rod $= (5000 - L_b)$ mm.

Area of bored tube, $A_b = \frac{\pi}{4}(25)^2 = 490.87 \text{ mm}^2$

Permissible total extension $= (1 + 0.25)\, 0.621 = 0.7763$ mm.

But $$\Delta = \frac{P}{E}\left[\frac{L}{A} + \frac{L_b}{A_b}\right]$$

$$\therefore \quad 0.7763 = \frac{50 \times 10^3}{2.05 \times 10^5}\left[\frac{5000 - L_b}{1963.5} + \frac{L_b}{490.87}\right]$$

or $$0.7763 = 0.2439\left[2.5465 - 5.0929 \times 10^{-4} L_b + 20.372 \times 10^{-4} L_b\right]$$

or $$15.2791 \times 10^{-4} L_b = 0.6363$$

From which L_b = **416.5 mm**

Example 2.11. *For the steel bar shown in Fig. 2.30 (a), determine the longitudinal force P and stress p at all cross-sections. Determine also the vertical displacement Δ of all cross-sections of the bar. Represent the result graphically by plotting P, p and Δ diagrams. Take* $E = 2 \times 10^5\ N/mm^2$. *Also, locate the section of zero displacement.*

Solution

Considering the overall equilibrium of the bar, we get

$$R - 50 + 25 = 0$$

or $R = 25$ kN (↓)

Force diagram

For the portion AB, force $P = R = 25$ kN (comp.)

For the portion BC, force $P = 50 - R = 50 - 25 = 25$ kN (tensile)

The force diagram is shown in Fig. 2.30 (*b*).

Stress diagram

For the portion AB, stress $p = P/A = 25 \times 10^3/250 = 100$ N/mm². (compressive)

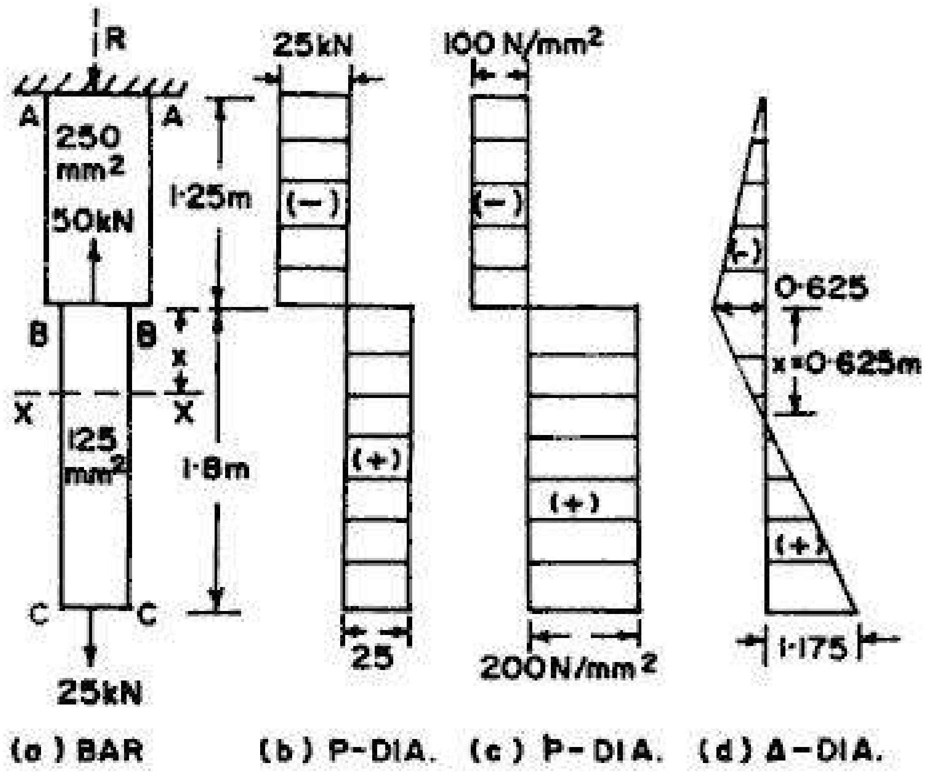


FIG. 2.30

For the portion CD, stress $p = P/A = 25 \times 10^3/1250 = 200$ N/mm² (tensile)

The stress diagram is shown in Fig. 2.30 (*c*).

Δ diagram

Point A is fixed in position. Hence $\Delta_A = 0$

Bar AB is subjected to a compressive force $P = 25$ kN

$$\therefore \quad \Delta_B = \frac{P_{AB} \cdot L_{AB}}{A_{AB} \cdot E} = -\frac{(25 \times 10^3)(1.25 \times 1000)}{250 \times 2 \times 10^5} = -0.625 \text{ mm}$$

Bar BC is subjected a tensile force $P = 25$ kN

$$\therefore \quad \Delta_C = \Delta_{AB} + \Delta_{BC} = \frac{P_{AB} \cdot L_{AB}}{A_{AB} \cdot E} + \frac{P_{BC} \cdot L_{BC}}{A_{BC} \cdot E}$$

$$= -\frac{(25 \times 10^3)(1.25 \times 1000)}{250 \times 2 \times 10^5} + \frac{(25 \times 10^3)(1.8 \times 1000)}{125 \times 2 \times 10^5}$$

$$= -0.625 + 1.8 = +1.175 \text{ mm}$$

The displacement diagram is shown in Fig. 2.30 (*d*). Let the section of zero displacement be section X-X, located at distance x m from BB.

$$\therefore \quad \Delta_{xx} = -0.625 + \frac{P_{xx} \cdot L_{xx}}{A_{xx} \cdot E} = -0.625 + \frac{(25 \times 1000)(x \times 1000)}{125 \times 2 \times 10^5} = 0$$

or $\quad -0.625 + 1x = 0$, From which $x =$ **0.625 m**

Example 2.12. *The piston of a steam engine is 200 mm diameter and the piston rod is of 30 mm diameter. The steam pressure is 1.2 N/mm². Find the stress in the piston rod and the elongation of a length of 750 mm, taking $E = 2 \times 10^5 N/mm^2$, when the piston is on the in stroke.*

Solution

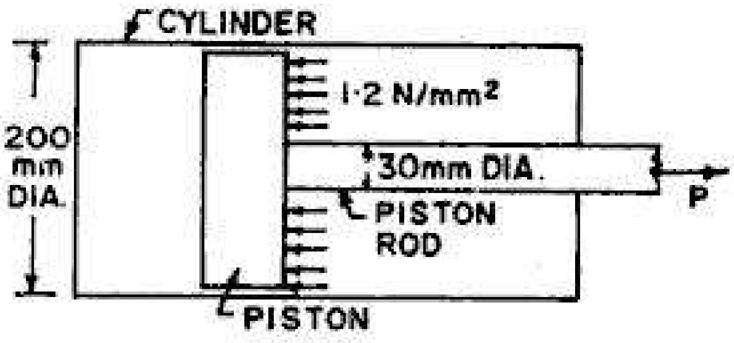


FIG. 2.31

$$\text{Net area of piston} = \frac{\pi}{4}(200^2 - 30^2) = 30709 \text{ mm}^2$$

$$\text{Load on piston rod} = 30709 \times 1.2 = 36851 \text{ N}$$

$$\text{Area of piston rod} = \frac{\pi}{4}(30)^2 = 706.86 \text{ mm}^2$$

$$\therefore \quad \text{Stress in piston rod} = \frac{36851}{706.86} = \mathbf{52.13 \ N/mm^2}$$

$$\text{Also, elongation of piston rod} = \frac{pL}{E} = \frac{52.13 \times 750}{2 \times 10^5} = \mathbf{0.196 \ mm}$$

Example 2.13. *A signal is being worked by steel wire 600 m long and 6.25 mm is diameter. Find the movement which must be given to the signal box end of wire, at a pull of 1.5 kN, if the movement of the signal end is to be 180 mm. Take $E = 2.05 \times 10^5 N/mm^2$.*

Solution (Fig. 2.32)

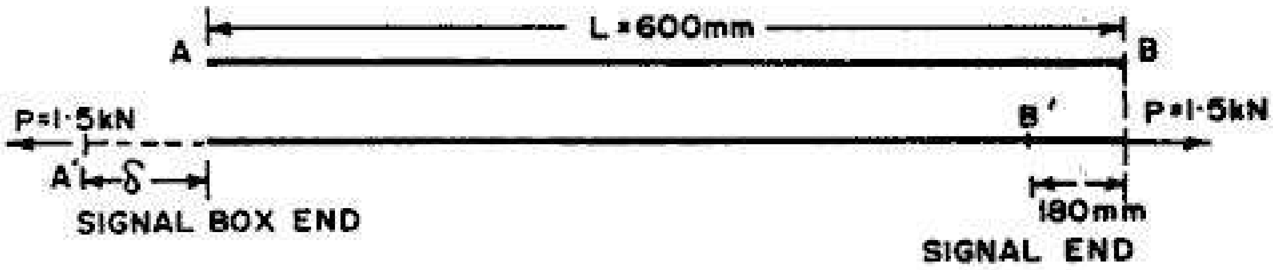


FIG. 2.32

AB (= 600 m) is the initial position of the wire, where A is the signal box end and B is the signal end. If the wire were rigid (*i.e.* stretch proof), the movement at end A at the signal box would have been the required movement of 180 mm at the signal end. But the wire extends by an amount Δ under a tensile force $P = 1.5$ kN. Hence the required movement δ at A will be equal to $(180 + \Delta)$, to move the signal end by 180 mm.

Thus, $\quad \delta = 180 + \Delta$

$$\text{Now, from Hooke's Law, } \Delta = \frac{PL}{AE} = \frac{1.5 \times 10^3 \times 600 \times 10^3}{\frac{\pi}{4}(6.25)^2 \times 2.05 \times 10^5} = 143.1 \text{ mm}$$

$$\text{Total movement } \delta = 180 + \Delta = 180 + 143.1 = \mathbf{323.1 \ mm}$$

Example 2.14. *In a tensile test on a specimen of mild steel, 12.5 mm diameter and gauge length 200 mm, the following results were recorded :*

Load (N)	*5000*	*10000*	*15000*	*20000*	*25000*	*30000*	*35000*	*40000*
Extension (mm)	*0.040*	*0.080*	*0.121*	*0.161*	*0.201*	*0.242*	*0.282*	*0.322*

Determine the value of Young's modulus of elasticity.

When the specimen was afterwards tested to destruction, the maximum load recorded was 58500 N, the diameter of the neck was 7.35 mm and the length between the gauge marks was 268.7 mm. Determine the ultimate tensile strength, percentage reduction of area and percentage elongation.

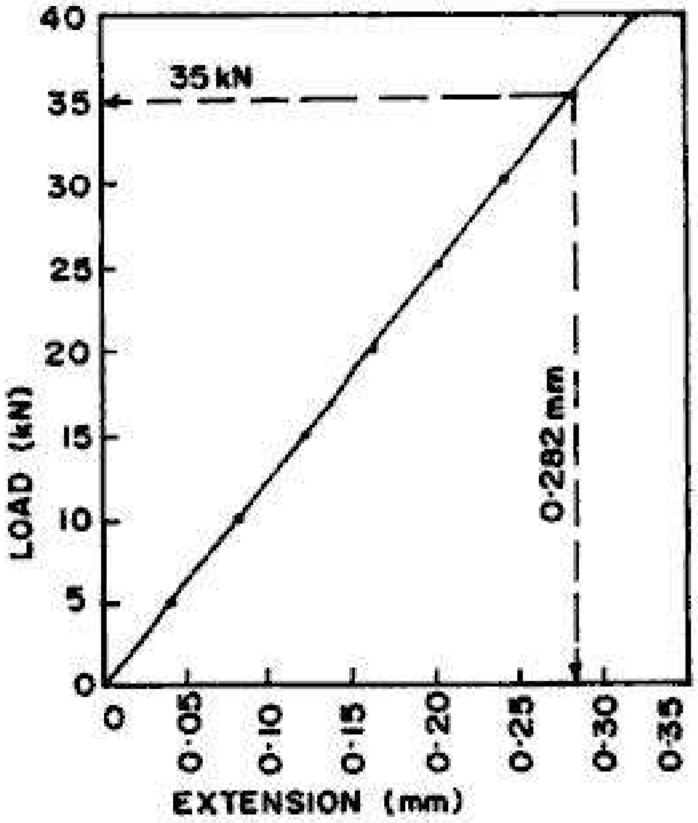


FIG. 2.33

Solution

Fig. 2.33 shows the plot between load and extension.

Slope of the graph

$$= \frac{P}{\Delta} = \frac{35000}{0.282} = 124113 \text{ N/mm}$$

Original $A = \frac{\pi}{4}(12.5)^2 = 122.718 \text{ mm}^2$

Original length $L = 200$ mm

Now $E = \frac{PL}{A\Delta} = \frac{L}{A}$ (slope of graph)

$$= \frac{200}{122.718} \times 124113$$

$$= \mathbf{2.023 \times 10^5 \ N/mm^2}$$

$$= \mathbf{202.3 \ kN/mm^2}$$

Ultimate tensile strength $= \dfrac{\text{Ultimate or maximum load}}{\text{original area of cross–section}} = \dfrac{58500}{122.718} = \mathbf{476.7 \ N/mm^2}$

Reduced area = area of neck $= \frac{\pi}{4}(7.35)^2 = 42.429 \text{ mm}^2$

$\therefore$ Percentage reduction in area $= \dfrac{122.718 - 42.429}{122.718} = \mathbf{65.43\%}$

Percentage elongation $= \dfrac{L_f - L_0}{L_f} = \dfrac{268.7 - 200}{200} = \mathbf{34.35\%}$

Example 2.15. *A tensile test on a bar having a cross-sectional area of 161.3 mm² and gauge length of 50.8 mm, gave the following results :*

Load (kN)	*2.5*	*5.00*	*7.5*	*10.0*	*12.5*	*15.0*	*17.5*	*20.0*	*22.5*
Extension (mm)	*0.013*	*0.025*	*0.038*	*0.051*	*0.064*	*0.082*	*0.102*	*0.128*	*0.162*
Load (kN)	*25.0*	*27.5*	*30.0*						
Extension	*0.203*	*0.263*	*0.383*						

Calculate for the material (i) the limit of proportionality stress (ii) the 0.1 percent proof stress.

Solution

Fig. 2.34 shows the graph between load and extension.

From the graph, we get

Load at limit of proportionality

$= 12.5$ kN

$\therefore$ Limit of proportionality stress

$$= \frac{12.5 \times 10^3}{161.3} = \mathbf{77.5\ N/mm^2}$$

0.1 percent of gauge length

$$= \frac{0.1}{100} \times 50.8 = 0.0508 \text{ mm}$$

By drawing a line parallel to the linear portion of the graph from the offset of 0.0508 mm, the intersection with the graph gives a proof load $= 22.4$ kN

$\therefore$ 0.1 percent proof stress

$$= \frac{22.4 \times 10^3}{161.3} = \mathbf{138.9\ N/mm^2}$$

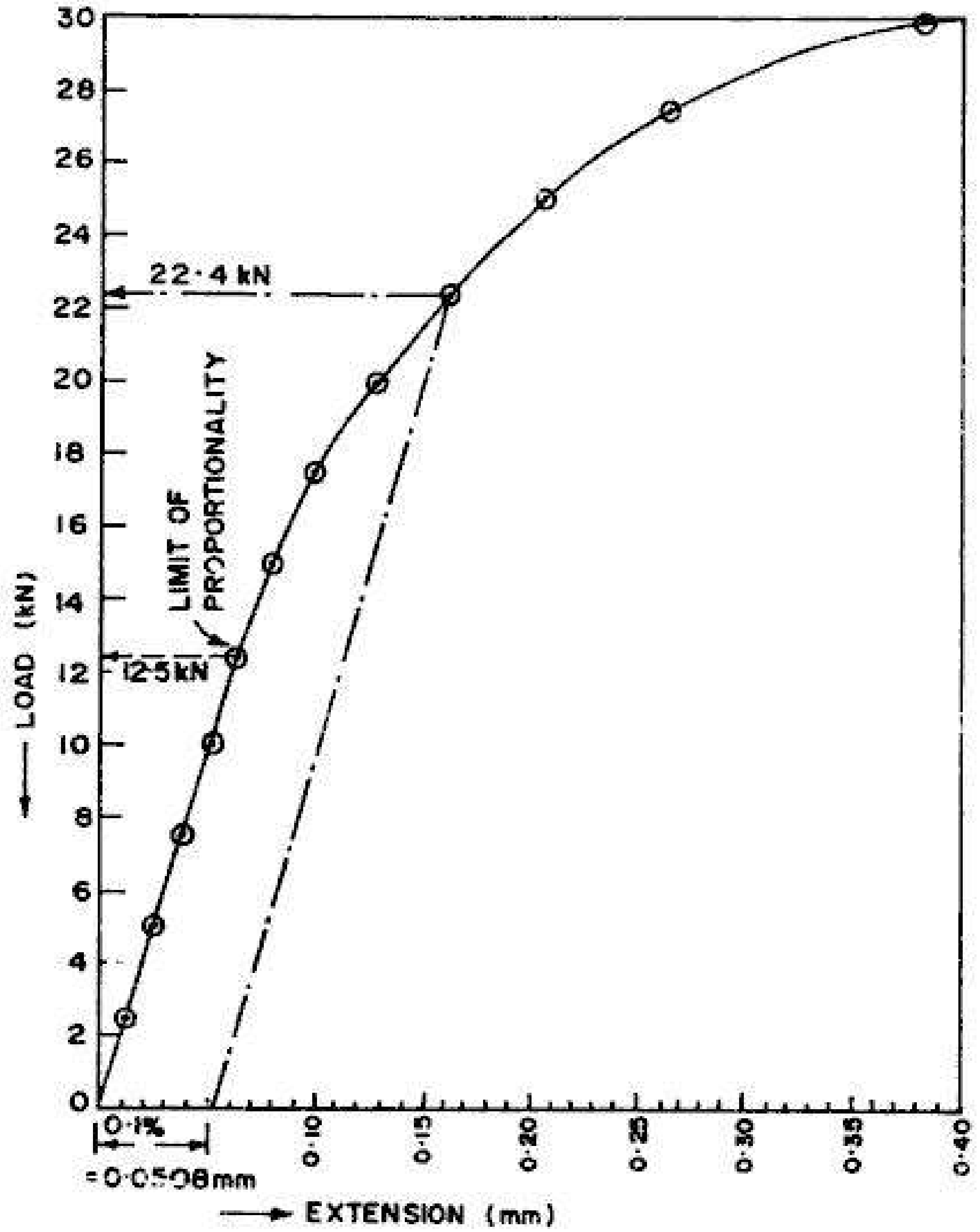


FIG. 2.34

Example 2.16. *A tensile test to destruction was conducted on two specimens A and B of the same material, and the following data was obtained*

Specimen	*Gauge length (mm)*	*Thickness (mm)*	*Width (mm)*	*% elongation*
A	*200*	*7.5*	*40*	*28*
B	*250*	*9*	*65*	*30*

Estimate the percentage elongation of a third specimen C of the same material, having a gauge length of 150 mm and a diameter of 20 mm.

Solution

From Unwin's formula,

Total elongation = Uniform elongation + Local extension

i.e. $$\Delta_T = a\,L_0 + b\,\sqrt{A_0} \qquad ...(2.9)$$

or $$\%\text{ elongation} = \frac{\Delta_T}{L_0} \times 100 = (a + \frac{b}{L_0}\sqrt{A_0})\,100$$

Substituting the values for specimens A and B, we get

$$28 = \left\{a + \frac{b}{200}\sqrt{7.5 \times 40}\right\} \times 100 \qquad ...(a)$$

and $$30 = \left\{a + \frac{b}{250}\sqrt{9 \times 65}\right\} \times 100 \qquad ...(b)$$

After simplifying these, we get

$$a + 0.0866\,b = 0.28 \qquad ...(i)$$

and $$a + 0.09675\,b = 0.30 \qquad ...(ii)$$

Solving (*i*) and (*ii*), we get $a = 0.1094$ and $b = 1.97$

Hence for the third specimen C, percentage elongation

$$= \left(a + \frac{b}{L_0}\sqrt{A_0}\right)100 = \left[0.1094 + \frac{1.97}{150}\sqrt{\frac{\pi}{4}(20)^2}\right]100$$

$$= \mathbf{34.22\ \%}$$

Example 2.17. *A short, hollow, circular, cast iron cylinder, shown in Fig. 2.35 is to support an axial compressive load of $P = 500$ kN. The ultimate stress in compression for the material is $p_u = 240\ N/mm^2$. Determine the minimum required outside diameter d of the cylinder of 25 mm wall thickness, if the factor of safety is to be 3.0 with respect to ultimate strength.*

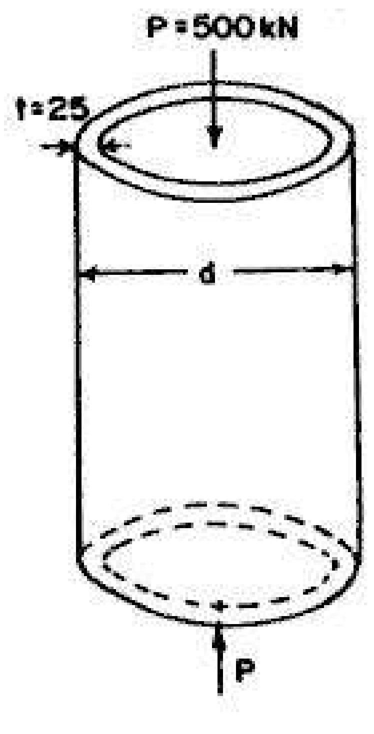


FIG. 2.35

Solution:

Allowable stress $p_{allow.} = \frac{p_u}{F} = \frac{240}{3} = 80\ \text{N/mm}^2$

$\therefore$ Required area of cross-section,

$$A = \frac{P}{p_{allow}} = \frac{500 \times 10^3}{80} = 6250\ \text{mm}^2$$

Now, for a hollow cylinder of outside diameter d and thickness t

$$A = \frac{\pi}{4}d^2 - \frac{\pi}{4}(d - 2\,t)^2 = \pi\,t\,(d - t)$$

or $$d = t + \frac{A}{\pi\,t} = 25 + \frac{6250}{\pi\,(25)}$$

$$= 104.58\ \text{mm} \approx \mathbf{105}\ \text{mm}$$

Example 2.18. *A steel bar of rectangular cross-section (12.5 × 50 mm) carries a load P and is attached to a support by means of round pin of 16 mm diameter, as shown in Fig. 2.36. Determine the maximum permissible value of load P if the allowable stresses for the bar in tension and the pin in shear are 140 N/mm^2 and 70 N/mm^2 respectively.*

Solution : Since a hole of 16 mm diameter has been made in the steel bar, to let the pin pass through it, the net area (A_{net}) will carry the tensile load P in the bar.

$$A_{net} = (b - d)\, t = (50 - 16)\, 12.5 = 425 \text{ mm}^2$$

Hence allowable load P_1 based on tension in the bar is

$$P_1 = A_{net} \cdot p_{allow} = 425 \times 140 \times 10^{-3} = 59.5 \text{ kN}$$

Again the pin tends to shear in two cross-sections (*i.e.* the pin is in double shear). Hence, allowable load P_2 based on shear in pin is

$$P_2 = (2A_p)\, \tau_{allow} = 2 \times \frac{\pi}{4}(16)^2 \times 70 \times 10^{-3} = 28.1 \text{ kN}$$

Comparing P_1 and P_2, we find that the pin governs the allowable load P.

$\therefore \quad P_{allow} = \mathbf{28.1\ kN}$

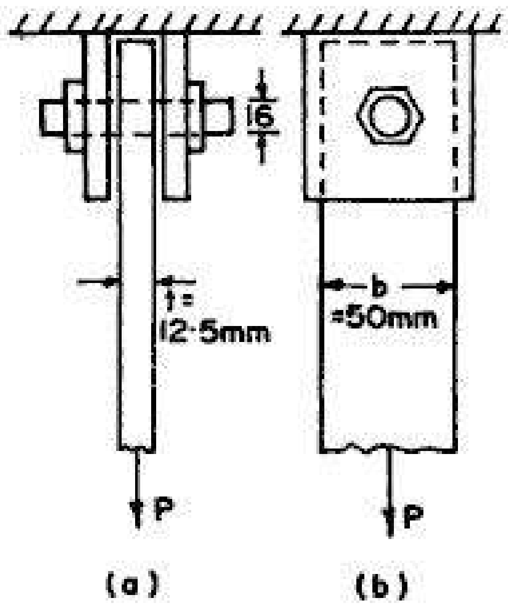


FIG. 2.36

2.9. UNIFORMLY TAPERING CIRCULAR BARS

Let us now consider a *uniformly tapering* circular bar, subjected to an axial force P, as shown in Fig. 2.37. The bar of length L has a diameter d_1 at one end and d_2 at the other end $(d_2 > d_1)$.

Consider a very short section XX of length δx and diameter d_x, situated at a distance x from end A.

Diameter $d_x = d_1 + \dfrac{d_2 - d_1}{L} x = d_1 + kx$ where $k = \dfrac{d_2 - d_1}{L}$.

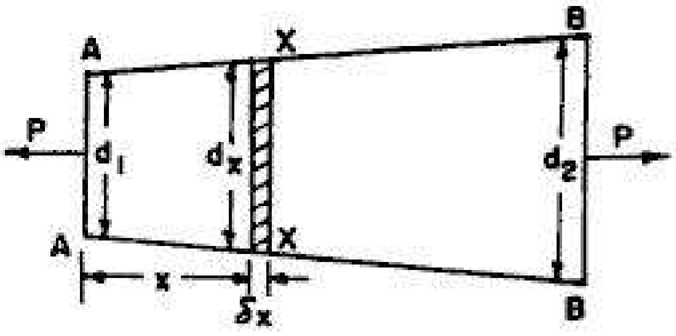


FIG. 2.37

Extension of the short length $= \delta\Delta = \dfrac{P \cdot \delta x}{\frac{\pi}{4} d_x^2 E}$

Hence the extension of the whole length of the rod is

$$\Delta = \sum_{x=0}^{x=L} \delta\Delta = \int_0^L \frac{4P\,dx}{\pi (d_1 + kx)^2 E} = -\frac{4P}{\pi E}\frac{1}{k}\left|\frac{1}{d_1 + kx}\right|_0^L$$

or $$\Delta = -\frac{4PL}{\pi E (d_2 - d_1)}\left(\frac{1}{d_1 + d_2 - d_1} - \frac{1}{d_1}\right) = \frac{4PL}{\pi E (d_2 - d_1)}\left\{\frac{1}{d_1} - \frac{1}{d_2}\right\}$$

or $$\Delta = \frac{4PL}{\pi E d_1 d_2} \qquad \text{...(2.16)}$$

When $d_1 = d_2 = d$, $\Delta = \dfrac{4PL}{\pi E d^2} = \dfrac{PL}{AE}$, which is the same as Eq. 2.11.

2.10. UNIFORMLY TAPERING RECTANGULAR BARS

Fig. 2.38 shows a uniformly tapering bar of rectangular cross-section, length L and thickness t. The width of the bar at one end is b_1 and the width at the other end is b_2, where $b_2 > b_1$. The bar is subjected to an axial force P.

Consider a very short section XX of length δx and width b_x, situated at distance x from end A.

Width $b_x = b_1 + \frac{b_2 - b_1}{L} x = b_1 + kx$

where $k = \frac{b_2 - b_1}{L}$

$\therefore$ Extension of the short length

$$= \delta\Delta = \frac{P \, . \, \delta x}{(b_1 + kx)\, t \, . \, E}$$

Hence the extension of the whole length of the rod is

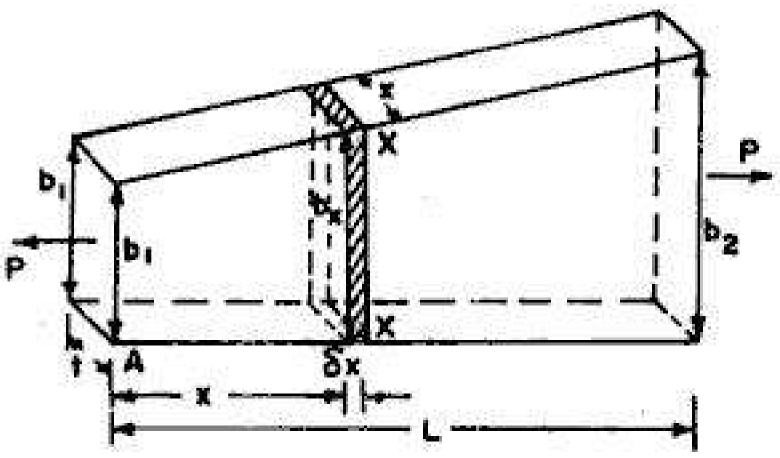


FIG. 2.38

$$\Delta = \sum_{x=0}^{x=L} \delta\Delta = \int_0^L \frac{P\,dx}{(b_1 + kx)\, t E} = \frac{P}{tE} \cdot \frac{1}{k} \Big[\log_e (b_1 + kx) \Big]_0^L$$

or $$\Delta = \frac{P}{k t E} \log_e \frac{b_1 + kL}{b_1} = \frac{P}{k t E} \log_e \frac{b_2}{b_1}$$

or $$\Delta = \frac{PL}{(b_2 - b_1)\, t E} \log_e \frac{b_2}{b_1} \qquad ...(2.17)$$

2.11. ELONGATION OF BAR OF UNIFORM SECTION DUE TO SELF WEIGHT

Fig. 2.39 shows a bar of uniform section, hanging freely under its own weight. The bar is either of circular section (Fig. 2.37*a*) or of rectangular section (Fig. 2.37 *b*). In each case, the area of cross-section, A, is constant along its whole length L. Let λ be the *specific weight* or *unit weight* of the material. In MKS units, λ is expressed in kg(f)/m^3, while in SI units, $\lambda = \rho g$ where ρ is the *specific mass* or *unit mass* or *density* of the material (expressed in kg/m^3) and g is the acceleration due to gravity which may be taken as 9.81 m/sec^2. In SI units λ is expressed in N/cm^3 or kN/m^3 since it is the gravitational weight developed by the body per unit volume.

Consider a small section of length δx, at a distance x from the free end. The deformation $\delta\Delta$ is given by

$$\delta\Delta = \frac{W_x \, . \, \delta x}{A \, . \, E}$$

where $W_x =$ weight of portion below the section $= A \, . \, x \, . \, \lambda$

$$\therefore \qquad \delta\Delta = \frac{(A\, x\, \lambda)\, \delta x}{A \, . \, E} = \frac{x \lambda\, \delta x}{E}$$

$\therefore$ Total deformation of the rod,

$$\Delta = \sum_{x=0}^{x=L} \delta\,\Delta = \int_0^L \frac{x\lambda}{E} dx$$

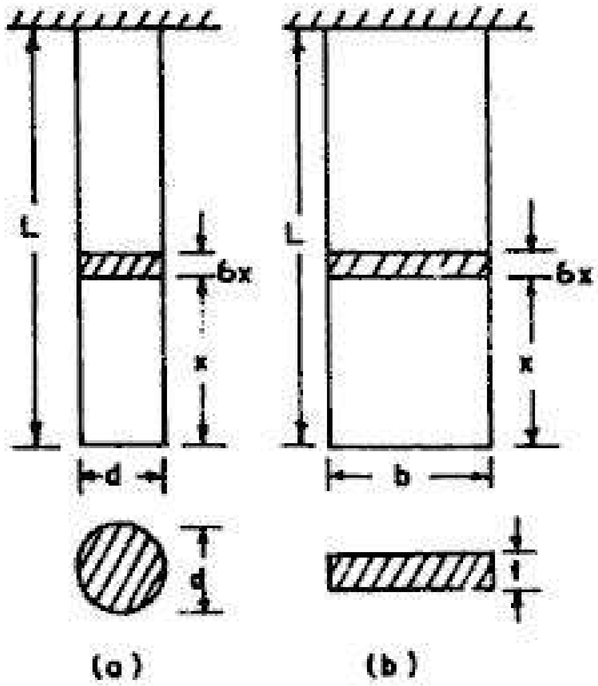


FIG. 2.39

or
$$\Delta = \left[\frac{x^2 \lambda}{2E}\right]_0^L = \frac{\lambda L^2}{2E} = \frac{\rho g L^2}{2E} \quad ...(2.18)$$

If W= total weight of the bar, we have $\lambda = \frac{W}{AL}$

Hence
$$\Delta = \frac{WL}{2AE} \quad ...(2.19)$$

From Eq. 2.19, it is clear that *the deformation of the bar of uniform section, under its own weight is equal to half the deformation of the bar under the axial load equal to the weight of the body.*

2.12. ELONGATION OF BAR OF UNIFORMLY TAPERING SECTION

Fig. 2.40 shows a bar of uniformly tapering section of length L, hanging freely under its own weight.

Consider an elementary section of length δx , at a distance x from the free end. Let A_x be the area of cross section of the elementary section. The extension of this elementary section is given by

$$\delta\Delta = \frac{W_x \,.\, \delta x}{A_x \,.\, E}$$

where W_x = weight of the portion below the section $= \frac{1}{3} A_x \,.\, x \,.\, \lambda$

(where λ is the specific weight or unit weight of the material)

$$\therefore \quad \delta\Delta = \frac{1}{3}\frac{A_x \,.\, x \,.\, \lambda\, \delta x}{A_x \,.\, E} = \frac{x \lambda\, \delta x}{3E}$$

Hence total extension of the bar, $\Delta = \sum_{x=0}^{x=L} \delta\Delta = \int_0^L \frac{x \lambda\, \delta x}{3E}$

which gives
$$\Delta = \frac{\lambda L^2}{6E} = \frac{\rho g L^2}{6E} \quad ...(2.20)$$

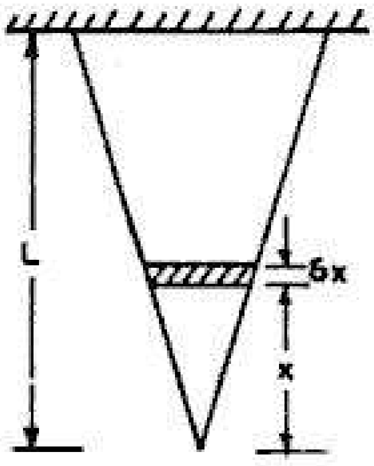


FIG. 2.40

If d is the diameter of bar at its uppermost section (*i.e.* at the support) total weight of the bar $= W = \frac{1}{3}\left(\frac{\pi}{4}d^2\right) L \lambda$

or
$$\lambda = \frac{12W}{\pi d^2 L}$$

Substituting this value of λ in Eq. 2.20, we get

$$\Delta = \frac{2WL}{\pi d^2 E} = \frac{WL}{2AE} \quad ...(2.21)$$

where A = area of cross-section of the bar at its support.

2.13. ELONGATION OF TRUNCATED CONE SHAPED BAR

Fig. 2.41 shows a solid bar in the form of a truncated cone, having a diameter d_1 at the free end and diameter d_2 at the support. The length of the bar is L and it is hanging freely.

Prolong AD and BC to meet at O. Let l' be the length of cone DCO and l be the length of the full cone ABO.

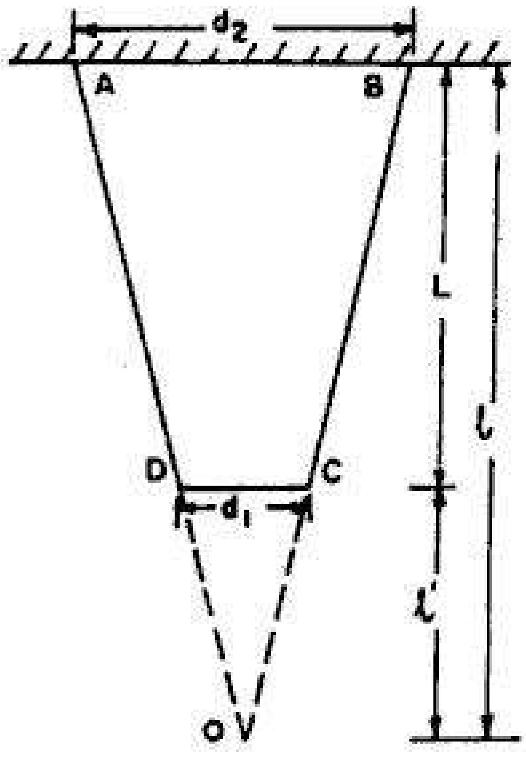


FIG. 2.41

From Eq. 2.20, the extension of the cone ABO under its own weight is given by

$$\Delta_c = \frac{\lambda l^2}{6E} \quad ...(1)$$

Similarly, the extension of cone DCO, under its own weight is

$$\Delta_c' = \frac{\lambda l'^2}{6E} \quad ...(2)$$

The weight of cone $DCO = W' = \frac{1}{3}\left(\frac{\pi}{4}d_1^2\right) l' \lambda = \frac{\pi}{12} d_1^2 l' \lambda$

Again, from Eq. 2.16, the extension of a tapering circular bar of length L, under an axial pull W' is given by

$$\Delta_{tc}' = \frac{4 W' L}{\pi E d_1 d_2} = \frac{4L}{\pi E d_1 d_2}\left[\frac{\pi}{12} d_1^2 l' \lambda\right]$$

or
$$\Delta_{tc}' = \frac{\lambda}{3}\frac{L l'}{E}\cdot\frac{d_1}{d_2} \quad ...(3)$$

Now extension of $ABCD$ = Extension of ABO − extension of DCO − extension of $ABCD$ under weight W'

or
$$\Delta_{tc} = \Delta_c - \Delta_c' - \Delta_{tc}'$$

or
$$\Delta_{tc} = \frac{\lambda l^2}{6E} - \frac{\lambda l'^2}{6E} - \frac{\lambda}{3}\frac{L l'}{E}\cdot\frac{d_1}{d_2} \quad ...(4)$$

In the above expression, l and l' are not directly known and hence these are to be eliminated.

From the geometry of the cone, $\frac{d_2}{d_1} = \frac{l}{l'} = \frac{L + l'}{l'} = \frac{L}{l'} + 1$

Hence
$$\frac{L}{l'} = \frac{d_2 - d_1}{d_1}, \text{ from which } l' = \frac{d_1 L}{d_2 - d_1} \quad ...(i)$$

and
$$l = \frac{d_2}{d_1} l' = \frac{d_2}{d_1} \times \frac{d_1 L}{d_2 - d_1} = \frac{d_2 L}{d_2 - d_1} \quad ...(ii)$$

Substituting these values in (4), we get

$$\Delta_{tc} = \frac{\lambda}{6E}\left(\frac{d_2 L}{d_2 - d_1}\right)^2 - \frac{\lambda}{6E}\left(\frac{d_1 L}{d_2 - d_1}\right)^2 - \frac{\lambda}{3 E d_2}\frac{L^2 d_1^2}{d_2 - d_1}$$

or
$$\Delta_{tc} = \frac{\lambda L^2}{6E}\left[\frac{d_2^3 - d_1^2 d_2 - 2 d_1^2 (d_2 - d_1)}{d_2 (d_2 - d_1)^2}\right]$$

or $$\Delta_{tc} = \frac{\lambda L^2}{6E}\left[\frac{d_2^3 + 2d_1^3 - 3d_1^2 d_2}{d_2(d_2 - d_1)^2}\right] \qquad ...(2.22)$$

This is the required expression.

When $d_1 = 0$ and $d_2 = d$ (*i.e.* case of § 2.12, Fig. 2.40), we get

$$\Delta = \frac{\lambda L^2}{6E}\left[\frac{d^3}{d^3}\right] = \frac{\lambda L^2}{6E}$$ which is the same as Eq. 2.20.

2.14. DEFORMATION OF BAR OF UNIFORM STRENGTH

For a bar to have *constant strength*, the stress at any section, due to external load P and the weight of the portion below it should be constant.

Consider an elementary section of a length δx, at a distance x from the lower end. The bar is subjected to a pull P at its free end. Let A be the area of cross-section of the free end (section CC). Let A_x be the area of cross-section of the bar at the lower end of the elementary section and $A_x + \delta A_x$ be the area of cross-section at the upper end of the elementary section.

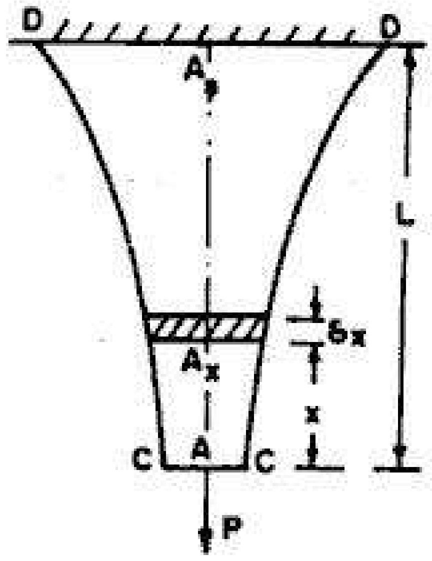


FIG. 2.42

For a bar of uniform section, $f = f_x =$ constant

Now $$f = \frac{P}{A}\ ;\ f_x = \frac{W_x}{A_x} \text{ and } f_{x+\delta x} = \frac{W_x'}{A_x + \delta A_x}$$

Hence $$f = \frac{P}{A} = \frac{W_x}{A_x} = \frac{W_x'}{A_x + \delta A_x} \qquad ...(1)$$

where $W_x = P +$ weight of bar below the elementary section $= P + W_1$ (say)

$$W_x' = P + W_1 + A_x \,.\, \lambda \,.\, \delta x$$

From (1), we get

$$f.A = P \qquad ...(i)$$

$$fA_x = P + W_1 \qquad ...(ii)$$

and $$f(A_x + \delta A_x) = P + W_1 + A_x \,.\, \lambda \,.\, \delta x \qquad ...(iii)$$

where $\lambda =$ specific weight (or unit weight) of the material $= \rho g$

$\rho =$ specific mass (or unit mass) of the material

Subtracting (*ii*) from (*iii*), we get

$$f \,.\, \delta A_x = A_x \,.\, \lambda \,.\, \delta x$$

or $$\frac{\delta A_x}{A_x} = \frac{\lambda}{f} \,.\, \delta x$$

Integrating, we get $$\int_A^{A_x} \frac{dA_x}{A_x} = \int_0^x \frac{\lambda}{f}\, dx$$

or $$\log_e \frac{A_x}{A} = \frac{\lambda}{f} x$$

From which $$\frac{A_x}{A} = e^{\lambda x / f}$$

or $$A_x = A \,.\, e^{\lambda x/f} = A\, e^{\rho g x/f} \qquad ...(2.23)$$

The above expression gives the area of cross-section (A_x) of the bar at a distance x from the free end. Knowing the values of A, λ and f, the shape of the bar can be determined from Eq. 2.22.

Also, area at the support, $A_s = A\, e^{\lambda L/f}$ ...(2.23 *a*)

Example 2.19. *A conical bar tapers uniformly from a diameter of 15 mm to a diameter of 40 mm in a length of 400 mm. Determine the elongation of the bar under an axial tensile force of 100 kN. Take $E = 2 \times 10^5$ N/mm^2.*

Solution

The deformation of a tapering conical bar is given by Eq. 2.16.

$$\Delta = \frac{4PL}{\pi E d_1 d_2}$$

Here $P = 100$ kN$= 100 \times 10^3$ N ; $L = 400$ mm; $d_1 = 15$ mm; $d_2 = 40$ mm

$$\therefore \qquad \Delta = \frac{4 \times 100 \times 10^3 \times 400}{\pi \times 2 \times 10^5 \times 15 \times 40} = \mathbf{0.424\ mm}$$

Example 2.20. *A flat steel plate is of trapezoidal form. The thickness of the plate is 15 mm and it tapers uniformly from a width of 60 mm to a width of 10 mm in a length of 300 mm. Determine the elongation of the plate under an axial force of 120 kN.*
Take $E = 2.04 \times 10^5$ N/mm^2.

Solution

The deformation of uniformly tapering rectangular bar is given by Eq. 2.17.

$$\Delta = \frac{PL}{(b_2 - b_1)\, t E} \log_e \frac{b_2}{b_1}$$

Here $P = 120$ kN $= 120 \times 10^3$ N; $L = 300$ mm
$b_1 = 10$ mm; $b_2 = 60$ mm; $t = 15$ mm

$$\therefore \qquad \Delta = \frac{120 \times 10^3 \times 300}{(60 - 10)\, 15 \times 2.04 \times 10^5} \log_e \frac{60}{10} = \mathbf{0.422\ mm}$$

Example 2.21. *A solid conical bar tapers uniformly from a diameter of 60 mm at the support to 20 mm at the end, in a length of 1 m. It is suspended vertically. Calculate the elongation of the bar due to self weight. Take unit weight of bar material as 78.5 kN/m^3 and E as 204 kN/mm^2.*

Solution

The extension of a truncated shaped bar is given by Eq. 2.22.

$$\Delta_{tc} = \frac{\lambda L^2}{6E} \left[\frac{d_2^3 + 2 d_1^3 - 3 d_1^2 d_2}{d_2 (d_2 - d_1)^2} \right]$$

Solution

Here $d_1 = 20$ mm; $d_2 = 60$ mm; $L = 1000$ mm

$\lambda = 78.5 \text{ kN/m}^3 = 78.5 \times 10^3/10^9 \text{ N/mm}^3 = 7.85 \times 10^{-5} \text{ N/mm}^3$

$E = 204 \text{ kN/mm}^2 = 204 \times 10^3 \text{ N/mm}^2$

$$\therefore \quad \Delta_{tc} = \frac{7.85 \times 10^{-5}(1000)^2}{6 \times 204 \times 10^3}\left[\frac{60^3 + 2 \times 20^3 - 3(20)^2 60}{60(60-20)^2}\right]$$

$$= \mathbf{1.07 \times 10^{-4} \text{ mm}}$$

Example 2.22. *If a tension test bar is found to taper from $(D + a)$ diameter to $(D - a)$ diameter, prove that the error involved in using the mean diameter to calculate the Young's modulus is $\left(\frac{10a}{D}\right)^2$ percent.* *(U.L.)*

Solution

Smaller diameter, $d_1 = D - a$ and larger diameter, $d_2 = D + a$

Let $E_0 =$ modulus of elasticity computed by mean diameter

$E =$ actual Young's modulus

From Eq. 2.16, we get $E = \dfrac{4PL}{\pi d_1 d_2 \Delta}$

$$\therefore \quad E = \frac{4PL}{\pi(D-a)(D+a)\Delta} = \frac{K}{D^2 - a^2} \quad \text{where } K = \frac{4PL}{\pi \Delta} \qquad \text{...(i)}$$

Also, based on the basis of mean diameter, we get

$$E_0 = \frac{PL}{A\Delta} = \frac{PL}{\frac{\pi}{4}D^2\Delta} = \frac{4PL}{\pi \Delta D^2} = \frac{K}{D^2} \qquad \text{...(ii)}$$

$$\text{Hence \% error} = \frac{E - E_0}{E} \times 100 = \frac{\dfrac{K}{D^2 - a^2} - \dfrac{K}{D^2}}{\dfrac{K}{D^2 - a^2}} \times 100$$

$$= \frac{D^2 - (D^2 - a^2)}{D^2} \times 100 = \frac{100a^2}{D^2}$$

$$= \left(\frac{10a}{D}\right)^2 \text{ percent}$$

Example 2.23. *A steel wire of 10 mm diameter and length 150 m is used to lift a weight of 2.5 kN at its lowest end. Calculate the total elongation of the wire if the unit mass (or mass density) of the wire is 7.95 kg/m^3 and $E = 2.04 \times 10^5$ N/mm^2.*

Solution

Area of cross-section of wire, $A = \frac{\pi}{4}(10)^2 = 78.54 \text{ mm}^2$

Extension of wire due to the load, $\Delta_1 = \dfrac{PL}{AE}$

$$= \frac{2.5 \times 10^3 \times 150 \times 10^3}{78.54 \times 2.04 \times 10^5} = 23.41 \text{ mm}$$

Extension of wire due to self weight, $\Delta_2 = \dfrac{\lambda L^2}{2E}$

Here, $\lambda = \rho g = 7.95 \times 9.81 = 77.99\ \text{kN/m}^3$

$= 77.99 \times 10^3/10^9\ \text{N/mm}^3 = 7.799 \times 10^{-5}\ \text{N/mm}^3$

$$\therefore \quad \Delta_2 = \frac{7.799 \times 10^{-5}\,(150 \times 1000)^2}{2 \times 2.04 \times 10^5} = 4.30\ \text{mm}$$

$\therefore$ Total elongation $= \Delta_1 + \Delta_2 = 23.41 + 4.30 =$ **27.71 mm**

Example 2.24. *A vertical tie of uniform strength is 10 m long and hangs from a ceiling. The area of the bar at the lower end is 400 mm². Find the area at the upper end when the tie is to carry a load of 60 kN. The material of the tie weighs 78 kN/m³.*

Solution

Area of the tie at the bottom $= A = 400\ \text{mm}^2$

$$\therefore \quad \text{Intensity of stress at the bottom} = f = \frac{60 \times 10^3}{400} = 150\ \text{N/mm}^2$$

The area at any section is given by Eq. 2.23

$A_x = A\,e^{\lambda x/f}$ where $\lambda = 78\ \text{kN/m}^3 = 78 \times 10^3/10^9\ \text{N/mm}^3 = 7.8 \times 10^{-5}\ \text{N/mm}^3$

At $\quad x = 10 \times 10^3$ mm, $A_S = 400\,e^{\frac{7.8 \times 10^{-5} \times 10 \times 10^3}{150}}$

or $\quad A_S = 400\,e^{5.2 \times 10^{-3}} = \mathbf{402.09\ mm^2}$

2.15. COMPOUND BARS : COMPOSITE SECTIONS

A structural member, composed of two or more elements of different materials rigidly connected together at their ends to form a parallel arrangement and subjected to axial loading is termed a *compound bar*. Such a section is also known as *composite section*. Such a problem is ***statically indeterminate***, since equation of statics alone based on conditions of equilibrium, will provide only one equation for the stresses in the individual sections. Other equation can be obtained from the consideration of the deformation of the whole structure.

Consider the effect of a compressive load P upon a composite bar consisting of a rod and enveloping tube having the same length, but made of different material. Let the end collars be rigid. Let us use suffix 1 for the rod and 2 for the tube.

From the conditions of equilibrium,

$$P_1 + P_2 = P \qquad \text{...}(i)$$

Since the members are of the same initial length and are shortened by the same amount under load, we have

$$\Delta_1 = \Delta_2$$

or

$$\frac{P_1 L_1}{A_1 E_1} = \frac{P_2 L_2}{A_2 E_2} \qquad \text{...}(ii)$$

The second equation is known as the *equation of compatibility*.

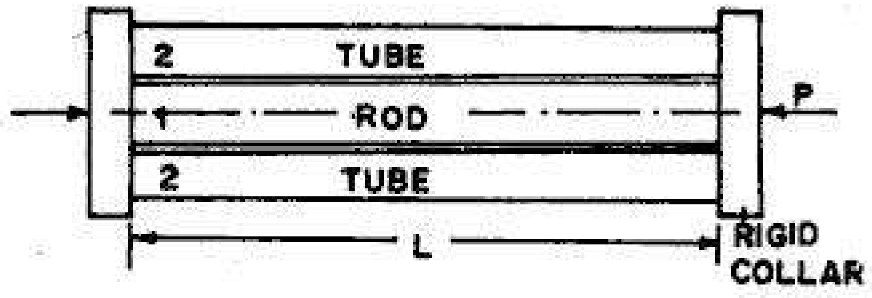


FIG. 2.43

Now from (*ii*), $$P_1 = P_2 \cdot \frac{A_1 E_1}{A_2 E_2} \qquad \text{...(iii)}$$

Substituting in (*i*), we get

$$P_2 \left[\frac{A_1 E_1}{A_2 E_2} + 1 \right] = P$$

From which $$P_2 = \frac{P}{1 + \frac{A_1 E_1}{A_2 E_2}} \qquad \text{...(2.24)}$$

Hence from (*iii*), $$P_1 = \frac{P}{1 + \frac{A_2 E_2}{A_1 E_1}} \qquad \text{...(2.24 } a\text{)}$$

Example 2.25. *A mild steel rod of 25 mm diameter and 400 mm long is encased centrally inside a hollow copper tube of external diameter 35 mm and inside diameter 30 mm. The ends of the rod and tube are rigidly attached, and the composite bar is subjected to an axial pull of 40 kN.*

If E for steel and copper is 200 GN/m² and 100 GN/m² respectively, find the stress developed in the rod and the tube. Find also the extension of the rod.

(Based on Cambridge University)

Solution

Let us use suffix 1 for the steel rod and suffix 2 for the copper tube.

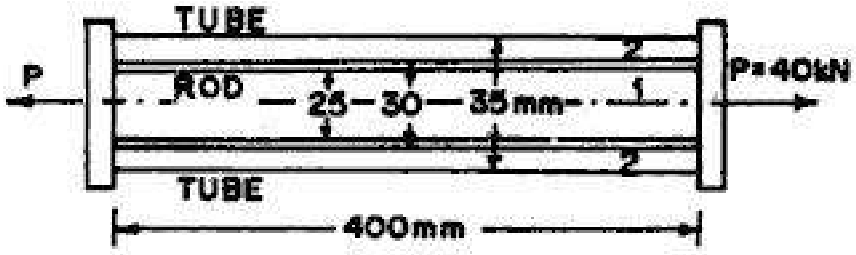


FIG. 2.44

$$A_1 = \frac{\pi}{4}(25)^2 = 490.87 \text{ mm}^2$$

$$A_2 = \frac{\pi}{4}\left[(35)^2 - (30)^2\right] = 255.25 \text{ mm}^2$$

From the equilibrium of the bar

$$p_1 A_1 + p_2 A_2 = P = 40 \times 10^3 \qquad \text{...(i)}$$

Also, from compatibility, $\dfrac{p_1 L}{E_1} = \dfrac{p_2 L}{E_2}$

or $$p_1 = p_2 \frac{E_1}{E_2} = p_2 \times \frac{200}{100} = 2 p_2 \qquad \text{...(ii)}$$

Substituting the value of p_1 in (*i*), we get

$$2 p_2 A_1 + p_2 A_2 = 40 \times 10^3$$

or $$2 p_2 (490.87) + p_2 (255.25) = 40 \times 10^3$$

From which, $p_2 = \mathbf{32.34\ N/mm^2}$

Hence $p_1 = 2 p_2 = \mathbf{64.68\ N/mm^2}$

$$\text{Also, extension of rod } = \frac{p_1 L}{E} = \frac{64.88 \times 4000}{2 \times 10^5} = \mathbf{1.294\ mm}$$

Example 2.26. *A steel rod of cross-sectional area 2000 mm² and two brass rods each of cross-sectional area of 1200 mm² together support a load of 60 kN, as shown in Fig. 2.45. Find the stresses in the rods. Take E for steel* $=2\times10^5\ N/mm^2$ *and E for brass* $= 1\times10^5\ N/mm^2$.

Solution

Let us use suffix 1 for steel rod and suffix 2 for brass rods.

Given :

$$A_1 = 2000\ \text{mm}^2 \text{ and } L_1 = 400\ \text{mm}$$

$$A_2 = 1200\ \text{mm}^2 \text{ and } L_2 = 300\ \text{mm}$$

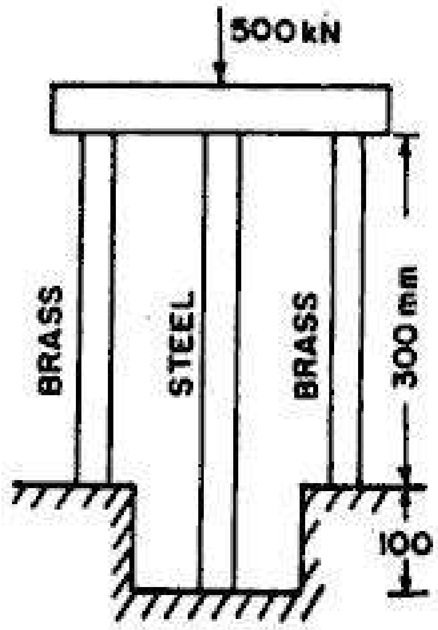


FIG. 2.45

From the equilibrium of the system,

$$p_1 A_1 + 2 p_2 A_2 = P$$

or $$2000\, p_1 + 2400\, p_2 = 500 \times 10^3 \quad \ldots(i)$$

Also, $$\Delta_1 = \Delta_2$$

$\therefore$ $$\frac{p_1 L_1}{E_1} = \frac{p_2 L_2}{E_2}$$

or $$\frac{p_1 \times 400}{2 \times 10^5} = \frac{p_2 \times 300}{1 \times 10^5}$$

$\therefore$ $$p_1 = 1.5\, p_2 \quad \ldots(ii)$$

Hence from (*i*), we get $2000\ (1.5\ p_2) + 2400\, p_2 = 500 \times 10^3$

From which $$p_2 = \mathbf{92.59\ N/mm^2}$$

$\therefore$ $$p_1 = 1.5 \times 92.59 \approx \mathbf{138.89\ N/mm^2}$$

Example 2.27. *A reinforced concrete column 450 mm × 450 mm has four steel rods of 25 mm diameter embedded in it. Find the stresses in steel and concrete when the total load on the column is 1000 kN. Find also the adhesive force between the steel and concrete. Take* $E_s = 205\ N/mm^2$ *and* $E_c = 13.6\ kN/mm^2$.

Solution

$$\text{Total area of column} = 450 \times 450 = 202500\ \text{mm}^2$$

$$\text{Area of steel rods} = 4\left(\frac{\pi}{4} 25^2\right) = 1963.5\ \text{mm}^2$$

$$\therefore \quad \text{Area of concrete, } A_c = 202500 - 1963.5 = 200536.5\ \text{mm}^2$$

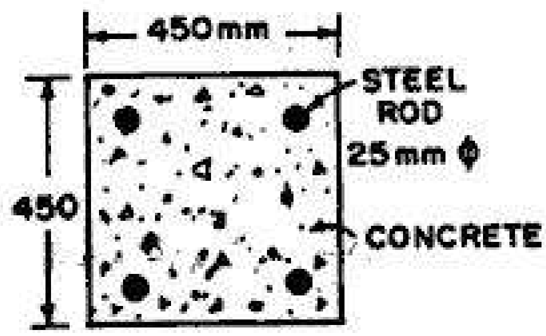


FIG. 2.46

From equilibrium, $p_s A_s + p_c A_c = 1000 \times 10^3 \quad \ldots(i)$

From compatibility, $\frac{p_s}{E_s} = \frac{p_c}{E_c}$

or $$p_s = p_c \cdot \frac{E_s}{E_c} = p_c \frac{205}{13.6} = 15.074\, p_c \qquad ...(2)$$

Substituting this value of p_s in (1) we get

$\therefore \quad 15.074\, p_c\,(1963.5) + p_c\,(200536.5) = 1000 \times 10^3$

From which, $p_c =$ **4.345 N/mm²**

Hence $p_s = 15.074 \times 4.345 =$ **65.501 N/mm²**

Now, average compressive stress $= \frac{1000 \times 10^3}{202500} = 4.938$ N/mm²

$\therefore$ Average load carried by concrete $= 4.938 \times 200536.5 \times 10^{-3} = 990.3$ kN

Actual load carried by concrete $= 4.345 \times 200536.5 \times 10^{-3} = 871.3$ kN

Adhesive force $= 990.3 - 871.3 =$ **119 kN**

Example 2.28. *In the preceding example, compute the maximum safe load P that may be applied if the allowable stresses in steel and concrete are 140 N/mm² and 5 N/mm² respectively.*

Solution

An unwary student may find the value of P by substituting the values of allowable stresses only in the equation of static equilibrium. This would give wrong result, since it would not be on the consideration of equal deformations of the two materials.

From consideration of equal deformations, we have $p_s = 15.074\, p_c$

Hence, if concrete were to be stressed to 6 N/mm², we get

$$p_s = 15.074 \times 6 = 90.444 \text{ N/mm}^2$$

On the contrary, if steel were to be stressed to 140 N/mm², we get

$$p_c = p_s/15.074 = 140/15.074 = 9.288 \text{ N/mm}^2$$

Thus, concrete will be overstressed if p_s is taken as 140 N/mm². Therefore, steel could not be stressed to 140 N/mm² without over stressing the concrete.

The actual *working stresses* could therefore be $p_c = 6$ N/mm² and $p_s = 90.444$ N/mm².

Substituting these values in the equation of static equilibrium, we get

$$P = p_s A_s + p_c A_c = (90.444 \times 1963.5 + 6 \times 200536.5)\, 10^{-3}$$

$$= \mathbf{1380.81 \text{ kN}}$$

Example 2.29. *The aluminium and steel pipes shown in Fig. 2.47 are fastened to rigid supports at one of their ends and to a rigid plate C at the other ends. Derive expressions for axial stresses in the two pipes. Hence find the numerical values if $P = 30$ kN, $A_a = 4000$ mm², $A_s = 400$ mm², $E_a = 0.7 \times 10^5$ N/mm² and $E_s = 2 \times 10^5$ N/mm².*

Solution

From the inspection of Fig. 2.47, we observe that steel pipe will carry tensile stress (p_s) while aluminium pipe will carry compressive stress (p_a) . If R_s and R_a are the reactions at the two ends, as marked in Fig. 2.47, we get from static equilibrium :

$$R_s + R_a = 2P$$

Since R_s causes tensile stress in steel pipe, $R_s = p_s . A_s$. Similarly, R_a causes compressive stress in aluminium pipe and hence $R_a = p_a A_a$.

$$p_s A_s + p_a A_a = 2P \qquad ...(i)$$

Also, extension of steel pipe = contraction of aluminium pipe

or $$\frac{p_s L}{E_s} = \frac{p_a (2L)}{E_a}$$

or $$p_s = p_a \frac{2L}{L} . \frac{E_s}{E_a} = 2 p_a . \frac{E_s}{E_a} \qquad ...(ii)$$

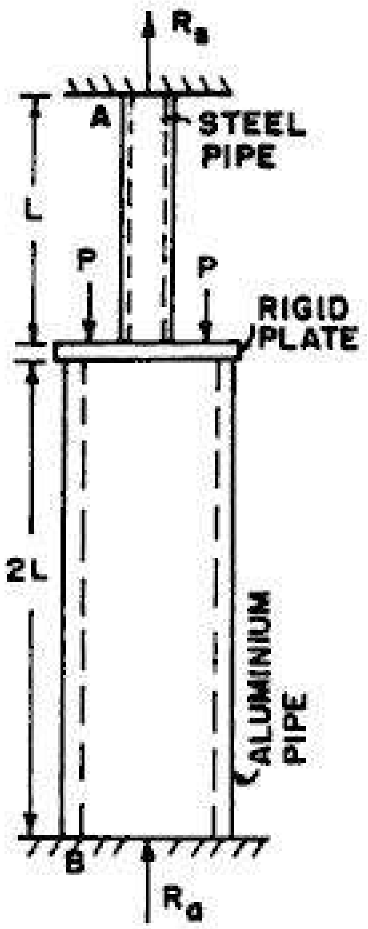


FIG. 2.47

Substituting in (*i*), we get

$$2 p_a \frac{E_s}{E_a} A_s + p_a A_a = 2P$$

From which $$p_a = \frac{2P}{2A_s . \frac{E_s}{E_a} + A_a}$$

Hence $$p_s = 4P . \frac{E_s/E_a}{2A_s \frac{E_s}{E_a} + A_a} = \frac{4P}{2A_s + A_a . \frac{E_a}{E_s}}$$

These are the required expressions.

Here, $E_s/E_a = 2 \times 10^5 / 0.7 \times 10^5 = 2.857$

and $E_a/E_s = 0.7 \times 10^5 / 2 \times 10^5 = 0.35$

$\therefore$ $$p_a = \frac{2 \times 30 \times 10^3}{2 \times 400 \times 2.857 + 4000} = \textbf{9.55 N/mm}^2 \text{ (compressive)}$$

and $$p_s = \frac{4 \times 30 \times 10^3}{2 \times 400 + 4000 \times 0.35} = \textbf{54.44 N/mm}^2\textbf{(tensile)}$$

Example 2.30. *A rigid bar ABC is supported by three rods of the same material and of equal diameter as shown in Fig. 2.48. Calculate the forces in the bars due to an applied force P, if the bar ABC remains horizontal after the load has been applied. Neglect the weight of the rigid bar.*

Solution

Let us use suffix 1 for outer bars and suffix 2 for the inner bar.

From statical equilibrium,

$$2P_1 + P_2 = P \quad ...(i)$$

From compatibility, $\Delta_1 = \Delta_2$

or $$\frac{P_1 L}{AE} = \frac{P_2(2L)}{AE},$$

from which $$P_1 = 2P_2 \quad ...(ii)$$

Substituting in (*i*), we get

$$2(2P_2) + P_2 = P \text{ , from which } P_2 = \mathbf{0.2}\,P$$

Hence $$P_1 = 2 \times 0.2\,P = \mathbf{0.4}\,P$$

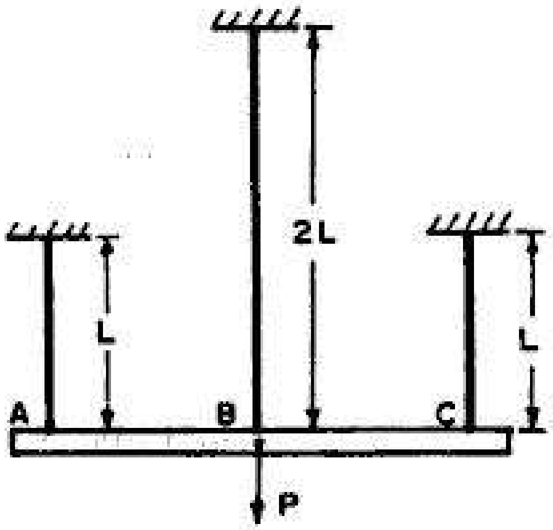


FIG. 2.48

Example 2.31. *Two horizontal rigid bars AB and CD are connected by two wires of aluminium and copper as shown in Fig. 2.49. Both the wires have length L. Vertical loads P are applied at mid points E and F of the two bars. Compute the increase Δ in the distance between points E and F when the loads are applied.*

Solution

Let us use suffix 1 for aluminium wire and suffix 2 for copper wire.

The extensions Δ_1 and Δ_2 of the two wires will be different. Hence the displacement Δ between points E and F is given by

$$\Delta = \frac{1}{2}(\Delta_1 + \Delta_2) \quad ...(i)$$

Also, from statical equilibrium, by taking moments about plane AC, we get

$$P_2 \times AB = P \times AE$$

$$\therefore \quad P_2 = P\frac{AE}{AB} = \frac{1}{2}P \quad ...(ii)$$

Hence $$P_1 = \frac{1}{2}P$$

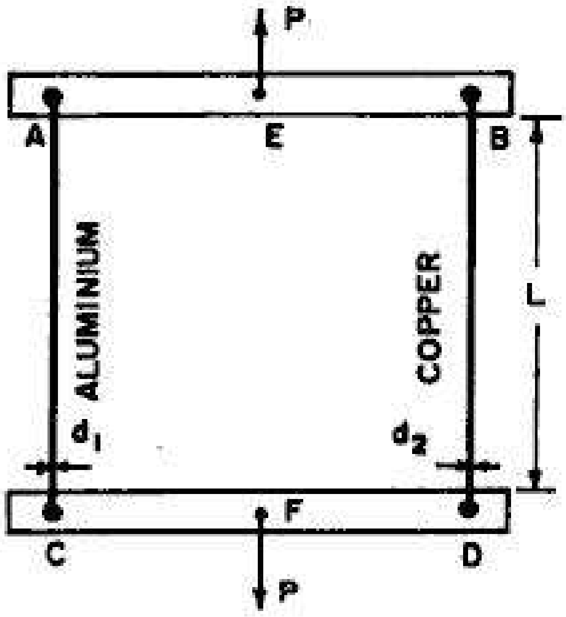


FIG. 2.49

Now $$\Delta_1 = \frac{P_1 L}{A_1 E_1} = \frac{\left(\frac{P}{2}\right) L}{\frac{\pi}{4} d_1^2 E_1} = \frac{2PL}{\pi d_1^2 E_1}$$

and $$\Delta_2 = \frac{P_2 L}{A_2 E_2} = \frac{(P/2) L}{\frac{\pi}{4} d_2^2 E_2} = \frac{2PL}{\pi d_2^2 E_2}$$

$$\therefore \quad \Delta = \frac{1}{2}(\Delta_1 + \Delta_2) = \frac{1}{2}\left[\frac{2PL}{\pi d_1^2 E_1} + \frac{2PL}{\pi d_2^2 E_2}\right]$$

or $$\Delta = \frac{PL}{\pi}\left[\frac{1}{d_1^2 E_1} + \frac{1}{d_2^2 E_2}\right],$$ which is the required expression.

If, however, both the wires are of the same material, $E_1 = E_2 = E$

$$\therefore \qquad \Delta = \frac{PL}{\pi E}\left[\frac{1}{d_1^2} + \frac{1}{d_2^2}\right]$$

2.16. EQUIVALENT MODULUS OF A COMPOUND BAR

Fig. 2.50 (*a*) shows a compound bar of length L, consisting of two materials of modulus E_1 and E_2 and having areas of cross-section A_1 and A_2. The *equivalent bar*, of length L and made up of a material having modulus E and cross-sectional area $A = A_1 + A_2$, is shown in Fig. 2.50 (*b*). Such equivalent bar should have the same deformation under the load as that of the compound bar.

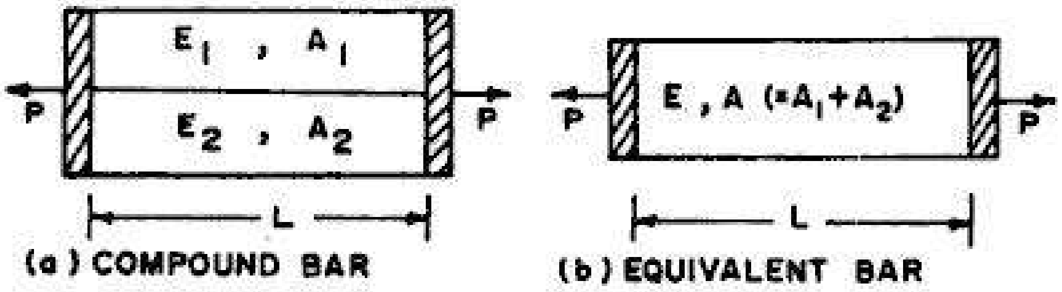


FIG. 2.50

For the equivalent bar,

Extension $$\Delta = \frac{PL}{AE} = \frac{PL}{(A_1 + A_2)E}$$

$$\therefore \qquad \frac{PL}{\Delta} = (A_1 + A_2)E \qquad ...(i)$$

For the compound bar

$$\frac{p_1}{E_1} = \frac{p_2}{E_2} \qquad \text{or} \qquad p_1 = p_2 \cdot \frac{E_1}{E_2}$$

and $$p_1A_1 + p_2A_2 = P$$

or $$\left(p_2\frac{E_1}{E_2}\right)A_1 + p_2A_2 = P$$

From which $$p_2 = \frac{P}{\dfrac{E_1}{E_2} \cdot A_1 + A_2}$$

Now $$\Delta = \frac{p_2L}{E_2} = \frac{PL}{A_1E_1 + A_2E_2}$$

or $$\frac{PL}{\Delta} = A_1E_1 + A_2E_2 \qquad ...(ii)$$

Equating (*i*) and (*ii*), we get

$$(A_1 + A_2)E = A_1E_1 + A_2E_2$$

From which $$E = \frac{A_1E_1 + A_2E_2}{A_1 + A_2} \qquad ...(2.25)$$

Example 2.32. *A compound bar is made by fastening one flat bar of steel between two similar bars of aluminium alloy. The dimensions of each bar are 40 mm wide × 8 mm thick, so that the cross-section of the composite bar measures 40 mm × 24 mm. If E for steel*

= 2.04 × 10⁵ N/mm² and E for alloy = 0.612 × 10⁵ N/mm², find the apparent value of E for the composite bar loaded in tension. If the respective elastic limits are 230 N/mm² and 50 N/mm², find the elastic limit of the compound bar.

Solution

Let us use suffix s for steel and a for aluminium alloy.

$\therefore$ $A_s = (40 \times 8) = 320 \text{ mm}^2$ and $A_a = 2 \times 40 \times 8 = 640 \text{ mm}^2$

$\therefore$ Total area $A = A_s + A_a = 320 + 640 = 960 \text{ mm}^2$

From Eq. 2.25, $E = \dfrac{A_a E_a + A_s E_s}{A_a + A_s} = \dfrac{640 \times 0.612 \times 10^5 + 320 \times 2.04 \times 10^5}{640 + 320}$

$$= \mathbf{1.088 \times 10^5 \ N/mm^2}$$

The *elastic limit* of the compound bar is the stress at which one of the members, either steel or aluminium alloy, is stressed to its elastic limit.

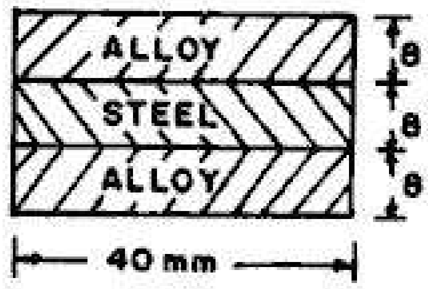


FIG. 2.51

Let P be the load at elastic limit of the compound bar, and p_a and p_s be the corresponding stresses in the two materials at the elastic limit of the compound bar.

Since the strains in the two materials are equal,

$$\frac{p_s}{E_s} = \frac{p_a}{E_a}, \text{ from which } p_s = p_a \frac{E_s}{E_a} = p_a \frac{2.04}{0.612} = 3.333\, p_a \quad ...(i)$$

Also, $\quad p_s A_s + p_a A_a = P$

$\therefore \quad 3.333\, p_a A_s + p_a A_a = P$

$\therefore \quad p_a \left[3.333 \times 320 + 640 \right] = P$

$$p_a = \frac{P}{1706.7}\ ; \text{ Hence } p_s = \frac{P}{512} \quad ...(ii)$$

Let $\quad p$ = elastic limit of the compound bar.

Hence $\quad p\,(A_a + A_s) = P$,

or $\quad 960\, p = P \quad ...(iii)$

But $\quad P = 1706.7\, p_a$, from (ii)

Hence $\quad 960\, p = 1706.7\, p_a$

From which $\quad p = \dfrac{1706.7}{960} p_a = 1.778\, p_a = 1.778 \times 50 = 88.89 \text{ N/mm}^2 \quad ...(I)$

Also, from Eq. (ii), $\quad p = 512\, p_s$

Hence from (iii), $\quad 960\, p = 512\, p_s$

or $\quad p = \dfrac{512}{960} p_s = \dfrac{512}{960} \times 230 = 122.67 \text{ N/mm}^2 \quad ...(II)$

The elastic limit (p) is the lower of the two values of p given by Eqs (I) and (II). Hence the elastic limit of the compound bar will be **88.89 N/mm²**.

2.17. STRESSES IN BOLTS AND NUTS

Fig. 2.52 shows a bolt passing through a tube. The bolt is tightened by turning the nut. When the nut is tightened by placing washers as shown, the nut will be easily turned initially, till the space between the two washers is exactly equal to the body of the bolt (and hence length of the tube) between the them. If the bolt is tightened further, the bolt will be subjected to tension while the washers and the tube will be subjected to compression. The following criteria apply to such an arrangement.

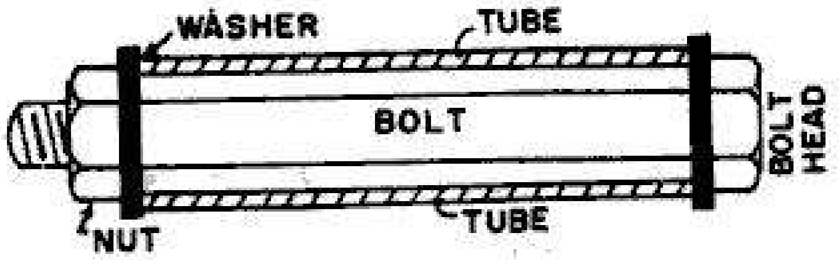


FIG. 2.52

(i) From statical equilibrium, the tensile load in the bolt will be equal to the compressive load in the tube

and (ii) From compatibility point of view, the *axial advancement* of the nut is equal to be sum of extension of the bolt and the contraction of the tube.

Example 2.33. *A steel bolt 650 mm²cross-sectional area passes centrally through a copper tube to 1200 mm²cross-sectional area. The tube is 500 mm long and is closed by rigid washers, which are fastened by the threads on the steel bolt.*

The nut is now tightened by 1/4 of a turn. Find the stress in the bolt and the tube if the pitch of the thread is 3 mm. Take $E_s = 2.05 \times 10^5$ N/mm² and $E_c = 1.1 \times 10^5$ N/mm².

(*Based on Oxford University*)

Solution

Let us use suffix s for steel bolt and c for copper tube.

$$\therefore \quad p_s A_s = p_c A_c$$

or
$$p_s\,(650) = p_c\,(1200), \text{ which gives } p_s = 1.846\,p_c \qquad \ldots(i)$$

Since the final length of the bolt and tube is the same, we have

$$\Delta_s + \Delta_c = \text{movement of the nut}$$

$$\therefore \quad \frac{p_s L}{E_s} + \frac{p_c L}{E_c} = \frac{1}{4} \times 3$$

or
$$\frac{p_s \times 500}{2.05 \times 10^5} + \frac{p_c \times 500}{1.1 \times 10^5} = 0.75 \qquad \ldots(ii)$$

Substituting the value of p_s from (i) we get

$$(1.846\,p_c)\,243.9 + p_c \times 454.5 = 0.75 \times 10^5$$

which gives p_c= **82.89 N/mm²** and $p_s = 1.846 \times 82.89 =$ **153 N/mm²**

Example 2.34. *Two steel plugs fit freely into the ends of a steel tubular distance piece 400 mm long and are drawn together by a steel bolt (500 mm long) and nut, the nut being tight fit in the beginning. The nut is further tightened by $\frac{1}{4}$ turn to draw the pieces together, the pitch of the bolt thread being 2 mm. The pieces are then subjected to forces of 50 kN tending to pull them apart. Calculate the stresses in the bolt and the tube. The area of cross-section of the bolt is 700 sq. mm and that of the tube is 500 sq. mm. Take E for steel as 200 GN/m².*

Solution

Let us consider both the effects separately.

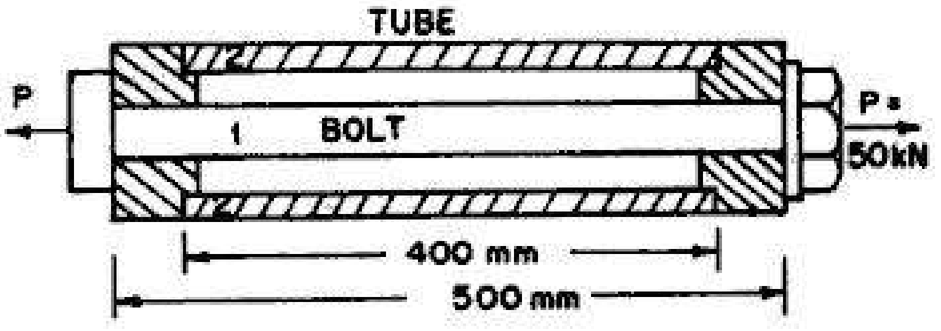


FIG. 2.53

(*a*) ***Effect of tightening the nut***

Due to tightening of the nut, tensile stress will be induced in the bolt and compressive stress will be induced in the tube. Let us use suffix 1 for the bolt and suffix 2 for the tube.

From statical equilibrium, $p_1 A_1 = p_2 A_2$ ...(*i*)

Also, $\Delta_1 + \Delta_2 =$ movement of the nut

$$\therefore \quad \frac{p_1 L_1}{E} + \frac{p_2 L_2}{E} = \frac{1}{4} \times 2 = 0.5$$

or
$$p_1 = \left(0.5 - \frac{p_2 L_2}{E}\right) \frac{E}{L_1} \qquad ...(ii)$$

Substituting this value of p_1 is (*i*), we get

$$p_2 = \left(0.5 - \frac{p_2 L_2}{E}\right) \frac{E}{L_1} \times \frac{A_1}{A_2}$$

or
$$p_2 \left(1 + \frac{L_2}{L_1} \times \frac{A_1}{A_2}\right) = 0.5 \frac{E}{L_1} \times \frac{A_1}{A_2}$$

Substituting the numerical values and noting that

$$E = 205 \text{ GN/m}^2 = 205 \times 10^9 \text{ N/m}^2 = 2.05 \times 10^5 \text{ N/mm}^2, \text{ we get}$$

$$p_2 \left(1 + \frac{500}{400} \times \frac{700}{500}\right) = 0.5 \times \frac{2.05 \times 10^5}{400} \times \frac{700}{500}$$

From which $p_2 = 130.45 \text{ N/mm}^2$ (Compressive)

Hence
$$p_1 = \left(0.5 - \frac{130.45 \times 500}{2.05 \times 10^5}\right) \frac{2.05 \times 10^5}{400} = 93.19 \text{ N/mm}^2 \text{ (tensile)}$$

(*b*) ***Effect of external load***

Let p_1' and p_2' be the additional stresses, both tensile, due to external tensile load of 50 kN.

$$\therefore \quad p_1' A_1 + p_2' A_2 = 50000 \qquad ...(iii)$$

Also, from compatibility, $\frac{p_1' L_1}{E} = \frac{p_2' L_2}{E}$

or $$p_1' = p_2' \frac{L_2}{L_1} = \frac{500}{400} p_2' = 1.25\, p_2' \quad ...(iv)$$

Hence from (*iii*) $1.25\, p_2' (700) + p_2' (500) = 50000$

From which $p_2' = 36.36\ \text{N/mm}^2$ (tensile), and $p_1' = 1.25 \times 36.36 = 45.45\ \text{N/mm}^2$ (tensile)

(c) ***Final stresses***

Total stresses in bolt $= 93.19 + 45.45 = \mathbf{138.64\ N/mm^2}$ (tensile)

and Total stress in tube $= 130.45 - 36.36 = \mathbf{94.09\ N/mm^2}$ (comp.)

Alternative solution

In the previous solution, both the effects have been considered separately and the final solution obtained by adding the effects together. This is a long and cumbersome procedure. However, the solution can be obtained directly, and in short, by considering both the effects together.

Let p_1(tensile) and p_2 (compressive) be the final stresses in the bolt and the tube respectively.

From static equilibrium, we have :

$$\therefore \quad p_1 A_1 - p_2 A_2 = 50{,}000$$

or $$700\, p_1 - 500\, p_2 = 50000$$

or $$p_1 - 0.714\, p_2 = 71.43 \quad ...(1)$$

Also, from compatibility,

(Total extension of bolt) + (total compression of tube) = movement of nut

$$\therefore \quad \frac{p_1 L_1}{E} + \frac{p_2 L_2}{E} = \frac{1}{4} \times 2 = 0.5$$

or $$p_1 \times 400 + p_2 \times 500 = 0.5 \times 2.05 \times 10^5$$

$$\therefore \quad p_1 + 1.25\, p_2 = 256.25 \quad ...(2)$$

Subtracting (1) from (2), we get $1.964\ p_2 = 184.82$

From which $p_2 = \mathbf{94.1\ N/mm^2}$ (comp.)

Hence $p_1 = 71.43 + 0.714 \times 94.1 = \mathbf{138.62\ N/mm^2}$ (tensile)

2.18. TEMPERATURE STRESSES IN UNIFORM BARS

When the temperature of an object changes, its dimensions are changed. Let us consider a block of homogeneous and isotropic material that is *free* to expand in all the directions. If this block is subjected to *uniform* changes in temperature (t, increase), the sides of the block will increase, and will take the shape shown by the dotted lines, with corner C taken as the reference point. If α is the *coefficient of thermal expansion* (or contraction), the *uniform thermal strain* (ε_t) is given by

$$\varepsilon_t = \alpha\,(t) \qquad ...(2.26)$$

The coefficient α is a property of the material and has a unit reciprocal of temperature change. In SI units, α has the dimensions of either 1/K (the reciprocal of kelvins) or 1/C (the reciprocal of degree Celsius), because the change in temperature is numerically the same in both kelvins and degrees Celsius. Common values of α are : 10×10^{-6} to $18 \times 10^{-6}/°C$ for steel, $17 \times 10^{-6}/°C$ for copper and $23 \times 10^{-6}/°C$ for aluminium and aluminium alloys. Ordinary materials expand when heated and contract when cooled. Increase in temperature produces a positive thermal strain while decrease in temperature cause negative thermal strains. Also, thermal strains are reversible in nature, meaning thereby that the member returns to its original shape when the temperature returns to its original value. However, there are some special materials, recently developed which do not behave in customary manner; these special metals decrease in the dimensions when heated and increase when cooled, over certain temperature range. Another example of *unusual material* is water which *expands* when heated above 4°*C* and also *expands* when cooled at temperature below 4°C, thus attaining its maximum density at 4°C .

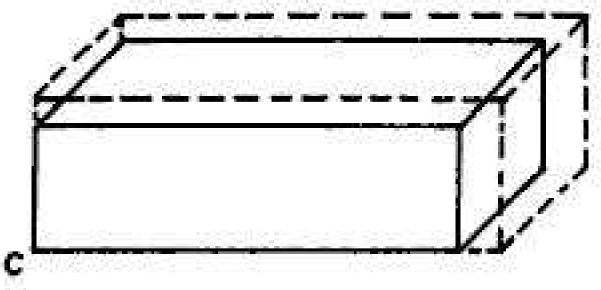


FIG. 2.54

Consider a bar of length L (Fig. 2.55 *a*), subjected to uniform temperature increase t, the bar being *free* to expand. Then the increase in the length of bar will be

$$\Delta_t = \varepsilon_t \,.\, L = \alpha\, t\, L \qquad ...(2.26\ a)$$

It should be clearly noted that since the bar is free to expand, there will be no *thermal stress* in the bar.

In contrast to this, consider a bar (Fig. 2.55 *b*) of length L, fixed at both ends between rigid supports. If the supports do not yield, the bar is not free to expand by the amount Δ_t and hence it will be subjected to *thermal stress* p_t given by

$$p_t = \frac{\Delta_t E}{L} = L\,\alpha\, t \,.\, \frac{E}{L} = E\,\alpha\, t \qquad ...(2.27)$$

This stress will be *compressive* when the change in temperature (t) is positive (*i.e.* increase) and *tensile* when the change in temperature is negative (*i.e.* decrease). *Note that the thermal stress* p_t *does not depend upon the cross-sectional area.*

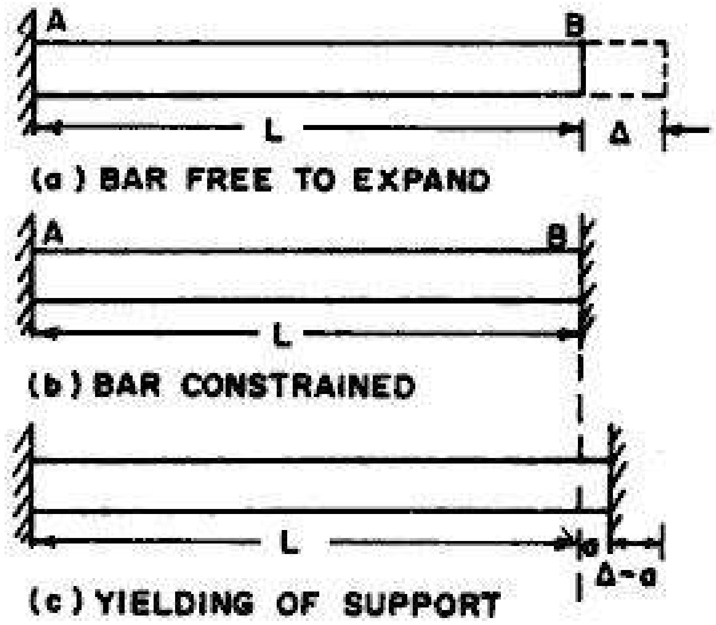


FIG. 2.55

Lastly, consider the case shown in Fig. 2.55 (*c*), in which one of the support *yields* by an amount a. In this case, the total amount of expansion *checked* will be $(\Delta_t - a)$. Hence the resulting temperature stress is

$$p_t = (\Delta_t - a)\,\frac{E}{L} = (L\,\alpha\, t - a)\,\frac{E}{L} \qquad ...(2.28)$$

2.19. TEMPERATURE STRESSES IN BARS OF TAPERING SECTION

Consider a bar of uniformly tapering section, shown in Fig. 2.25, rigidly fixed between two end supports. When the temperature is raised by t, compressive force P will be induced since the bar is not free to expand. This force is the same for all cross-sections, and hence maximum stress will be induced at section AA where diameter is d_1.

If the bar were free to expand, we have

$$\Delta_t = L\,\alpha\,t \qquad ...(i)$$

The force induced in the bar will be a compressive force P which is required to prevent free expansion of Δ given by (i).

Now, for an element of length dx, the deformation due to P is

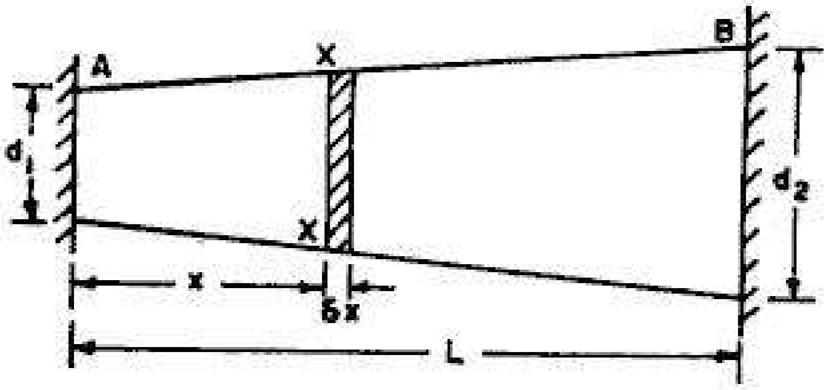


FIG. 2.56

$$\delta\Delta = \frac{P\,.\,dx}{A_x E}$$

$\therefore$ Total deformation $\quad \Delta = \int_0^L \frac{P\,dx}{A_x E}$

But $\quad A_x = \frac{\pi}{4}\left(d_1 + \frac{d_2 - d_1}{L}x\right)^2 = \frac{\pi}{4}(d_1 + k\,x)^2$ where $x = \frac{d_2 - d_1}{L}$

$\therefore$
$$\Delta = \frac{4P}{\pi E}\int_0^L \frac{dx}{(d_1 + b\,x)^2} = -\frac{4P}{\pi k E}\left[\frac{1}{d_1 + kx}\right]_0^L$$

or
$$\Delta = -\frac{4PL}{\pi E\,(d_2 - d_1)}\left[\frac{1}{d_1 + d_2 - d_1} - \frac{1}{d_1}\right] = \frac{4PL}{\pi E\,d_1 d_2} \qquad ...(ii)$$

Equating (i) and (ii), we get, $\Delta_t = \Delta$

or
$$L\,\alpha\,t = \frac{4PL}{\pi E\,d_1 d_2} \qquad ...(iii)$$

From which
$$P = \frac{\pi E}{4} d_1 d_2\,.\,\alpha\,t \qquad ...(2.29)$$

Maximum stress induced at section AA is given by

$$p_{t,\,\max} = \left(\frac{\pi E}{4} d_1 d_2\,\alpha\,t\right) \div \left(\frac{\pi}{4} d_1^2\right) = E\,\alpha\,t\,\frac{d_2}{d_1} \qquad ...(2.30)$$

If $d_1 = d_2$, $p_t = E\,\alpha\,t$, which is the same as Eq. 2.27.

Again, if the support yields by an amount a, the amount of expansion *checked* will be $(\Delta_t - a)$. Hence

$$\Delta_t - a = \Delta$$

or
$$(L\,\alpha t - a) = \frac{4PL}{\pi E\,d_1 d_2}$$

From which $$P = \frac{\pi E}{4} d_1 d_2 \left(\frac{L \alpha t - a}{L} \right) \qquad ...(2.31)$$

2.20. TEMPERATURE STRESSES IN COMPOSITE BAR

When a composite bar, consisting of two materials having different coefficients of thermal expansion, is subjected to temperature change, opposite kinds of stresses (*i.e.* tensile and compressive) will be set up in the two materials. Consider a composite bar of steel (S) and copper (C), subjected to increase in temperature.

The free expansion (Δ_c^t) of copper due to temperature rise will be more than the free expansion (Δ_s^t) of steel, since α ($= \alpha_c$) for copper is more than $\alpha (= \alpha_s)$ of steel. In order to have common expansion (Δ), tensile stress will be developed in steel and compressive stress in copper. From statics, we get :

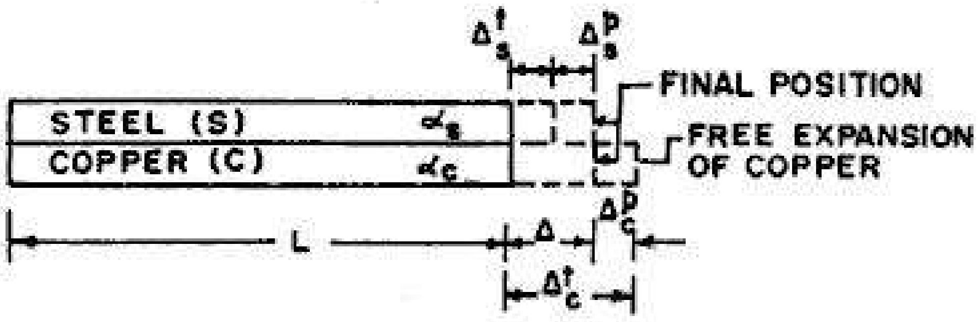


FIG. 2.57

Tensile force in steel = compressive force in copper

$$\therefore \qquad P_s = P_c = P \text{ (say)} \qquad ...(i)$$

From compatibility equation,

Final extension of copper = Final extension of steel

$$\therefore \qquad \Delta_c = \Delta_s = \Delta \text{ (say)} \qquad ...(ii)$$

Also, from geometry of Fig. 2.57, we get

$$\Delta_s = \Delta_s^t + \Delta_s^p \quad \text{and} \quad \Delta_c = \Delta_c^t - \Delta_c^p \qquad ...(iii)$$

where $\Delta_s = \Delta_c = \Delta$ = Final expansion

Δ_s^t = Free expansion of steel, due to temperature rise

Δ_c^t = Free expansion of copper, due to temperature rise

Δ_s^p = Expansion of steel due to temperature stress (tensile)

Δ_c^p = Compression of copper due to Temperature stress (comp.)

From (*ii*), we get

$$\Delta_s^t + \Delta_s^p = \Delta_c^t - \Delta_c^p$$

or $$\Delta_s^p + \Delta_c^p = \Delta_c^t - \Delta_s^t$$

or $$\Delta_s^p + \Delta_c^p = L\,t\,(\alpha_c - \alpha_s) \qquad ...(iv)$$

But $\Delta_s^p = \dfrac{PL}{A_s E_s}$ and $\Delta_c^p = \dfrac{PL}{A_c E_c}$, where P is the final force in the bar.

$$\therefore \quad PL\left(\frac{1}{A_s E_s} + \frac{1}{A_s E_s}\right) = L\,t\,(\alpha_c - \alpha_s) \qquad ...(v)$$

From which $$P = \frac{t\,(\alpha_c - \alpha_s)}{\dfrac{1}{A_s E_s} + \dfrac{1}{A_c E_c}} \qquad \text{...(2.32)}$$

Knowing the value of P, the individual stresses in steel and copper are :

$$p_s = P/A_s \text{ and } p_c = P/A_c$$

Eq. (v) can also be written as

$$\frac{P}{A_s E_s} + \frac{P}{A_c E_c} = t\,(\alpha_c - \alpha_s)$$

or $$\frac{p_s}{E_s} + \frac{p_c}{E_c} = t\,(\alpha_c - \alpha_s) \qquad \text{...(2.33 } a)$$

or $$e_s + e_c = (\alpha_c - \alpha_s)\,t \qquad \text{...(2.33)}$$

The above equation relates the strains in the two materials.

Also, from Eqs. (ii) and (iii), the final elongation of the composite bar is given by

$$\Delta = \Delta_s = \Delta_s^t + \Delta_s^p$$

or $$\Delta = L\,\alpha_s\,t + \frac{PL}{A_s E_s}$$

Substituting the value of P from Eq. 2.32.

$$\Delta = L\,\alpha_s\,t + \frac{L}{A_s E_s}\left[\frac{t\,(\alpha_c - \alpha_s)}{\dfrac{1}{A_s E_s} + \dfrac{1}{A_c E_c}}\right]$$

From which $$\Delta = \frac{(\alpha_s A_s E_s + \alpha_c A_c E_c)\,L\,t}{A_s E_s + A_c E_c} \qquad \text{...(2.34)}$$

Example 2.35. *Two parallel walls, 8 m apart, are stayed together by a steel rod of 20 mm diameter passing through metal plates and nuts at each end. The nuts are screwed upto the plates while the bar is at a temperature of 400 K. Find the pull exerted by the bar after it has cooled to 300 K, (a) if the ends do not yield (b) if the total yielding at the two ends is 5 mm. Take α for steel $= 12 \times 10^{-6}$ per K and $E = 2 \times 10^5$ N/mm^2.*

Solution

Change in temperature (fall) = 400 − 300 = 100 K

Area of cross-section of rod $= \frac{\pi}{4}(20)^2 = 314.16\text{ mm}^2$

(a) Non-yielding walls

$$p_t = \alpha\,t\,E = 12 \times 10^{-6} \times 100 \times 2 \times 10^5 = 240\text{ N/mm}^2$$

$\therefore$ Pull $P = p_t . A = 240 \times 314.16 = 75398$ N $=$ **75.398 kN**

(b) Walls yielding by 5 mm

$$\frac{p_t L}{E} = \Delta = (L\,\alpha t - a) \text{ where } a = 5 \text{ mm}$$

$$\therefore \qquad p_t = \frac{E}{L}(L\,\alpha t - 5) = E\,\alpha t - \frac{5E}{L}$$

$$= 240 - \frac{5 \times 2 \times 10^5}{8000} = 240 - 125 = 115 \text{ N/mm}^2$$

$\therefore$ Pull $P = p_t \, . A = 115 \times 314.16 = 36128 \text{ N} = \mathbf{36.128 \text{ kN}}$

Example 2.36. *A railway is laid so that there is no stress in the rails at 8°C. Calculate (a) the stress on the rails at 50°C if there is no allowance for expansion, (b) the stress in the rails at 50°C if there is an expansion allowance of 8 mm per rail. (c) the expansion allowance if the stress in the rail is to be zero when the temperature is 50° C (d) the maximum temperature to have no stress in the rails if the expansion allowance is 12 mm per rail. The rails are 30 m long. Take $\alpha = 12 \times 10^{-6}$ per °C and $E = 2 \times 10^5$ N/mm².*

Solution

Change in temperature (rise) = 50 − 8 = 42°C

(a) *No expansion allowance*

$$\text{Stress } p_t = \alpha\, t\, E = 12 \times 10^{-6} \times 42 \times 2 \times 10^5 = \mathbf{100.8 \text{ N/mm}^2}$$

(b) *Expansion allowance of 8 mm*

$$\frac{p_t \, . L}{E} = \Delta = (L\,\alpha t - a)$$

$$\therefore \qquad p_t = E\,\alpha\, t - \frac{aE}{L} = 100.8 - \frac{5 \times 2 \times 10^5}{30000} = \mathbf{67.5 \text{ N/mm}^2}$$

(c) *Value of a if p_t is to be zero*

$$p_t = E\,\alpha t - \frac{aE}{L}$$

For p_t to be zero, $\quad a = L\alpha\, t = 30000 \times 12 \times 10^{-6} \times 42 = \mathbf{15.12 \text{ mm}}$

(d) *Value of temperature for p_t to be zero*

$$p_t = \frac{E}{L}(L\,\alpha\, t - a) = 0$$

$$\therefore \qquad L\,\alpha\, t = a$$

Hence $\quad t = \dfrac{a}{L\,\alpha} = \dfrac{12}{30000 \times 12 \times 10^{-6}} = 33.33°$

$\therefore$ Allowable temperature = 8 + 33.33 = **41.33°**

Example 2.37. *A circular section tapering bar is rigidly fixed at both the ends. The diameter change from 75 mm at one end to 150 mm at the other end, in a length of 1.2 m. Compute the maximum stress in the bar if the temperature is raised by 32°C. Take $E = 2 \times 10^5$ N/mm² and $\alpha = 12 \times 10^{-6}$ per 1°C.*

Solution

For Eq. 2.30, $\quad p_{t,\max} = E\,\alpha\, t \, . \dfrac{d_2}{d_1} = 2 \times 10^5 \times 12 \times 10^{-6} \times 32 \times \dfrac{150}{75}$

$$= \mathbf{153.6 \text{ N/mm}^2}$$

Example 2.38. *A gun metal rod screwed at the ends passes through a steel tube. The tube has 25 mm external diameter and 20 mm internal diameter. The diameter of the rod is 16 mm. The assembly is heated to 400 K and the nuts on the rod are then screwed tightly home on the ends of the tube. Find the intensity of stress in the rod and in the tube when the common temperature falls to 300 K.*

Coefficient of expansion per K for steel $= 12 \times 10^{-6}$

Coefficient of expansion per K for gun metal $= 20 \times 10^{-6}$

Young's modulus for gun metal $= 0.91 \times 10^5\ N/mm^2$

Young's modulus for steel $= 2 \times 10^5\ N/mm^2$

Solution

Let us use suffix s for steel and g for gun metal. From statics, we get

$$p_s A_s = p_g \cdot A_g \qquad \text{...}(i)$$

Also $\Delta_s = \Delta_g = \Delta =$ net reduction in length.

Fall in temperature $= 400 - 300 = 100$ K.

Since $\alpha_g > \alpha_s$, gun metal will shrink more than steel. Hence compressive stress will induced in steel and tensile stress will be induced in gun metal.

Also, From Eq. 2.33(a), $\dfrac{p_s}{E_s} + \dfrac{p_g}{E_g} = (\alpha_g - \alpha_s)\, t$ ...(ii)

Now $A_s = \dfrac{\pi}{4}(25^2 - 20^2) = 176.7\ \text{mm}^2$ and $A_g = \dfrac{\pi}{4}(16)^2 = 201.06\ \text{mm}^2$

Hence from (i), $p_g = p_s \dfrac{A_s}{A_g} = p_s \times \dfrac{176.71}{201.06} = 0.879\, p_s$

Substituting in (ii), we get

$$p_s \left[\frac{1}{E_s} + \frac{0.879}{E_g} \right] = (20-12)\ 10^{-6} \times 100$$

or
$$p_s \left[\frac{1}{2 \times 10^5} + \frac{0.879}{0.91 \times 10^5} \right] = 800 \times 10^{-6}$$

or
$$p_s\,(0.5 + 0.966)\,\frac{1}{10^5} = 800 \times 10^{-6}$$

From which $p_s =$ **54.57 N/mm²** (tension)

Hence $p_g = 0.879 \times 54.57 =$ **47.97 N/mm²** (comp.)

Example 2.39. *A copper bar 25 mm diameter is completely enclosed in a steel tube, 25 mm internal diameter and 40 mm external diameter. A pin, 10 mm in diameter is fitted transversely to the axis of the bar near each end, to secure the bar to the tube. Calculate the intensity of shear stress induced in the pin when the temperature of the whole is raised by 40 K.*

Take $E_c = 1 \times 10^5\ N/mm^2;\ E_s = 2 \times 10^5\ N/mm^2.$

$$\alpha_c = 18 \times 10^{-6} \text{ per } K;\ \alpha_s = 12 \times 10^{-6} \text{ per } K.$$

Solution

Area of cross-section of copper bar, $A_c = \frac{\pi}{4}(25)^2 = 490.87 \text{ mm}^2$

Area of cross-section of steel tube, $A_s = \frac{\pi}{4}(40^2 - 25^2) = 765.76 \text{ mm}^2$

From compatibility, $\quad \frac{p_s}{E_s} + \frac{p_c}{E_c} = (\alpha_c - \alpha_s)\, t$

or $$\frac{p_s}{2 \times 10^5} + \frac{p_c}{1 \times 10^5} = (18 - 12)\, 10^{-6} \times 40 = 240 \times 10^{-6}$$

or $$p_s + 2p_c = 240 \times 10^{-6} \times 2 \times 10^5 = 48 \quad \text{...(i)}$$

From statics, $\quad p_s A_s = p_c A_c$

$$\therefore \quad p_s = p_c \cdot \frac{A_c}{A_c} = p_c \times \frac{490.87}{765.76} = 0.641\, p_c \quad \text{...(ii)}$$

Substituting in (*i*), we get

$$0.651\, p_c + 2p_c = 48$$

From which $\quad p_c = 18.17 \text{ N/mm}^2$

$\therefore$ Force P between steel tube and copper bar $= p_c A_c = 18.17 \times 490.87 = 8921$ N

Area of cross-section of pin $= \frac{\pi}{4}(10)^2 = 78.54 \text{ mm}^2$

Since the pin is fitted transversely, and passes through the tube and the rod, it will be in *double shear.*

$$\therefore \text{ Shear stress in pin} = \frac{8921}{2 \times 78.54} = \mathbf{56.79 \text{ N/mm}^2}$$

Example 2.40. *A composite bar made up of aluminium and steel is held between two supports as shown in Fig. 2.58. The bars are stress free at a temperature of 42°C. What will be the stresses in the two bars when the temperature drops to 24°C, if (a) the supports are un-yielding, (b) the supports come nearer to each other by 0.1 mm. The cross-sectional area of steel bar is 160 mm² and that of aluminium bar is 240 mm².*

Take E for aluminium as 0.7×10^5 N/mm² and that for steel as 2×10^5 N/mm². The coefficients of thermal expansion for aluminium and steel are 24×10^{-6} per °C and 12×10^{-6} per °C respectively.

Solution

Using suffix s for steel and a for aluminium, the free contraction due to drop of temperature ($t = 42 - 24 = 18$°C) is given by:

$$\Delta = (L_s \alpha_s + L_a \alpha_a)\, t = (500 \times 12 \times 10^{-6} + 250 \times 24 \times 10^{-6})\, 18 = 0.216 \text{ mm} \quad \text{...(i)}$$

However, the above free contraction is checked by the supports, either fully or partially. Hence tensile stresses will be set up in both steel and aluminium bars. However, force in steel bar and aluminium bars will be the same. Hence

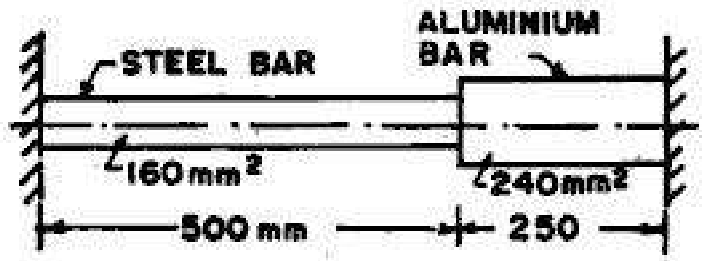


FIG. 2.58

$$A_s p_s = A_a p_a$$

or $$160\, p_s = 240\, p_a$$

From which $$p_s = 1.5\, p_a \qquad \text{...}(ii)$$

Let Δ_s^p = elongation of steel bar due to temperature stress

$$= \frac{p_s}{E_s} L_s = \frac{500}{2 \times 10^5} \times p_s = 250 \times 10^{-5} p_s \qquad \text{...}(ii)$$

and Δ_a^p = elongation of aluminium bar due to temperature stress

$$= \frac{p_a}{E_a} L_a = \frac{250}{0.7 \times 10^5} \times p_a = 357.1 \times 10^{-5} p_a \qquad \text{...}(iv)$$

Case (a) : Supports non-yielding

$$\Delta_s^p + \Delta_a^p = \text{Free contraction } \Delta \qquad \text{...}(v)$$

$\therefore \quad 250 \times 10^{-5} p_s + 357.1 \times 10^{-5} p_a = 0.216$

But $\quad p_s = 1.5\, p_a$ from (ii)

Hence $250 \times 10^{-5} \times 1.5\, p_a + 357.1 \times 10^{-5} p_a = 0.216$

From which $\quad p_a =$ **29.50 N/mm²** (tensile)

Hence $\quad p_s = 1.5 \times 29.5 =$ **44.26 N/mm²** **(tensile)**

Case (b) : Supports yielding by a = 0.1 mm

$$\Delta_s^p + \Delta_a^p = \Delta - a \quad (vi)$$

$\therefore \quad 250 \times 10^{-5} p_s + 357.1 \times 10^{-5} p_a = 0.216 - 0.1$

Substituting $\quad p_s = 1.5\, p_a$ from (ii), we get

$250 \times 10^{-5} \times 1.5\, p_a + 357.1 \times 10^{-5} p_a = 0.116$

From which, $\quad p_a =$ **15.84 N/mm²** (tensile)

Hence $\quad p_s = 1.5 \times 15.84 =$ **23.77 N/mm²** (tensile)

Example 2.41. *A composite bar is made up by connecting a steel member and a copper member rigidly fixed at their ends as shown in Fig. 2.59. The cross-sectional area of steel member is A mm² while that of the copper member is 2A mm² for half the length and A mm² for the other half of the length. The coefficients of expansion of steel and copper are α and 1.25 α respectively while the elastic moduli are E and 0.5 E respectively. Estimate the stresses induced in the members due to a temperature rise of t degrees.* *(Based on U.L.)*

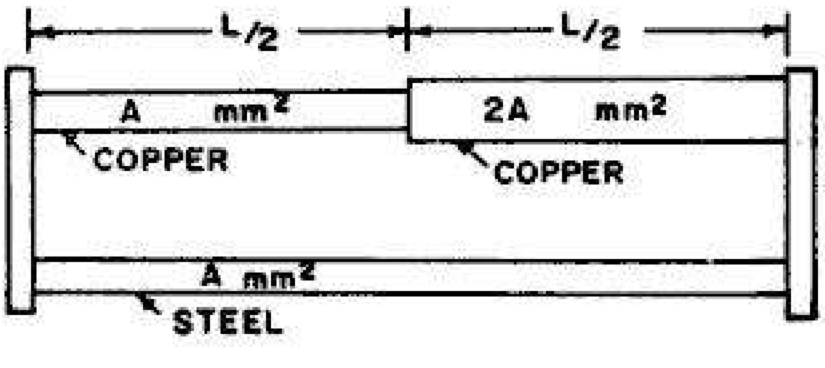


FIG. 2.59

Solution

When the temperature rises, both copper and steel bars will expand. However, copper will expand more than steel, and hence compressive stress will be induced in copper and tensile stress will be induced in steel.

Let p_c = compressive stress in copper bar of A mm^2 area of cross-section

$\therefore$ Stress in copper bar of $2A$ mm^2 area $= \frac{p_c A}{2A} = 0.5\,p_c$

Hence, compression of copper bar due to temperature stress is given by

$$\Delta_c^p = \frac{(p_c)}{E_c}\cdot\frac{L}{2} + \frac{(0.5\,p_c)}{E_c}\cdot\frac{L}{2} = \frac{3}{4}\frac{p_c L}{E_c} = \frac{3}{2}\frac{p_c L}{E} \qquad \ldots(i)$$

Also, expansion of steel bar due to temperature stress $= \Delta_s^p = \frac{p_s L}{E}$ ...(*ii*)

Now, from compatibility, $\Delta_s^p + \Delta_c^p = L\,t\,(\alpha_c - \alpha_s)$

$$\therefore \qquad \frac{3}{2}\frac{p_c L}{E} + \frac{p_s L}{E} = L\,t\,(1.25\,\alpha - \alpha)$$

or
$$p_s + 1.5\,p_c = 0.25\,E\,\alpha\,t \qquad \ldots(a)$$

Since there is no external force acting,

$$p_c A = p_s A$$

or
$$p_c = p_s \qquad \ldots(b)$$

Hence from (*a*) and (*b*), we get.

$$p_c = p_s = \mathbf{0.1\,E\,\alpha\,t}$$

Example 2.42. *Two steel rods, one of 70 mm diameter and the other of 50 mm diameter and joined end to end by means of a turn buckle, as shown in Fig. 2.60. The other end of each rod is rigidly fixed and there is initially a little tension in the rods. The effective length of each rod is 4 m. Calculate the increase in the tension when the turn buckle is tightened by one quarter of a turn if there are 2 threads per cm on the bigger diameter and 3 threads per cm on the other end. Neglect the extension of turn buckle. Also, find what rise in temperature could nullify the increase in tension. Take* $E = 2 \times 10^5$ *N/mm*^{2} *and* $\alpha = 12 \times 10^{-6}$ *per* °*C.*

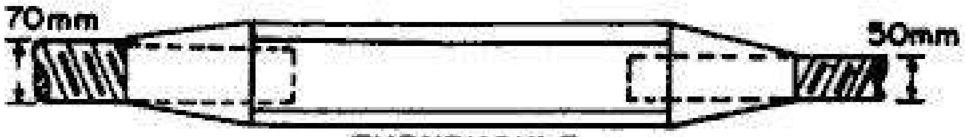


FIG. 2.60

Solution

Use suffix 1 for large diameter rod and 2 for smaller diameter rod.

$$\therefore \qquad A_1 = \frac{\pi}{4}(70)^2 = 3848.5 \text{ mm}^2 \text{ and } A_2 = \frac{\pi}{2}(50)^2 = 3927 \text{ mm}^2$$

When the turn-buckle is turned by one quarter of a turn, Extension of larger diameter rod, $\Delta_1 = \frac{1}{4} \cdot \frac{10}{3} = 0.833$ mm

Extension of smaller diameter rod $\Delta_2 = \frac{1}{4} \cdot \frac{10}{2} = 1.25$ mm.

Total extension of both the rods, $\Delta = \Delta_1 + \Delta_2 = 0.833 + 1.25 = 2.083$ mm ...(*i*)

Let P Newtons be the tensile force in each bar when the turn buckle is tightened.

Hence from Hooke's law, $$\Delta = \frac{Pl_1}{A_1 E} + \frac{Pl_2}{A_2 E} = \frac{P \times 4000}{2 \times 10^5}\left[\frac{1}{3848.5} + \frac{1}{3927}\right]$$

$$= 1.029 \times 10^{-5} P \text{ mm} \quad ...(ii)$$

Equating (*i*) and (*ii*), we get $P = 202434$ N $=$ **202.434 kN**

Now, in order to nullify this tension, the rise of temperature of the two rods must be such that the total thermal expansion of the two rods is equal to 2.083 mm.

$\therefore$ $l\alpha t = 2.083$

or $(2 \times 4000)\, t \times 12 \times 10^{-6} t = 2.083$

From which $t =$ **21.7°C**

2.21. STATICALLY INDETERMINATE PROBLEMS

In the preceding articles, we considered axially loaded bars and other *simple* structures which could be analysed by the simple equations of statics by considering static equilibrium. The axial forces in the members and the reactions at the supports could be found by drawing the free body diagrams and solving equilibrium equations. Such problems are usually called *statically determinate problems*. Now we shall consider some simple *statically indeterminate* problems which can not be solved by the simple equations of statics *alone*. To solve such problems, the deformation characteristics of the structure will have to be taken into account, in addition to the equations of statical equilibrium. We shall illustrate the solution of several such problems through solved examples.

Example 2.43. *A bar of uniform section is fixed at both the ends and is loaded with a force P shown in Fig. 2.61. Determine the reactions at both the ends and the extension of AC.*

Solution

Let R_1 and R_2 be the two reactions, obviously in the directions shown.

From statics, $R_1 + R_1 = P$ ...(1)

Since the ends are fixed in position,

Extension of AB = compression of BC

$$\therefore \quad \frac{R_1 a}{AE} = \frac{R_2 b}{AE} \quad ...(2)$$

or $a R_1 = b R_2$

$$\therefore \quad R_1 = \frac{b}{a} R_2$$

Substituting in (1), we get

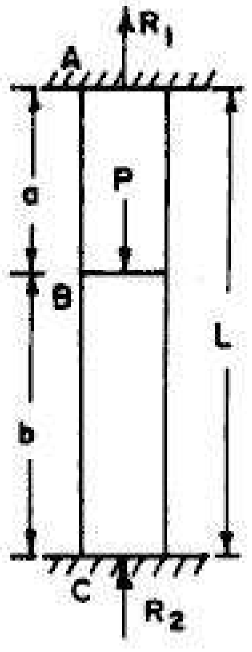


FIG. 2.61

$$R_2\left(\frac{b}{a}+1\right)=P$$

From which $$R_2=\frac{Pa}{a+b}=\frac{Pa}{L}$$

Hence $$R_1=\frac{b}{a}\cdot\frac{Pa}{a+b}=\frac{Pb}{L}$$

These are thus the values of reactions R_1 and R_2

$$\text{Now extension of } AB=\Delta_1=\frac{R_1 a}{AE}=\frac{Pab}{LAE}$$

Example 2.44. *A rigid bar AB is supported by three equidistant rods in the same vertical plane, and is loaded as shown in Fig. 2.62. All the three rods have the same area of cross-section. Determine the forces in the three rods, if the bars 1 and 3 are of steel and bar 2 is of brass. Take $E_s/E_b=2$.*

Solution

From statics, $P_1+P_2+P_3=P$ ...(1)

Also, taking moments about B, $P_3(2a)+P_2(a)=P\times 1.5a$

or $$2P_3+P_2=1.5P \qquad ...(2)$$

There are three unknown forces (*i.e.* P_1, P_2 and P_3), while we have only two equations obtained from statical equilibrium.

The third equation is obtained from the deformation pattern of the structure. The dotted lines show the *deformed position* of the structure, where Δ_1, Δ_2 and Δ_3 are the extensions of the three rods. From compatibility, we obtain

$$\Delta_2=\frac{\Delta_1+\Delta_3}{2}$$

or $$\frac{P_2}{A_2E_2}=\frac{P_1}{2A_1E_1}+\frac{P_3}{2A_3E_3}$$

Here, $A_1=A_2=A_3$.

Also, $E_1=E_3=2E_2=2E$ (say)

$$\therefore \quad \frac{P_2}{E}=\frac{P_1}{2(2E)}+\frac{P_3}{2(2E)}$$

or $$4P_2=P_1+P_3 \qquad ...(3)$$

Substituting this value of (P_1+P_3) in (1), we get

$$4P_2+P_2=P$$

From which $$P_2=\frac{P}{5}=\mathbf{0.2\,P}$$

Hence from (2), $2P_3+0.2P=1.5P$

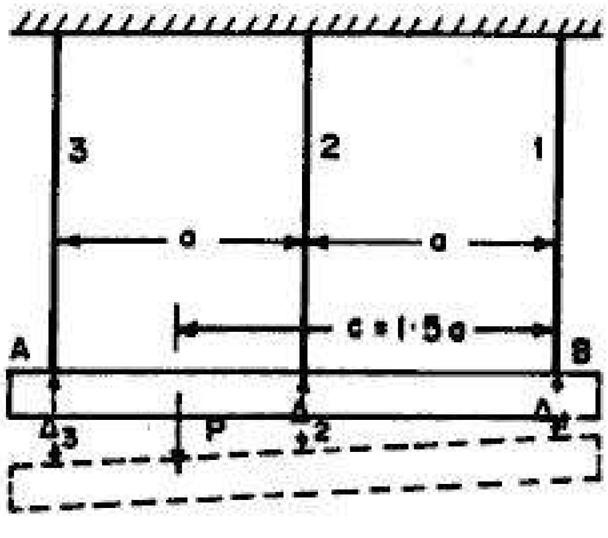


FIG. 2.62

From which $P_3 = \mathbf{0.65}\, \boldsymbol{P}$

Substituting these values of P_2 and P_3 in (1), we get

$$P_1 + 0.2\,P + 0.65\,P = P$$

From which $P_1 = \mathbf{0.15}\,\boldsymbol{P}$

Example 2.45. *A load P is supported by a system of three rods shown in Fig. 2.63. The outer rods are identical and are of the same material while the inner rod is of different material. Determine the forces in the rods. Hence if P = 30 kN and θ = 30°, determine value of these forces when the outer bars are of steel and the central bar is of brass. Take the ratio of moduli of elasticity of steel and brass as 2.*

Solution

From statics, $2\,P_1 \cos\theta + P_2 = P$ ...(1)

The dotted lines show the deformed shape of the structure, in which point O takes the position O'. Assuming that θ does not change appreciably, we get.

$$\Delta_1 = \Delta_2 \cos\theta$$

or
$$\frac{P_1 L_1}{A_1 E_1} = \frac{P_2 L_2}{A_2 E_2}\cos\theta \quad ...(2)$$

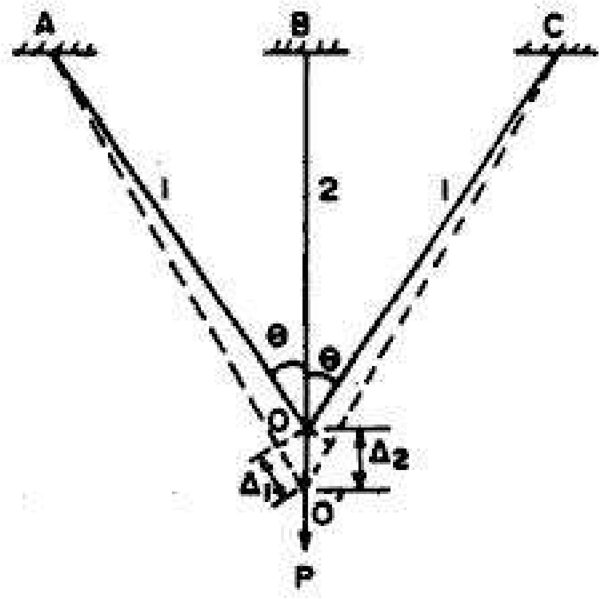


FIG. 2.63

But, from geometry, $L_2/L_1 = \cos\theta$.

Hence, from (2),

$$P_1 = P_2 \cdot \frac{A_1 E_1}{A_2 E_2} \cdot \frac{L_2}{L_1}\cos\theta = P_2 \cdot \frac{A_1 E_1}{A_2 E_2}\cos^2\theta \quad ...(a)$$

Substituting this value of P_1 in (2), we get

$$2\,P_2 \frac{A_1 E_1}{A_2 E_2}\cos^3\theta + P_2 = P$$

From which
$$P_2 = \frac{P}{1 + \dfrac{2 A_1 E_1}{A_2 E_2}\cos^3\theta} \quad ...(i)$$

Hence
$$P_1 = \left(\frac{P}{1 + \dfrac{2A_1E_1}{A_2E_2}\cos^3\theta}\right) \cdot \frac{A_1E_1}{A_2E_2}\cos^2\theta = \frac{P}{\dfrac{A_2E_2}{A_1E_1\cos^2\theta} + 2\cos\theta} \quad ...(ii)$$

Taking the values : $P = 30$ kN, $\theta = 30°$, $A_1 = A_2$, $E_1/E_2 = 2$, we get

$$P_2 = \frac{P}{1 + 2 \times 2\cos^3 30°} = 0.278\,P = \mathbf{8.34\ kN}$$

and
$$P_1 = \frac{P}{\dfrac{1}{2\cos^2 30°} + 2\cos 30°} = 0.417\,P = \mathbf{12.5\ kN}$$

Example 2.46. *Three identical pin-connected bars are arranged as shown in Fig. 2.64, and support a load P. Find the axial force in each bar and also the vertical displacement of the point of application of load. Neglect any possibility of lateral buckling of the bar. All the bars are of the same length and same areas of cross-section.*

Solution

Let us use suffix 1 for bars AO and BO and suffix 2 for CO. Obviously, force P_1 in bars AO and BO will be tensile while force P_2 in bar OC will be compressive.

From statics, $2P_1 \cos 60° + P_2 = P$

From which $P_1 + P_2 = P$ ...(*i*)

The dotted lines show the deformed position of the structure. Assuming that the angles between the rods do not change appreciably, we get

$$\Delta_1 = \Delta_2 \cos 60° = \frac{1}{2}\Delta_2$$

or

$$\frac{P_1 L}{A E} = \frac{1}{2}\frac{P_2 L}{AE}$$

From which $P_2 = 2P_1$ ...(*ii*)

Substituting in (*i*), we get $P_1 + 2P_1 = P$

From which $P_1 = \frac{P}{3}$.

Hence $P_2 = 2P_1 = \frac{2P}{3}$

Also, $\Delta_2 = \frac{P_2 L}{AE} = \frac{2PL}{3AE}$

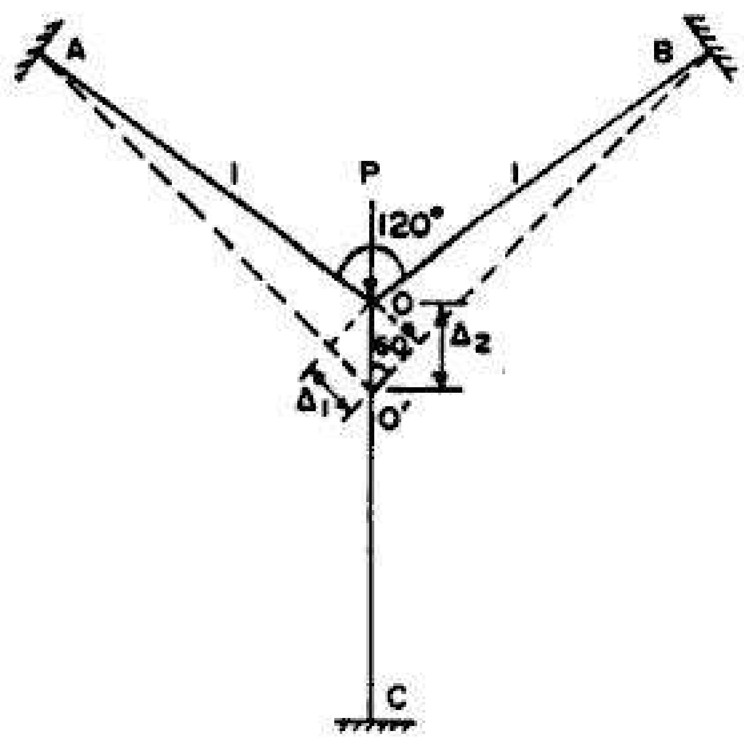


FIG. 2.64

Example 2.47. *A circular bar ABCD is rigidly fixed at A and D and is subjected to axial forces as shown in Fig. 2.65. Determine the reactions, the forces in each portion of the bar and the displacement of point B and C. Take E = 200 kN/mm².*

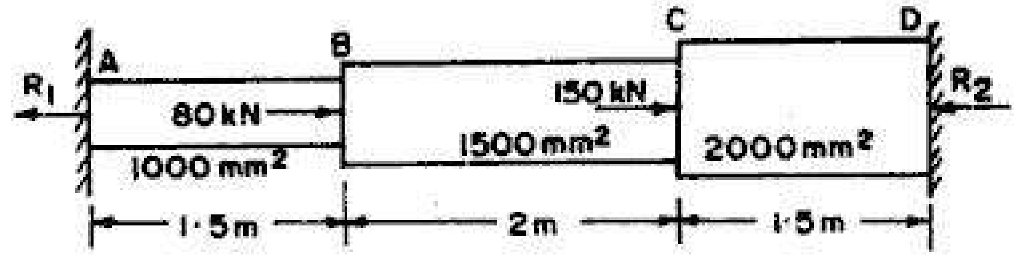


FIG. 2.65

Solution

Let R_1 and R_2 be the end reactions, obviously in the directions shown in Fig. 2.65. The free body diagrams for the three portions AB, BC and CD of the bar are shown in Fig. 2.66. Obviously portion AB will be in tension while portions BC and CD will be in compression.

For AB, force $P_1 = R_1$; $L_1 = 1500$ mm ; $A_1 = 1000 \text{ mm}^2$

For BC, force $P_2 = 80 - R_1 = R_2 - 150$; $L_2 = 2000$ mm; $A_2 = 1500 \text{ mm}^2$

For CD, force $P_3 = R_2$; $L_3 = 1500$ mm ; $A_3 = 2000 \text{ mm}^2$

From statics, $R_1 + R_2 = 80 + 150 = 230$ kN ...(1)

From compatibility, extension of AB = compression of BC and CD

$$\therefore \quad \Delta_1 = \Delta_2 + \Delta_3$$

$$\frac{P_1 L_1}{A_1 E} = \frac{P_2 L_2}{A_2 E} + \frac{P_3 L_3}{A_3 E}$$

$$\frac{R(1500)}{1000} = \frac{(80-R_1)2000}{1500} + \frac{R_2(1500)}{2000}$$

or $R_1 = 0.889\,(80 - R_1) + 0.5\,R_2$

or $R_1 - 0.2647\,R_2 = 37.647$...(2)

From (1) and (2), we get

R_2 = **152.09 kN**

and R_1 = **77.91 kN**

Hence $P_1 = R_1$ = **77.91 kN** (tension)

$P_2 = 80 - R_1$
$= R_2 - 150$
= **2.09 kN** (compression)

$P_3 = R_2$ = **152.09 kN** (compression)

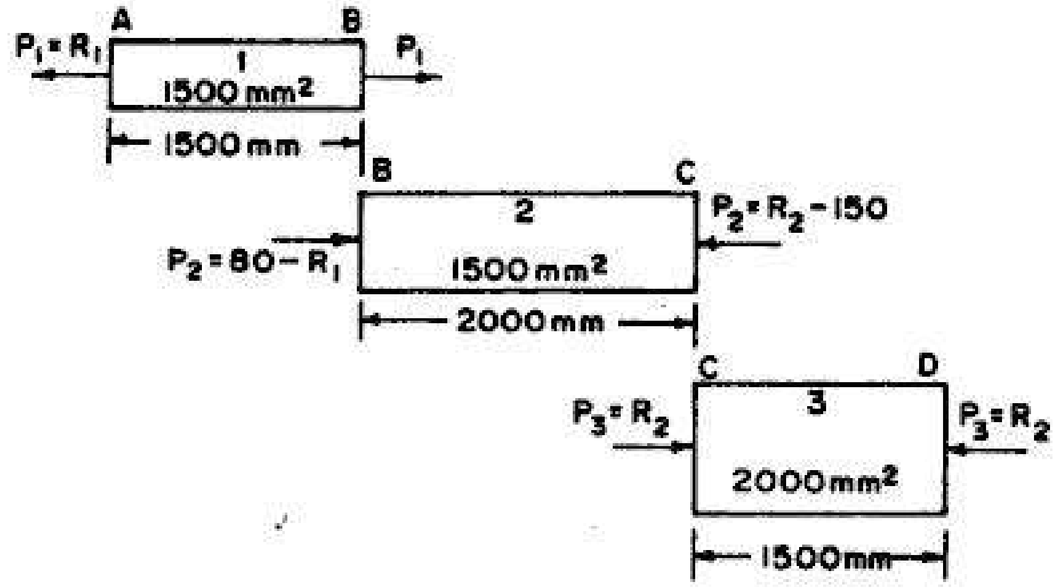


FIG. 2.66

$$\text{Displacement of } B = \Delta_1 = \frac{P_1 L_1}{A_1 E} = \frac{77.91 \times 1500}{1000 \times 200} = \mathbf{0.584 \text{ mm}} \; (\rightarrow)$$

$$\text{Displacement of } C = \Delta_1 + \Delta_2 \; \left(\text{where } \Delta_2 = \frac{2.09 \times 2000}{1500 \times 200} = 0.014 \text{ mm} \leftarrow\right)$$

$$= 0.584 - 0.014 = \mathbf{0.570 \text{ mm}} \; (\rightarrow)$$

Check : $$\text{Displacement of } C = \Delta_3 = \frac{152.09 \times 1500}{2000 \times 200} = \mathbf{0.570} \; (\rightarrow)$$

Example 2.48. *A load of 10 kN is suspended by the ropes shown in Fig. 2.67 (a) and (b). In both the cases, the cross sectional area of the ropes is 100 mm² and the values of E is 200 kN/mm². In case (a), the rope ABC is continuous over a smooth pulley from which the load is suspended. In case (b), the ropes AB and CB are separate ropes joined by a rigid block from which the load is suspended in such a way that both the ropes are stretched by the same amount. Determine, for both the cases, the stresses in the ropes and the deflections of the pulley and the block due to the load.* *(Based on U.L.)*

Solution : Let us use suffix 1 for rope AB and 2 for rope CB. Load = 10 kN ; length of $AB = l_1 = 6$ m; length of $CB = l_2 = 8$ m. Area of rope $= 200 \text{ mm}^2$.

Case (a): Since the load is applied on a smooth pulley, the external load (=10 kN) is shared *equally* by both the ropes.

$\therefore$ P in each rope $= \dfrac{10}{2} = 5$ kN.

(Note that reactions at A and C, each are equal to 5 kN)

Hence stress $\quad p = \dfrac{5 \times 1000}{200} = \mathbf{25\ N/mm^2}$

Also, $\Delta = \dfrac{Pl}{AE} = \dfrac{P(l_1 + l_2)}{AE} = \dfrac{5\,(6000 + 8000)}{100 \times 200} = \mathbf{3.50\ mm}$

Case (b): Load is suspended is such a way that :

$$\Delta_1 = \Delta_2$$

or
$$\frac{p_1 l_1}{E} = \frac{p_2 l_2}{E}$$

or
$$p_1 = p_2 \frac{l_2}{l_1} = p_2 \times \frac{8}{6} = \frac{4}{3} p_2 \qquad ...(1)$$

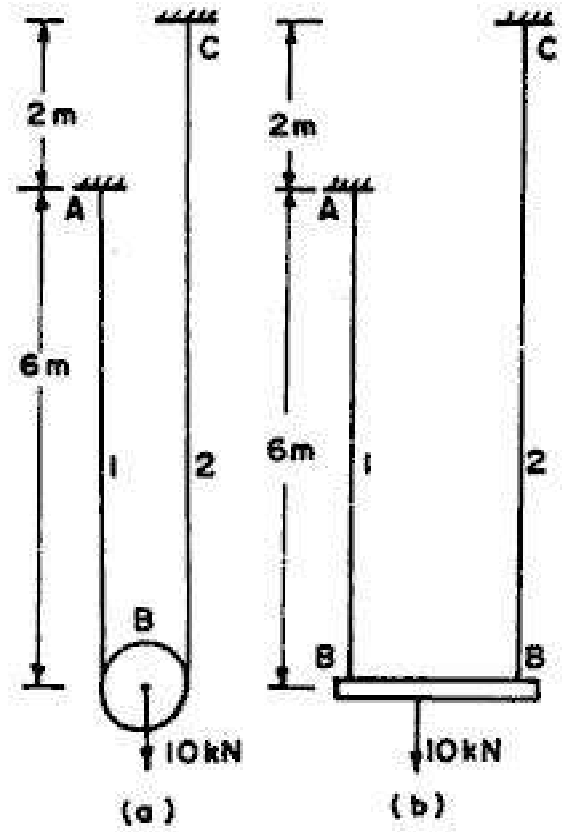


FIG. 2.67.

Also, from statics, $\quad p_1 A_1 + p_2 A_2 = P$

or
$$100\,(p_1 + p_2) = 10 \times 1000$$

or
$$p_1 + p_2 = 100 \qquad ...(2)$$

Substituting the value of p_1 in (2), we get

$$\frac{4}{3} p_2 + p_2 = 100$$

From which $\quad p_2 = \mathbf{42.86\ N/mm^2}$

Hence, $\quad p_1 = 42.86 \times \dfrac{4}{3} = \mathbf{57.14\ N/mm^2}$

Also, $\quad \Delta = \dfrac{p_1 l_1}{E} = \dfrac{57.14 \times 6000}{200 \times 1000} = \mathbf{1.71\ mm}$

Example 2.49. *Two vertical wires are suspended at a distance of 500 mm apart as shown in Fig. 2.68. Their upper ends are firmly secured and their lower ends support a rigid horizontal bar, which carries a load W. The left hand wire has a diameter of 1.6 mm and is made of copper, and the right hand wire has a diameter of 0.8 mm and is made of steel. Both the wires, initially, are 4 m long. Determine : (a) The position of line of action of W, if due to W both the wires extend by the same amount. (b) The slope of the wire if a load of 500 N is hung at the centre of the bar.*

Neglect the weight of the bar and take E for steel $= 2 \times 10^5\ N/mm^2$ *and E for copper* $= 1.2 \times 10^5\ N/mm^2$.

Solution

Let us use suffix c for copper wire and s for steel wire.

$$A_c = \frac{\pi}{4}(1.6)^2 = 2.011 \text{ mm}^2;\ A_s = \frac{\pi}{4}(0.8)^2 = 0.503 \text{ mm}^2$$

(a) Position of W for equal extension

Let W be applied at x mm from the copper wire.

From statics, $P_c + P_s = W$ ...(*i*)

Taking moments about copper wire, $P_s = \dfrac{Wx}{500}$ ...(*ii*)

Also, Taking moments about steel wire, $P_c = \dfrac{W(500-x)}{500}$ ...(*iii*)

Also, $\Delta_c = \Delta_s$

or $$\frac{P_c\, l}{A_c E_c} = \frac{P_s\, l}{A_s E_s}$$

or $$\frac{P_c}{P_s} = \frac{A_c}{A_s}\cdot\frac{E_c}{E_s} = \frac{2.011}{0.503} \times \frac{1.2 \times 10^5}{2 \times 10^5} = 2.399 \qquad ...(iv)$$

From (*ii*) and (*iii*), $\dfrac{P_c}{P_s} = \dfrac{500-x}{x}$ ...(*v*)

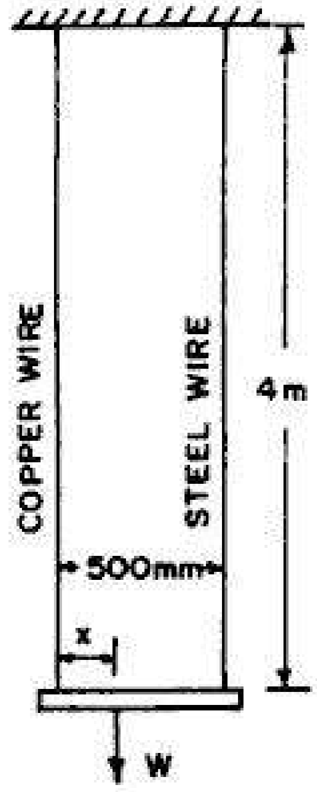


FIG. 2.68.

Hence from (*iv*) and (*v*), we get $\dfrac{500-x}{x} = 2.399$

From which $x = \mathbf{147.11}$ **mm**

(b) Slope of wire if W is hung in the middle of bar

Since W is hung in the middle (*i.e.* $x = 250$ mm), $P_s = P_c = \dfrac{W}{2} = 250$ N

Now $$\Delta_c = \frac{P_c\, l}{A_c E_c} = \frac{250 \times 4000}{2.011 \times 1.2 \times 10^5} = 4.144 \text{ mm}$$

and $$\Delta_s = \frac{P_s\, .\, l}{A_s E_s} = \frac{250 \times 4000}{0.503 \times 2 \times 10^5} = 9.940 \text{ mm}$$

If θ is the slope of the bar, we get

$$\tan\theta = \frac{9.940 - 4.144}{500} = 0.011592$$

From which $\theta = 0.6641° = \mathbf{0°\ 39.85'}$ (clockwise)

Example 2.50. *A rigid bar AB is hinged at A and supported by a bronze rod GD of length 2L and a steel rod FC of length L. A load P is applied at the end B as shown in Fig. 2.69. Calculate the load carried by each rod and the reaction at A. Area of steel rod is equal to 1.5 times the area of bronze rod. The modulus of elasticity for steel is 2.5 times that of bronze.*

Solution :

Let us use suffix s for steel and b for bronze. There are three unknown forces : *(i)* force P_s in steel rod, *(ii)* force P_b in bronze rod and *(iii)* reaction R at A. Hence from statics, we get

$$R + P_s + P_b = P \qquad ...(i)$$

Taking moments about A,

$$P_s \cdot L + P_b\,(3L) = P \cdot (4L)$$

or $$P_s + 3\,P_b = 4\,P \qquad ...(ii)$$

Fig. 2.69 (b) shows the deformation diagram, from which we get

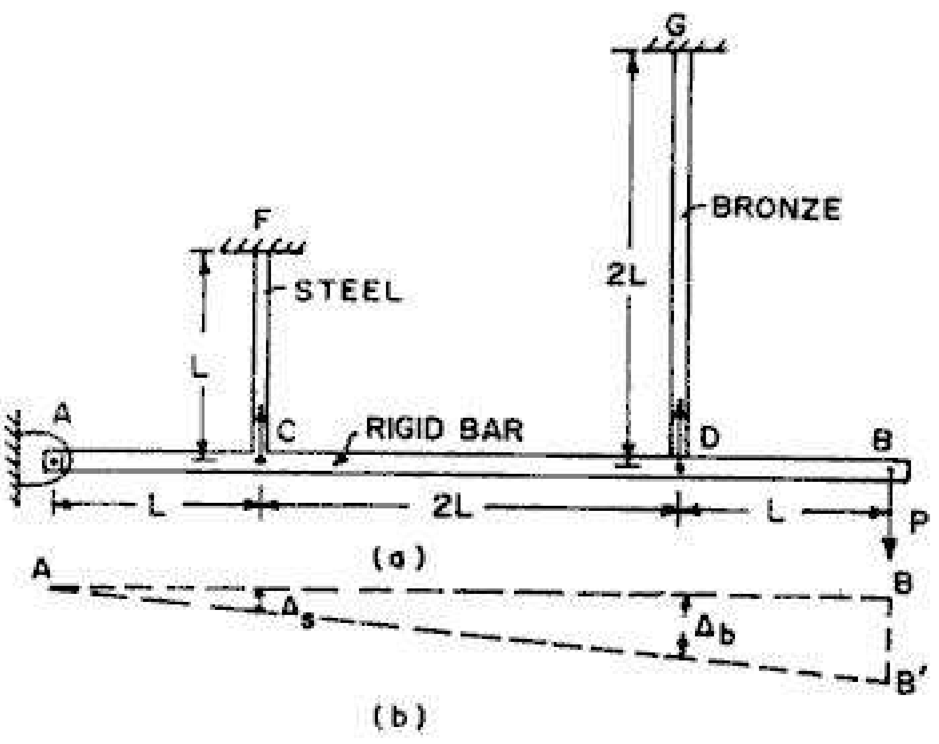


FIG. 2.69

$$\frac{\Delta_b}{\Delta_s} = \frac{3\,L}{L} = 3 \text{ or } \Delta_b = 3\,\Delta_s$$

or $$\frac{P_b\,(2L)}{A_b\,E_b} = 3\,\frac{P_s\,(L)}{A_s\,E_s}$$

or $$P_b = \frac{3}{2}\,P_s \cdot \frac{A_b}{A_s} \cdot \frac{E_b}{E_s}$$

But $$\frac{A_b}{A_s} = \frac{1}{1.5} \text{ and } \frac{E_b}{E_s} = \frac{1}{2.5}$$

$$\therefore \qquad P_b = \frac{3}{2}\,P_s\left(\frac{1}{1.5}\right)\left(\frac{1}{2.5}\right) = \frac{P_s}{2.5} = \frac{2}{5}\,P_s \qquad ...(iii)$$

Substituting in (ii), $P_s + 3\left(\frac{2}{5}P_s\right) = 4\,P$

From which $$P_s = \frac{\mathbf{20}}{\mathbf{11}}\,\boldsymbol{P}$$

Hence $$P_b = \frac{2}{5}\,P_s = \frac{2}{5} \times \frac{20}{11}\,P = \frac{\mathbf{8}}{\mathbf{11}}\,\boldsymbol{P}$$

Substituting the values of P_s and P_b in (i),

$$R = P - (P_s + P_b) = P\left[1 - \frac{20}{11} - \frac{8}{11}\right] = -\frac{17}{11}\,P$$

Hence $$R = \frac{\mathbf{17}}{\mathbf{11}}\,\boldsymbol{P}\ (\downarrow)$$

Example 2.51. *A composite bar ABC, rigidly fixed at upper support at A and hanging 1 mm above the lower support D, is loaded as shown in Fig. 2.70. Determine the reaction at the two supports and stresses in the two sections. Take* $E = 2 \times 10^5\ N/mm^2$. *The diameter of upper portion AB is 12 mm and that of lower portion BC is 16 mm.*

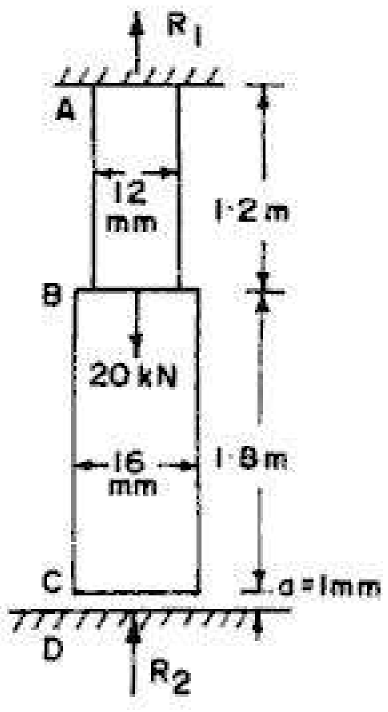


FIG. 2.70

Solution :

Let R_1 be the reaction at the upper support A. Similarly, let R_2 be the reaction at the lower support D when the bar touches it.

Area $A_{AB} = \frac{\pi}{4}\,(12)^2 = 113.1\ \text{mm}^2$ and $A_{BC} = \frac{\pi}{4}\,(16)^2 = 201.1\ \text{mm}^2$

When the bar finally touches the lower support, we have

$$R_1 + R_2 = 20{,}000 \qquad ...(i)$$

Now, force P_{AB} in bar $AB = R_1 = 20000 - R_2$ (tensile)
Also, force P_{BC} in bar $BC = R_2$ (compressive)

$\therefore$ Extension of AB, $$\Delta_{AB} = \frac{(20000 - R_2)\,1200}{113.1 \times 2 \times 10^5} = \frac{20000 - R_2}{18850} \text{ mm}$$

Contraction of BC, $$\Delta_{BC} = \frac{R_2 \times 1800}{201.1 \times 2 \times 10^5} = \frac{R_2}{22344} \text{ mm}$$

In order that end C rests on the lower support to induce a reaction R_2, we have the mpatibility equation :

$$\Delta_{AB} - \Delta_{BC} = a$$

or $$\frac{20000 - R_2}{18850} - \frac{R_2}{22344} = 1$$

or $$1.061 - 5.305 \times 10^{-5} R_2 - 4.475 \times 10^{-5} R_2 = 1$$

From which $R_2 = 6238\,N = \mathbf{6.238\ kN}$

$\therefore$ $R_1 = 20000 - 6238 = 13762\text{ N} = \mathbf{13.762\ kN}$

$\therefore$ Stress in bar $AB = p_{AB} = \dfrac{R_1}{A_{AB}} = \dfrac{13762}{113.1} = \mathbf{121.7\ N/mm^2}$ (tensile)

Stress in bar $BC = p_{BC} = \dfrac{R_2}{A_{BC}} = \dfrac{6238}{201.1} = \mathbf{31\ N/mm^2}$ (comp.)

Example 2.52. *A rigid block ABC, weighing 180 kN is supported by three rods symmetrically placed as shown in Fig. 2.71 (a). Assuming the block to remain horizontal, determine the stress in each rod after a temperature rise of 25° C. The lower ends of the rods are assumed to have been at the same level before the block was attached and the temperature changed. Given :*
Area of steel rod 800 mm²;
Area of bronze rod 1400 mm²;
E for steel 200 kN/mm² ;
E for bronze 80 kN/mm²;
α for steel 12×10^{-6} per °C and
α for bronze 20×10^{-6} per °C.

What will be the stress in each rod if the weight of the block is 120 kN only ?

Solution :

Let us use suffix s for steel rods and b for bronze rod. Fig. 2.71 (b) shows the deformations of the rods at *different stages*. The ends of three rods are at the same level before the increase

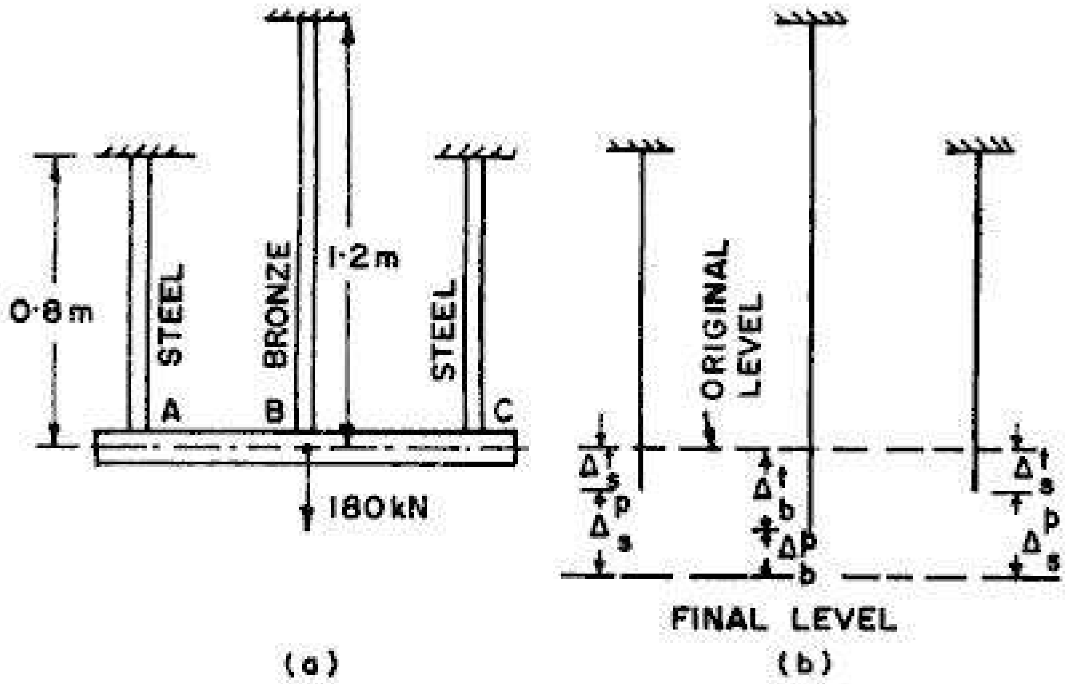


FIG. 2.71

in temperature and attachment of the block. Let the temperature rise, with the block *detached*. Then the ends of the rods will be free to expand and the extensions of steel and bronze bars due to this temperature rise t will be Δ_s^t and Δ_b^t respectively, where $\Delta_b^t > \Delta_s^t$. Now, after the temperature is increased, let us attach the block weighing 160 kN to the three rods. In doing so, the expanded ends of the three rods will be pulled further through load deformations Δ_s^p and Δ_b^p, so that finally all the ends are at the same level. From the study of Fig. 2.71 (*b*), it is clear that the sum of the two deformations of steel rod is equal to the sum of the two deformations of the bronze rod.

$$\therefore \quad \Delta_s^t + \Delta_s^p = \Delta_b^t + \Delta_b^p$$

or
$$L_s \alpha_s t + \frac{P_s L_s}{A_s E_s} = L_b \alpha_b t + \frac{P_b . L_b}{A_b E_b}$$

where P_s is the load carried by each steel rod, and P_b is the load carried by the bronze rod.

Substituting the numerical values, we get

$$800 \times 12 \times 10^{-6} \times 25 + \frac{P_s \times 800}{800 \times 2 \times 10^5} = 1200 \times 20 \times 10^{-6} \times 25 + \frac{P_b\,(1200)}{1400 \times 0.8 \times 10^5}$$

or
$$P_s - 2.143\,P_b = 72000 \qquad ...(1)$$

Also, from statics, $2\,P_s + P_b = 180000$ ...(2)

From (1) and (2), we get $P_b = 6810$ N and $P_s = 86595$ N

Hence the stresses are : $p_b = \dfrac{6810}{1400} = \mathbf{4.86\ N/mm^2}$ (tension)

and
$$p_s = \frac{86595}{800} = \mathbf{108.24\ N/mm^2} \text{ (tension)}$$

(*b*) If the weight of the block is 120 kN only, we have

$$2\,P_s + P_b = 120000 \qquad ...(3)$$

Hence from (1) and (3), we get $P_s = 62260$ N and $P_b = -4520$ N

The minus sign with P_b indicates that the bronze bar will be in compression.

The stresses are : $p_b = \dfrac{4520}{1400} = \mathbf{3.23\ N/mm^2}$ (compression)

and
$$p_s = \frac{62260}{800} = \mathbf{77.83\ N/mm^2} \text{ (tension)}$$

Example 2.53. *A rigid beam ABC is hinged at D and supported by two springs A and B as shows in Fig. 2.72. The beam carries a vertical load W at a point C at a distance of b from the hinged end. The flexibilities (deflection/unit load) of the springs at A and B are d_1 and d_2 respectively. Determine the forces in the springs and also the reaction at the hinged support.*

Solution :

The dotted line in Fig. 2.72 shows the deformed shape of the rigid beam. Let P_A and P_B be the loads carried by springs A and B.

For spring A, extension $\Delta_A = P_A \,.\, d_1$

For spring B, extension $\Delta_B = P_B \,.\, d_2$

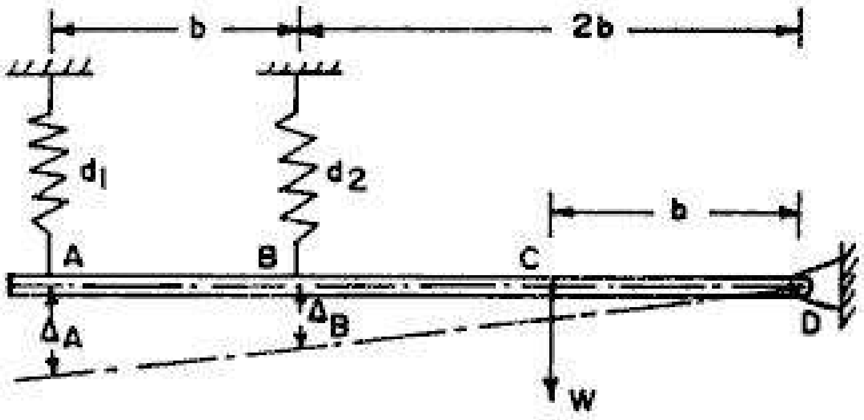


FIG. 2.72

Now, from compatibility, $\dfrac{\Delta_A}{b+2b} = \dfrac{\Delta_B}{2b}$

or $\qquad \Delta_A = \dfrac{3}{2}\Delta_B$

Substituting the values of Δ_A and Δ_B,

$$P_A \,.\, d_1 = \frac{3}{2} P_B \,.\, d_2$$

or $$P_A = \frac{3}{2}\frac{d_2}{d_1} \,.\, P_B \qquad ...(1)$$

Also, by taking moments about hinge D, we get

$$P_A \times 3b + P_B \times 2b = W \,.\, b$$

or $$3P_A + 2P_B = W \qquad ...(2)$$

Substituting the value of P_A from (1) in (2), we get

$$3\left(\frac{3}{2}\frac{d_2}{d_1} P_B\right) + 2P_B = W$$

From which $$P_B = \frac{2d_1}{4d_1 + 9d_2} W$$

and $$P_A = \frac{3d_2}{4d_1 + 9d_2} W$$

Example 2.54. *Three wires, each of 6 mm diameter are used to lift a load W=6900 N, as shown in Fig. 2.73. Their unstressed lengths are 19.994 m, 19.997 m and 20.000 m.*

(*a*) *What stress exists in the longest wire ?*

(*b*) *Determine the stress in the shortest wire if W=2200 N.*

Take $\qquad E = 2 \times 10^5\ N/mm^2.$

Solution :

Area of cross section of each wire $= \dfrac{\pi}{4}(6)^2 = 28.27\ \text{mm}^2$

The difference in length between the shortest wire (first wire) and the second wire $= 19.997 - 19.994 = 0.003$ m. Hence when the stretching of the shortest wire reaches value of 0.003 m, load taken by it is

$$P_1 = \frac{\Delta AE}{L} = \frac{0.003 \times 28.27 \times 2 \times 10^5}{19.994} = 848.5\ \text{N} < 6900\ \text{N}.$$

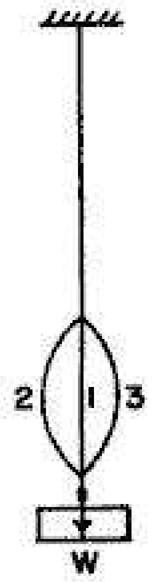


FIG. 2.73

Hence the second wire will also be stretched. The difference in the length of second wire and the third (i.e. longest) wire

$= 20.000 - 19.997 = 0.003$ m. Hence when the second wire gets stretched by an amount 0.003 m, the first wire gets a total stretch of $0.003 + 0.003 = 0.006$ m.

Thus, $\Delta_1 = 0.006$ m and $\Delta_2 = 0.003$ m.

$$\therefore \quad P_1 = \frac{0.006 \times 28.27 \times 2 \times 10^5}{19.994} = 1696.7 \text{ N}$$

and

$$P_2 = \frac{0.003 \times 28.27 \times 2 \times 10^5}{19.997} = 848.2 \text{ N}$$

$\therefore$ Total $P = P_1 + P_2 = 1696.7 + 848.2 = 2544.9 \text{ N} < 6900$ N.

Hence the third wire will also get stretched.

Now, at the equilibrium position, we get the following relations :

$$P_1 + P_2 + P_3 = P = 6900 \quad ...(i)$$

$$\Delta_2 = \Delta_3 + 0.003 \quad ...(ii)$$

and

$$\Delta_1 = \Delta_3 + 0.006 \quad ...(iii)$$

From (*ii*), we get $\dfrac{P_2 L_2}{AE} = \dfrac{P_3 L_3}{AE} + 0.003$

or

$$\frac{P_2 \times 19.997}{28.27 \times 2 \times 10^5} = \frac{P_3 \times 20.000}{28.27 \times 2 \times 10^5} + 0.003$$

or

$$P_2 \approx P_3 + 848.23 \quad ...(iv)$$

Similarly, from (*iii*), $\dfrac{P_1 L_1}{AE} = \dfrac{P_3 L_3}{AE} + 0.006$

or

$$\frac{P_1 \times 19.994}{28.27 \times 2 \times 10^5} = \frac{P_3 \times 20.000}{28.27 \times 2 \times 10^5} + 0.006$$

$$P_1 \approx P_3 + 1696.71 \quad ...(v)$$

Substituting these values of P_1 and P_2 in (*i*), we get

$$(P_3 + 1696.71) + (P_3 + 848.23) + P_3 = 6900$$

From which $P_3 = 4355.1$ kN. Hence $P_3 = \dfrac{4355.1}{28.27} = \mathbf{154.05\ N/mm^2}$

(*b*) When $W = 2200$ N, then as the second wire gets stretched by an amount of 0.003 m,

$$P_1 + P_2 = 2544.9\,N > 2200\,N$$

Hence the second wire does not get fully stretched. The final equilibrium equations are:

$$P_1 + P_2 = 2200 \text{ N} \quad ...(a)$$

and

$$\Delta_1 = \Delta_2 + 0.003 \text{ m} \quad ...(b)$$

From Eq. (*b*), $\dfrac{P_1 \times 19.994}{28.27 \times 2 \times 10^5} = \dfrac{P_2 \times 19.997}{28.27 \times 2 \times 10^5} + 0.003$

or $$P_2 = P_1 - 861.1 \qquad ...(c)$$

Substituting in Eq.(a), we get $P_1 + P_1 - 861.1 = 2200$

From which $$P_1 = 1530.6 \text{ N}$$

$$\therefore \qquad p_1 = \frac{1530.6}{28.27} = \mathbf{54.14\ N/mm^2}$$

2.22. STRESSES DUE TO LACK OF FIT OR PRE-STRAIN

A framed structure consists of an assemblage of a number of individual members; these individual members are to be of definite length as per the requirements of the geometry of structure. If, however, any one member of the structure is not of correct length, stresses will be set up in all the members of the structure when that member is fitted into the structure. Such a stress is known as *stress due to lack of fit*, and the strain set up in the member is known as *pre-strain*.

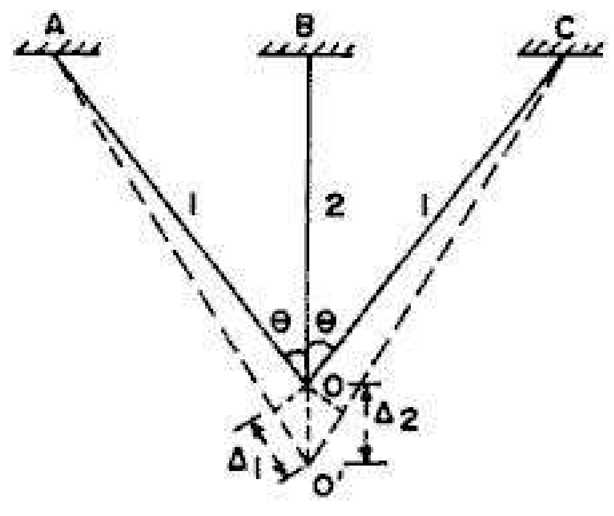


FIG. 2.74

To analyse such cases, let us take the case of a simple structure consisting of an assemblage of three bars shown in Fig. 2.74, in which the central member BO is excess in length by an amount λ (say). When this member is fitted, compressive stress is set up in OB while tensile stresses are setup in OA and OC. The dotted lines show the deformed shape of the structure when the central member having a lack of fit (λ) is fitted.

Let us use suffix 1 for outer bars and 2 for central bar. From statics, we have :

$$2P_1 \cos\theta = P_2 \qquad ...(i)$$

From compatibility, we have $$\Delta_1 = \Delta_2 \cos\theta \qquad ...(ii)$$

where $$\Delta_1 = \frac{P_1 L_1}{A_1 E_1} = \frac{P_1 L_2}{A_1 E_1 \cos\theta}$$

and $$\Delta_2 = \lambda - \frac{P_2 L_2}{A_2 E_2}$$

Hence from (ii), $$\frac{P_1 L_2}{A_1 E_1 \cos\theta} = \left(\lambda - \frac{P_2 L_2}{A_2 E_2}\right)\cos\theta$$

From which $$P_1 = \frac{A_1 E_1}{L_2}\left(\lambda - \frac{P_2 L_2}{A_2 E_2}\right)\cos^2\theta \qquad ...(iii)$$

Substituting in (i), we get $$2\frac{A_1 E_1}{L_2}\left(\lambda - \frac{P_2 L_2}{A_2 E_2}\right)\cos^3\theta = P_2$$

From which $$P_2 = \frac{2A_1 E_1 \lambda \cos^3\theta}{L_2\left(1 + \frac{2A_1 E_1}{A_2 E_2}\cos^3\theta\right)} \qquad ...(2.35\ a)$$

Substituting this value of P_2 Eq. (*i*), we get

$$P_1 = \frac{P_2}{2\cos\theta} = \frac{A_1 E_1 \lambda \cos^2\theta}{L_2\left(1 + \frac{2A_1E_1}{A_2E_2}\cos^3\theta\right)} \quad ...(2.35\ b)$$

If, however, all the bars have the same area of cross-section (A) and are of the same material, we have :

$$P_2 = \frac{2AE\lambda\cos^3\theta}{L\,(1 + 2\cos^3\theta)} \quad ...(2.35\ c)$$

and

$$P_1 = \frac{AE\lambda\cos^2\theta}{L\,(1 + 2\cos^3\theta)} \quad ...(2.35\ d)$$

where $L = L_2 =$ geometrical length of the central bar.

Temperature Stresses : If instead of a lack of fit λ, the central bar of the system is heated through a temperature rise t, the temperature stresses set up in the members can be found in the same manner as explained above, by substituting $\lambda = L\,\alpha\,t$.

Example 2.55. *The central member of the system shown in Fig. 2.74 is 1.5 mm excess in length. Determine the stresses set up in all the members, if all the members are of the same material and have diameter of 10 mm. Take $\theta = 30°$ and $E = 200$ kN/mm². The geometrical length of central member is 900 mm.*

Solution :

All the bars have the same area of cross-section, and are of the same material. Hence from Eq. 2.35 (*c*)

$$P_2 = \frac{2AE\lambda\cos^3\theta}{L\,(1 + 2\cos^3\theta)}$$

Here $A = \frac{\pi}{4}(10)^2 = 78.54\ \text{mm}^2$; $\lambda = 1.5$ mm , $L = 300$ mm and $\theta = 30°$

$$\therefore \quad P_2 = \frac{2 \times 78.54 \times 200 \times 1.5 \times \cos^3 30°}{900\,(1 + 2\cos^3 30)} = 14.793\ \text{kN}$$

$$\text{Hence} \quad p_2 = \frac{14.793 \times 10^3}{78.54} = \mathbf{188.4\ N/mm^2}\ \text{(comp.)}$$

$$\text{Also,} \quad P_1 = \frac{P_2}{2\cos\theta} = \frac{14.793}{2\cos 30°} = 8.54\ \text{kN}$$

$$\therefore \quad p_1 = \frac{8.54 \times 10^3}{78.54} = \mathbf{108.7\ N/mm^2}\ \text{(tension)}$$

Example 2.56. *In the system of bars shown in Fig. 2.74, the temperature of central bar is reduced by 80° F. Compute the stresses set up in the three rods if the outer rods are 150 mm² in area and of steel, and the central rod is of brass having area of cross-section of 200 mm². The length of central bar is 800 mm and angle $\theta = 30°$.*

Take $E_s = 2 \times 10^5$ N/mm², $E_b = 1 \times 10^5$ N/mm² and $\alpha = 10 \times 10^{-6}$ per F.

Solution :

Decrease in length of central bar, $\lambda = L\alpha t = 800 \times 10 \times 10^{-6} \times 80 = 0.64$ mm

Now, from Eq. 2.35 (*a*) $P_2 = \dfrac{2A_1E_1\lambda\cos^3\theta}{L_2\left(1 + \dfrac{2A_1E_1}{A_2E_2}\cos^3\theta\right)}$

Here, $A_1 = 150\ \text{mm}^2$; $A_2 = 200\ \text{mm}^2$; $E_1 = 2 \times 10^5$ and $E_2 = 1 \times 10^5\ \text{N/mm}^2$

$$\therefore \quad P_2 = \frac{2 \times 150 \times 2 \times 10^5 \times 0.64\cos^3 30^\circ}{800\left(1 + \dfrac{2 \times 150 \times 2 \times 10^5}{200 \times 1 \times 10^5}\cos^3 30^\circ\right)} = 10574\ \text{N (tension)}$$

and $$P_1 = \frac{P_2}{2\cos 30} = 6105\ \text{N (Compression)}$$

Hence $$p_1 = \frac{6105}{150} = \mathbf{40.7\ N/mm^2}\ \text{(comp.)}$$

and $$p_2 = \frac{10574}{200} = \mathbf{52.87\ N/mm^2}\ \text{(tensile)}$$

2.33. STRESS DUE TO SHRINKING ON

Some times, tyre of steel (or other metal) is shrunk on a wheel. In such cases, the diameter of the tyre is kept slightly smaller than that of wheel so that it does not come out easily. While putting on the tyre to the wheel, the tyre is heated up so that it expands. On cooling, the tyre shrinks and gets fitted tightly on to the wheel.

Let D be the diameter of the wheel and d be the diameter of the tyre ($D > d$). Let us increase the temperature of the tyre by t° so that its diameter increases from d to D. When the tyre is slipped on to the wheel and the temperature falls, the metal tyre will try to come to its original diameter, and in doing so hoop stress (tensile) will be set up in it.

Tensile strain induced in tyre $= e = \dfrac{\pi D - \pi d}{\pi d} = \dfrac{D - d}{d}$

$\therefore$ Circumferential tensile stress or hoop stress induced $= eE = \left(\dfrac{D-d}{d}\right)E$ ...(2.36)

Example 2.57. *A thin tyre of steel is to be shrunk on to a rigid wheel of 900 mm diameter. Compute (a) internal diameter of the tyre if the hoop stress is limited to 120 N/mm², and (b) the least temperature to which the tyre must be heated above that of the wheel before it could be slipped on. For the tyre, take $\alpha = 12 \times 10^{-6}$ per °C and $E = 2 \times 10^5\ N/mm^2$.*

Solution :

Hoop stress $$p = \frac{E(D-d)}{d} = 120\ \text{(given)}$$

$$\therefore \quad \frac{D-d}{d} = \frac{120}{2 \times 10^5} = 6 \times 10^{-4}$$

or $$\frac{D}{d} = (1 + 6 \times 10^{-4})$$

$\therefore$ $$\frac{d}{D} = (1 + 6 \times 10^{-4})^{-1} \approx (1 - 6 \times 10^{-4})$$

$\therefore$ $$d = 900\,(1 - 6 \times 10^{-4}) = 900 - 0.54 = \mathbf{899.46\ mm}$$

Also, $$\pi D = \pi d\,(1 + \alpha t) \text{ or } \alpha t = \frac{D - d}{d} = 6 \times 10^{-4}$$

$\therefore$ $$t = \frac{6 \times 10^{-4}}{12 \times 10^{-6}} = \mathbf{50°C}$$

2.24. ADDITIONAL ILLUSTRATIVE EXAMPLES*

Some additional illustrative examples of advanced nature, extremely useful for *competitive examinations* are given here, with short solutions.

Example 2.58. *Fig. 2.75 shows a deep well pump rod operated by a crank. The pump rod made of steel having specific weight of 78 kN/m³ has a diameter of 15 mm and length of 100 m. The resistance encountered by the piston during the downstroke is 1 kN and during the upstroke is 10 kN. Considering only the resistance forces and the weight of the rod, determine the maximum tensile and compressive stresses in the pump rod.*

Solution :

(*i*) During down stroke, the resistance of piston creates a compressive force (C) of 1 kN, uniform throughout the length of the rod.

(*ii*) During upstroke, the resistance of piston creates a tensile force (T) of 10 kN, uniform throughout the length of the rod.

(*iii*) The weight of the rod produces a tensile force which varies from zero at the lower end to a maximum weight force W at the upper end.

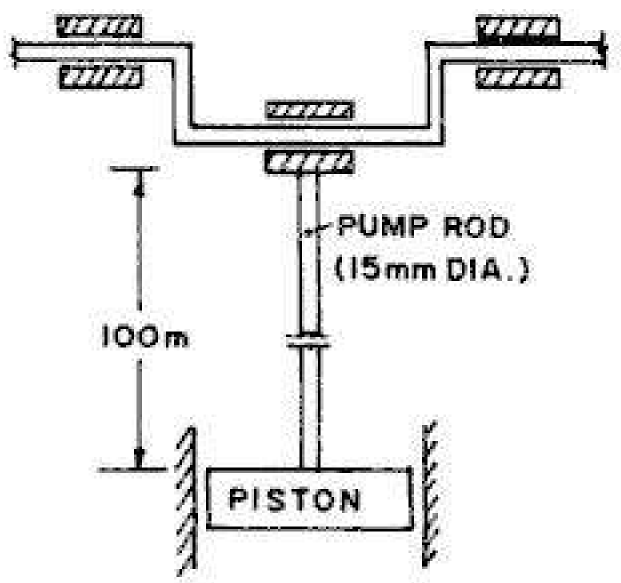


FIG. 2.75

Area of cross-section of piston rod, $A = \frac{\pi}{4}(15)^2 = 176.71\ \text{mm}^2$

Max. weight force $W = \gamma L A = 78 \times 100 \times (176.71 \times 10^{-6}) = 1.378$ kN

Maximum tensile force occurs at the upper end of piston rod, during the upstroke, and its value is equal to $(T + W) = 10 + 1.378 = 11.378$ kN

$\therefore$ Max. tensile stress, $p_t = \dfrac{11.378 \times 10^3}{176.71} = \mathbf{64.38\ N/mm^2}$

Maximum compressive force occurs at the lower end of piston rod, during the down stroke, its value being equal to $C = 1$ kN

$\therefore$ Max compressive stress, $p_c = \dfrac{1 \times 10^3}{176.71} = \mathbf{5.66\ N/mm^2}$

* CAN BE LEFT OUT BY JUNIOR STUDENTS.

Example 2.59. *A steel bar of uniform cross-sectional area is loaded as shown in Fig. 2.76. Find the magnitude of force P_1 so that the lower end D does not move vertically when the load are applied.*

Solution :

Given $\Delta_D = 0$

But $\Delta_D = \Sigma \frac{PL}{AE} = \frac{1}{AE} \Sigma PL$

In order that point D does not move vertically, $P_1 > (P_2 + P_3)$, so that AB is subjected to compressive load. Evidently, AB is subjected to a compressive force equal to R ($= P_1 - 10$), BC will be subjected to a tensile force equal to $10 + 6 = 16$ kN ($= P_1 - R$) while CD is subjected to a tensile load of 6 kN.

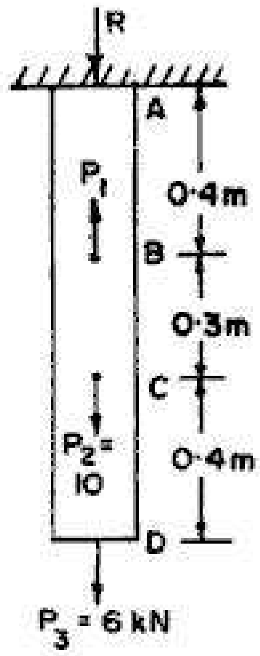


FIG. 2.76

$$\therefore \quad \Delta_D = \frac{1}{AE}\left[-(P_1 - 16) \times 0.4 + 16 \times 0.3 + 6 \times 0.4\right] = 0$$

or $\quad 0.4\,(P_1 - 16) = 4.8 + 2.4 = 7.2$

From which $\quad P_1 = \mathbf{34\ kN}$

Example 2.60. *A copper bar AB, carrying a tensile load of 400 kN hangs from a pin supported by two steel pillars, as shown in Fig. 2.77. The copper bar has a diameter of 100 mm and each steel pillar has a diameter of 50 mm. Taking E for steel = 200 kN/mm² and E for copper as 105 kN/mm², determine the vertical displacement of point B.*

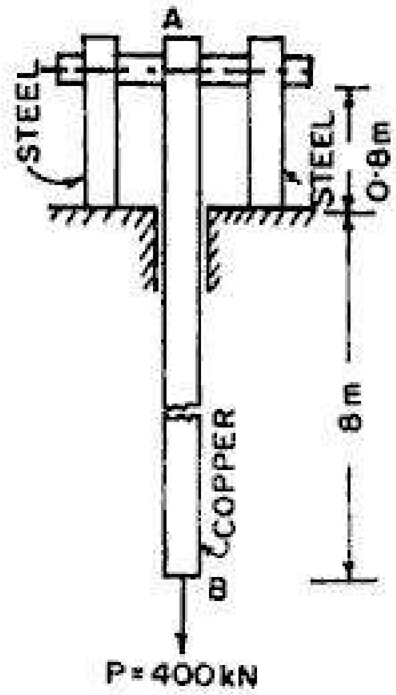


FIG. 2.77

Solution :

Area of cross section of copper bar $= \frac{\pi}{4}(100)^2 \approx 7854\ \text{mm}^2$

Area of cross-section of each steel pillar $= \frac{\pi}{4}(50)^2 = 1963.5\ \text{mm}^2$

Tensile force in copper bar = 400 kN, while compressive force in each steel pillar = 400/2 = 200 kN.

Now Δ_B= Extension of copper bar + compression of steel pillars.

$$\therefore \quad \Delta_B = \Delta_C + \Delta_S = \frac{400 \times 8800}{7854 \times 105} + \frac{200 \times 800}{1963.5 \times 200}$$

$$= 4.268 + 0.407 = \mathbf{4.675\ mm}$$

Example 2.61. *A steel rod, having a diameter of 40 mm is loaded as shown in Fig. 2.78. Taking E = 200 kN/mm², determine (a) the deformation of the free end (b) the distance x from the left hand support to a point at which the deformation is zero.*

Solution :

$$A = \frac{\pi}{4}(40)^2 = 1256.6\ \text{mm}^2$$

Reaction $R = 12 + 10 - 10 = 12$ kN ($\leftarrow$)

A little consideration will show that portion AB is subjected to a tensile load $= R$ $(= 12 \text{ kN})$, portion BC is subjected to a load $= 10 - 10 = 0$ (or $= R - 12 = 0$), while CD is subjected to a compressive load of 10 kN.

$$\Delta_D = \Sigma \frac{PL}{AE} = \frac{1}{AE}\left[P_{AB} L_{AB} + P_{BC} . L_{BC} + P_{CD} . L_{CD}\right]$$

$$= \frac{1}{1256.6 \times 200}\left[12 \times 1000 + 0 - 10 \times 200\right] = \mathbf{-0.0318 \text{ mm}}$$

Minus sign shows that point D moves to the left.

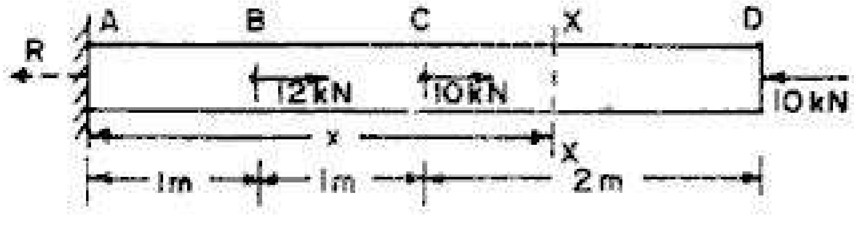


FIG. 2.78

Again $\Delta_x = \frac{1}{AE}[P_{AB} L_{AB} + P_{BC} L_{BC} + P_{CD}(x - 2000)] = 0$

$\therefore \quad 12 \times 1000 + 0 - 10(x - 2000) = 0$

or $\quad x - 2000 = 1200$

which gives $\quad x = 3200 \text{ mm} = \mathbf{3.2 \text{ m}}$

Example 2.62. *A sample of a metal is tested in tension, and it is found that there is a strain of 0.0080 at a corresponding stress of 450 N/mm². On the removal of the load a permanent strain of 0.0015 is found to be present. Compute the value of modulus of elasticity for the metal.*

Solution :

Fig. 2.79 shows the loading and unloading diagram.

Elastic recovery $= 0.0080 - 0.0015 = 0.0065$

Stress $= 450 \text{ N/mm}^2$

$$\therefore \quad E = \frac{\text{Stress}}{\text{elastic recovery}} = \frac{450}{0.0065} = \mathbf{0.692 \times 10^5 \text{ N/mm}^2}$$

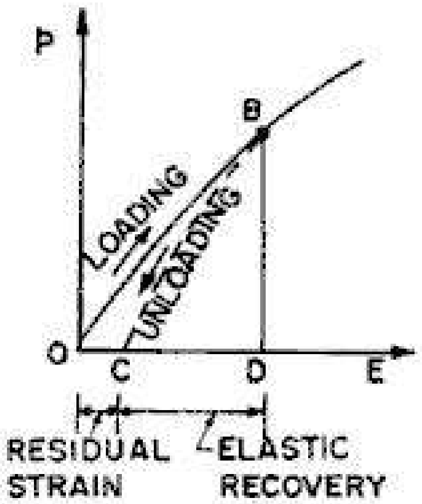


FIG. 2.79

Example 2.63. *A pile of uniform section is embedded in soil by a depth H. The pile supports a structural load P at its top which is transferred to the soil entirely by friction as shown in Fig. 2.80 (a). The variation of friction (f) along the depth of the pile is given by $f_y = k y^2$, where y is the elevation above the bottom of the pile. Determine the total shortening of the pile.*

Solution :

The variation of frictional resistance (f) along the depth is shown in Fig. 2.80 (b).

Total frictional resistance,

$$F = \int_0^H f_y . dy = \int_0^H k y^2 . dy = \frac{k H^3}{3}$$

$$\therefore \quad F = P = \frac{k H^3}{3}, \text{ from which } k = \frac{3P}{H^3} \quad ...(i)$$

The compressive force on pile varies with the depth. At any height y above the bottom of the pile, the total compressive force will be equal to the frictional resistance (F_y) of clay for the bottom height y.

$\therefore$ Total compression,

$$P_{cy} = \int_0^y f_y \, . \, dy = \int_0^y k y^2 \, . \, dy = \frac{k y^3}{3}$$

or
$$P_{cy} = \frac{3P}{H^3} \cdot \frac{y^3}{3} = \frac{P y^3}{H^3} \qquad ...(ii)$$

The variation of P_{cy} along the depth of the pile is shown in Fig. 2.80 (c),

Now, shortening of small length dy of the pile

$$= \frac{P_{cy} \, . \, dy}{AE}$$

$$\therefore \text{ Total shortening of pile} = \sum_{y=H}^{y=0} \frac{P_{cy} \, dy}{AE} = \int_0^H \frac{P y^3}{H^3} \cdot \frac{dy}{AE}$$

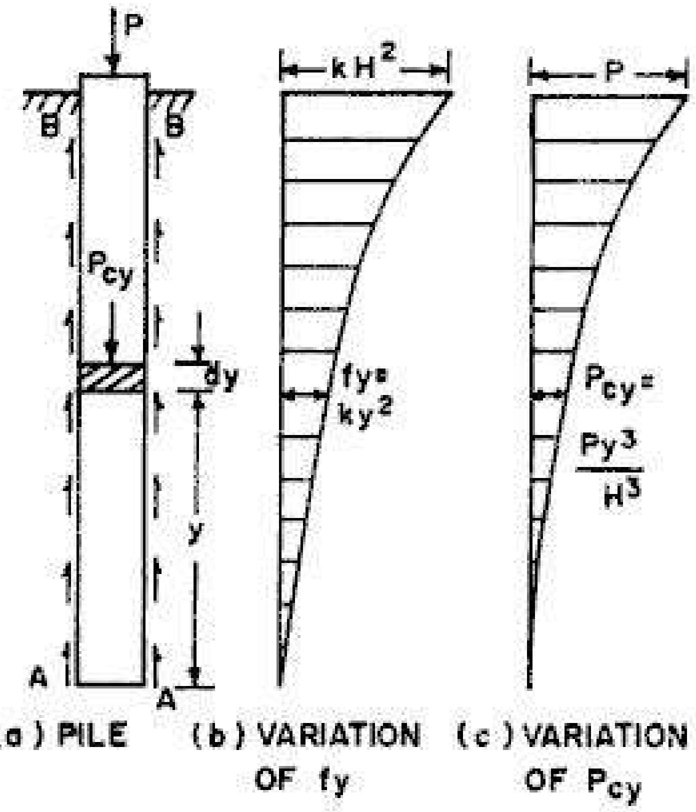


FIG. 2.80

$$= \frac{P}{AEH^3} \int_0^H y^3 \, dy = \frac{P}{AEH^3} \left(\frac{H^4}{4} \right) = \frac{\boldsymbol{PH}}{\boldsymbol{4AE}}$$

Note: If the pile were a point bearing pile (*i.e.* transferring the load to the solid strata at the end), the shortening of the pile would have been equal to $\frac{PH}{AE}$.

Example 2.64. *Fig. 2.81 shows a bimetallic thermometer, made of a tungsten bar AB and a magnesium bar CD the upper ends of which are attached to a pointer BDP. Derive an expression for the vertical displacement of the pointer in terms of a temperature increase t and the dimension shown.*

Solution :

The coefficient of thermal expansion of magnesium is more than that of tungsten. Hence the pointer P will move up (*i.e.* +ve vertical displacement) for rise of temperature and vice-versa.

Now $\qquad \Delta_B = L_1 \alpha_T t$ and $\Delta_D = L_1 \alpha_M t$

$$\therefore \qquad (\Delta_D - \Delta_B) = L_1 t \, (\alpha_M - \alpha_T)$$

Fig. 2.81 (b) shows the displacement diagram, from which :

$$\Delta_P = \Delta_B + \frac{\Delta_D - \Delta_B}{L_1} (L_2 + L_3)$$

$$= L_1 \alpha_T t + \frac{L_1 \alpha_M t - L_1 \alpha_T t}{L_2} (L_2 + L_3)$$

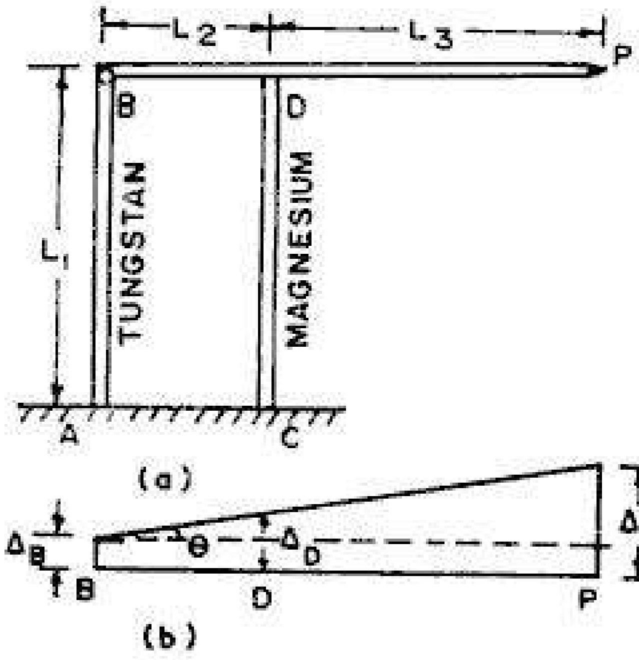


FIG. 2.81

or $$\Delta_P = L_1 \alpha_T t + L_1 t\,(\alpha_M - \alpha_T)\left(1 + \frac{L_3}{L_2}\right)$$

or $$\Delta_P = L_1 t\left[\alpha_M + (\alpha_M - \alpha_T)\frac{L_3}{L_2}\right]$$

Example 2.65. *A solid cylinder of steel is placed inside a copper tube. The assembly is compressed between rigid plates by forces P. Find the value of increase in temperature so that all the load is carried by the copper tube.*

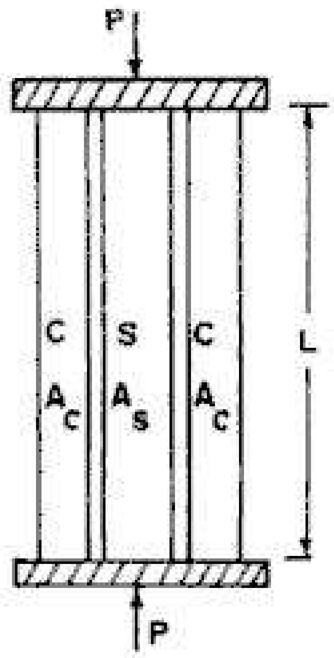


FIG. 2.82

Solution :

Let the increase in temperature be t. Due to this increase, copper will expand more than steel.

Difference between the two expansions $= \Delta_t = L\,(\alpha_c - \alpha_s)\,t$

If the load P is to be borne entirely by the copper tube, the shortening (Δ_c) of the copper tube due to P should be equal to Δ_t

$$\therefore \quad \Delta_c = \Delta_t$$

or $$\frac{P\,L}{A_c\,E_c} = L\,(\alpha_c - \alpha_s)\,t$$

or $$t = \frac{P}{A_c\,E_c\,(\alpha_c - \alpha_s)}$$

Example 2.66. *A steel bar AB of length L is fixed at the ends and is subjected to a non-uniform increase in temperature represented by the expression* $t_x = t\,.\,\frac{x^2}{L^2}$ *, as shown in Fig. 2.83. Determine the value of compressive stress set up in the bar.*

Solution :

Consider a short length dx, at a distance x from end A.

Temperature $\quad t_x = t\,\frac{x^2}{L^2}$

$\therefore$ Increase in length of dx due to temp. rise

$= \Delta_{tx} = dx\,.\,\alpha\,t_x = \alpha\,t\,\frac{x^2}{L^2}\,dx$

Hence total increase in length $= \Delta_t$

$$= \Sigma\Delta_{tx} = \int_0^L \alpha t\,\frac{x^2}{L^2}\,dx = \frac{\alpha t}{L^2}\,.\,\frac{L^3}{3} = \frac{\alpha t L}{3}$$

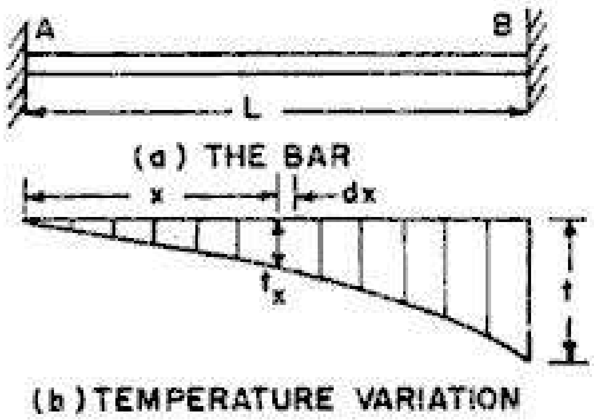


FIG. 2.83

Since this increase in length is restricted by end supports, compressive stress will be set up in the bar, the value of which is given by.

$$p_c = \Delta_t\,.\,\frac{E}{L} = \frac{\alpha\,t\,L}{3}\left(\frac{E}{L}\right) = \frac{E\,\alpha\,t}{3}$$

Example 2.67. *Three steel bars A, B and C, having the same axial rigidity EA support a horizontal rigid beam ABC as shown in Fig. 2.84. Determine the distance a between bars A and B in order that the rigid beam will remain horizontal when a load P is applied at its mid-point.*

***Solution* :**

Let us use suffix 1, 2 and 3 for bars A, B and C respectively.

From statics, $P_1 + P_2 + P_3 = P$ ...(1)

Taking moments about A,

$$P_2 . a + P_3 L = P . \frac{L}{2} \quad ...(2)$$

Since the beam ABC remains horizontal,

$$\Delta_1 = \Delta_2 = \Delta_3$$

or
$$\frac{P_1 (2b)}{AE} = \frac{P_2 (1.5\,b)}{AE} = \frac{P_3 . b}{AE}$$

or
$$2 P_1 = \frac{3}{2} P_2 = P_3 \quad ...(3)$$

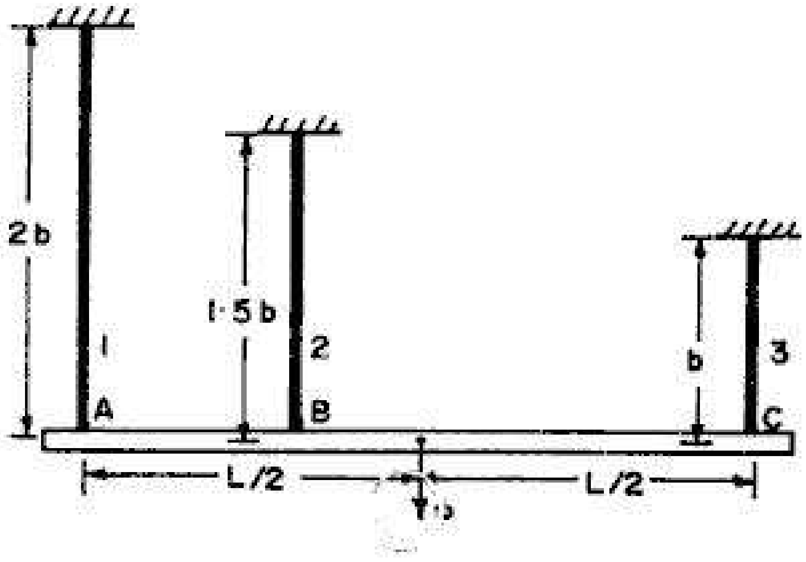


FIG. 2.84

Substituting in (1), we get

$$P_1 + \frac{4}{3} P_1 + 2 P_1 = P \text{ , from which } P_1 = \frac{3}{13} P \quad ...(4)$$

Also, from (2), $\left(\frac{4}{3} P_1\right) a + 2 P_1 L = P . \frac{L}{2}$

or
$$\left(\frac{4}{3} a + 2L\right) P_1 = P . \frac{L}{2}$$

or
$$\left(\frac{4}{3} a + 2L\right) \frac{3}{13} P = \frac{PL}{2}$$

From which we get $\quad a = \dfrac{L}{8}$

Example 2.68. *Two rigid bars AB and CD are connected by linear elastic springs of stiffness k and are supported at A and D by hinged supports as shown in Fig. 2.85 (a). When no loads are acting, the bars are horizontal and the springs are unstressed. Determine the vertical deflection at point C when the point load P is applied.*

***Solution* :**

Let P_1 be the force in spring 1 at C and P_2 be the force in spring 2. Fig. 2.85 (b) shows the deformation diagram while Fig. 2.85 (c) shows the free body diagrams for bars AB and CD. A careful study of these diagrams will reveal that spring 1 (at C) carries a tensile force (P_1) and spring 2 (at B) carries a compressive force P_2.

Taking moments about A (Fig. 2.85 c), we get $P_1 = 3 P_2$

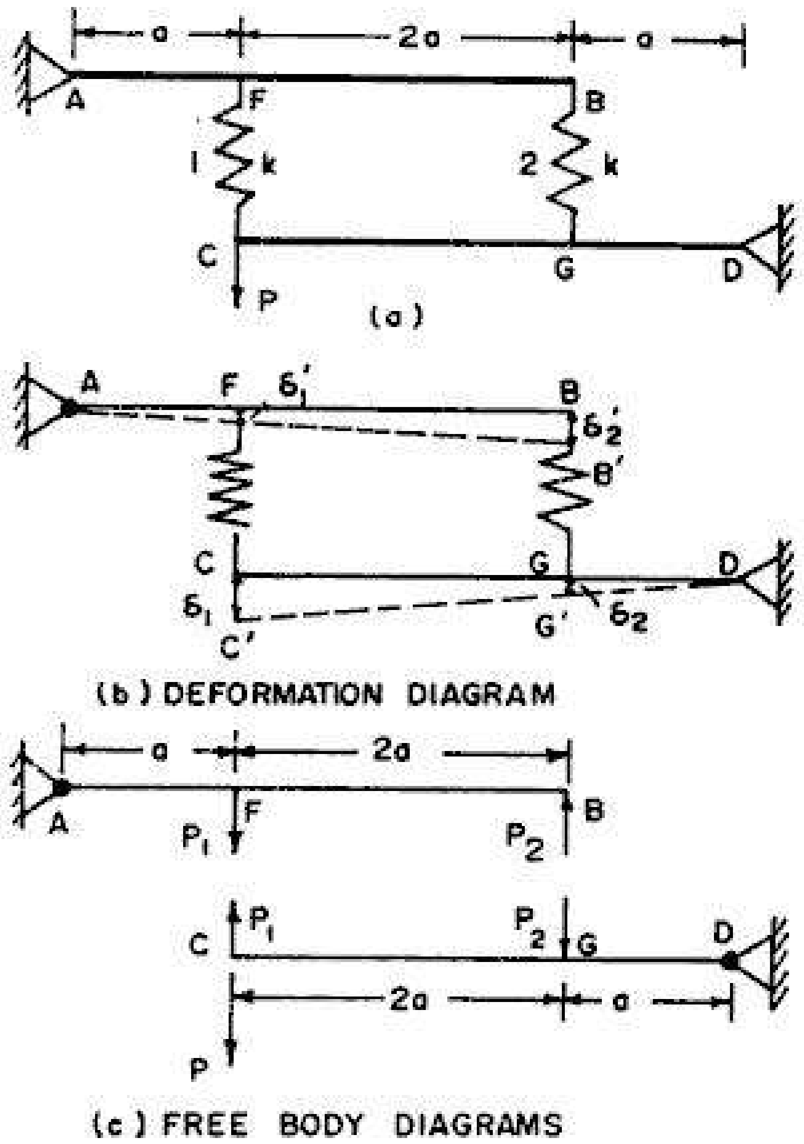


FIG. 2.85

Taking moments about D (Fig. 2.85 c), we get $(P_1 - P)\,3a = P_2\,a$

or $(P_1 - P)\,3 = \frac{P_1}{3}$, from which $P_1 = \frac{9}{8}P$ and $P_2 = \frac{3}{8}P$

Net deformation of spring $1 = \delta_1 - \delta_1' = \Delta_1 = \frac{P_1}{k}$ (extension)

Net deformation of spring $2 = \delta_2' - \delta_2 = \Delta_2 = \frac{P_2}{k}$ (compression)

But $\frac{\delta_1}{\delta_2} = \frac{3a}{a}$, from which $\delta_1 = 3\,\delta_2$ ...(i)

Also, $\frac{\delta_2'}{\delta_1'} = \frac{3a}{a}$, from which $\delta_2' = 3\,\delta_1'$ ...(ii)

Now $\frac{P_1}{k} = \delta_1 - \delta_1' = \delta_1 - \frac{\delta_2'}{3}$

or $\frac{3\,P_1}{k} = 3\,\delta_1 - \delta_2'$ ...(a)

and $\frac{P_2}{k} = \delta_2' - \delta_2 = \delta_2' - \frac{\delta_1}{3}$

or $\frac{P_2}{k} = -\frac{\delta_1}{3} + \delta_2'$ ...(b)

Eliminating δ_2' from (a) and (b), we get $\delta_1 = \frac{3}{8\,k}(3\,P_1 + P_2)$ ...(1)

Substituting the values of P_1 and P_2, we get

$$\delta_1 = \frac{3}{8\,k}\left(3 \times \frac{9}{8}P + \frac{3}{8}P\right) = \frac{45\,P}{32\,k}$$

Example 2.69. *Fig. 2.86 shows a bimetallic thermal control, made of a brass bar of 8 mm diameter and a magnesium bar of 12 mm diameter. The bars are so arranged that the gap between their ends are 0.1 mm at room temperature. Find the temperature increase above the room temperature at which the two bars just come in contact. Also find the compressive stress in magnesium bar when the temperature increase in 320° F.*

Take $\alpha_b = 10 \times 10^{-6}$ *per° F*, $\alpha_m = 15 \times 10^{-6}$ *per ° F*; $E_b = 100\,kN/mm^2$ *and* $E_m = 40\,kN/mm^2$.

Solution :

Let t be the rise in temperature. Using suffix b for brass and m for magnesium, we have

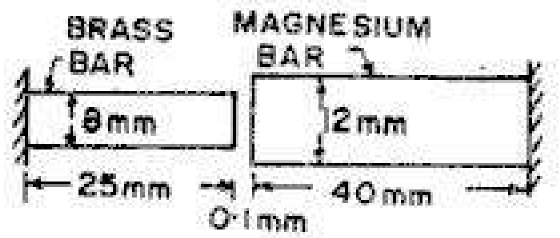


FIG. 2.86

$\Delta_{bt} = L_b\,\alpha_b\,t = 25 \times 10 \times 10^{-6}\,t$ and

$\Delta_{mt} = L_m\,\alpha_m\,t = 40 \times 15 \times 10^{-6}\,t$

Hence $\Delta_{bt} + \Delta_{mt} = 0.1 = 250 \times 10^{-6}\,t + 600 \times 10^{-6}\,t$

From which $t = \mathbf{117.65°F}$

$$A_b = \frac{\pi}{4}(8)^2 = 50.27\text{ mm}^2 \text{ and } A_m = \frac{\pi}{4}(12)^2 = 113.1\text{ mm}^2$$

When $t = 320°F$, $\Delta_{bt} = 25 \times 10 \times 10^{-6} \times 320 = 0.08$ mm

and $\Delta_{mt} = 40 \times 15 \times 10^{-6} \times 320 = 0.192$ mm

Now expansion restricted, $\Delta_t = \left(\Delta_{bt} + \Delta_{mt}\right) - 0.1 = (0.08 + 0.192) - 0.1 = 0.172$ mm ...(*i*)

If P is the compressive force induced, compression (Δ_c) of the two bars is

$$\Delta_c = \frac{P\,L_b}{A_b\,E_b} + \frac{P\,L_m}{A_m\,E_m} = P\left[\frac{25}{50.27 \times 100} + \frac{40}{113.1 \times 40}\right] = 0.0138\,P \quad ...(ii)$$

Equating (*i*) and (*ii*), we get $0.0138\,P = 0.172$ from which $P = 12.45$ kN

$$\therefore \quad p_m = \frac{P}{A_m} = \frac{12.45}{113.1} \times 1000 = \mathbf{110.1\ N/mm^2} \text{ (comp.)}$$

Example 2.70. *A vertical bar ABC is attached to a rigid bar BDF, as shown in Fig. 2.87(a). Find the ratio of loads P_2 and P_1 so that the vertical deflection of point C is zero.*

Solution :

From the moment equilibrium of the rigid bar *BDF*, we find that a vertical force P (thrust) $= P_2 \times \frac{1.5\,L}{L} = 1.5\,P_2$ is transmitted to the vertical bar, through pin *B*. Hence the vertical bar *ABC* is subjected to forces as shown in Fig. 2.87 (*b*). The reaction $R_A = P - P_1$. Thus, the portion *AB* is subjected to a compressive force $= R_A\ (= P - P_1)$ and portion *BC* is subjected to a tensile force P_1.

Hence $\delta_c = \Sigma\frac{PL}{AE} = \frac{P_1\,(L)}{aE} - \frac{(P - P_1)\,(2\,L)}{2\,a\,E}$

Substituting the value of P and noting that δ_c has to be zero, we have

$$\frac{P_1\,L}{a\,E} - \frac{(1.5\,P_2 - P_1)\,2\,L}{2\,a\,E} = 0$$

or $P_1 - 1.5\,P_2 + P_1 = 0$ or $2\,P_1 = \frac{3}{2}P_2$

From which $\frac{P_2}{P_1} = \frac{4}{3}$

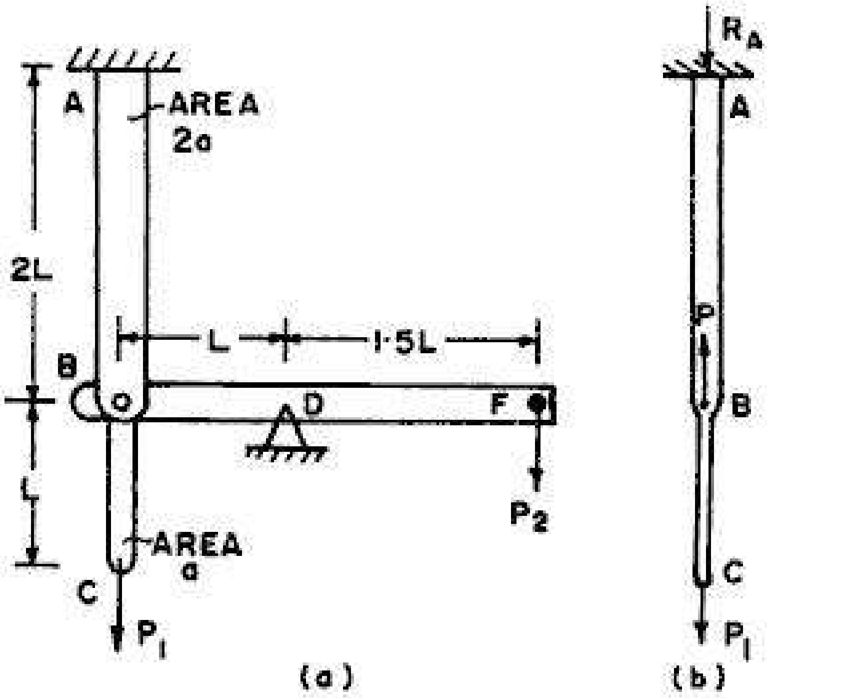


FIG. 2.87

Example 2.71. *A rigid beam AB is hinged at A and is connected to an aluminium alloy bar BC at the other end. The beam is also supported on a steel post DE through a rigid bearing at D, as shown in Fig .2.88. At no load, the beam is horizontal and there is an initial clearance $\delta = 0.1$ mm between the beam and rigid bearing. Determine the maximum permissible load W which may be applied at B if the axial stresses are not to exceed 120 N/mm² for steel and 140 N/mm² for the aluminium alloy. All the connections may be treated as smooth pins. Take $E = 2 \times 10^5$ N/mm² for steel and $E = 0.8 \times 10^5$ N/mm² for aluminium alloy.*

***Solution* :**

Let us use suffix a for aluminium and suffix s for steel. There are four unknowns R_A, P_S, P_a and W as shown in Fig. 2.89 (a). Let p_s be the axial stress (comp.) in steel post and p_a be the axial stress (tensile) in aluminium bar. One of these axial stresses will be equal to permissible value (given) while the other one will be lesser than the allowable value.

Fig. 2.89 (b) shows the deformed position of the bars, from which we have :

$$\frac{\Delta_s + \delta}{0.1} = \frac{\Delta_a}{0.4}$$

or $$4\,(\Delta_s + \delta) = \Delta_a$$

or $$4\left[\frac{p_s \times 200}{2 \times 10^5} + 0.1\right] = \frac{p_a \times 400}{0.8 \times 10^5}$$

From which we get

$$p_a = 0.8\,p_s + 80 \quad ...(1)$$

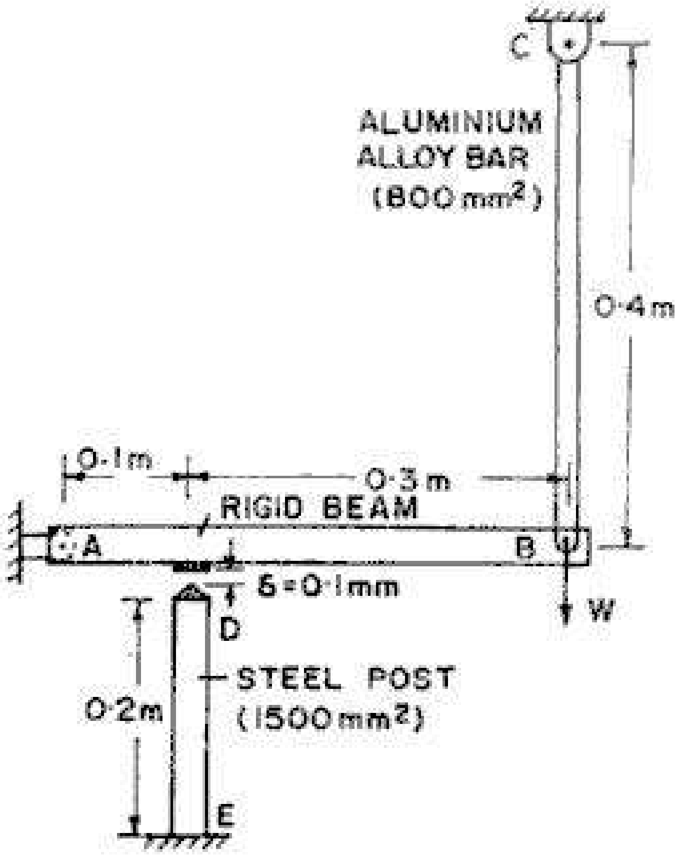


FIG. 2.88

If we have $p_s = 120\ \text{N/mm}^2$, we get $p_a = 0.8 \times 120 + 80 = 176\ \text{N/mm}^2$ which is more than the permissible value of $140\ \text{N/mm}^2$. Hence adopt $p_a = 140\ \text{N/mm}^2$ (the permissible value), corresponding to which, we obtain from (1)

$$p_s = \frac{1}{0.8}(p_a - 80)$$

$$= \frac{1}{0.8}(140 - 80)$$

$$= 75\ \text{N/mm}^2$$

$$\therefore\ P_a = 140 \times 800 \times 10^{-3}$$

$$= 112\ \text{kN} \text{ and}$$

$$P_s = 75 \times 1500 \times 10^{-3}$$

$$= 112.5\ \text{kN}$$

Now, taking moments about A (Fig. 2.89 a), we get

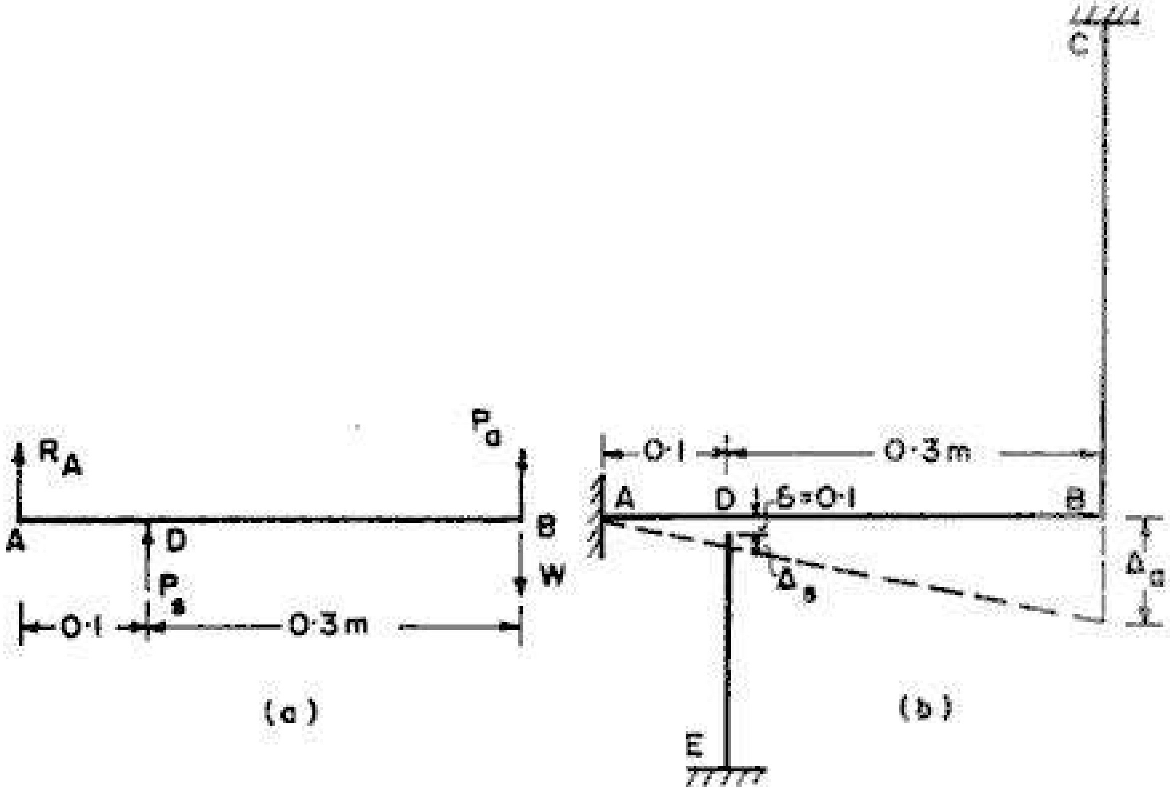


FIG. 2.89

$$112 \times 0.4 + 112.5 \times 0.1 = 0.4\,W$$

From which we get $\quad W = \mathbf{140.125\ kN}$

Example 2.72. *If the mechanism of Fig. 2.88 is loaded with force $W=50$ kN and the temperature increases by 25°F, calculate the axial stresses in BC and DE. Take coefficient of thermal expansion per degree F is 6.5×10^{-6} for steel and 13×10^{-6} for aluminium alloy.*

Solution :

Fig. 2.90 shows various deformations. For proper understanding, let us imagine that temperature rises by 25°F when W does not act. In that case, both ED and CB will expand, point D going to D_1, and point B going to B_1.

Thus, $DD_1 = \Delta_s^t =$ free expansion of steel bar due to temperature rise

and $BB_1 = \Delta_a^t =$ free expansion of aluminium bar due to temperature rise.

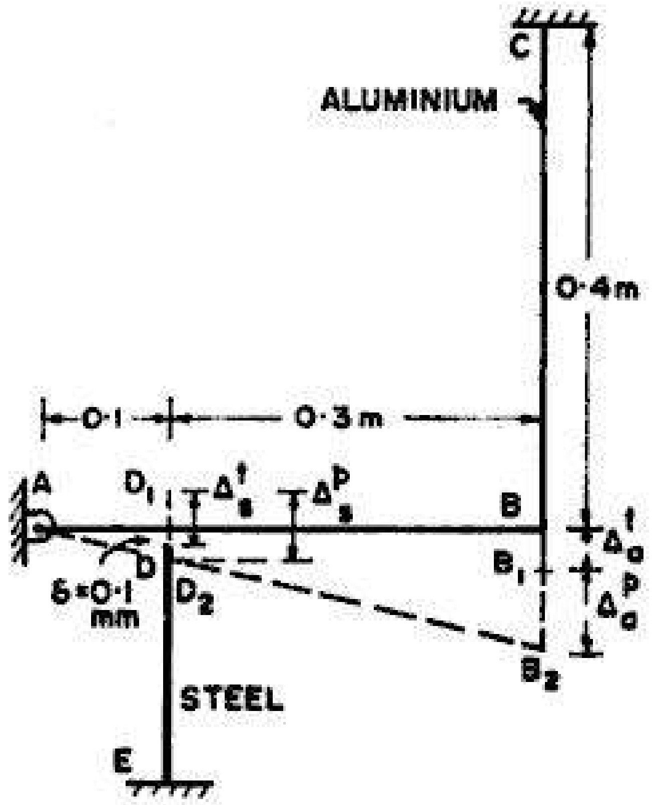


FIG. 2.90

Now, when the load W is applied at B, the rod CB extends further from B_1 to B_2 while the bar DE is *compressed* from D_1 to D_2 such that AD_2B_2 is the final position of the rigid bar.

Thus $\quad B_1B_2 = \Delta_a^p =$ expansion of aluminium bar due to external load.

and $\quad D_1D_2 = \Delta_s^p =$ contraction of steel bar, due to external load

The compatibility equation is

$$\frac{\Delta_s^p - \Delta_s^t + 0.1}{0.1} = \frac{\Delta_a^t + \Delta_a^p}{0.4}$$

or

$$4\,(\Delta_s^p - \Delta_s^t + 0.1) = \Delta_a^t + \Delta_a^p \qquad ...(i)$$

Now

$$\Delta_s^p = \frac{p_s L_s}{E_s} = \frac{p_s\,(200)}{2 \times 10^5} = p_s \times 10^{-3}$$

$$\Delta_s^t = L_s \alpha_s t = 200 \times 6.5 \times 10^{-6} \times 25 = 0.0325 \text{ mm}$$

$$\Delta_a^p = \frac{p_a L_a}{E_a} = \frac{p_a\,(400)}{0.8 \times 10^5} = 5 \times 10^{-3} p_a$$

$$\Delta_a^t = L_a \alpha_a t = 400 \times 13 \times 10^{-6} \times 25 = 0.13 \text{ mm}$$

Hence from (*i*), $4\,(p_s \times 10^{-3} - 0.0325 + 0.1) = 5 \times 10^{-3} p_a + 0.13$

From which $\quad p_s \doteq 1.25\,p_a - 35 \qquad ...(a)$

Also, taking moments about A (Fig. 2.89 *a*), $0.1\,P_s + 0.4\,P_a = 0.4\,W$

or $\quad 0.1 \times 1500\, p_s + 0.4 \times 800\, p_a = 0.4 \times 50000$

which gives $\quad p_s + 2.133\, p_a = 133.33 \quad$ ...(b)

From (a) and (b) we get

$p_a =$ **49.76 N/mm²** (tensile) and $p_s =$ **27.20 N/mm²** (compressive)

Example 2.73. *A rigid bar ABC is hinged at A and attached to brass bar BF (length 0.2 m, area 4000 mm²) and steel bar CD (length 0.3 m and area 250 mm²). The temperature of brass bar BF is lowered by 30 K and that of bar CD is raised by 30 K. Neglecting any possibility of lateral bucking, find the normal stresses in the brass and steel. Take $E = 0.9 \times 10^5$ N/mm² and $\alpha = 20 \times 10^{-6}$ per K for brass, and $E = 2 \times 10^5$ N/mm² and $\alpha = 12 \times 10^{-6}$ per K for steel.*

Solution :

Free expansion of steel bar

$= \Delta_S^t = 300 \times 12 \times 10^{-6} \times 30 = 0.108$ mm

Free contraction of brass bar

$= \Delta_B^t = 350 \times 20 \times 10^{-6} \times 30 = 0.21$ mm

For no temperature stresses to be developed, $\dfrac{\Delta_B^t}{AB} = \dfrac{\Delta_S^t}{AC}$

$$\therefore \quad \Delta_B^t = \Delta_S^t \cdot \frac{AB}{AC} = \Delta_S^t \times \frac{300}{700} = 0.4286\, \Delta_S^t$$

$$= 0.4286 \times 0.108 = 0.0463 \text{ mm}$$

But $\quad \Delta_B^t = 0.21$ mm

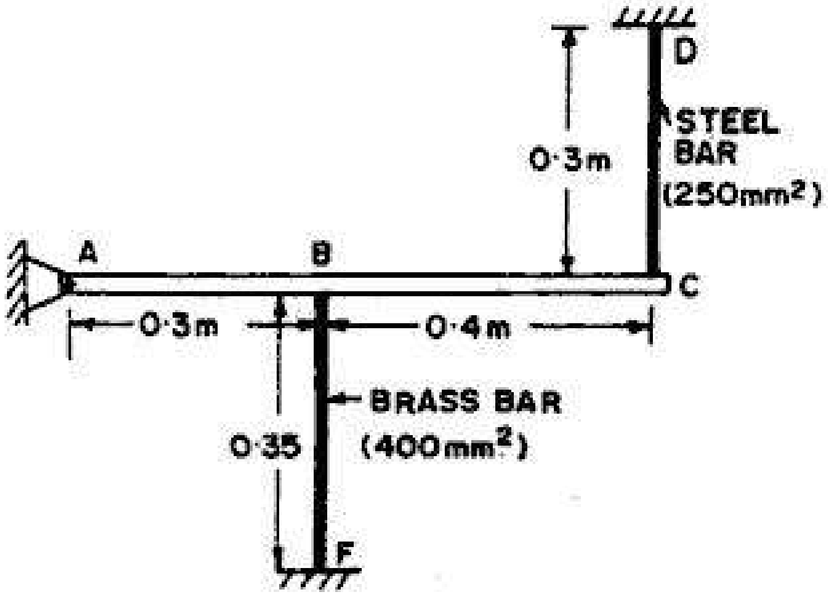


FIG. 2.91

Hence in the equilibrium position attained due to rigidity of bar *ABC*, tensile stress will be developed in the brass bar, and consequently, tensile stress will also be developed in steel bar.

If Δ_B and Δ_S are the *final deformations* in the two bars at the *equilibrium position*, we have

$$\frac{\Delta_B}{AB} = \frac{\Delta_S}{AC}$$

$$\therefore \quad \Delta_B = \Delta_S \frac{AB}{AC} = \frac{0.3}{0.7} \Delta_S = 0.4286\, \Delta_S \qquad ...(i)$$

Now $\quad \Delta_B = \Delta_B^t - \Delta_B^p = L_B\, \alpha_B\, t_B - \dfrac{p_B L_B}{E_B} = 0.21 - \dfrac{360\, p_B}{0.9 \times 10^5}$

an $\quad \Delta_S = \Delta_S^t + \Delta_S^p = L_S\, \alpha_S t_S + \dfrac{p_S L_S}{E_S} = 0.108 + \dfrac{300\, p_S}{2 \times 10^5}$

Hence from (i), $\quad 0.21 - \dfrac{350\, p_B}{0.9 \times 10^5} = 0.4286 \left(0.108 + \dfrac{300\, p_S}{2 \times 10^5} \right)$

which on simplification gives $p_S + 6.049\, p_B = 254.64 \qquad$ ...(a)

Also, taking moments about A, $p_B . A_B \times 350 - p_S A_S \times 700 = 0$

or $\quad p_B \times 400 \times 350 = p_S \times 250 \times 700$

which gives $\quad p_B = 1.25\, p_S$ ...(b)

Hence from (a) and (b), we get

$$p_S = \mathbf{29.74\ N/mm^2} \text{ (tension) and } p_B = \mathbf{37.18\ N/mm^2} \text{ (tension)}$$

Example 2.74. *A composite bar made up of aluminium bar and steel bar is firmly held between two unyielding supports as shown in Fig. 2.92. An axial load of 200 kN is applied at B at 20°C. Find the stresses in each material, when the temperature is 70°C. Take E for aluminium and steel as 0.7×10^5 N/mm² and 2×10^5 N/mm² respectively and coefficients of expansions for aluminium and steel as 24×10^{-6} per °C and 12×10^{-6} per °C respectively.*

Solution :

Let us consider both the effects *separately.*

(a) Effect of 200 kN load

Let R_A and R_B be the two reactions. From statics, we have

$$R_A = 200 - R_B \quad ...(i)$$

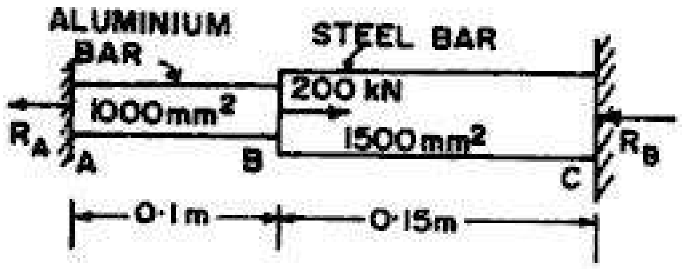


FIG. 2.92

Portion AB is subjected to a tensile load $= R_A = 200 - R_B$ while BC is subjected to a compressive load R_B ($= 200 - R_A$).

From compatibility, extension (Δ_{AB}) of AB= contraction (Δ_{BC}) of BC.

$$\therefore \quad \frac{(200 - R_B) \times 1000 \times 100}{1000 \times 0.7 \times 10^5} = \frac{R_B \times 1000 \times 150}{1500 \times 2 \times 10^5}$$

From which $\quad R_B = 148.15$ kN

Hence force in AB, $\quad P_{AB} = 200 - R_B = 200 - 148.15 = 51.85$ kN (tension)

and force in BC, $\quad P_{BC} = R_B = 148.15$ kN (compression)

(b) Effect of temperature rise : $t = 70 - 20 = 50°\,C$

Since the temperature strain in restricted, both the bars will be subjected to equal temperature force P_t of compressive nature.

$$\text{Free expansion of } AB = \Delta_{AB,t} = 100 \times 24 \times 10^{-6} \times 50 = 0.12 \text{ mm}$$

$$\text{Free expansion of } BC = \Delta_{BC,t} = 150 \times 12 \times 10^{-6} \times 50 = 0.09 \text{ mm}$$

$$\therefore \quad \text{Total expansion, } \Delta_t = 0.12 + 0.09 = 0.21 \text{ mm}$$

Since this expansion is completely restricted, we have

$$\Delta_t = P_t \left[\frac{L_{AB}}{A_{AB}\, E_{AB}} + \frac{L_{BC}}{A_{BC}\, E_{BC}} \right]$$

or $$0.21 = P_t \left[\frac{100}{1000 \times 0.7 \times 10^5} + \frac{150}{1500 \times 2 \times 10^5}\right] = 0.1929 \times 10^{-5} P_t$$

From which $P_t = 1.08889 \times 10^5$ N $=$ **108.89 kN (compression)**

(c) ***Final Stresses***

Final force in $AB = 108.89 - 51.85 = 57.04$ kN (compression)

and Final force in $BC = 108.89 + 148.15 = 257.04$ kN (compression)

$\therefore$ Stress in AB, $p_{AB} = \dfrac{57.04 \times 1000}{1000} =$ **57.04 N/mm²** (compression)

Stress in BC, $p_{BC} = \dfrac{257.04 \times 1000}{1500} =$ **171.36 N/mm²** (compression)

Example 2.75. *A steel bar is sandwiched between two copper bars each having the same area and length as the steel bar, at an initial temperature of 10°C. These are rigidly connected together at both the ends. When the temperature is raised to 260°C, the length of the bars increases by 1.0 mm. Determine the original length and the final stresses in the bars. Take the following values :*

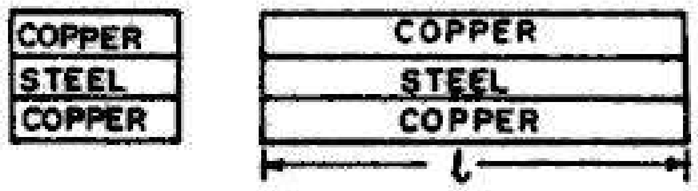


FIG. 2.93

$$E_s = 2 \times 10^5 \text{ N/mm}^2; \; E_c = 1 \times 10^5 \text{ N/mm}^2$$

$$\alpha_s = 12 \times 10^{-6} \text{ per °C} \; ; \; \alpha_c = 18 \times 10^{-6} \text{ per °C}$$

Solution :

Let A (mm²) be the cross-sectional area of steel component. Hence cross-sectional area of copper component will be= $2A$ (mm²)

If δ is the actual expansion of the composite bar, we have $\alpha_s t\, l < \delta < \alpha_c t\, l$. Hence steel will be in tension and copper will be in compression.

For equilibrium, $f_c A = f_c (2A)$

or $$f_s = 2 f_c \qquad ...(1)$$

Also, final expansion of steel = final expansion of copper.

$$\therefore \quad \alpha_s t\, l + \frac{f_s}{E_s} l = \alpha_c t\, l - \frac{f_c}{E_c} l$$

Here $t = 260 - 10 = 250°\,C$

$$\therefore \quad 12 \times 10^{-6} \times 250 + \frac{f_s}{2 \times 10^5} = 18 \times 10^{-6} \times 250 - \frac{f_c}{1 \times 10^5} \qquad ...(2)$$

But $f_s = 2 f_c$. Hence from (2), we get.

$$3000 \times 10^{-6} + \frac{2 f_c}{2 \times 10^5} = 4500 \times 10^{-6} - \frac{f_c}{1 \times 10^5}$$

From which we get $f_c =$ **75 N/mm² (compression)**

$\therefore \qquad f_s = 2 \times 75 = \mathbf{150\ N/mm^2}$ (tension)

Now $\qquad \Delta_s = \alpha_s\, t\, l + \dfrac{f_s}{E_s}\, l$ (where $\Delta_s = 1.0$ mm, given)

$$\therefore \qquad 1.0 = l\left(12 \times 10^{-6} \times 250 + \frac{150}{2 \times 10^5}\right) = 3.75 \times 10^{-3}\, l$$

From which $\qquad l = \mathbf{266.7\ mm}$

Example 2.76. *A flat bar of aluminium alloy, 30 mm wide and 8 mm thick is placed between two steel bars each 30 mm wide and 10 mm thick, to form a composite bar 30 mm × 28 mm, as shown in Fig. 2.94. The three bars are fastened together at their ends when the temperature is 12°C. Find the stress in each of the material when the temperature of the whole assembly is raised to 42°C. If, at the new temperature, a compressive load of 30 kN is applied to the composite bar, what are the final stresses in steel and aluminium ?*
Take $E_s = 2 \times 10^5\ N/mm^2$, $E_a = 0.7 \times 10^5\ N/mm^2$; $\alpha_s = 12 \times 10^{-6}$ per °C and $\alpha_a = 23 \times 10^{-6}$ per °C.

Solution :

Let us use suffix a for aluminium and s for steel.

$A_s = 20 \times 30 = 600\ \text{mm}^2$ and $A_a = 8 \times 30 = 240\ \text{mm}^2$

We shall consider both the effects separately.

(a) Effect of temperature rise :

Due to temperature rise, steel will expand lesser than aluminium. Hence steel bar will be in tension and aluminium bar will be in compression.

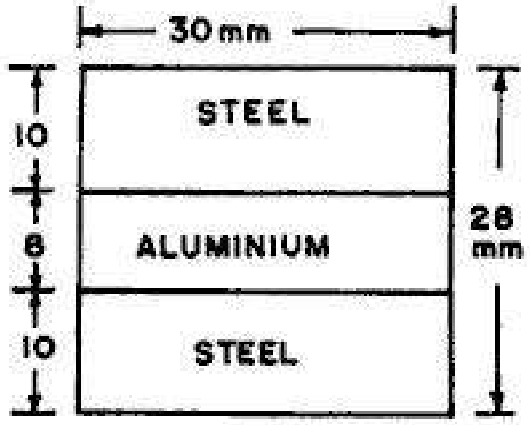


FIG. 2.94

For equilibrium,

$$p_s A_s = p_a A_a$$

or $\qquad p_s \times 600 = p_a \times 240$ from which $p_a = 2.5\, p_s \qquad$ ...(1)

Also, $\qquad \Delta_s = \Delta_a$

$$\text{or} \qquad \alpha_s\, t\, l + \frac{p_s}{E_s}\, l = \alpha_a\, t\, l - \frac{p_a\, l}{E_a}$$

$$\therefore \quad 12 \times 10^{-6} \times 30 + \frac{p_s}{2 \times 10^5} = 23 \times 10^{-6} \times 30 - \frac{p_a}{0.7 \times 10^5}$$

$$\text{or} \qquad 3.6 \times 10^{-4} + \frac{p_s}{2 \times 10^5} = 6.9 \times 10^{-4} - \frac{2.5\, p_s}{0.7 \times 10^5}$$

From which $p_s = 42\ \text{N/mm}^2$ (tension) and $p_a = 105\ \text{N/mm}^2$ (comp.)

(b) Effect of external load of 30 kN

$$f_s A_s + f_a A_a = P$$

$\therefore \qquad f_s \times 600 + f_a \times 240 = 30000 \qquad$ ...(*i*)

Also, $$\frac{f_s}{E_s} = \frac{f_a}{E_a}$$

$\therefore$ $$f_s = f_a \frac{E_s}{E_a} = \frac{2}{0.7} f_a = 2.857 f_a \qquad \text{...}(ii)$$

Substituting in (1) we get

$$600\,(2.857 f_a) + 240 f_a = 30000$$

From which $f_a = 15.4 \text{ N/mm}^2$ (comp.)

Hence $f_s = 43.9 \text{ N/mm}^2$ (comp.)

(c) Final Stresses

Stress in aluminium $= 105 + 15.4 = \mathbf{120.4 \text{ N/mm}^2}$ (comp.)

Stress in steel $= -42 + 43.9 = \mathbf{1.9 \text{ N/mm}^2}$ (tension)

Example 2.77. *A rigid bar ABCD of negligible mass is pinned at B and is attached to two vertical rods CF and DG. Rod CF is of steel having $L = 0.8$ m and $A = 800$ mm^2, while rod DG is of bronze have $L = 1$ m and $A = 250$ mm^2. Assuming that the rods were initially stress free, what maximum load P can be applied without exceeding stresses of 160 N/mm^2 in steel rod and 80 mm^2 in bronze bar. Take $E_s = 2 \times 10^5$ N/mm^2 and $E_b = 0.83 \times 10^5$ N/mm^2.*

Solution :

Let p_s be the stress in steel rod (limited to 160 N/mm^2) and p_b be the stress in bronze rod (limited to 80 N/mm^2). For moment balancing about B,

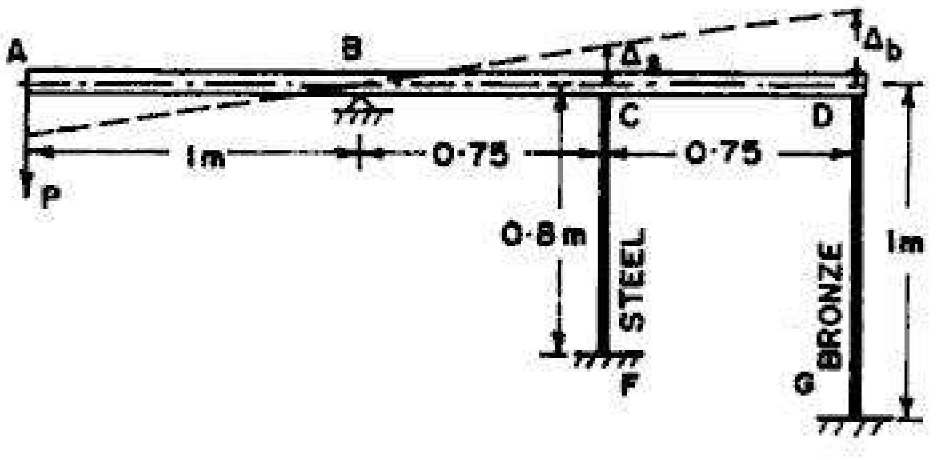


FIG. 2.95

$$P \times 1 = p_s A_s \times 0.75 + p_b A_b \times 1.5 \qquad \text{...}(1)$$

Also, from compatibility,

$$\frac{\Delta_s}{0.75} = \frac{\Delta_b}{1.5}$$

or $$\Delta_s = \frac{0.75}{1.5} \Delta_b = 0.5\,\Delta_b$$

or $$\frac{p_s l_s}{E_s} = 0.5 \frac{p_b l_b}{E_b}$$

$\therefore$ $$p_s = 0.5 \frac{l_b}{l_s} \cdot \frac{E_s}{E_b} p_b = 0.5 \times \frac{1}{0.8} \times \frac{2}{0.83} p_b = 1.506\, p_b \qquad \text{...}(2)$$

or $$p_b = 0.664\, p_s \qquad \text{...}(3)$$

If p_b is kept equal to its maximum permissible value of 80 N/mm^2, we get $p_s = 1.506 \times 80 = 120.48$ N/mm^2, which is less than its maximum permissible value of 160 N/mm^2. On the other hand, if p_s is kept equal to its maximum permissible value of 160 N/mm^2, we get $p_b = 0.664 \times 160 = 106.24$ N/mm^2 which exceeds the permissible value of 80 N/mm^2.

Hence the governing values of the stress is $p_b = 80\ \text{N/mm}^2$, corresponding to which $p_s = 120.48\ \text{N/mm}^2$. Substituting these values in Eq. (1) we get

$$P = 120.48 \times 800 \times 0.75 + 80 \times 250 \times 1.5 = 102288\ N \approx \mathbf{102.288\ kN.}$$

Example 2.78. *Fig. 2.96 shows a round steel rod supported in a recess and surrounded by a co-axial brass tube. The upper end of the rod is 0.1 mm below that of the tube and an axial load is applied to a rigid plate resting on the top of the tube.*

(a) Determine the magnitude of the maximum permissible load if the compressive stress in rod is not to exceed 120 N/mm² and that in the tube is not to exceed 90 N/mm².

(b) Find the amount by which the tube will be shortened by the load if the compressive stress in the tube is the same as that in the rod.

Take $E_s = 200\ \text{kN/mm}^2$ and $E_b = 100\ \text{kN/mm}^2$.

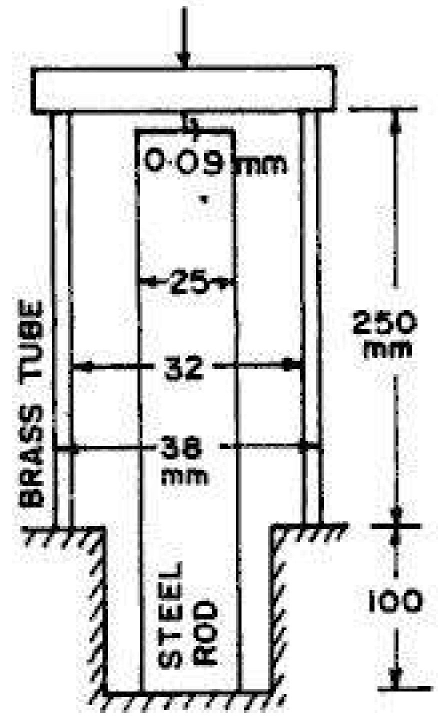


FIG. 2.96

Solution :

$$A_s = \frac{\pi}{4}(25)^2 = 490.87\ \text{mm}^2\ ;\ A_b = \frac{\pi}{4}(38^2 - 32^2) = 329.87\ \text{mm}^2$$

Part (a) : The initial compressive stress (p_{bi}) in the tube, before the steel rod is compressed is

$$p_{bi} = \frac{\Delta E_b}{L_b} = \frac{0.09 \times 100 \times 1000}{250} = 36\ \text{N/mm}^2 < 90\ .$$

$$\therefore \qquad W_1 = p_{bi} . A_b = 36 \times 329.87 \approx \mathbf{11875\ \ N}$$

Now let W_2 be the additional load to compress further both the bar and the tube.

$$\therefore \qquad p_b A_b + p_s A_s = W_2 \qquad ...(i)$$

Also, $\Delta_b = \Delta_s$

or
$$\frac{p_b \times 250}{E_b} = \frac{p_s \times 350}{E_s}$$

$$\therefore \qquad p_s = p_b \cdot \frac{E_s}{E_b} \cdot \frac{250}{350} = p_b \frac{200}{100} \times \frac{250}{350} = 1.4286\, p_b \qquad ...(ii)$$

Now limiting the stress in brass to $90\ \text{N/mm}^2$, we have

$$p_b = 90 - p_{bi} = 90 - 36 = 54\ \text{N/mm}^2$$

and corresponding $p_s = 1.4286 \times 54 = 77.14\ \text{N/mm}^2 < 120\ \text{N/mm}^2$

Hence the governing value are : $p_b = 54\ \text{N/mm}^2$ and $p_s = 77.14\ \text{N/mm}^2$

Hence from (*ii*), $W_2 = 54 \times 329.87 + 77.14 \times 490.87 \approx 55679\ \ \text{N}$

$$\therefore \text{ Total max. load} = W_1 + W_2 = 11875 + 55679 = 67554\ \text{N} = \mathbf{67.554\ kN}$$

Part a : Let the *additional* shortening be $= \Delta_b = \Delta_s$

$\therefore$ Total shortening of the tube $= (\Delta_b + 0.09)$ mm

$$\text{Hence stress in brass} = \frac{E_b(\Delta_b + 0.09)}{L_b} = \frac{100 \times 10^3}{250}(0.09 + \Delta_b) \quad ...(a)$$

$$\text{and} \qquad \text{stress in steel} = \frac{E_s \Delta_s}{L_s} = \frac{200 \times 10^3}{350}\Delta_b \quad ...(b)$$

$$\text{Equating the two,} \quad \frac{100 \times 10^3}{250}(0.09 + \Delta_b) = \frac{200 \times 10^3}{350}\Delta_b$$

or $$0.09 + \Delta_b = 1.4286\,\Delta_b$$

From which $$\Delta_b = 0.21 \text{ mm}$$

Hence total deformation of brass tube $= 0.09 + 0.21 =$ **0.3 mm**

Example 2.79. *A rigid member AD of length 2L is supported at ends A and C and at its mid-point B by three vertical wires DA, EB and FC, as shown in Fig 2.97. The wires DA, EB and FC extend α_1, α_2 and α_3 respectively under unit load. Calculate the load in each wire if the member AC weighs w per unit length.*

Solution :

Total downward load $= 2\,w\,L$

Let the loads in the three bars be P_1, P_2 and P_3 respectively.

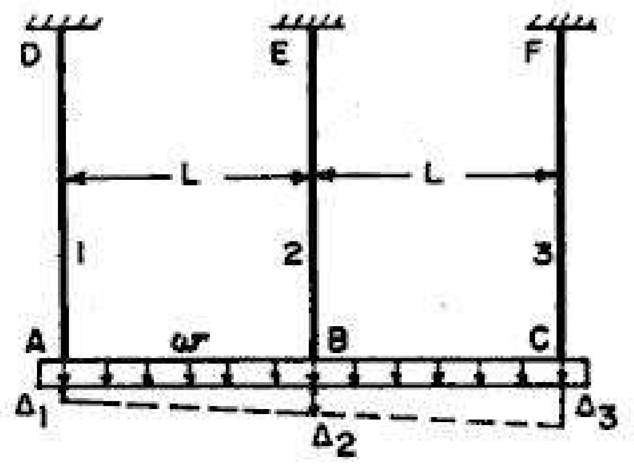


FIG. 2.97

$$\therefore \qquad P_1 + P_2 + P_3 = 2\,w\,L \quad ...(1)$$

Taking moments about AD,

$$P_2 L + P_3 2L = 2\,w\,L \times L$$

or $$P_2 + 2P_3 = 2\,w\,L \quad ...(2a)$$

Also, by taking moments about FC,

$$P_1 \times 2L + P_2 L = 2\,w\,L\,.\,L$$

or $$2P_1 + P_2 = 2\,w\,L \quad ...(2b)$$

From 2 (a) and 2 (b), we observe that $P_1 = P_3$ ...(2)

From compatibility, $$\Delta_2 = \frac{\Delta_1 + \Delta_3}{2}$$

or $$P_2\alpha_2 = \frac{1}{2}\left(P_1\alpha_1 + P_3\alpha_3\right)$$

or $$P_2\alpha_2 = \frac{1}{2}\left(P_1\alpha_1 + P_1\alpha_3\right) = \frac{P_1}{2}\left(\alpha_1 + \alpha_3\right)$$

Hence $$P_2 = \frac{P_1}{2\alpha_2}\left(\alpha_1 + \alpha_3\right) \quad ...(3)$$

Substituting in (1), we get

$$P_1 + \frac{P_1}{2\alpha_2}\left(\alpha_1 + \alpha_3\right) + P_1 = 2wL$$

or
$$P_1\left[2 + \frac{\alpha_1 + \alpha_3}{2\alpha_2}\right] = 2wL$$

or
$$P_1\left[\frac{4\alpha_2 + (\alpha_1 + \alpha_3)}{2\alpha_2}\right] = 2wL$$

From which
$$P_1 = \frac{4wL\alpha_2}{\alpha_1 + 4\alpha_2 + \alpha_3} = P_3 \text{ Ans.}$$

Hence from (3),
$$P_2 = \frac{\alpha_1 + \alpha_3}{2\alpha_2}\left[\frac{4wL\alpha_2}{\alpha_1 + 4\alpha_2 + \alpha_3}\right]$$

or
$$P_2 = \frac{2wL(\alpha_1 + \alpha_3)}{\alpha_1 + 4\alpha_2 + \alpha_3} \textbf{ Ans.}$$

PROBLEMS

1. A steel bar *ABCD* of varying sections is subjected to the axial forces as shown in Fig. 2.98. Find the value of P necessary for equilibrium. If $E = 210\,\text{kN/mm}^2$, determine the total elongation of the bar.

2. A member is formed by connecting a steel bar to an aluminium bar as shown in Fig. 2.99. Assuming that the bars are prevented from buckling sideways, calculate the magnitude of axial force P that will cause the total length of member to decrease by 0.3 mm.

Take $E_s = 2 \times 10^5\,\text{N/mm}^2$ and $E_a = 0.7 \times 10^5\,\text{N/mm}^2$.

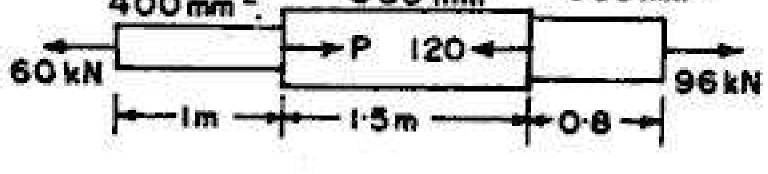


FIG. 2.98

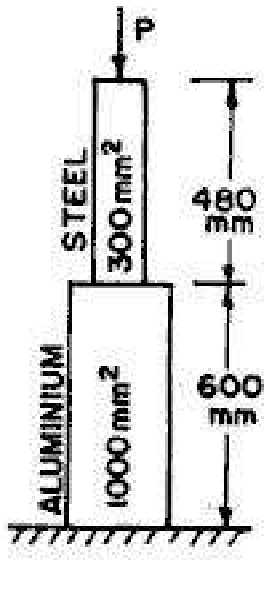


FIG. 2.99

3. A steel wire of 10 mm diameter is used for lifting a load of 1 kN at its lowest end, the length of the wire hanging vertically being 200 m.

Taking the unit weight of steel $= 78\,\text{kN/m}^3$ and $E = 2 \times 10^5\,\text{N/mm}^2$, calculate the total elongation of the wire.

4. The bar shown in Fig. 2.100 is subjected to a tensile load of 100 kN. Find diameter of the middle portion if the stress there is to be limited to 150 N/mm^2. Find also the length of the middle portion if the elongation of the bar is limited to 0.2 mm.
Take $E = 2 \times 10^5$ N/mm^2.

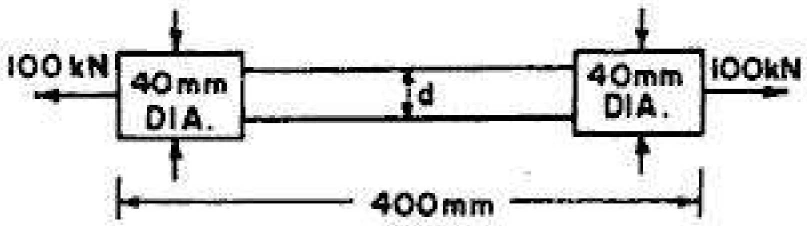


FIG. 2.100

5. A steel flat plate of 10 mm thickness tapers uniformly from 100 mm to 50 mm width in a length of 400 mm. From first principles, determine the elongation of the plate if the axial tensile force is 100 kN. Take $E = 2 \times 10^5$ N/mm^2.

6. A mild steel rod of 20 mm diameter and 300 mm long is enclosed centrally inside a hollow copper tube of external diameter 30 mm and internal diameter 25 mm. The ends of the rod and tube are brazed together, and the composite bar is subjected to an axial pull of 50 kN.

If E for steel and copper is 200 GN/m^2 and 100 GN/m^2 respectively, find the stresses developed in the rod and the tube. Also find the extension of the rod.

7. A reinforced concrete column 500 mm diameter has four steel rods of 30 mm diameter embedded in it and carries a load of 680 kN. Find the stresses in steel and concrete. Take E for steel $= 2.04 \times 10^5$ N/mm^2 and E for concrete $= 0.136 \times 10^5$ N/mm^2. Find also the adhesive force between steel and concrete.

8. A vertical rod of uniform section fixed at both its ends is axially loaded at intermediate sections by forces as shown in Fig. 2.101. Find the end reactions.

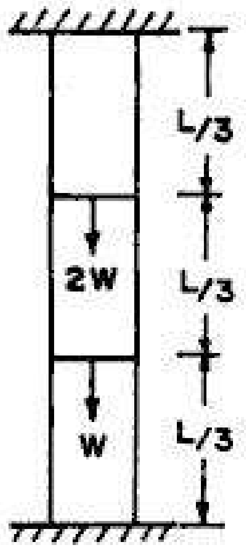


FIG. 2.101

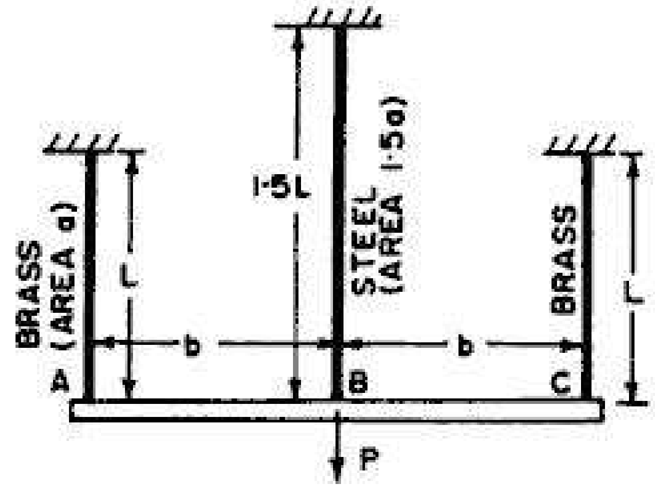


FIG. 2.102

9. A rigid bar ABC is supported by three rods in the same vertical plane and equidistant, as shown in Fig. 2.102. The outer rods are of brass and of length L and area a. The central rod is of steel of length $1.5\,L$ and area $1.5\,a$. Calculate the forces in the bars due to an applied force of P, if the bar AC remains horizontal after the load has been applied. Take $\frac{E_s}{E_b} = 2$.

10. A rigid beam carrying a load of W is supported by three bars as shown in Fig. 2.103. All the bars have the same length and same area of cross-section. Taking $E_s = 2\,E_b$, find the load carried by each bar.

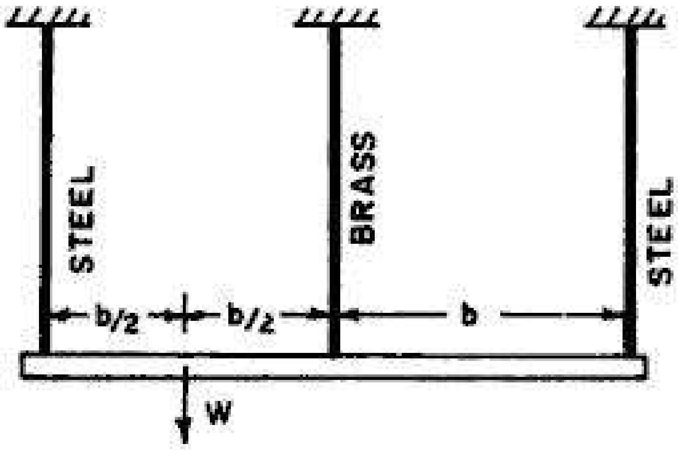


FIG. 2.103

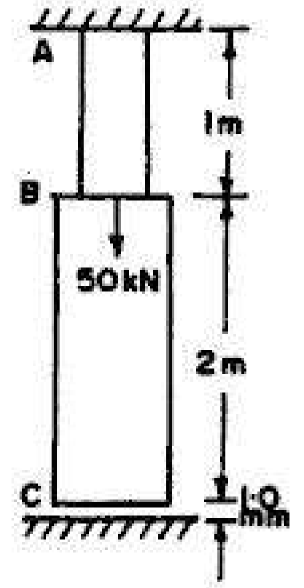


FIG. 2.104

11. A composite bar ABC, rigidly fixed at A and 1 mm above the lower support is loaded as shown in Fig. 2.104. If the cross- sectional area of section AB is 100 mm^2 and that of section BC is 200 mm^2, find the reactions at the ends, and the stresses in the two sections. Take $E = 2 \times 10^5 \text{ N/mm}^2$.

12. A compound bar consists of a central steel strip 20 mm wide and 5 mm thick placed between two strips of brass each 20 mm wide and t mm thick. The strips are firmly fixed together to form a compound bar of rectangular section 20 mm wide and $(2t+5)$ mm thick. Determine (a) the thickness of the brass strips which will make the apparent modulus of elasticity of compound bar $1.6 \times 10^5 \text{ N/mm}^2$, and ($b$) the maximum axial pull the bar can then carry if the stress is not to exceed 150 N/mm^2, in either brass or steel. Take the values of E for steel and brass as $2 \times 10^5 \text{ N/mm}^2$ and $1 \times 10^5 \text{ N/mm}^2$.

13. Three bars, made of copper, zinc and aluminium are of equal length and have cross-section of 500, 750 and 1000 sq. mm respectively. They are rigidly connected at their ends, as shown in Fig. 2.105. If this compound member is subjected to a longitudinal pull of 200 kN, estimate the proportion of load carried by each rod and the induced stresses. Take $E_c = 1.3 \times 10^5 \text{ N/mm}^2$, $E_z = 1 \times 10^5 \text{ N/mm}^2$ and $E_a = 0.8 \times 10^5 \text{ N/mm}^2$.

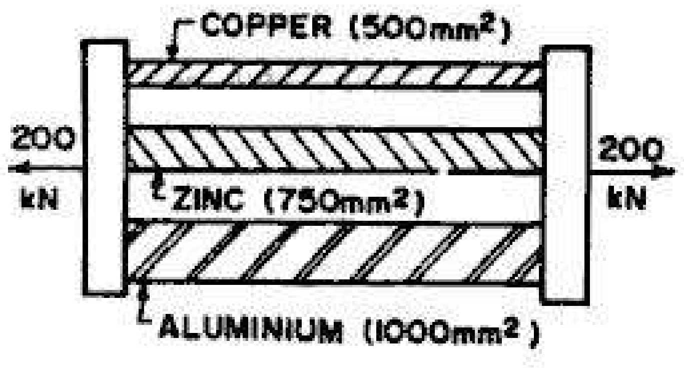


FIG. 2.105

14. A composite bar, made up of aluminium and steel, is held between two supports as shown in Fig. 2.106. The bars are stress free at a temperature of 40°C. What will be the stresses in the two bars when the temperature is 23°C, if (a) the supports are non-yielding, and (b) the supports come nearer to each other by 0.1 mm.

Take $E_s = 200 \text{ kN/mm}^2$, $E_a = 70 \text{ kN/mm}^2$, $\alpha_s = 12 \times 10^{-6}$ per °C and $\alpha_a = 24 \times 10^{-6}$ per °C

(Based on Oxford University)

15. A composite bar is made up by connecting a steel member and a copper member, rigidly fixed at their ends as shown in Fig. 2.107. The cross sectional area of steel member is A mm² for half of the length and $2A$ mm² for the other half of the length; while that for the copper member is A mm². The coefficient of expansion for steel and copper are α and $1.3\,\alpha$, while elastic moduli are E and $0.5\,E$ respectively. Determine the stresses induced in both the members when the composite bar is subjected to a rise of temperature of t degrees.

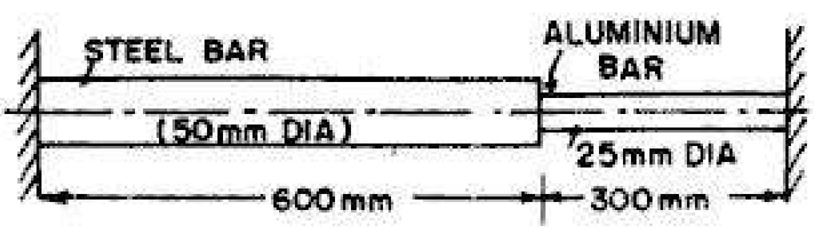


FIG. 2.106

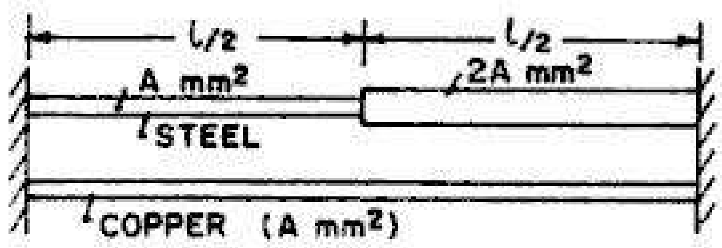


FIG. 2.107

16. Fig. 2.108 shows a bar of varying sections, rigidly fixed at A and D and subjected axial forces. Determine the force in each portion of the bar and the displacements of point B and C. Portion AB, of length 1 m has area of cross-section of 2000 mm^2, portion BC, of length 1.5 m has area of cross-section of 3000 mm^2, while portion CD has a length of 2 m and area of cross-section of 4000 mm^2. Take $E = 200$ kN/mm^2.

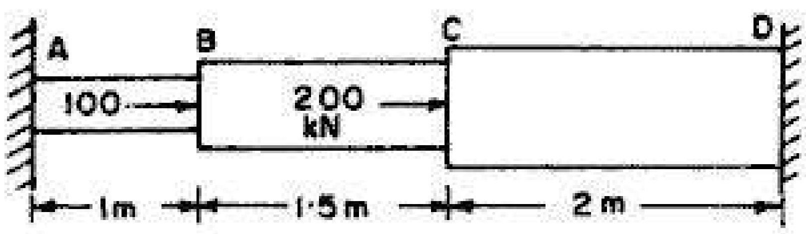


FIG. 2.108

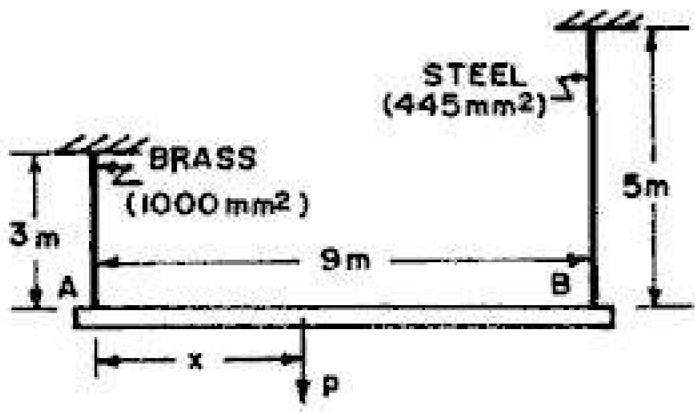


FIG. 2.109

17. A rigid bar AB is suspended by two vertical rods at A and B, as shown in Fig. 2.109. At what distance x from A may a vertical load P be applied if the bar is to remain horizontal after the load is applied ? Take $E_s = 2 \times 10^5$ N/mm^2 and $E_b = 1 \times 10^5$ N/mm^2.

18. A hollow steel cylinder of length 300 mm, inside diameter 150 mm and uniform wall thickness of 3 mm is filled with concrete and compressed between two rigid parallel plates at the ends by a load of 600 kN. Find the compressive stress in each material and the total shortening of the cylinder. Take $E_s = 2 \times 10^5$ N/mm^2 and $E_c = 0.2 \times 10^5$ N/mm^2.

19. A rigid bean $ABCD$ is hinged at D and supported by two springs A and B as shown in Fig. 2.110. The beam carries a vertical load W at the point C at a distance of c from the hinged end. The

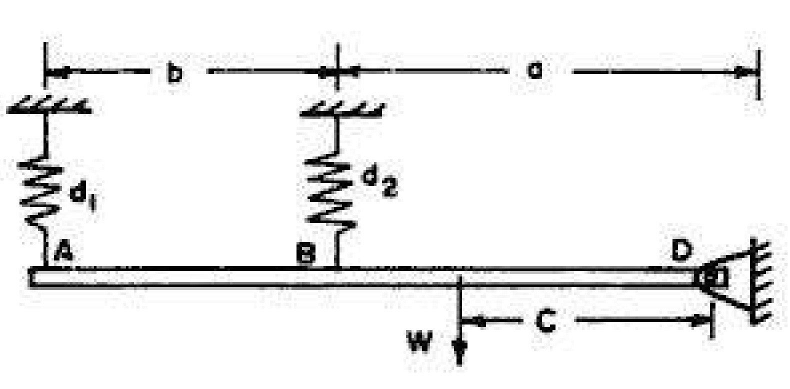


FIG. 2.110

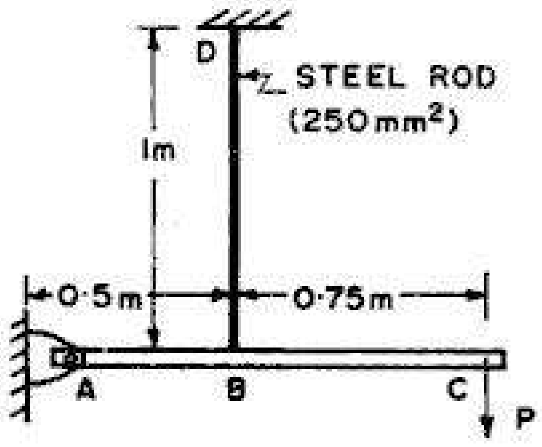


FIG. 2.111

flexibilities (deflection/unit load) of the springs at A and B are d_1 and d_2 respectively. Determine the forces in the springs and also the reaction at the hinged support.

20. A rigid bar shown in Fig. 2.111 is hinged at A and supported by a steel rod at B. Determine the largest load P that can be applied at C if the stress in rod is limited to 150 N/mm^2 and the vertical movement of end C is not to exceed (*a*) 1.8 mm (*b*) 2.2 mm. Take $E=2\times10^5$ N/mm^2 for the steel rod.

21. A long bar of rectangular cross-section hangs vertically and carries a load P at its lower end, in addition of its own weight, as shown in Fig. 2.112. The bar has constant thickness, but its width b varies along its length in such a way that the tensile stress σ_t is constant throughout. Find an expression of width b_x at x from the free end. Also find the width b_1 and b_2 and the volume V of the bar. Take γ as the specific weight.

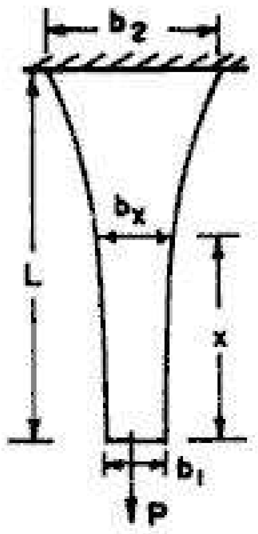


FIG. 2.112

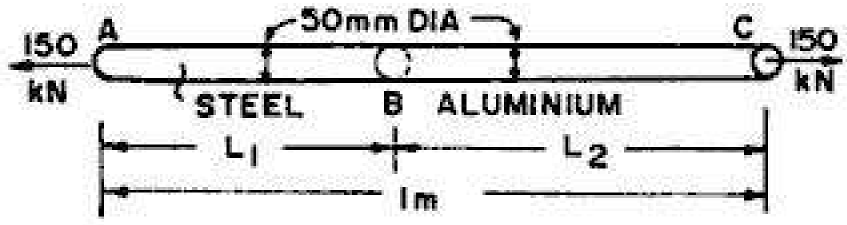


FIG. 2.113

22. A bar ABC, of 1000 mm length and 50 mm diameter is composed of two materials : part AB of steel and part BC of aluminium. If it is subjected to a tensile force of 150 kN, determine the lengths L_1 and L_2 for the steel and aluminium parts, respectively, in order that both the parts have the same elongation.
Also, determine the total elongation of the bar. Take $E_s = 210$ kN/mm^2 and $E = 70$ kN/mm^2.

23. Determine the radius r_x of a pillar of circular cross-section and height H in order that the volume of the pillar will be a minimum , if the pillar supports a compressive load P at the top, in addition

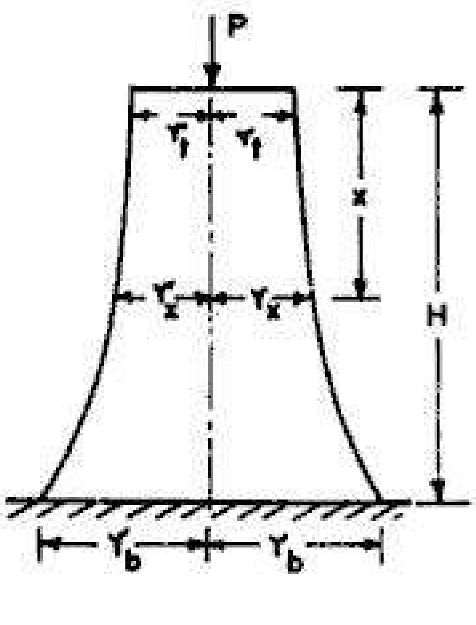


FIG. 2.114

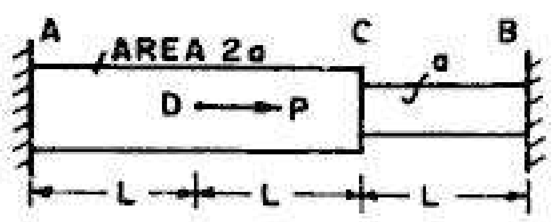


FIG. 2.115

to its own weight. Take f_c as the allowable stress in compression and γ as the specific weight of the material. Also determine the volume of the pillar.

24. An axially loaded bar AB, shown in Fig. 2.115 is held between rigid supports. The bar has cross-sectional area $2a$ from A to C and a from C to B. Determine the displacement δ_d of point D where the load P acts. Also determine the reactions at A and B.

25. A light rigid bar $ABCD$ pinned at B and connected to two vertical rods, is shown in Fig. 2.116. Assuming that the bar was initially horizontal and the rods stress free, determine the stress in each rod after the load $P=20$ kN is applied.
Take E for steel $= 200\ \text{kN/mm}^2$ and E for aluminium $= 100\ \text{kN/mm}^2$.

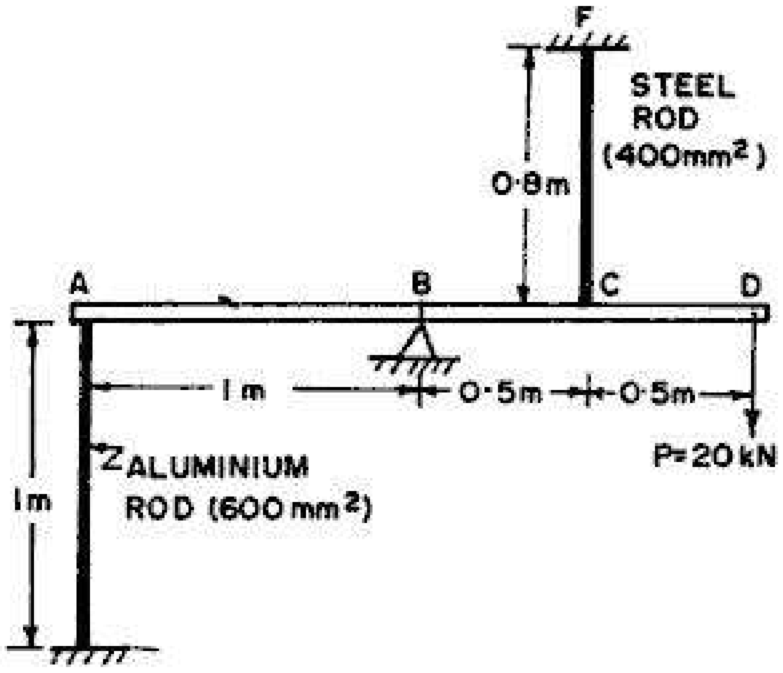


FIG. 2.116

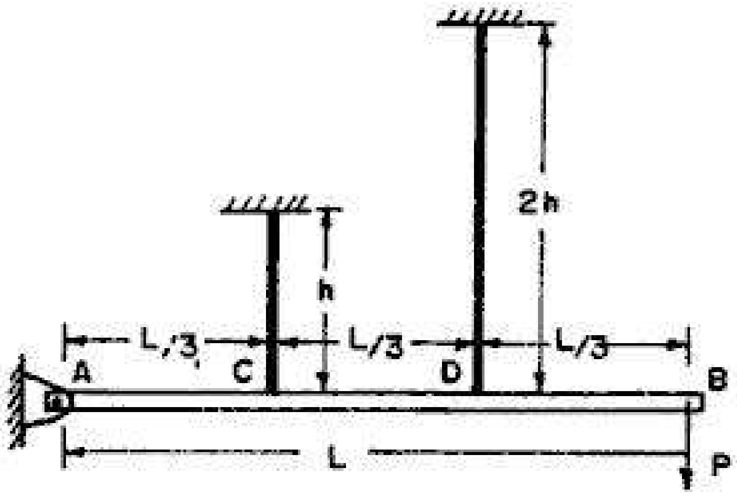


FIG. 2.117

26. A rigid bar AB of length L is hinged to a wall at A and supported by two vertical wires attached at point C and D, as shown in Fig. 2.117. Both the wires have the same area of cross-section and are made of the same material. Determine the forces in the wires.

ANSWERS

1. $P = 84$ kN ; 1.11 mm
2. 196.875 kN
3. 20.53 mm
4. 29.13 mm ; 116 mm
5. 0.28 mm
6. $p_s = 118.44\ \text{N/mm}^2$; $59.22\ \text{N/mm}^2$; 0.178 mm
7. $p_s = 43.24\ \text{N/mm}^2$; $p_c = 2.88\ \text{N/mm}^2$; 112.475 kN
8. $\frac{5}{3}W$; $\frac{4}{3}W$
9. $P_b = \frac{1}{4}P$; $P_s = \frac{1}{2}P$
10. 0.65 W ; 0.20 W ; 0.15 W

11. $R_A = 35$ kN ; $R_C = 15$ kN ; $p_{AB} = 350$ N/mm^2 (tension); $p_{BC} = 75$ N/mm^2 (comp.)

12. (*a*) 2.5 mm (*b*) 22.5 kN

13. $P_c = 59.091$ kN ; $P_z = 68.178$ kN ; $P_a = 72.729$ kN

$p_c = 118.18$ N/mm^2 ; $p_z = 90.90$ N/mm^2 ; $p_a = 72.729$ N/mm^2

14. (a) $p_a = 48.61$ N/mm^2; $p_s = 12.15$ N/mm^2 (b) $p_a = 28.75$ N/mm^2 ; $p_s = 7.19$ N/mm^2

15. $p_s = \frac{6}{55}\alpha\, t\, E$; $p_s' = \frac{3}{55}\alpha\, t\, E$; $p_c = \frac{6}{55}\alpha\, t\, E$

16. $P_{AB} = 133.33$ kN (tension); $P_{BC} = 33.33$ kN (comp.) ; $P_{CD} = 166.67$ kN (comp.)

$\Delta_B = 0.333$ mm ($\rightarrow$) ; $\Delta_c = 0.417$ mm ($\rightarrow$)

17. 3.133 m

18. $p_c = 18.70$ N/mm^2 ; $p_s = 186.97$ N/mm^2 ; 0.28 mm

19. $F_A = \dfrac{W.c\,(a+b)\,d_2}{d_1(a+b)^2 + d_1 a^2}$; $F_B = \dfrac{W.c.a.d_1}{d_2(a+b)^2 + d_1 d^2}$

$$R_D = W\left[\frac{ad_1(a-c) + (a+b)\,d_2\,(a+b-c)}{a^2 d_1 + (a+b)^2 d_2}\right]$$

20. (*a*) 14.4 kN (*b*) 15 kN

21. $b_x = \dfrac{P}{\sigma_t . t} e^{\gamma x/\sigma_t}$

22. $L_1 = 750$ mm ; $L_2 = 250$ mm ; $\Delta = 0.546$ mm

23. $r_x = \left(\dfrac{P.e^{\gamma x/fc}}{f_c.\pi}\right)^{1/2}$; $V = \dfrac{P}{\gamma}\left(e^{\gamma h/fc} - 1\right)$

24. $\Delta_d = \dfrac{3PL}{8aE}$; $R_A = \dfrac{3}{4}P$; $R_B = \dfrac{1}{4}P$

25. $p_s = 29.41$ N/mm^2 ; $p_a = 23.53$ N/mm^2

26. $P_c = P_D = P$

3

Elastic Constants

3.1. ELASTIC CONSTANTS

Elastic constants are those *factors* which determine the deformations produced by a given *stress system* acting on a material. These factors (i.e. elastic constants) are *constant* within the limits for which Hooke's laws are obeyed. Various elastic constants are :

(*i*) Modulus of elasticity (E)

(*ii*) Poisson's ratio (μ or $1/m$)

(*iii*) Modulus of rigidity (G or N)

and (*iv*) Bulk modulus (K)

3.2. LONGITUDINAL STRAIN : MODULUS OF ELASTICITY

As stated in §2.3, when an axial stress p (say, tensile) is applied along the longitudinal axis of a bar, the length of the bar will be increased. This change in the length (usually called deformation) per unit length of the bar, is termed as *longitudinal strain* (ε) or *primary strain*. The ratio of stress to strain, within elastic limits, is called the *modulus of elasticity* (E):

Thus, modulus of elasticity $E = \frac{p}{\varepsilon}$ ...(3.1)

The modulus of elasticity (also called Young's modulus of elasticity) is the *constant of proportionality* which is defined as the intensity of stress that causes unit strain.

Table 3.1. gives the values of *modulus of elasticity* (E) for some common materials.

3.3. LATERAL STRAIN : POISSON'S RATIO

When an axial force (say, tensile) is applied along the longitudinal axis of a bar, the length of the bar will *increase*, but at the same time its lateral dimensions (i.e. either the width and breadth or the diameter) will be *decreased*. This phenomenon is shown by dotted lines in Fig. 3.1.

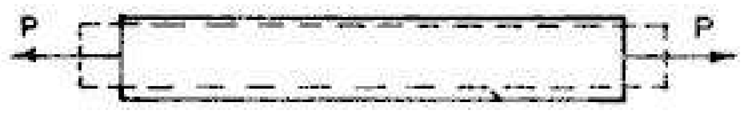


FIG. 3.1

TABLE 3.1. VALUES OF MODULUS OF ELASTICITY

S.No	*Material*	*Modulus of elasticity E (kN/mm² or GPa)*
1.	Aluminium (Pure)	70
2.	Aluminium alloys	70 – 79
3.	Brass	96 – 110
4.	Bronze	96 – 120
5.	Cast iron	83 – 170
6.	Concrete (compression)	18 – 30
7.	Copper (Pure)	110 – 120
8.	Rubber	0.0007 – 0.004
9.	Steel	190 – 210
10.	Wrought iron	190

Thus, any direct stress produces a strain in its own direction (called longitudinal strain) and an opposite kind of strain (called lateral strain) in every direction at right angles to this. The ratio of the *lateral strain* to the *longitudinal strain* is *constant* for a given material. This constant is known as *Poisson's ratio.*

Thus, $$\frac{\text{Lateral strain}}{\text{Longitudinal strain}} = \text{constant} = \frac{1}{m} = \mu \qquad ...(3.2)$$

The *constant* is named after French mathematician Poisson, who first predicted its existence and found its value for many isotropic materials. The value of Poisson's ratio varies from 0.25 to 0.42 for most of metals. For rubber, its value range from 0.45 to 0.50. Table 3.2 gives the value of Poisson's ratio for some common materials. The value of Poisson's ratio is the same in tension and compression.

TABLE 3.2. VALUES OF POISSON'S RATIO

S.No	*Material*	*Poisson's Ratio $\frac{1}{m}$ or μ*
1.	Aluminium (Pure)	0.33
2.	Aluminium alloys	0.33
3.	Brass	0.34
4.	Bronze	0.34
5.	Cast iron	0.2 – 0.3
6.	Concrete (compression)	0.1 – 0.2
7.	Copper (Pure)	0.33 – 0.36
8.	Rubber	0.45 – 0.50
9.	Steel	0.27 – 0.30
10.	Wrought iron	0.3

3.4. VOLUMETRIC STRAIN DUE TO SINGLE DIRECT STRESS

Fig 3.2 shows a rectangular bar of length L, width b and thickness t subjected to single direct stress (p) acting along its longitudinal axis. Let this stress be tensile in nature.

Then longitudinal strain, $e_1 = \frac{p}{E}$ (tensile)

Lateral strain, $e_2 = -\frac{1}{m} e_1 = -\frac{1}{m}\frac{p}{E} = \frac{p}{mE}$ (compressive) ...(3.3)

Now, for the *rectangular bar*, $V = L \,.\, b \,.\, t$ ...(*i*)

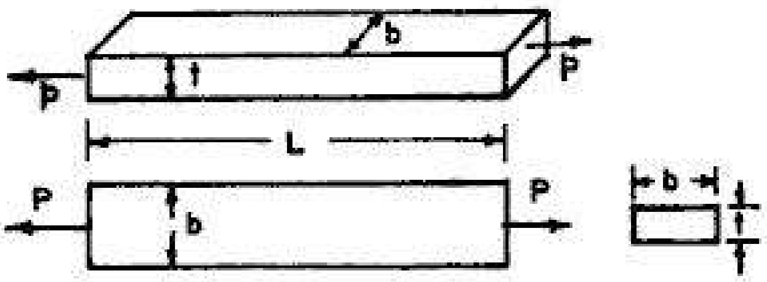


FIG. 3.2

If δL, δb and δt denote the *change* in the corresponding dimension, the final volume $(V + \delta V)$ is given by

$$(V + \delta V) = (L + \delta L)(b + \delta b)(t + \delta t)$$

or $$(V + \delta V) = Lbt\left(1 + \frac{\delta L}{L}\right)\left(1 + \frac{\delta b}{b}\right)\left(1 + \frac{\delta t}{t}\right)$$

or $$(V + \delta V) = V\left[1 + \frac{\delta L}{L} + \frac{\delta b}{b} + \frac{\delta t}{t} + \frac{\delta L}{L}\cdot\frac{\delta b}{b} + \frac{\delta L}{L}\frac{\delta t}{t} + \frac{\delta b}{b}\cdot\frac{\delta t}{t} + \frac{\delta L}{L}\cdot\frac{\delta b}{b}\cdot\frac{\delta t}{t}\right]$$

Ignoring the *products* of small quantities, we have

$$(V + \delta V) \approx V\left[1 + \frac{\delta L}{L} + \frac{\delta b}{b} + \frac{\delta t}{t}\right]$$

or $$\delta V \approx V\left(\frac{\delta L}{L} + \frac{\delta b}{b} + \frac{\delta t}{t}\right)$$

or $$\frac{\delta V}{V} = \frac{\delta L}{L} + \frac{\delta b}{b} + \frac{\delta t}{t} \qquad \text{...}(ii)$$

or $$\frac{\delta V}{V} = e_v = e_1 + e_2 + e_3 = e_x + e_y + e_z \qquad \text{...(3.4)}$$

The above equation could also be obtained directly by taking logarithm and partially differentiating both the sides of Eq. (*i*). Thus $\log V = \log L + \log b + \log t$

$$\therefore \quad \frac{\delta V}{V} = \frac{\delta L}{L} + \frac{\delta b}{L} + \frac{\delta t}{t} = e_1 + e_2 + e_3 \qquad \text{...(3.4)}$$

Now, $$\frac{\delta L}{L} = e_1 = \frac{p}{E};\ \frac{\delta b}{b} = e_2 = -\frac{p}{mE} \text{ and } \frac{\delta t}{t} = e_3 = -\frac{p}{mE}$$

$\therefore$ Volumetric strain, $$e_v = \frac{\delta V}{V} = \frac{p}{E} - \frac{p}{mE} - \frac{p}{mE} = \frac{p}{E}\left(1 - \frac{2}{m}\right) \qquad \text{...(3.5)}$$

$\therefore$ Again, for the *circular bar* of diameter d, we have

$$V = \frac{\pi}{4} d^2 L$$

Hence $$\frac{\delta V}{V} = \frac{\delta L}{L} + \frac{2\,\delta d}{d}$$

Here, $$\frac{\delta L}{L} = e_1 = \frac{p}{E} \quad \text{and} \quad \frac{\delta d}{d} = e_2 = -\frac{p}{mE}$$

$\therefore$ Volumetric strain, $e_v = \frac{\delta V}{V} = \frac{p}{E} - \frac{2p}{m} = \frac{p}{E}\left(1 - \frac{2}{m}\right)$ ...(3.5)

3.5. VOLUMETRIC STRAIN DUE TO THREE MUTUALLY PERPENDICULAR STRESS SYSTEM

Fig. 3.3 shows a parallelopiped subjected to three tensile stresses p_1, p_2 and p_3 in the three mutually perpendicular directions. As obtained in the previous article,

$$\frac{\delta V}{V} = e_1 + e_2 + e_3$$

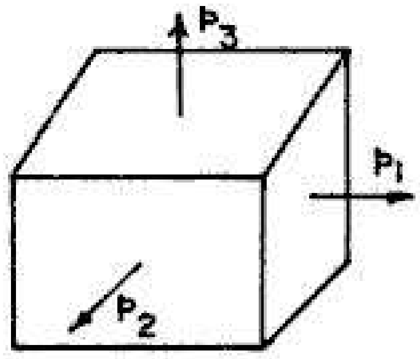


FIG. 3.3

Since any direct stress produces a strain in its own direction and an opposite kind of strain in every direction at right angles to this we have,

Longitudinal strain $e_1 = \frac{p_1}{E} - \frac{p_2}{mE} - \frac{p_3}{mE} = \frac{p_1}{E} - \frac{p_2 + p_3}{mE}$ ...(3.7 *a*)

Similarly, $e_2 = \frac{p_2}{E} - \frac{p_3}{mE} - \frac{p_1}{mE} = \frac{p_2}{E} - \frac{p_3 + p_1}{mE}$ ...(3.7 *b*)

and $e_3 = \frac{p_3}{E} - \frac{p_1}{mE} - \frac{p_2}{mE} = \frac{p_3}{E} - \frac{p_1 + p_2}{mE}$ ...(3.7 *c*)

Eqs. 3.7 are known as *general equations of Hooke's law* or *generalised Hooke's law*.

Adding the three expressions of Eqs. 3.7, we get.

$$e_v = \frac{\delta V}{V} = e_1 + e_2 + e_3 = \left(1 - \frac{2}{m}\right)\left(\frac{p_1 + p_2 + p_3}{E}\right) \quad ...(3.8)$$

The sum of three strains (e_1, e_2, e_3) is known as *dilatation* and, for small strains, it represents change in volume per unit volume.

If some of the stresses are of opposite sign (i.e., compressive) necessary changes in the algebraic signs of the above expressions will have to be made.

3.6. UPPER LIMIT OF POISSON'S RATIO

From Eq. 3.8 we have

$$e_1 + e_2 + e_3 = \left(1 - \frac{2}{m}\right)\left(\frac{p_1 + p_2 + p_3}{E}\right)$$

For the case of hydrostatic tension, we have $p_1 = p_2 = p_3 = p$ and the resulting strains are $e_1 = e_2 = e_3 = e$.

Hence $$3e = \left(1 - \frac{2}{m}\right)\left(\frac{3p}{E}\right)$$

or $$Ee = p\left(1 - \frac{2}{m}\right) \quad ...(3.9)$$

Since Ee and p in the above expression are positive numbers, $\left(1-\frac{2}{m}\right)$ must also be positive. This limits $\frac{2}{m}$ to a maximum of 1 or the Poisson's ratio $\frac{1}{m}$ to 0.5. *No material is known to have a higher value for Poisson's ratio although* $\frac{1}{m}$ *for materials like rubber approaches this value.*

Example 3.1. *A bar of steel has rectangular cross-section 30 mm × 20 mm. Find the dimensions of the sides and percentage decrease of area of cross-section, when it is subjected to a tensile force of 120 kN in the direction of its length. Take* $E = 2 \times 10^5$ N/mm^2 *and* $m = 10/3$.

Solution

Strain in the direction of pull $e_1 = \frac{P}{AE} = \frac{120 \times 10^3}{30 \times 20 \times 2 \times 10^5} = 10 \times 10^{-4}$

Lateral strain $= -\frac{e_1}{m} = -\frac{3}{10} \times 10 \times 10^{-4} = 3 \times 10^{-4}$

Hence 30 mm side is decreased by $30 \times 3 \times 10^{-4} = 0.009$ mm

and 20 mm side is decreased by $20 \times 3 \times 10^{-4} = 0.006$ mm

Hence dimension of 30 mm side $= 30 - 0.009 \approx 29.991$ mm

and dimension of 20 mm side $= 20 - 0.006 = 19.994$ mm.

New area of cross-section $= (30 - 0.009)(20 - 0.006) \approx 600 - 0.36$

% decrease of area of cross-section $= \frac{0.36}{600} \times 100 =$ **0.06%**

Example 3.2. *A steel bar, 300 mm long and 30 mm × 30 mm cross-section, is subjected to a tensile force of 150 kN in the direction of its length. Determine the change in volume, taking* $E = 2 \times 10^5$ N/mm^2 *and* $1/m = 0.3$.

Solution :

Longitudinal strain $e_1 = \frac{150 \times 10^3}{30 \times 30 \times 2 \times 10^5} = 8.333 \times 10^{-4}$

$\therefore$ Lateral strain $e_2 = e_3 = -\frac{1}{m} e_1 = -0.3 \times 8.333 \times 10^{-4} = -2.5 \times 10^{-4}$

Now $\frac{\delta V}{V} = e_1 + e_2 + e_3 = [8.333 - 2 \times 2.5]\, 10^{-4} = 3.333 \times 10^{-4}$

$V = 300 \times 30 \times 30 = 270000 \text{ mm}^3 = 27 \times 10^4 \text{ mm}^3$

$\therefore$ $\delta V = (27 \times 10^4) \times (3.333 \times 10^{-4}) = \mathbf{90\ mm^3}$

Example 3.3. *A metal bar, 40 mm × 40 mm section, is subjected to a tensile load of 320 kN. The extension of a 200 mm gauge length is found to be 0.2 mm and the decrease in thickness 0.012 mm. Find the value of Young's modulus and Poisson's ratio.*

Solution :

$A = 40 \times 40 = 1600 \text{ mm}^2$; $\delta L = 0.2$ mm; $\delta b = 0.012$ mm

Linear strain, $e_1 = \frac{\delta L}{L} = \frac{0.2}{200} = 10 \times 10^{-4}$

Lateral strain $e_2 = \frac{\delta b}{b} = \frac{0.012}{40} = 3 \times 10^{-4}$

From Hooke's law, $\delta L = \frac{PL}{AE}$

$$\therefore \quad E = \frac{PL}{A \,.\, \delta L} = \frac{320 \times 10^3 \times 200}{40 \times 40 \times 0.2} = \mathbf{2 \times 10^5\ N/mm^2}$$

Now, $$\frac{1}{m} = \frac{\text{Lateral strain}}{\text{linear strain}} = \frac{e_2}{e_1} = \frac{3 \times 10^{-4}}{10 \times 10^{-4}} = \mathbf{0.3}$$

Example 3.4. *A steel cube block of 50 mm side is subjected to a force of 10 kN (tension), 12.5 kN (compression) and 7.5 kN (tension) along x, y and z directions respectively. Determine the change in the volume of the block. Take $E = 200\ kN/mm^2$ and $1/m = 0.3$.*

Solution Area of each side = $50 \times 50 = 2500\ mm^2$

Stress in x direction $= p_x = \frac{10 \times 10^3}{2500} = 4\ N/mm^2$ (tensile)

Stress in y direction $= p_y = \frac{12.5 \times 10^3}{2500}$
$= 5\ N/mm^2$ (comp.)

Taking tension as positive and compression as negative, we have

$$e_x = \frac{p_x}{E} + \frac{p_y}{mE} - \frac{p_z}{mE} = \frac{4}{E} + \frac{5 \times 0.3}{E} - \frac{3 \times 0.3}{E} = \frac{4.6}{E}$$

$$e_y = -\frac{p_y}{E} - \frac{p_z}{mE} - \frac{p_x}{mE} = -\frac{5}{E} - \frac{3 \times 0.3}{E} - \frac{4 \times 0.3}{E} = -\frac{7.1}{E}$$

$$e_z = \frac{p_z}{E} - \frac{p_x}{mE} + \frac{p_y}{mE} = \frac{3}{E} - \frac{4 \times 0.3}{E} + \frac{5 \times 0.3}{E} = \frac{3.3}{E}$$

Now $$\frac{\delta V}{V} = e_x + e_y + e_z$$

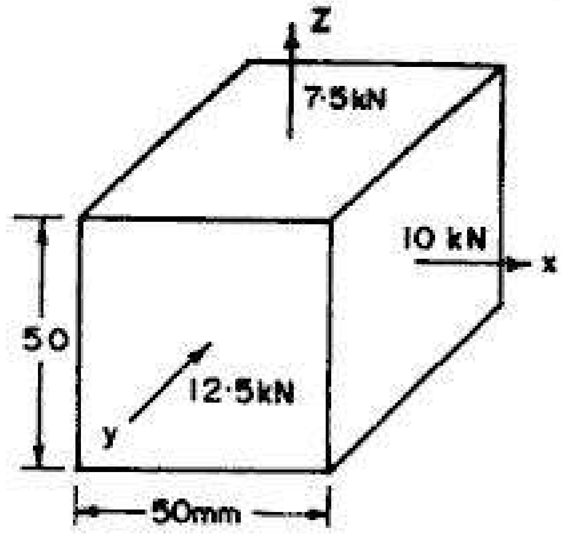


FIG. 3.4

$$\therefore \quad \frac{\delta V}{(50)^3} = \frac{4.6}{E} - \frac{7.1}{E} + \frac{3.3}{E} = \frac{0.8}{E}$$

$$\therefore \quad \delta V = \frac{0.8}{2 \times 10^5} \times (50)^3 = \mathbf{0.5\ mm^3}$$

Example 3.5. *The plates of a cylindrical boiler, 1.6 metre diameter and 2.5 m long are subjected to a tensile stress of 70 N/mm² in the direction of circumference and tensile stress of*

35 N/mm² in the axial direction. Neglecting the compressive stress due to steam pressure on the inner surface, determine the increase in the internal capacity.

Take $E = 2 \times 10^5$ N/mm² and $1/m = 0.3$.

Solution

Let us use suffix x for axial direction and suffix y for circumferential direction.

$$\therefore \qquad e_x = \frac{p_x}{E} - \frac{p_y}{mE} = \frac{35}{E} - \frac{0.3 \times 70}{E} = +\frac{14}{E}$$

and

$$e_y = \frac{p_y}{E} - \frac{p_x}{mE} = \frac{70}{E} - \frac{0.3 \times 35}{E} = +\frac{59.5}{E}$$

Since the diameter of the boiler increases or decreases in direct proportion to its circumference, e_y also denotes the *diametrical strain*. Hence the strains are : $e_1 = +\frac{14}{E}$ parallel to the axis and $e_2 = e_3 = +\frac{59.5}{E}$ along any two perpendicular radii.

$$\therefore \qquad \frac{\delta V}{V} = e_1 + e_2 + e_3 = \frac{14}{E} + \frac{59.5}{E} + \frac{59.5}{E} = \frac{133}{E}$$

Now

$$V = \frac{\pi}{4} d^2 L = \frac{\pi}{4}(1.6)^2 \times 2.5 = 5.0265 \text{ m}^3 = 5.0265 \times 10^9 \text{ mm}^3$$

$$\therefore \qquad \delta V = \frac{133}{E} \times V = \frac{133}{2 \times 10^5} \times 5.0265 \times 10^9 = 3343 \times 10^3 \text{ mm}^3$$

$$\mathbf{= 3343\ cm^3 = 3.343\ litres}$$

3.7. SHEAR MODULUS OR MODULUS OF RIGIDITY

The shear modules or modulus of rigidity (also called the *modulus of transverse elasticity*) expresses the relation between shear stress and shear strain. It has been found experimentally that, within elastic limit, shear stress (q) is proportional to the shear strain (φ)

Thus,

$$q \propto \varphi \qquad \text{...(3.10 } a)$$

or

$$q = N\varphi \qquad \text{...(3.10 } b)$$

or

$$\frac{q}{\varphi} = N \qquad \text{...(3.10)}$$

where N = modulus of rigidity

(also sometimes denoted by symbol C or G)

φ = Shear strain (in radians)

(also sometimes denoted by the symbol γ)

Table 3.3 gives the values of *modulus of rigidity* for some common engineering materials.

TABLE 3.3. VALUES OF MODULUS OF RIGIDITY

S.No	*Material*	*Modulus of Rigidity N (kN/mm^2 or GPa)*
1.	Aluminium (Pure)	26
2.	Aluminium alloys	26 - 30
3.	Brass	36 - 41
4.	Bronze	36 - 44
5.	Cast iron	32 - 69
6.	Copper (Pure)	40 - 47
7.	Rubber	0.0002 - 0.001
8.	Steel	75 - 80
9.	Wrought iron	75

3.8. COMPLIMENTARY SHEAR STRESS

Fig. 3.5 (*a*) shows an infinitely small block *ABCD*, under shear stress intensity q. Such a state of shear will have a tendency to rotate the block in the clockwise direction. Since there is no other force acting on the block, there will be no equilibrium. The block can, however, attain equilibrium only if a couple is applied in such a way that it has the tendency to rotate the block in the counter-clockwise direction. This can be achieved by having a shear stress intensity q' on faces AD and BC in the direction shown in Fig. 3.5 (*b*). Considering unit thickness of the block perpendicular to the Fig., we have

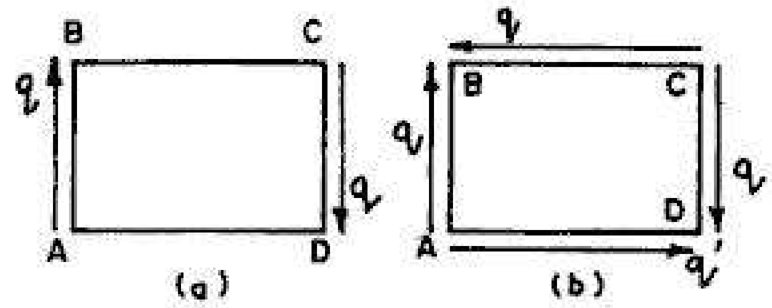


FIG. 3.5

Moment of given couple = Force × lever arm $= (q \,.\, AB)\, AD$

Moment of balancing couple $= (q' \,.\, AD)\, AB$.

Equating the two to attain equilibrium, we have

$$(q'\, AD)\, AB = (q \,.\, AB)\, AD.$$

From which $$q' = q \qquad ...(3.11)$$

The stress q' is known as the *complimentary shear stress* and consequently, we have the following *rule* of complimentary shear stress:

"A shear stress in a given direction cannot exist without a balancing shear stress of equal intensity in a direction at right angles to it".

3.9. STATE OF SIMPLE SHEAR

The state of stress indicated in Fig. 3.5 (*b*) is known as the *state of simple shear*, in which no other stresses are acting. In order to study the effect of such a state of simple shear, let us consider a square block under the state of simple shear (Fig. 3.6 *a*).

Let $$AB = BC = CD = DA = b$$

$\therefore$ Length of diagonals $AC = BD = b\sqrt{2}$. Consider unit thickness of the block, perpendicular to the Fig.

Consider the equilibrium of the triangular portion ABC (Fig. 3.6 b). Resolved sum of q perpendicular to the diagonal

$$= 2\,(q \times b \times 1)\cos 45° = \frac{2qb}{\sqrt{2}} = \sqrt{2}.qb$$

If p is the tensile stress so induced on the diagonal, we have

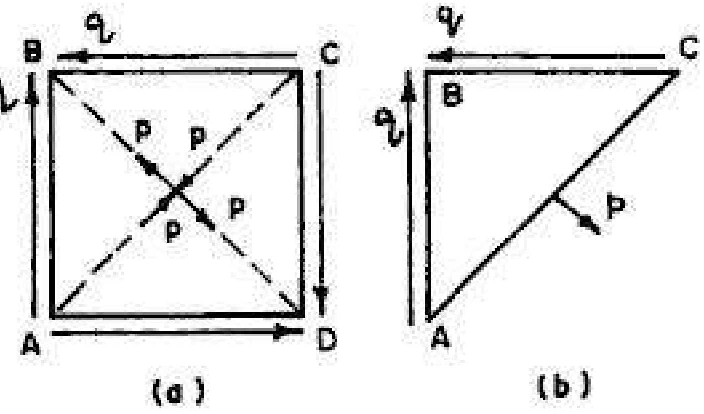


FIG. 3.6. STATE OF SIMPLE SHEAR

$$p\,(AC \times 1) = \sqrt{2}\,q\,.\,b$$

or $$p\left(\sqrt{2}\,b\right) = \sqrt{2}\,q\,b$$

From which $$p = q \text{ (tensile)} \qquad ...(3.12)$$

Resolving the shear stresses q along the diagonal AC, the resultant is zero.

Similarly, it can be shown that the intensity of *compressive stress* p on plane BD is numerically equal to q. Hence we have the following rule :

"A state of simple shear produces pure tensile and compressive stresses across planes inclined at 45° to those of pure shear, and the intensities of these direct stresses are each equal to the intensity of the pure shear stress".

3.10. LINEAR STRAIN OF DIAGONAL DUE TO SHEAR

Fig. 3.7 (a) shows a square block $ABCD$ subjected to a state of simple shear q. Fig. 3.7 (b) shows the resulting distorted shape of the block, wherein the total change in each corner angles is $\pm\left(\frac{\varphi}{2}+\frac{\varphi}{2}\right) = \pm\,\varphi$. For the purpose of convenience in computations, Fig. 3.7 (c) shows the distorted shape of the block in which the direction of one side AD has been kept fixed, while keeping the shearing strain φ still the same.

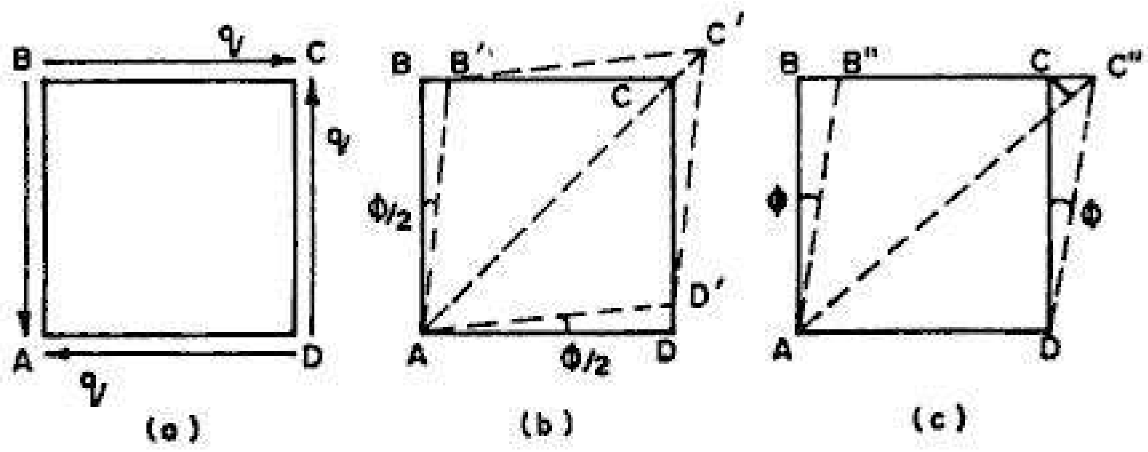


FIG. 3.7. LINEAR STRAIN DUE TO SHEAR

The shearing strain φ is extremely small. Hence we can assume BB'' as an arc with A as centre and AB as radius.

Then $$\varphi = \frac{BB''}{AB} = \frac{CC''}{CD} \qquad ...(i)$$

Draw CF perpendicular to AC''. Considering CF also an arc with A as centre and AC as radius, the elongation of diagonal AC can be nearly taken equal to FC''.

$$\therefore \text{ Linear strain } (e) \text{ of diagonal} = \frac{FC''}{AC} = \frac{CC'' \cos 45°}{CD \sec 45°} \qquad ...(ii)$$

$$\therefore \qquad e = \frac{1}{2}\frac{CC''}{CD} = \frac{1}{2}\varphi = \frac{q}{2N} \qquad ...(3.13)$$

Hence linear strain of the diagonal is equal to half the shear strain φ.

3.11. RELATION BETWEEN E AND N

From Eq. 3.13, we have seen that linear strain of a diagonal is given by

$$e = \frac{1}{2}\varphi = \frac{q}{2N} \qquad ...(i)$$

Also, from § 3.9, we find that state of simple shear produces tensile and compressive stresses along the diagonal planes, and the value of each such direct stress is numerically equal to q. Thus, along diagonal AC (Fig. 3.7 c), there is a tensile stress $p\ (= q)$ and along diagonal BD, there is compressive stress $p\ (= q)$.

Hence, the linear strain e of the diagonal AC, due to two mutually perpendicular direct stresses is given by.

$$e = \frac{p}{E} - \left(-\frac{p}{mE}\right) = \frac{p}{E}\left(1 + \frac{1}{m}\right) = \frac{q}{E}\left(1 + \frac{1}{m}\right) \qquad ...(ii)$$

Equating (i) and (ii), we get

$$\frac{q}{2N} = \frac{q}{E}\left(1 + \frac{1}{m}\right)$$

or

$$E = 2N\left(1 + \frac{1}{m}\right) \qquad ...(3.14)$$

Also,

$$N = \frac{mE}{2(m+1)} . \qquad ...(3.14\ a)$$

These are the desired relations between E and N.

Example 3.6. *A shaft is subjected to a twisting moment which produces a shearing stress at the surface of $100\ N/mm^2$ in planes perpendicular to the axis of the shaft. A small square is scratched on the surface of the shaft with two of its sides parallel to the axis of the shaft. Find the change in the angle of the corners of the square. Take $N = 80\ N/mm^2$.*

Solution :

Let us assume that limit of proportionality is not exceeded.

$$\therefore \qquad q = N\varphi$$

or

$$\varphi = \frac{q}{N} = \frac{100}{80 \times 1000} = 1.25 \times 10^{-3} \text{ radian}$$

But 1 radian = 206265 seconds.

$$\therefore \qquad \varphi \text{ (in seconds)} = 1.25 \times 10^{-3} \times 206265 \approx 258''$$

$\therefore$ Change in angle $\varphi = \mathbf{4' \, 18''}$

Example 3.7. *A hole is to be punched through a steel plate of 8 mm thickness. Find the least diameter of hole which can be punched, if (i) steel punch can be worked to a compressive stress of 800 N/mm² and (ii) the ultimate shear strength is 300 N/mm².*

Solution :

Let d be the diameter of the hole, t be the thickness of the plate, q be the shear stress and p be the compressive stress.

$$\text{Compressive force exerted} = \frac{\pi}{4} d^2 p \quad ...(i)$$

$$\text{Shear force required} = (\pi \, d \, . \, t) \, q \quad ...(ii)$$

Equating the two, $\frac{\pi}{4} d^2 p = \pi \, d \, t \, q$

$$\therefore \qquad d = \frac{4 \, t \, q}{p} = \frac{4 \times 8 \times 300}{800} = \mathbf{12 \ mm}$$

Example 3.8. *A lever is keyed to a shaft of 100 mm diameter. The width of the key is 12 mm and the length is 50 mm. Find the load that can be applied at a radius of 1.2 m, if the shear stress in the key is not to exceed 100 N/mm².*

Solution :

Let W (in N) be the load, applied at a lever arm of 1.2 m.

$\therefore$ Torque (T) produced at the shaft $= 1200 \, W$ N-mm.

Shear force in the plane AB

$$= \frac{T}{r} = \frac{1200 \, W}{50} = 24 \, W$$

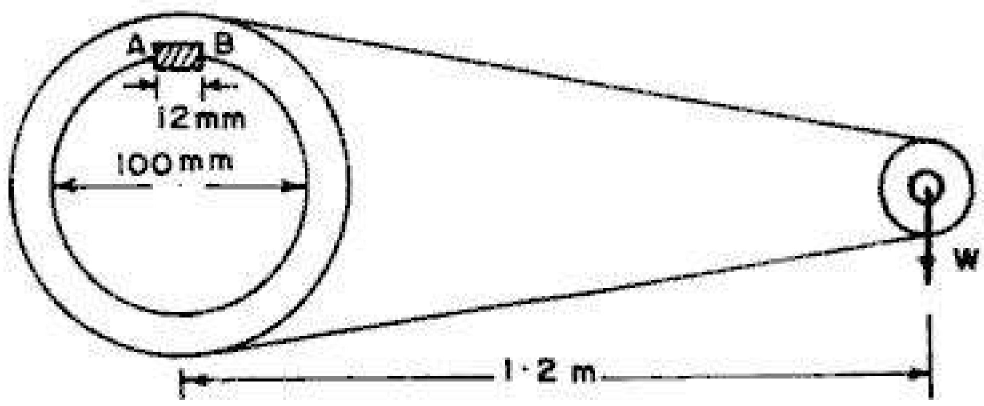


FIG. 3.8

Area of key $= 12 \times 50 = 600 \text{ mm}^2$

$$\text{Shear stress in key} = \frac{24 \, W}{600} = 0.04 \, W \text{ N/mm}^2.$$

But this is not to exceed 100 N/mm².

$$\therefore \qquad 0.04 \, W = 100$$

or
$$W = \frac{100}{0.04} = 2500 \text{ N} = \mathbf{2.5 \ kN}$$

Example 3.9. *A bronze specimen has a modulus of elasticity of 1.1×10^5 N/mm² and modulus of rigidity of 0.41×10^5 N/mm². Determine the Poisson's ratio for the material.*

Solution :

Given : $E = 1.1 \times 10^5 \text{ N/mm}^2$ and $N = 0.41 \times 10^5 \text{ N/mm}^2$.

From Eq. 3.14, $E = 2N\left(1 + \frac{1}{m}\right)$

$\therefore$ $\frac{1}{m} = \frac{E}{2N} - 1 = \frac{1.1 \times 10^5}{2 \times 0.41 \times 10^5} - 1 = \mathbf{0.341}$

3.12. BULK MODULUS

When a body is subjected to three mutually perpendicular like stresses of equal intensity (p), the ratio of direct stress (p) to the corresponding volumetric strain (e_v) is defined as the *bulk modulus* K for the material of the body.

Thus $$K = \frac{\text{Direct stress}}{\text{Volumetric strain}} = \frac{p}{e_v} \qquad ...(3.15)$$

3.13. RELATION BETWEEN E AND K

Let a cube of side L be subjected to three mutually perpendicular like compressive stresses of equal intensity q (Fig. 3.9).

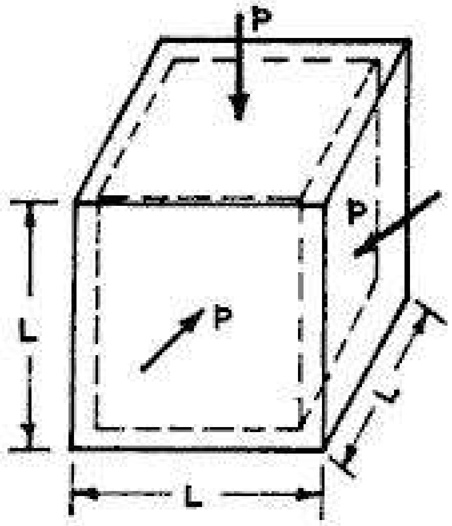


FIG. 3.9

Total linear strain of each side $= e = \frac{p}{E} - \frac{p}{mE} - \frac{p}{mE}$

Hence $$\frac{\delta L}{L} = e = \frac{p}{E}\left(1 - \frac{2}{m}\right) \qquad ...(i)$$

Now $$V = L^3$$

or $$\delta V = 3L^2 \delta L$$

$\therefore$ $$\frac{\delta V}{V} = e_v = \frac{3\,\delta L}{L} = 3e = \frac{3p}{E}\left(1 - \frac{2}{m}\right) \qquad ...(ii)$$

Also, by definition of bulk modulus, $\frac{p}{e_v} = K$

or $$e_v = \frac{\delta V}{V} = \frac{p}{K} \qquad ...(iii)$$

Equating (*ii*) and (*iii*)

$$\frac{p}{K} = \frac{3p}{E}\left(1 - \frac{2}{m}\right)$$

or $$E = 3K\left(1 - \frac{2}{m}\right) \qquad ...(3.16)$$

or $$K = \frac{mE}{3(m-2)} \qquad ...(3.16\ a)$$

This is the desired relation between E and K in terms of m.

3.14. RELATION BETWEEN E, N, K AND m

Equating Eq. 3.14 and 3.16

$$E = 2N\left(1 + \frac{1}{m}\right) = 3K\left(1 - \frac{2}{m}\right) \qquad ...(3.17)$$

Eliminating E from Eq. 3.17, we get $\frac{1}{m} = \frac{3K - 2N}{6K + 2N}$ ...(3.18)

Eliminating m from Eq. 3.17, we get $E = \frac{9KN}{N + 3K}$ ...(3.19)

Example 3.10. *A bar of 25 mm diameter is subjected to a pull of 40 kN. The measured extension on gauge length of 200 mm is 0.085 mm and the change in diameter is 0.003 mm. Calculate the Poisson's ratio and the values of the three moduli.*

Solution :

$$A = \frac{\pi}{4}(25)^2 = 490.87 \text{ mm}^2.$$

(a) Value of E

Linear strain $e_1 = \frac{P}{AE} = \frac{40 \times 10^3}{490.87 \times E} = \frac{81.49}{E}$ ...(*i*)

But linear strain $e_1 = \frac{\delta L}{L} = \frac{0.085}{200} = 4.25 \times 10^{-4}$ ...(*ii*)

Equating the two, $\frac{81.49}{E} = 4.25 \times 10^{-4}$

From which $E = \mathbf{1.917 \times 10^5\ N/mm^2}$

(b) Value of m

$$\text{Lateral strain} = e_2 = \frac{\delta d}{d} = \frac{0.003}{25} = 1.2 \times 10^{-4}$$

Now $\frac{\text{Lateral strain}}{\text{Longitudinal strain}} = \frac{1}{m}$

$\therefore$ Poisson's ratio $\frac{1}{m} = \frac{1.2 \times 10^{-4}}{4.25 \times 10^{-4}} = \frac{1}{\mathbf{3.542}}$

or $m = \mathbf{3.542}$

(c) Value of N

Now, from Eq. 3.14 (*a*), $N = \frac{mE}{2(m+1)}$

$\therefore$ $N = \frac{3.542 \times 1.917 \times 10^5}{2(3.542 + 1)} = \mathbf{0.748 \times 10^5\ N/mm^2}$

(d) Value of K

From Eq. 3.16 (*a*), $K = \frac{mE}{3(m-2)}$

$\therefore$ $K = \frac{3.542 \times 1.917 \times 10^5}{3(3.542 - 2)} = \mathbf{1.468 \times 10^5\ N/mm^2}$

Example 3.11. *Calculate the modulus of rigidity and bulk modulus of a cylindrical bar of diameter 25 mm and of length 1.2 m if the longitudinal strain in a bar during a tensile test is four times the lateral strain. Find the change in volume when the bar is subjected to a hydrostatic pressure of 120 N/mm². Take E = 1.2 × 10⁵ N/mm².*

Solution :

Volume of bar, $V = \frac{\pi}{4} d^2 L = \frac{\pi}{4} (25)^2 \times 1200 = 589049 \text{ mm}^3$.

(a) Value of m

Given : Longitudinal strain = 4 × Lateral strain.

$$\therefore \qquad \frac{1}{m} = \frac{\text{lateral strain}}{\text{longitudinal strain}} = \frac{\mathbf{1}}{\mathbf{4}} = 0.25$$

and $\qquad m = \mathbf{4}$

(b) Value of N

From Eq. 3.14 (*a*), $N = \frac{m E}{2(m+1)} = \frac{4 \times 1.2 \times 10^5}{2(4+1)} = \mathbf{0.48 \times 10^5\ N/mm^2}$

(c) Value of K

From Eq. 3.16 (*a*), $K = \frac{mE}{3(m-2)} = \frac{4 \times 1.2 \times 10^5}{3(4-2)} = \mathbf{0.8 \times 10^5\ N/mm^2}$

(d) Change in volume of bar

By definition $\qquad K = \frac{p}{e_v}$

$$\therefore \qquad e_v = \frac{\delta V}{V} = \frac{p}{K} = \frac{120}{0.8 \times 10^5}$$

Hence $\qquad \delta V = \frac{120}{0.8 \times 10^5} \times 589049 = \mathbf{884\ mm^3}$

Example 3.12 (*a*). *Determine the percentage change in volume of a steel bar 40 mm square in section and 1 m long when subjected to an axial compressive load of 15 kN. (b) What change in volume would a 100 mm cube of steel suffer at a depth of 4 km in sea water ?*

Take E = 2 × 10⁵ N/mm² and N = 0.81 × 10⁵ N/mm².

Solution :

(a) Volume of bar, $\qquad V = b^2 . L$

$$\therefore \qquad \frac{\delta V}{V} = 2\frac{\delta b}{b} + \frac{\delta L}{L} = 2 e_b + e_L$$

or $\qquad \frac{\delta V}{V} = -\frac{2p}{mE} + \frac{p}{E} = \frac{p}{E}\left(1 - \frac{2}{m}\right) \qquad$ ...(1)

Now, $\qquad E = 2N\left(1 + \frac{1}{m}\right) \qquad$ ...Eq. 3.14

$$\therefore \qquad \frac{1}{m} = \frac{E}{2N} - 1 = \frac{2 \times 10^5}{2 \times 0.81 \times 10^5} - 1 = 0.2346$$

or $\quad m = 4.263$

Hence from (1), $\frac{\delta V}{V} = \frac{p}{E}(1 - 2 \times 0.2346) = \frac{15000}{1600\,(2 \times 10^5)} \times 0.5309 = 2.488 \times 10^{-5}$

$\therefore$ % reduction in volume$= \frac{\delta V}{V} \times 100 = 2.488 \times 10^{-5} \times 100 = \mathbf{0.00249}$

(*b*) On the cube, $p = wh$

Here, $w = 10080\ \text{N/mm}^3$ (for sea water) and $h = 4\ \text{km} = 4000\ \text{m}$.

$\therefore \quad p = 10080 \times 4000 = 40.32 \times 10^6\ \text{N/m}^2 = 40.32\ \text{N/mm}^2$.

Now, from Eq. 3.16 (*a*) $K = \frac{m\,E}{3\,(m-2)} = \frac{4.263 \times 2 \times 10^5}{3\,(4.263 - 2)} = 1.256 \times 10^5\ \text{N/mm}^2$

Now $\quad e_v = \frac{\delta V}{V} = \frac{p}{K}$ (by definition)

$\therefore \quad \delta V = \frac{p}{K} \times V = \frac{40.32}{1.256 \times 10^5} \times (100)^3 = \mathbf{321\ mm^3}$

Example 3.13. *The modulus of rigidity for a material is $0.5 \times 10^5\ N/mm^2$. A 12 mm diameter rod of the material was subjected to an axial pull of 14 kN and the change in diameter was observed to be 3.6×10^{-3} mm. Calculate Poisson's ratio and the modulus of elasticity.*

Solution :

$$A = \frac{\pi}{4}(12)^2 = 113.1\ \text{mm}^2$$

$$p = \frac{14 \times 10^3}{113.1} = 123.79\ \text{N/mm}^2$$

Now, lateral strain $= \frac{\delta d}{d} = \frac{3.6 \times 10^{-3}}{12} = 3 \times 10^{-4}$

But lateral strain $= \frac{p}{m\,E}$

$$\therefore \quad \frac{p}{m\,E} = 3 \times 10^{-4}$$

or $$mE = \frac{p}{3 \times 10^{-4}} = \frac{123.79}{3 \times 10^{-4}} = 41.26 \times 10^4 \qquad \text{...}(i)$$

Now $$E = 2N\left(1 + \frac{1}{m}\right)$$

or $$m\,E = 2N\,(m+1)$$

or $$41.26 \times 10^4 = 2 \times 0.5 \times 10^5\,(m+1)$$

From which $\quad m = 3.126$ and $\frac{1}{m} \approx \mathbf{0.32}$

$$\therefore \qquad E = \frac{41.26 \times 10^4}{m} = \frac{41.26 \times 10^4}{3.126} = \mathbf{1.32 \times 10^5\ N/mm^2}$$

Also,
$$K = \frac{E}{3\left(1 - \frac{2}{m}\right)} = \frac{1.32 \times 10^5}{3\left(1 - \frac{2}{3.126}\right)} = \mathbf{1.22 \times 10^5\ N/mm^2}$$

Alternatively, from Eq. 3.17,
$$K = \frac{2N\left(1 + \frac{1}{m}\right)}{3\left(1 - \frac{2}{m}\right)} = \frac{2 \times 0.5 \times 10^5\,(1 + 0.32)}{3\,(1 - 2 \times 0.32)}$$
$$= 1.22 \times 10^5\ \text{N/mm}^2$$

Example 3.14. *A rectangular block 250 mm × 100 mm × 80 mm is subjected to axial loads as follows :*

480 kN tensile in the direction of its length,

1000 kN compressive on the 250 mm × 100 mm faces,

and

900 kN tensile on 250 × 80 mm faces.

Assuming Poisson's ratio as 0.25, find in terms of modulus of elasticity of the material E, the strains in the direction of each force.

If E = 2 × 10⁵ N/mm², find the values of the modulus of rigidity and bulk modulus for the material of the block. Also, calculate the change in volume of the block due to loading specified above.

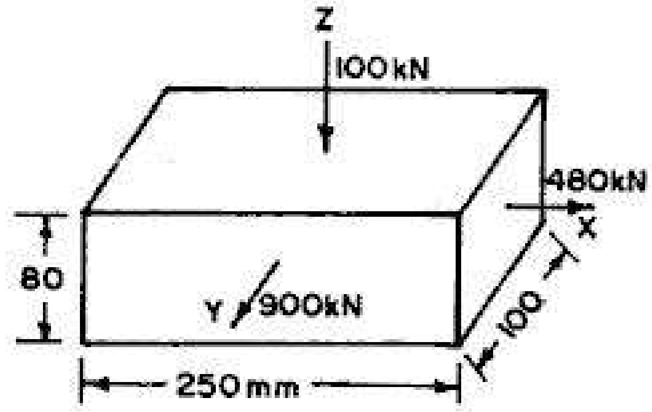


FIG. 3.10

(Based on Cambridge University)

Solution :

Volume $V = 250 \times 100 \times 80 = 2 \times 10^6\ \text{mm}^3$; $\frac{1}{m} = 0.25$; $m = 4$

(a) Strains in each direction

Stress in x-direction, $\quad p_x = \frac{P_x}{A_x} = \frac{480 \times 10^3}{100 \times 80} = 60\ \text{N/mm}^2$ (tension)

Stress in y-direction, $\quad p_y = \frac{P_y}{A_y} = \frac{900 \times 10^3}{250 \times 80} = 45\ \text{N/mm}^2$ (tension)

Stress in z-direction, $\quad p_z = \frac{P_z}{A_z} = \frac{1000 \times 10^3}{250 \times 100} = 40\ \text{N/mm}^2$ (comp.)

Taking tension as +ve and compression as negative, we have

$$e_x = \frac{p_x}{E} - \frac{p_y}{mE} + \frac{p_z}{mE} = \frac{60}{E} - \frac{45}{4E} + \frac{40}{4E} = \frac{\mathbf{58.75}}{\mathbf{E}}$$

$$= \frac{58.75}{2 \times 10^5} = \mathbf{2.9375 \times 10^{-4}}$$

$$e_y = \frac{p_y}{E} + \frac{p_z}{mE} - \frac{p_x}{mE} = \frac{45}{E} + \frac{40}{4E} - \frac{60}{4E} = \frac{\mathbf{40}}{\boldsymbol{E}}$$

$$= \frac{40}{2 \times 10^5} = \mathbf{2 \times 10^{-4}}$$

and
$$e_z = -\frac{p_z}{E} - \frac{p_x}{mE} - \frac{p_y}{mE} = -\frac{40}{E} - \frac{60}{4E} - \frac{45}{4E} = -\frac{\mathbf{66.25}}{\boldsymbol{E}}$$

$$= -\frac{66.25}{2 \times 10^5} = \mathbf{-3.3125 \times 10^{-4}}$$

(b) Change in volume

$$\frac{\delta V}{V} = e_x + e_y + e_z = 2.9375 \times 10^{-4} + 2 \times 10^{-4} - 3.3125 \times 10^{-4} = 1.625 \times 10^{-4}$$

$$\therefore \quad \delta V = 1.625 \times 10^{-4} \times (2 \times 10^6) = \mathbf{325\ mm^3}$$

(c) Modulus of rigidity

$$N = \frac{mE}{2(m+1)} = \frac{4 \times 2 \times 10^5}{2(4+1)} = \mathbf{0.8 \times 10^5\ N/mm^2}$$

(d) Bulk Modulus

$$K = \frac{mE}{3(m-2)} = \frac{4 \times 2 \times 10^5}{3(4-2)} = \mathbf{1.333 \times 10^5\ N/mm^2}$$

Example 3.15. *A metallic bar 250 mm × 100 mm × 50 mm is loaded as shown in Fig. 3.11.*

Find the change in volume. Take E = 200 kN/mm² and Poisson's ratio = 0.25. Also find the change that should be made in the 4000 kN load, in order that there should be no change in the volume of the bar. *(Based in U.L.)*

Solution :

$$V = 250 \times 100 \times 50 = 1250 \times 10^3\ \text{mm}^3;$$

$$\frac{1}{m} = 0.25\ ; m = 4$$

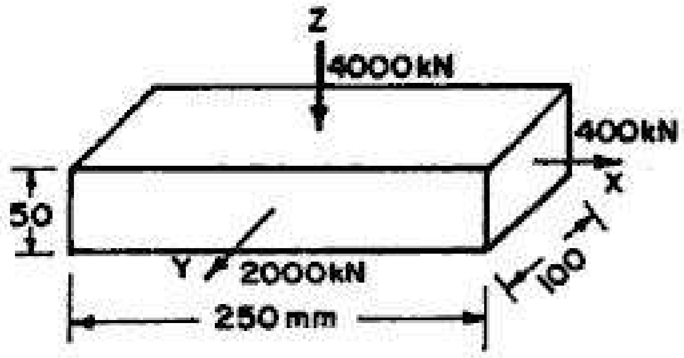


FIG. 3.11

(a) Change in volume

$$p_x = \frac{P_x}{A_x} = \frac{400 \times 10^3}{100 \times 50} = 80\ \text{N/mm}^2\ \text{(tension)}$$

$$p_y = \frac{P_y}{A_y} = \frac{2000 \times 10^3}{250 \times 50} = 160\ \text{N/mm}^2\ \text{(tension)}$$

$$p_z = \frac{P_z}{A_z} = \frac{4000 \times 10^3}{250 \times 100} = 160\ \text{N/mm}^2\ \text{(comp.)}$$

Considering tension as positive and compression as negative, we have

$$e_x = \frac{p_x}{E} - \frac{p_y}{mE} + \frac{p_z}{mE} = \frac{80}{E} - \frac{160}{4E} + \frac{160}{4E} = \frac{80}{E}$$

$$e_y = \frac{p_y}{E} + \frac{p_z}{mE} - \frac{p_x}{mE} = \frac{160}{E} + \frac{160}{4E} - \frac{80}{4E} = \frac{180}{E}$$

$$e_z = -\frac{p_z}{E} - \frac{p_x}{mE} - \frac{p_y}{mE} = -\frac{160}{E} - \frac{80}{4E} - \frac{160}{4E} = -\frac{220}{E}$$

Now $$\frac{\delta V}{V} = e_x + e_y + e_z = \frac{80}{E} + \frac{180}{E} - \frac{220}{E} = \frac{40}{E}$$

$\therefore$ $$\delta V = \frac{40}{E} \times V = \frac{40}{200 \times 1000} \times 1250 \times 10^3 = \mathbf{250\ mm^3}$$

(b) Change in 4000 kN load

Let P_z kN be the load, in place of 4000 kN load.

$\therefore$ $$p_z = \frac{P_z \times 10^3}{250 \times 100} = 0.04\, P_z \text{ N/mm}^2 \text{ (comp.)}$$

Hence $$e_x = \frac{p_x}{E} - \frac{p_y}{mE} + \frac{p_z}{mE} = \frac{80}{E} - \frac{160}{4E} + \frac{0.04\, P_z}{4E} = \frac{40}{E} + \frac{0.01\, P_z}{E}$$

$$e_y = \frac{p_y}{E} + \frac{p_z}{mE} - \frac{p_x}{mE} = \frac{160}{E} + \frac{0.04\, P_z}{4E} - \frac{80}{4E} = \frac{140}{E} + \frac{0.01\, P_z}{E}$$

and $$e_z = -\frac{p_z}{E} - \frac{p_x}{mE} - \frac{p_y}{mE} = -\frac{0.04\, P_z}{E} - \frac{80}{4E} - \frac{160}{4E} = -\frac{0.04\, P_z}{E} - \frac{60}{E}$$

In order that $\frac{\delta V}{V} = e_x + e_y + e_z$ should be equal to zero, we have

$$e_x + e_y + e_z = 0$$

$$\therefore \quad \left(\frac{40}{E} + \frac{0.01\, P_z}{E}\right) + \left(\frac{140}{E} + \frac{0.01\, P_z}{E}\right) - \left(\frac{0.04\, P_z}{E} + \frac{60}{E}\right) = 0$$

or $$\frac{120}{E} - \frac{0.02\, P_z}{E} = 0$$

From which $$P_z = \frac{120}{0.02} = \mathbf{6000}\text{ kN}$$

Hence P_z should be increased from its present value of 4000 kN to a new value of 6000 kN, in order that volumetric strain is zero.

Example 3.16. *A bar of steel is 40 mm in diameter and 500 mm long. A tensile load of 120 kN is found to stretch the bar by 0.24 mm. The same bar, when subjected to a torque of 1.3 kN-m is found to twist through 1.92°. Find the values of the four elastic constants.*

Solution :

(a) Value of E

As a bar, $\Delta = \dfrac{PL}{AE} = \dfrac{4\,PL}{\pi\, d^2 E}$

$\therefore$ $$E = \frac{4\,PL}{\pi\, d^2 \Delta} = \frac{4 \times 120 \times 10^3 \times 500}{\pi\,(40)^2 \times 0.24} = \mathbf{1.989 \times 10^5\ N/mm^2}$$

(b) Value of N

As a shaft, $N = \dfrac{TL}{J\theta} = \dfrac{32\,TL}{\pi\, d^4 \theta}$

where $\theta = \dfrac{\pi}{180} \times 1.92 = 0.0335$ radian, and $T = 1.3$ kN-m $= 1.3 \times 10^6$ N-mm

$\therefore$ $$N = \frac{32 \times 1.3 \times 10^6 \times 500}{\pi\,(40)^4 \times 0.0335} = \mathbf{0.772 \times 10^5\ N/mm^2}$$

(c) Value of $\frac{1}{m}$

Again $E = 2N\left(1 + \dfrac{1}{m}\right)$

$\therefore$ $$\frac{1}{m} = \frac{E}{2N} - 1 = \frac{1.989 \times 10^5}{2 \times 0.772 \times 10^5} - 1 = \mathbf{0.288}$$

(d) Value of K

$$K = \frac{E}{3\left(1 - \frac{2}{m}\right)} = \frac{1.989 \times 10^5}{3\,(1 - 2 \times 0.288)} = \mathbf{1.565 \times 10^5\ N/mm^2}$$

3.15. ADDITIONAL ILLUSTRATIVE EXAMPLES*

Some additional illustrative examples of advanced nature extremely useful for *competitive examinations* are given here with short solutions.

Example 3.17. *A bar is stretched in such a manner that all the lateral strain is prevented. Determine the modified value of modulus of elasticity and modified value of Poisson's ratio in terms of original values of E and μ respectively.*

Solution :

Let p_1 be the applied axial stress. In order to have e_2 and e_3 zero, corresponding stresses p_2 and p_3, each of the same sign as that of p_1, will be induced is the two lateral directions.

Hence $$e_1 = \frac{p_1}{E} - \frac{\mu}{E}\left(p_2 + p_3\right) \qquad ...(i)$$

$$e_2 = \frac{p_2}{E} - \frac{\mu}{E}\left(p_3 + p_1\right) = 0 \qquad ...(ii)$$

and $$e_3 = \frac{p_3}{E} - \frac{\mu}{E}\left(p_1 + p_2\right) = 0 \qquad ...(iii)$$

Adding (*ii*) and (*iii*), $p_2 + p_3 = 2p_1 \frac{\mu}{1-\mu}$

Substituting in (*i*), we get

$$e_1 = \frac{p_1}{E} - \frac{\mu}{E} \cdot \frac{2p_1\mu}{1-\mu} = \frac{p_1}{E}\left(1 - \frac{2\mu^2}{1-\mu}\right)$$

$$= \frac{p_1}{E}\left(\frac{1-\mu-2\mu^2}{1-\mu}\right) = \frac{p_1}{E}\frac{(1+\mu)(1-2\mu)}{1-\mu}$$

$\therefore$ Modified modulus, $E' = \frac{p_1}{e_1} = \frac{E(1-\mu)}{(1+\mu)(1-2\mu)}$

Also, modified Poisson's ratio, $\mu' = \frac{e_3}{e_1} = 0$

Example 3.18. *A bar of rectangular section is subjected to an axial stress of p_1. No constraint is exerted on one pair of sides but on the other, external pressure restrict the lateral strain to one-half of what it would be if there were no restraint. Determine the modified value of modulus of elasticity.*

Solution **:**

Let us represent axial direction by suffix 1 and the two lateral directions by suffix 2 and 3. Let stress p_3, of the same sign as that of p_1, be the induced stress in the direction of restraint. Here p_2 will be zero since there is no restraint in the second direction.

Hence $$e_1 = \frac{p_1}{E} - \frac{\mu}{E}p_3 \qquad ...(i)$$

$$e_2 = -\frac{\mu}{E}(p_1 + p_3) \qquad ...(ii)$$

$$e_3 = \frac{p_3}{E} - \frac{\mu}{E}p_1 \qquad ...(iii)$$

If there were no restraint in the third direction, $e_3' = -\frac{\mu}{E}p_1$

As per given condition, $e_3 = \frac{1}{2}e_3'$

Hence $$\frac{p_3}{E} - \frac{\mu}{E}p_1 = -\frac{1}{2}\frac{\mu}{E}p_1$$

From which $$p_3 = \frac{1}{2}\mu p_1$$

Substituting in (*i*), we get $e_1 = \frac{p_1}{E} - \frac{\mu}{E}\left(\frac{1}{2}\mu p_1\right) = \frac{p_1}{E}\left(1 - \frac{\mu^2}{2}\right) = \frac{p_1}{2E}(2-\mu^2)$

$\therefore$ Modified modulus of elasticity, $E' = \frac{p_1}{e_1} = \frac{2E}{2-\mu^2}$

Example 3.19. *If two pieces of materials A and B have the same bulk modulus, but the value of E for B is 1% greater than that for A, find the value of N for the material B in terms of E and N for material A.*

Solution :

Given $$K_A = K_B = K;\ E_B = 1.01\,E_A$$

From Eq. 3.19, we have $E = \dfrac{9\,KN}{N + 3\,K}$

or $$EN + 3\,EK = 9\,KN$$

or $$3\,K\,(3\,N - E) = EN, \text{ from which } K = \frac{E\,N}{3\,(3\,N - E)}$$

Hence $$\frac{E_A\,N_A}{3\,(3\,N_A - E_A)} = K = \frac{E_B\,N_B}{3\,(3\,N_B - E_B)}$$

$\therefore$ $$E_A\,N_A\,(3\,N_B - E_B) = E_B\,N_B\,(3\,N_A - E_A)$$

or $$3\,E_A\,N_A\,N_B - E_A\,N_A\,E_B = 3\,E_B\,N_B\,N_A - E_A\,E_B\,N_B$$

or $N_B\,(\,3\,E_A\,N_A - 3\,E_B\,N_A + E_A\,E_B) = E_A\,N_A\,E_B$

From which $$N_B = \frac{E_A\,N_A\,E_B}{3\,E_A\,N_A - 3\,E_B\,N_A + E_A\,E_B} = \frac{1.01\,E_A\,N_A\,E_A}{3\,E_A\,N_A - 3 \times 1.01\,E_A\,N_A + 1.01\,E_A.E_A}$$

or $$N_B = \frac{1.01\,E_A\,N_A}{1.01\,E_A - 3\,(1.01 - 1)\,N_A} = \frac{101\,E_A\,N_A}{101\,E_A - 3\,N_A}$$

Example 3.20. *Show that if E is assumed to be correct, an error of 1.5% in the determination of N will involve an error of about 6.5% in the calculation of Poisson's ratio when its correct value is 0.3.*

Solution :

Let E, N and μ be the correct values of the constants and μ' be the calculated value of Poisson's ratio.

From Eq. 3.14 $$E = 2\,N\,(1 + \mu) \qquad ...(i)$$

If, due to an error of 1.5%, N is increased to 1.015 N, the calculated value μ' of Poisson's ratio is given by

$$E = 2 \times 1.015\,N\,(1 + \mu') \qquad ...(ii)$$

Equating (*i*) and (*ii*), $2\,N\,(1 + \mu) = 2 \times 1.015\,N\,(1 + \mu')$

or $$1 + \mu = 1.015 + 1.015\,\mu'$$

or $$\mu' - \mu = -\,0.015 - 0.015\,\mu' = -\,0.015\,(1 + \mu')$$

Hence % error is μ is

$$\frac{\mu' - \mu}{\mu} \times 100 = -\,0.015\,\frac{1 + \mu'}{\mu} \times 100 \approx -\,1.5\,\frac{1 + \mu}{\mu}$$

(Taking $\mu' \approx \mu$)

$$\therefore \quad \frac{\mu' - \mu}{\mu} \times 100 \approx -1.5\left(\frac{1+0.3}{0.3}\right) \approx \mathbf{-6.5\ \%}$$

Alternative Solution :

$$E = 2N(1+\mu)$$

Differentiating partially and noting that $\delta E = 0$ since E does not vary,

$$\delta E = 0 = 2\,\delta N(1+\mu) + 2N\delta\mu$$

or

$$\delta\mu = -\frac{\delta N}{N}(1+\mu)$$

$$\therefore \quad \%\text{ error} = \frac{\delta\mu}{\mu} \times 100 = -\frac{\delta N}{N} \cdot \frac{1+\mu}{\mu} \times 100$$

$$= -\frac{1.5}{100} \times \frac{1+0.3}{0.3} \times 100 = \mathbf{-6.5\ \%}$$

Example 3.21. *The values of E and N, determined experimentally were found to be 1.9×10^5 N/mm² and 0.75×10^5 N/mm². Calculate Poisson's ratio and the bulk modulus. If both the moduli are liable for an error of ± 1 percent, find the maximum percentage error in the derived value of Poisson's ratio.*

Solution :

We have $\quad E = 2N(1+\mu)$

From which $\quad \mu = \dfrac{E}{2N} - 1 = \dfrac{1.9 \times 10^5}{2 \times 0.75 \times 10^5} - 1 = \mathbf{0.267}$

Also, $\quad K = \dfrac{E}{3(1-2\mu)} = \dfrac{1.9 \times 10^5}{3(1 - 2 \times 0.267)} = \mathbf{1.357 \times 10^5\ N/mm^2}$

For maximum percentage error in the derived value of μ, the error in the values of E and N should be of different sign. Let % error in E be +1 and that in N be –1.

Now $\quad \mu = \dfrac{E}{2N} - 1$

Hence $\quad \mu' = \dfrac{E'}{2N'} - 1,$

where $\quad E'$ = incorrect value of E = $(1.9 \times 10^5) \times 1.01$

N' = incorrect value of N = $(0.75 \times 10^5) \times 0.99$

and $\quad \mu'$ = computed incorrect value of μ.

$$\therefore \quad \mu' = \frac{1.9 \times 10^5 \times 1.01}{2(0.75 \times 10^5 \times 0.99)} - 1 = 0.2923$$

$$\therefore \quad \%\text{ error in } \mu = \frac{\mu' - \mu}{\mu} \times 100 = \frac{0.2923 - 0.267}{0.267} \times 100 = \mathbf{9.46\ \%}$$

Example 3.22. *A bar of elastic material is subjected to a direct compressive stress of p_1 in the longitudinal direction. Suitable lateral compressive stress p_2 is applied along the other two lateral directions to limit the net strain in each of the lateral directions to one third the magnitude that could be under p_1 acting alone. Find the magnitude of p_2 and the net strain in the longitudinal direction.*

Solution **: When p_2 is applied along the two lateral directions,**

$$e_2' = e_3' = \frac{p_2}{E} - \frac{\mu p_2}{E} - \frac{\mu p_1}{E}$$

However, if p_2 is not applied,

$$e_2 = -\frac{\mu p_1}{E}$$

$$\therefore \qquad \frac{p_2}{E} - \frac{\mu p_2}{E} - \frac{\mu p_1}{E} = -\frac{1}{3}\frac{\mu p_1}{E}$$

Hence $$p_2 = \frac{2}{3}\frac{\mu}{1-\mu}p_1 \text{ (Ans.)}$$

Net strain in the longitudinal direction :

$$e_1' = \frac{p_1}{E} - \frac{\mu p_2}{E} - \frac{\mu p_2}{E} = \frac{p_1}{E}\left[1 - 2\mu \times \frac{2}{3}\frac{\mu}{1-\mu}\right]$$

or $$e_1' = \frac{p_1}{E}\left[\frac{3 - 3\mu - 4\mu^2}{3(1-\mu)}\right] \text{ compressive.}$$

Substituting $\mu = \frac{1}{m}$, the above expression reduces to

$$e_1' = \frac{p_1}{E}\left[\frac{3m^2 - 3m - 4}{3m(m-1)}\right]$$

Example 3.23. *A vertical rod of length L and diameter d is fixed at its upper end and carries a load W at its lower end. The extension of the rod due to W is δ. If a torque T_1, applied in a horizontal plane at the lower end of the rod, gives an angle of twist θ radians, show that Poisson's ratio is given by*

$$\mu = \left(\frac{W d^2 \theta}{16 \times T\delta} - 1\right)$$

Solution **:**

As a vertical rod under the action of W, the extension δ is given by

$$\delta = \frac{WL}{AE} = \frac{WL}{\frac{\pi}{4}d^2 E} = \frac{4WL}{\pi d^2 E}$$

or $$E = \frac{4WL}{\pi d^2 \delta} \qquad ...(i)$$

As a shaft under the action of torque T, we have

$$\frac{T}{J} = \frac{N\theta}{L}$$

or
$$N = \frac{TL}{J\theta} = \frac{TL}{\frac{\pi}{32} d^4 \theta} = \frac{32\,TL}{\pi d^4 \theta} \qquad ...(2)$$

(where J=Polar moment of inertia $= \frac{\pi d^4}{32}$)

Now
$$E = 2N(1+\mu)$$

From which
$$1+\mu = \frac{E}{2N} = \frac{4\,WL}{\pi d^2 \delta} \cdot \frac{1}{2} \times \frac{\pi d^4 \theta}{32\,TL} = \frac{W d^2 \theta}{16\,T\delta}$$

Hence
$$\mu = \left(\frac{W d^2 \theta}{16\,T\delta} - 1\right)$$

Example 3.24. *A square frame ABCD consisting of five steel bars of 400 mm² cross-sectional area is subjected to the action of two forces P = 40 kN in the direction of the diagonal, as shown in Fig. 3.12. Determine the changes of the angles at A and C due to the deformation of the frame. Take $E = 2 \times 10^5$ N/mm².*

Solution **:**

Since loads P are applied at pin joints D and B, the diagonal DB will take *completely* these loads.

Assuming hinges D and the direction of diagonal DB as fixed, point B will move to point B_1 through a distance δ given by

$\delta = \frac{PL}{AE}$, where L is the length and A is the area of DB.

Due to the distortion of the frame, hinge C will move to its new position C_1. Draw C_1C' and CC' parallel to DC and DB respectively. The small triangle CC_1C' is right angled at C_1, wherein $CC_1 = \frac{\delta}{\sqrt{2}}$. The angle of rotation of the bar DC due to the deformation of the frame is

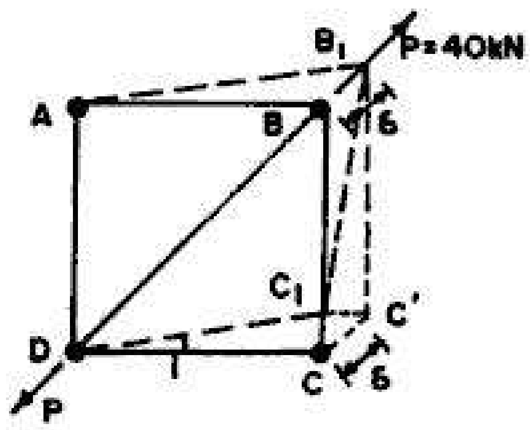


FIG. 3.12

$$i = \frac{CC_1}{DC} = \frac{\delta}{\sqrt{2}} \div \frac{L}{\sqrt{2}} = \frac{\delta}{L}$$

Substituting the value of δ, we get

$$i = \frac{PL}{AE} \cdot \frac{1}{L} = \frac{P}{AE} = \frac{40 \times 10^3}{400 \times 2 \times 10^5} = \frac{1}{2000} \text{ radian.}$$

Hence increase in the angle at C (or decrease in angle at A) will be

$$= 2 \times \frac{1}{2000} = \frac{1}{1000} \text{ radian}$$

Example 3.25. *A rectangular parallelopiped is subjected to tension in two perpendicular directions as shown in Fig. 3.13. Find the unit elongation ε in direction OC.*

***Solution* :**

Let a and b be the co-ordinates of point C. If ε_x and ε_y are the *unit elongations* in x and y directions, these co-ordinates of C, after deformation will be $a(1+\varepsilon_x)$ and $b(1+\varepsilon_y)$. Hence the length (L) of OC will become equal to L' after deformation, the value of which is given by

$$L' = \sqrt{a^2(1+\varepsilon_x)^2 + b^2(1+\varepsilon_y)^2} \approx \sqrt{a^2(1+2\varepsilon_x) + b^2(1+2\varepsilon_y)}$$

$$\approx \sqrt{a^2+b^2}\left(1 + \frac{a^2\varepsilon_x}{a^2+b^2} + \frac{b^2\varepsilon_y}{a^2+b^2}\right)$$

Initial length $\quad L = \sqrt{a^2+b^2}$

Unit elongation $\quad \dfrac{L'-L}{L} = \varepsilon = \dfrac{a^2\varepsilon_x}{a^2+b^2} + \dfrac{b^2\varepsilon_y}{a^2+b^2}$

But $\quad \dfrac{a^2}{a^2+b^2} = \cos^2\alpha$ and $\dfrac{b^2}{a^2+b^2} = \sin^2\alpha$

$$\therefore \quad \varepsilon = \varepsilon_x\cos^2\alpha + \varepsilon_y\sin^2\alpha$$

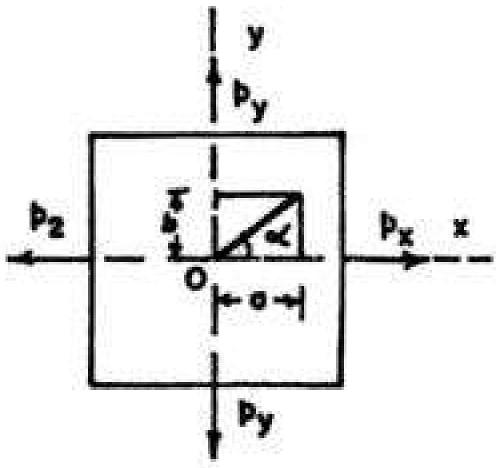


FIG. 3.13

PROBLEMS

1. (*a*) Determine the percentage change in volume of a steel bar 50 mm square in section and 1 m long when subjected to an axial compressive load of 20 kN.

(*b*) What change in volume would a 100 mm cube of steel suffer at a depth of 5 km in sea water ?

Take $E = 2.05 \times 10^5$ N/mm^2 and $N = 0.82 \times 10^5$ N/mm^2.

2. A bar 30 mm in diameter was subjected to a tensile load if 54 kN and the measured extension on 300 mm gauge length was 0.112 mm and change in diameter was 0.00366 mm. Calculate Poisson's ratio and values of three moduli.

3. For a given material, Young's modulus is 1.0×10^5 N/mm^2 and modulus of rigidity 0.4×10^5 N/mm^2. Find the bulk modulus and lateral contraction of a round bar of 50 mm diameter and 2.5 m long, when stretched 2.5 mm. Take Poisson's ratio as $\frac{1}{4}$.

4. A rectangular bar is subjected to axial stresses p_1, p_2 and p_3 on its sides. Show that the volumetric strain

$$\frac{\delta V}{V} = (p_1 + p_2 + p_3) \times \frac{1}{E}\left(1 - \frac{2}{m}\right)$$

5. A metal specimen is compressed in the direction of its axis and means are employed to reduce lateral expansion to one third of what it would be if free to expand. Calculate the modified value of the elastic constant. Prove that its value will be $\frac{9}{8}E$ if $m=4$.

6. Tensile stresses f_1 and f_2 act at right angles to one another on an element of isotropic elastic material. The strain in the direction of f_1 is twice that in the direction of f_2. Find the value of the ratio $\frac{f_1}{f_2}$. If E for the material is $120\,kN/mm^2$ and $\mu = 0.3$, find the value of $\frac{f_1}{f_2}$.

7. A bar of elastic material is subjected to direct stress in a longitudinal direction, and its strains in the two directions at right angles are reduced to one-half and one-third respectively to those which normally occur in an ordinary tension member. Find the value of modified elastic constant. If $E = 200\,kN/mm^2$ and $m=4$, what is the value of the elastic constant ?

8. A determination of E and N gives values of $2 \times 10^5\,N/mm^2$ and $0.79 \times 10^5\,N/mm^2$. Calculate Poisson's ratio and the bulk modulus.

If both moduli are liable to an error of ± 2 %, find the maximum percentage error in the derived value of Poisson's value.

9. Show that if E is assumed to be correct, an error of 1% in the determination of N will involve an error of about 5% in the calculation of Poisson's ratio when its correct value is 0.25.

10. A bar of steel is 38 mm in diameter and 450 mm long. A tensile load of 100 kN is found to stretch the bar by 0.20 mm. The same bar, when subjected to a torque of 1.24 kN-m is found to twist through 1.982°. Find the values of the four elastic constants.

11. A bar of elastic material is subjected to a direct compressive stress of p_1 in the longitudinal direction. Suitable lateral compressive stress p_2 is applied along the other two lateral directions to limit the net strain in each of the lateral directions to half the magnitude that could be under p_1 acting alone. Find the magnitude of p_2 and the net strain in the longitudinal direction.

12. Derive the relationship between modulus of elasticity, modulus of rigidity and Poisson's ratio of an elastic body. A bronze specimen has a modulus of elasticity of $120\,kN/mm^2$ and a modulus of rigidity of $47\,kN/mm^2$. Determine Poisson's ratio of the material.

ANSWERS

1. (a) 0.00195 (b) 369 mm^3
2. 0.327 ; $E = 2.046 \times 10^5\,N/mm^2$; $N = 0.771 \times 10^5\,N/mm^2$; $K = 1.971 \times 10^5\,N/mm^2$
3. $0.667 \times 10^5\,N/mm^2$; 0.125 mm
5. $\dfrac{3m(m-1)}{3m^2 - 3m - 4}E$
6. $\dfrac{2+\mu}{1+2\mu}$; 1.438
7. $\dfrac{6m(m^2-1)}{6m^3 - 13m - 7}E$; 221.54 kN/mm^2
8. 0.266 ; $1.423 \times 10^5\,N/mm^2$; 19.44%
10. $E = 1.984 \times 10^5\,N/mm^2$; $N = 0.788 \times 10^5\,N/mm^2$
    $\mu = 0.259$; $K = 1.372 \times 10^5\,N/mm^2$
11. $\frac{1}{2}\left(\frac{\mu}{1-\mu}\right)p_1$; $\frac{p_1}{E}\left(\frac{1-\mu-\mu^2}{1-\mu}\right)$
12. 0.277

4

Analysis of Stress : Principal Stresses

4.1. INTRODUCTION

In Chapter 2, we have considered only the normal stresses acting on cross-sections, such as on cross-section *mn* of a bar *AB* (Fig 4.1*a*) when the bar is subjected to either a tensile force or a compressive force acting along the axis of the bar. In this chapter, we shall analyse the stresses induced on inclined sections such as section *pq* (Fig 4.1*a*). In actual practice, all the three types of stresses (*i.e.* tension, compression and shear) may act simultaneously on appropriate planes passing through a point in the strained material. We shall therefore, analyse such a general stress system (Fig 4.1*b*) and find the *resultant stress* on *critical inclined planes* which may carry greater stresses than the applied ones.

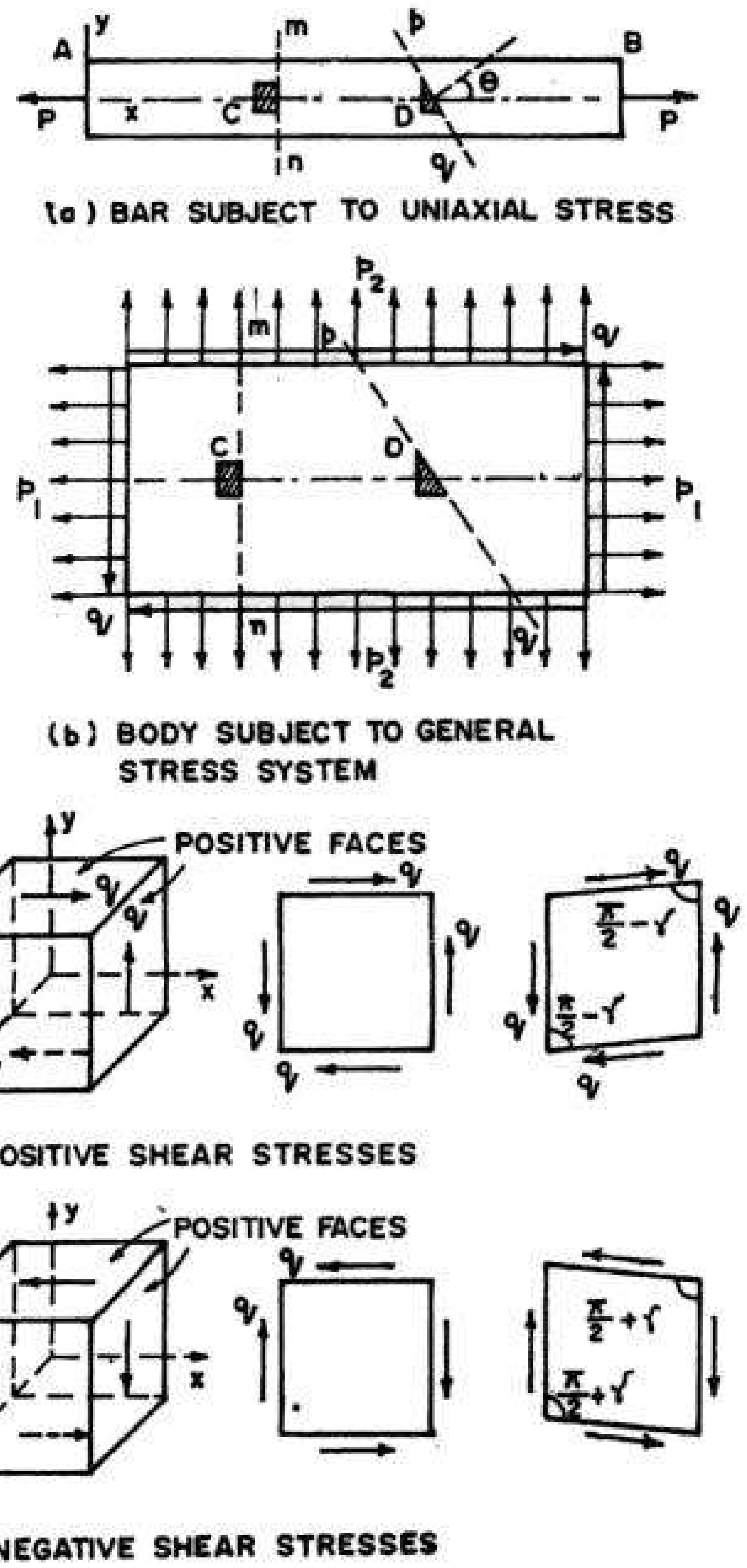


FIG. 4.1

Sign Conventions. We shall follow the following sign conventions :

(i) Tensile stress is considered positive and compressive stress is considered negative.

(ii) Angle θ is considered positive when it is in the anticlockwise direction.

*(iii) **Sign Convention for Shear** :* A shear stress acting on a positive face of an element is considered positive if it acts in the positive direction of one of the coordinate axes and negative if it acts in the negative direction of the axes. Similarly a shear stress acting

on a negative face of an element is positive if it acts in the negative direction of the axis and negative if it acts in the positive direction. Thus, all the shear stresses shown in Fig. 4.1(c) are positive. Such positive shear stresses tend to reduce the angle between two positive (or negative) faces. Similarly all shear stresses shown in Fig. 4.1(d) are negative.

4.2. STRESSES INDUCED DUE TO UNIAXIAL STRESS

(a) Stresses on normal cross-section *mn* (Fig 4.1 *a*)

Consider cross-section mn of a bar AB (Fig. 4.1 a) subjected to uniaxial tensile force P. Let us assume that the stress distribution is uniform over the entire cross-sectional area. This is possible when the following conditions exist : *(i)* the bar is prismatic, *(ii)* the material is homogeneous and isotropic *(iii)* the force acts at the centroid of the cross-sectional area and *(iv)* the cross-section is away from the ends of the bar where high localised stresses may exist (see Fig 2.2). Let us assume that bar of Fig. 4.1(a) meets all these conditions, so that stress distribution across the section mn is uniform. In that case, stress $p = \frac{P}{A}$, as shown in the stress element at point C (Fig 4.2) which has the shape of a rectangular parallelopiped the right hand face of which is in cross section mn. The dimensions of such an element are assumed to be infinitesimally small.

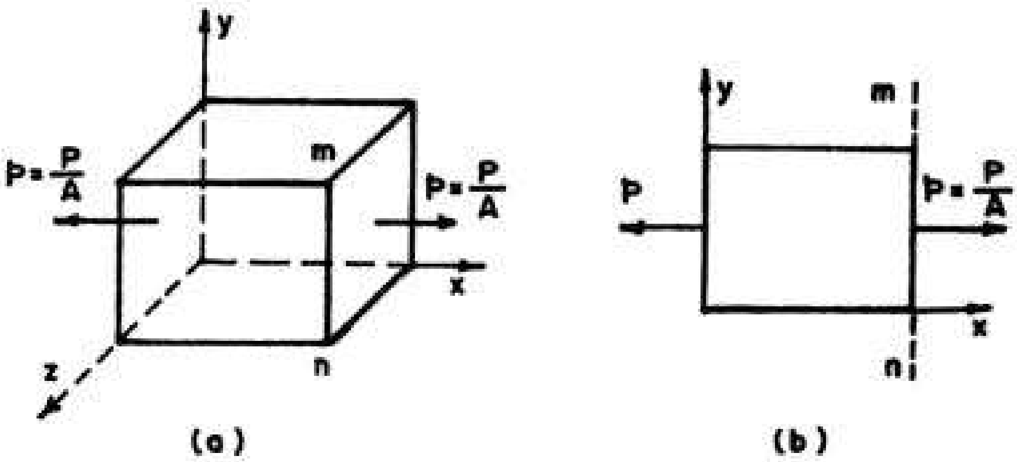


FIG. 4.2. STRESS ELEMENT AT POINT C

(b) Stresses on inclined section *pq* (Fig 4.1 *a* and 4.3 *a*)

Consider an inclined section pq of a bar AB (Fig. 4.1a and 4.3a) subjected to uniaxial tensile force P. This inclined section pq is cut through the bar at an angle θ between the x-axis and the normal to the plane pq.

Since all the parts of the bar have the same axial strains, the stresses acting on the cross-section pq are uniformly distributed, as shown in Fig. 4.3 (b). The resultant of these stresses will be force equal in magnitude to the axial force P. This resultant force acting on section pq may be resolved into two components (Fig 4.3 c):

(i) a normal force N acting normal to the plane pq

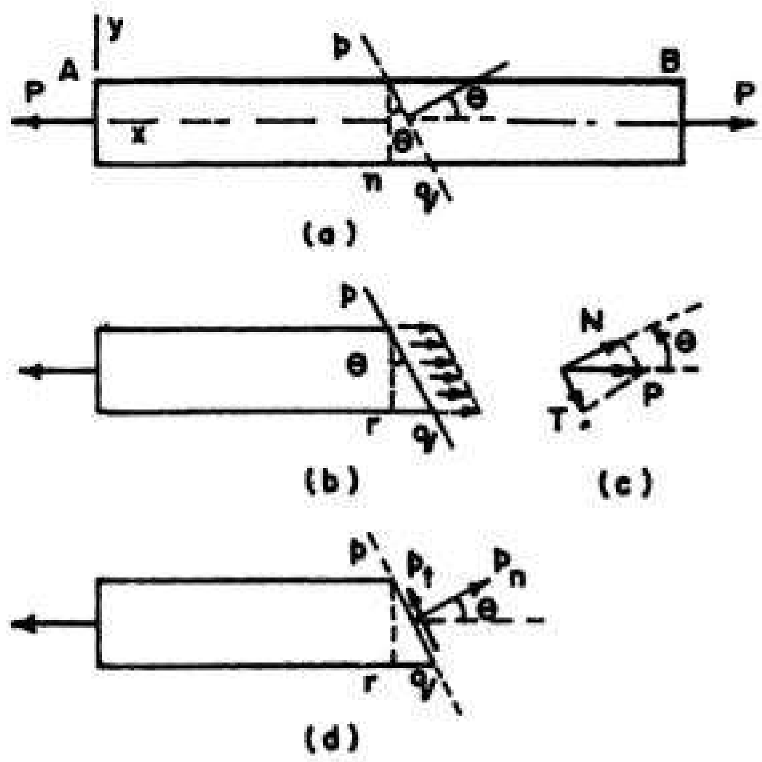


FIG. 4.3. STRESSES ON INCLINED SECTION pq

and (*ii*) a tangential force T acting tangential to the plane pq.

Obviously, $N = P \cos\theta$ and $T = P \sin\theta$ ...(4.1)

These force components N and T give rise to normal stress p_n and tangential stress p_t respectively (Fig. 4.3*d*) which are uniformly distributed over the inclined section pq. These stresses are shown acting in the positive directions (Fig. 4.3*d*), that is, p_n is positive in tension and p_t is positive when it tends to produce counterclockwise rotation of the material.

The area of section (A_1) of pq is $A/\cos\theta$ ($= A \sec\theta$) where A is the area of cross-section of the normal section mn.

Hence $$p_n = \frac{N}{A_1} = \frac{P\cos\theta}{A\sec\theta} = \frac{P}{A}\cos^2\theta = p\cos^2\theta \qquad ...(4.2\ a)$$

and $$p_t = -\frac{T}{A_1} = -\frac{P\sin\theta}{A\sec\theta} = -\frac{P}{A}\sin\theta\cos\theta = -p\sin\theta\cos\theta = -\frac{p}{2}\sin 2\theta \qquad ...(4.2\ b)$$

The resultant stress $$p_r = \sqrt{p_n^2 + p_t^2} = \sqrt{p^2\cos^4\theta + p^2\sin^2\theta\cos^2\theta}$$
$$= p\cos\theta\sqrt{\cos^2\theta + \sin^2\theta} = p\cos\theta \qquad ...(4.2\ c)$$

The above equations can also be derived by considering the wedge shaped stress element at point D (Fig. 4.1 *a*), having one face along the inclined section pq, as shown in Fig. 4.4.

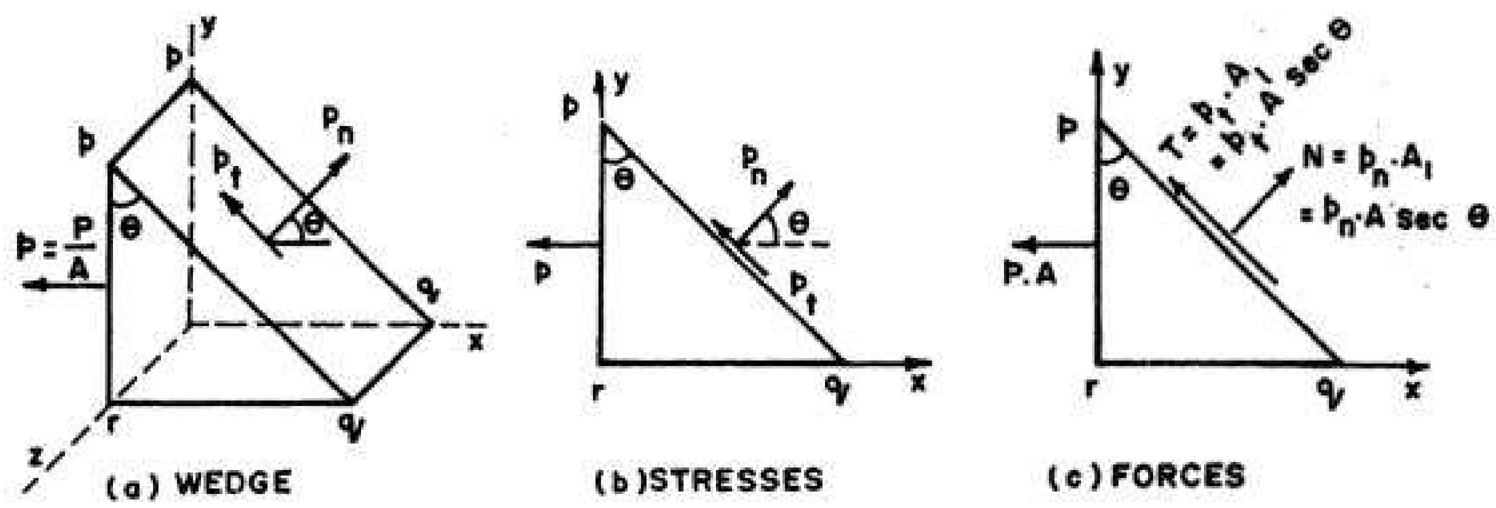


FIG. 4.4. STRESS ELEMENT AT POINT D

In Fig. 4.4 (*b*), the two dimensional view of the wedge shaped element is shown, where the stress on the left hand face pr is equal to stress $p\left(=\frac{P}{A}\right)$ while the stresses on the inclined face pq are p_n and p_t. Fig 4.4 (*c*) shows the forces on the two dimensional triangular element. The force on the left hand portion will be equal to $p.A$. The area of the inclined face is $A\sec\theta$. Hence the normal and tangential forces on this face are $p_n A\sec\theta$ and $p_t A\sec\theta$ respectively. Again the force $P\,(=pA)$ on the left face can be resolved into perpendicular (or normal) component $N(=pA\cos\theta)$ and parallel or tangential component $T\,(=pA\sin\theta)$. Considering the equilibrium of the element and summing the forces perpendicular to the inclined face (*i.e.* in the direction of p_n), we get

$$p_n A\sec\theta - pA\cos\theta = 0$$

or $$p_n = p\cos^2\theta \qquad ...(4.2\ a)$$

Similarly, summing up the forces in tangential direction

$$p_t A \sec\theta + p A \sin\theta = 0$$

or
$$p_t = -p\sin\theta\cos\theta = -\frac{p}{2}\sin 2\theta \qquad \text{...(4.2 } a)$$

which are the same as obtained earlier.

Variation of p_n, p_t and p_r with θ

Fig 4.5 shows the variation of p_n, p_t and p_r, as θ varies from a value $-90°$ to $+90°$. When $\theta = 0°$, plane pq becomes a cross-section mn and hence the graph gives $p_n = p$, as expected. As θ increases, p_n decreases; when $\theta = 90°$, $p_n = 0$, indicating that there are no normal stresses on a plane cut parallel to the longitudinal axis. Hence

$$\left(p_n\right)_{max} = p = \frac{P}{A} \qquad \text{...(4.3)}$$

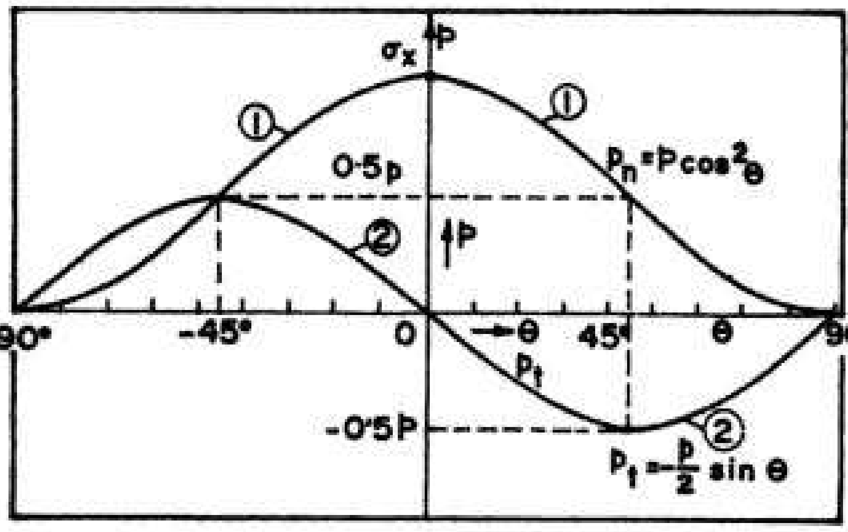


FIG. 4.5. VARIATION OF p_n, p_t AND p_r WITH θ.

Similarly, at cross-sections, where $\theta = 0$, $p_t = 0$. Also, on longitudinal sections, where $\theta = \pm 90°$, $p_t = 0$. The maximum positive value of p_t is obtained at $\theta = -45°$ and largest negative value at $\theta = +45°$. Thus numerically, largest shear stresses are

$$\left(p_t\right)_{max} = \frac{p}{2} \qquad \text{...(4.4)}$$

Hence all surfaces inclined at 45° to the axis of the pull are subjected to maximum shear stress equal to $\frac{P}{2A}\left(=\frac{p}{2}\right)$.

The complete state of stress on sections cut at 45° to axis of the pull is represented by the stress element shown in Fig. 4.6.

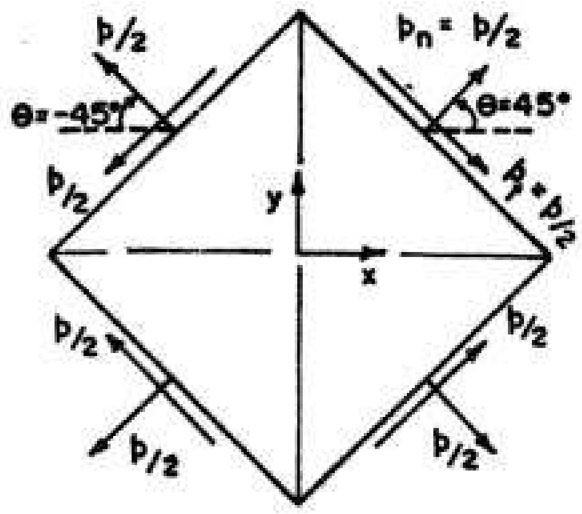


FIG. 4.6. STRESS ELEMENT AT $\theta = 45°$ FOR BAR IN TENSION

Thus in the uniaxial stress system, where the bar is subjected only to simple tension or compression, the two most important orientations of stress elements are $\theta = 0$ and 45°; the former orientation has maximum normal stress $p_{n,\,max}$ (Fig. 4.2) and the latter orientation has maximum shear stress $p_{t,\,max}$ (Fig. 4.6).

4.3. STRESSES INDUCED BY STATE OF SIMPLE SHEAR

Fig 4.7 (a) shows the state of simple shear or pure shear (see §3.8). Fig. 4.7 (b) and 4.7 (c) show wedge shaped stress element. It is required to find stresses p_n and p_t on inclined plane. Both the stresses p_n and p_t have been shown in their positive directions. Let the area of AB be A. Then area of $AC = A\sec\theta$.

Considering unit thickness of the element and balancing forces in direction normal to AC, we have (Fig 4.7 c) :

$$p_n \,.\, AC = q AB \sin\theta + q\, BC \cos\theta$$

or $p_n = q.\dfrac{AB}{AC}\sin\theta + q\dfrac{BC}{AC}\cos\theta$

or $p_n = q\cos\theta\sin\theta$

$+ \, q\sin\theta\cos\theta + q\sin 2\theta$ ...(4.5)

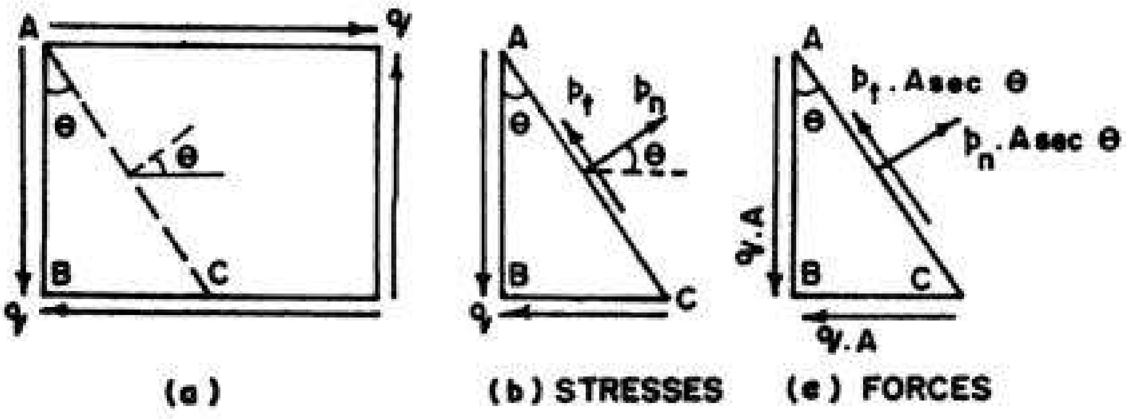


FIG. 4.7. STRESSES DUE TO STATE OF SIMPLE SHEAR

Similarly, resolving forces along tangential direction, we have (Fig. 4.7 c)

$$p_t \,.\, AC = -q \,.\, BC\sin\theta + q\, AB\cos\theta$$

or $$p_t = -q\,.\,\frac{BC}{AC}\sin\theta + q\frac{AB}{AC}\cos\theta = -q\sin^2\theta + q\cos^2\theta$$

or $$p_t = q\left(\cos^2\theta - \sin^2\theta\right) = q\cos 2\theta \qquad \text{...(4.6)}$$

From Eq. 4.5, we observe that $\left(p_n\right)_{max} = q$ when $\theta = 45°$, and for this position of the plane, p_t is zero.

4.4. STRESSES DUE TO STATE OF BIAXIAL STRESS

Fig. 4.8 (a) shows the state of biaxial stress, in which both p_1 and p_2 are tensile. Fig 4.8(b) shows the stresses on the wedge shaped stress element (both marked in the positive direction) while Fig 4.8 (c) shows forces on the stress element.

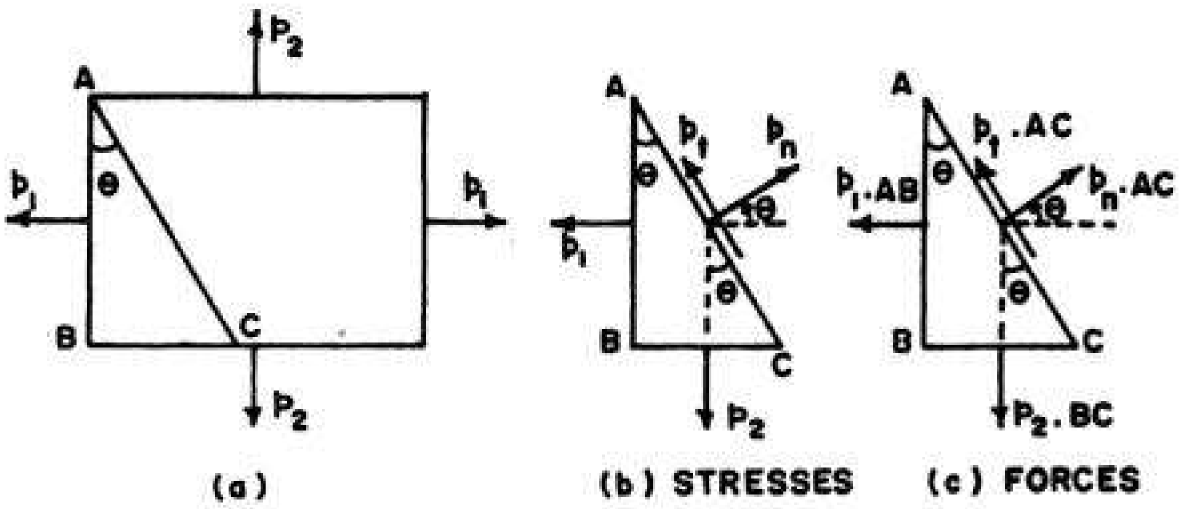


FIG. 4.8. STATE OF BIAXIAL STRESS

Taking unit thickness of the stress element and resolving the forces perpendicular to AC, we have

$$p_n \,.\, AC = p_1 AB\cos\theta + p_2 BC\sin\theta$$

or $$p_n = p_1\frac{AB}{AC}\cos\theta + p_2\frac{BC}{AC}\sin\theta = p_1\cos^2\theta + p_2\sin^2\theta \qquad \text{...(4.7)}$$

Similarly, resolving the forces along AC, we have

$$p_t \,.\, AC = -p_1 AB\sin\theta + p_2 BC\cos\theta$$

or $$p_t = -p_1 \frac{AB}{AC} \sin\theta + p_2 \frac{BC}{AC} \cos\theta = -p_1 \cos\theta \sin\theta + p_2 \sin\theta \cos\theta$$

or $$p_t = -\frac{p_1 - p_2}{2} \sin 2\theta \qquad ...(4.8)$$

(The negative sign shows that it acts in reversed direction than shown in Fig 4.8(c))
Maximum shear stress occurs at $\theta = 45°$:

$$\left(p_t\right)_{max} = -\frac{p_1 - p_2}{2} \qquad ...(4.9)$$

For this value of θ, normal stress is given by

$$p_n = p_1 \cos^2 45° + p_2 \sin^2 45° = \frac{p_1 + p_2}{2} \qquad ...(4.10)$$

The resultant stress p_r is given by

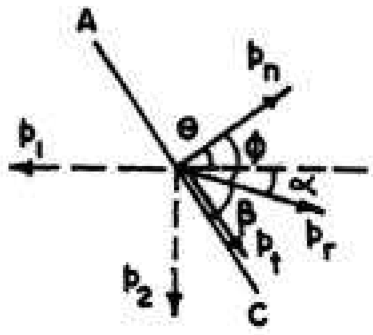


FIG. 4.9.

$$p_r = \sqrt{p_n^2 + p_t^2}$$

$$= \sqrt{\left(p_1 \cos^2\theta + p_2 \sin^2\theta\right)^2 + \left(-p_1 \cos\theta \sin\theta + p_2 \sin\theta \cos\theta\right)^2}$$

$$= \sqrt{p_1^2 \cos^4\theta + p_2^2 \sin^4\theta + 2p_1p_2 \sin^2\theta \cos^2\theta + \left(p_1^2 + p_2^2 - 2p_1p_2\right) \sin^2\theta \cos^2\theta}$$

$$= \sqrt{p_1^2 \cos^2\theta \left(\cos^2\theta + \sin^2\theta\right) + p_2^2 \sin^2\theta \left(\sin^2\theta + \cos^2\theta\right)}$$

or $$p_r = \sqrt{p_1^2 \cos^2\theta + p_2^2 \sin^2\theta} \qquad ...(4.11)$$

$$\tan\varphi = \frac{p_t}{p_n} = \frac{(p_1 - p_2)\sin\theta\cos\theta}{p_1 \cos^2\theta + p_2 \sin^2\theta} \qquad ...(4.12)$$

where φ is the angle made by p_r with p_n

Also, $$\tan\alpha = \frac{P_2}{P_1} = \frac{p_2 BC}{p_1 AB} = \frac{p_2}{p_1} \tan\theta \qquad ...(4.13)$$

where α is the angle made by p_r with p_1.

When stresses are not alike

A special case of biaxial stresses may arise when p_1 and p_2 may not be like, *i.e.* when p_1 may be tensile and p_2 may be compressive. In that case, p_n, p_t and p_r may be obtained by changing the sign of p_2 :

Thus, $$p_n = p_1 \cos^2\theta - p_2 \sin^2\theta \qquad ...(4.7\ a)$$

$$p_t = -\frac{p_1 + p_2}{2} \sin 2\theta \qquad ...(4.8\ a)$$

$$\left(p_t\right)_{max} = -\frac{p_1 + p_2}{2} \qquad ...(4.9\ a)$$

4.5. TWO PERPENDICULAR NORMAL STRESSES ACCOMPANIED WITH STATE OF SIMPLE SHEAR

Let us now consider a state of plane stress in which two perpendicular normal stresses are accompanied by the state of simple shear, as shown in Fig. 4.10 (*a*), (*b*). Such stress conditions are commonly encountered in axially loaded bars, shafts in torsion and beams is bending.

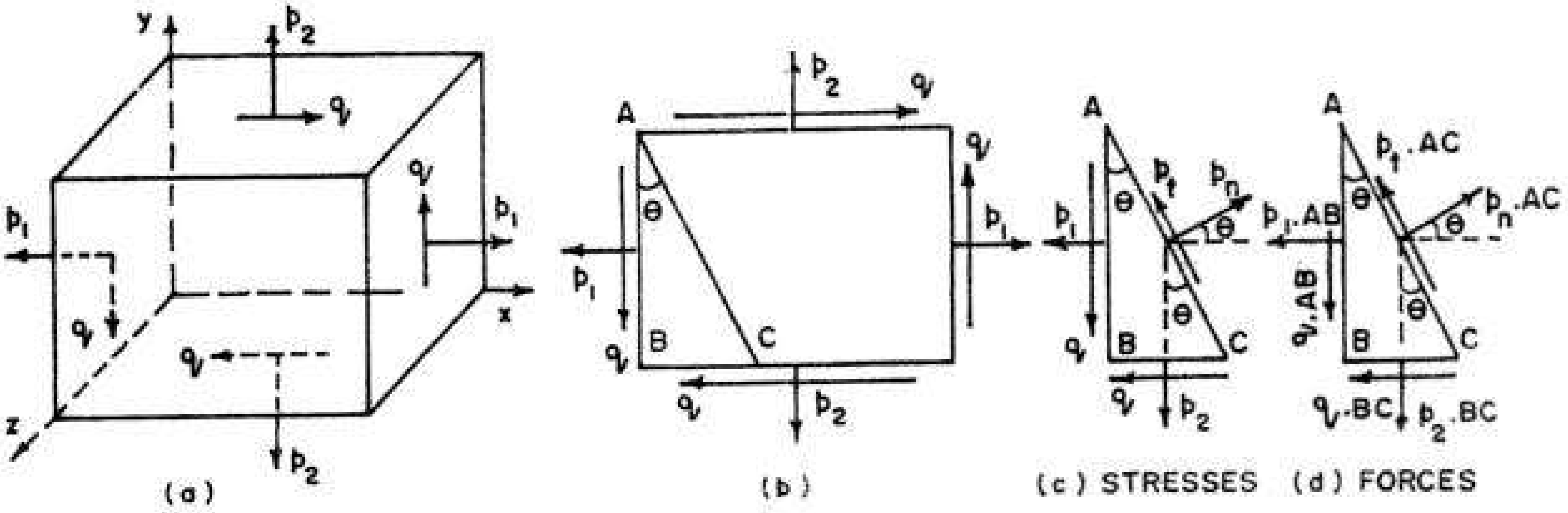


FIG. 4.10. GENERALISED CASE OF PLANE STRESS

Fig. 4.10 (*c*) and (*d*) show the wedge shaped stress elements in terms of stresses and forces respectively. Considering unit thickness and resolving forces perpendicular to *AC*, we get

$$p_n AC = p_1 . AB \cos\theta + p_2 . BC \sin\theta + q\, AB \sin\theta + q\, . BC \cos\theta$$

or $$p_n = p_1 . \frac{AB}{AC}\cos\theta + p_2 . \frac{BC}{AC}\sin\theta + q\, . \frac{AB}{AC}\sin\theta + q\, . \frac{BC}{AC}\cos\theta$$

or $$p_n = p_1 \cos^2\theta + p_2 \sin^2\theta + q \cos\theta \sin\theta + q \sin\theta \cos\theta$$

or $$p_n = p_1 \cos^2\theta + p_2 \sin^2\theta + q \sin 2\theta \qquad ...(4.14)$$

Eq. 4.14 can be expressed in an useful alternative form by using the following trigonometrical identities :

$$\cos^2\theta = \frac{1}{2}\left(1 + \cos 2\theta\right) ; \sin^2\theta = \frac{1}{2}\left(1 - \cos 2\theta\right) \text{ and } \sin\theta\cos\theta = \frac{1}{2}\sin 2\theta.$$

Then, Eq. 4.14 is reduced to the following form :

$$p_n = \frac{1}{2}\left(p_1 + p_2\right) + \frac{1}{2}\left(p_1 - p_2\right)\cos 2\theta + q \sin 2\theta \qquad ...(4.15)$$

Similarly, resolving the forces along *AC*, are get

$$p_t . AC = -p_1 . AB \sin\theta + p_2 . BC \cos\theta + q\, . AB \cos\theta - q\, . BC \sin\theta$$

or $$p_t = -p_1 . \frac{AB}{AC}\sin\theta + p_2\frac{BC}{AC}\cos\theta + q . \frac{AB}{AC}\cos\theta - q . \frac{BC}{AC}\sin\theta$$

or $$p_t = -p_1\cos\theta\sin\theta + p_2\sin\theta\cos\theta + q . \cos^2\theta - q . \sin^2\theta$$

or $$p_t = -\frac{p_1 - p_2}{2}\sin 2\theta + q\cos 2\theta \qquad ...(4.15)$$

If p_1 and p_2 are unlike (*i.e.* p_1 tensile and p_2 compressive), similar expressions for p_n and p_t can be obtained, by simply changing the sign of p_2. Thus, we have

$$p_n = p_1\cos^2\theta - p_2\sin^2\theta + q\sin 2\theta \qquad ...(4.14a)$$

. nd $$p_t = -\frac{p_1 + p_2}{2}\sin 2\theta + q\cos 2\theta \qquad ...(4.16\ a)$$

4.6. TRANSFORMATION EQUATIONS FOR PLANE STRESS*

Fig. 4.11 (*a*) (*b*) show a state of plane stress in which two perpendicular normal stresses are accompanied by the state of simple shear. Fig. 4.11 (*c*) show the stresses acting on an inclined section. As found in § 4.5, the values of normal and tangential stresses of this inclined plane are given by :

$$p_n = p_1' = \frac{1}{2}\left(p_1 + p_2\right) + \frac{1}{2}\left(p_1 - p_2\right)\cos 2\theta + q\sin 2\theta \qquad ...(4.15)$$

and $$p_t = q' = -\frac{p_1 - p_2}{2}\sin 2\theta + q\cos 2\theta \qquad ...(4.16)$$

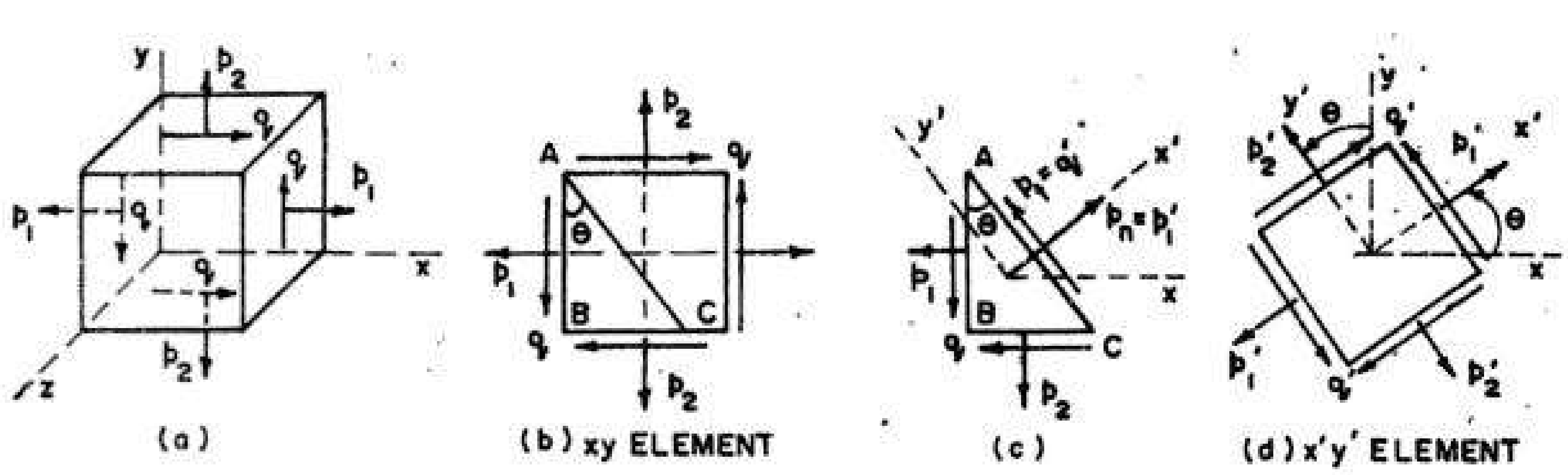


FIG. 4.11. ROTATION OF PLANE STRESS ELEMENT

The above equations for p_1' and q' are known as the transformation equations for plane stress because they transform the stress components from one set of axes (say x, y axes) to another set of axes (say x', y' axes) rotated through an angle θ. However, the intrinsic state of stress at the point under consideration is the same whether represented by the stresses on the xy element (Fig 4.11 *a*) or by the rotated $x'y'$ element (Fig. 4.11 *d*).

In a similar way, the normal stress p_2' acting on the y' face of the rotated element (Fig 4.11 *d*) can be obtained by Eq. 4.15 by substituting $\theta + 90°$ for θ. Thus, we have

$$\left(p_n\right)_{90+\theta} = p_2' = \frac{1}{2}\left(p_1 + p_2\right) - \frac{1}{2}\left(p_1 - p_2\right)\cos 2\theta - q\sin 2\theta \qquad ...(4.15\ a)$$

Summing Eqs. 4.15 and 4.15 (*a*), we have

$$p_1' + p_2' = p_1 + p_2$$

Hence the sum of normal stresses acting on perpendicular faces of a plane stress elements is constant and independent of the angle θ.

Example 4.1. *A tie bar is subjected to a uniform tensile stress of 100 N/mm². Find the intensity of normal stress, shear stress and resultant stress on a plane the normal to which is inclined 30° to the axis of the bar. Also estimate the maximum shear stress in the bar.*

Solution

$$p_n = p\cos^2\theta = 100\cos^2 30^\circ = \mathbf{75\ N/mm^2}$$

$$p_t = -\frac{p}{2}\sin 2\theta = -\frac{100}{2}\sin 60^\circ = -\mathbf{43.3\ N/mm^2}$$ (*i.e.* in the clockwise direction)

$$p_r = \sqrt{p_n^2 + p_t^2} = \sqrt{(75)^2 + (43.3)^2} = \mathbf{86.6\ N/mm^2}$$

$$(p_t)_{max} = -\frac{p}{2} = -\frac{100}{2} = -\mathbf{50\ N/mm^2} \text{ at } \theta = +45^\circ$$

Example 4.2. *A tension member is formed by connecting with glue two wooden scantlings each 100 mm × 200 mm at their ends which are cut at an angle of 60° as shown is Fig 4.12. The member is subjected to a pull P. Calculate the safe value of P if the permissible normal and shear stress in the glue are 2 N/mm² and 1 N/mm² respectively.*

Solution

Area $A = 100 \times 200 = 20000\ \text{mm}^2$

Angle of cut with horizontal = 60°

$\therefore$ Angle of cut (θ) with the vertical = 90° − 60° = 30°

Given : Permissible p_n = 2 N/mm² and permissible p_t = 4 N/mm².

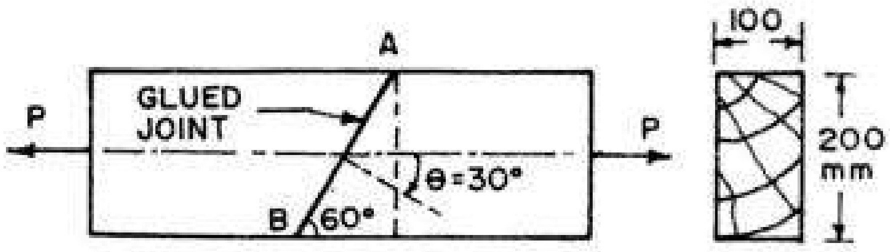


FIG. 4.12

But $p_n = p\cos^2\theta$

From which $p = p_n \sec^2\theta = 2\sec^2 30^\circ = 2.667\ \text{N/mm}^2$ ...(*i*)

Also, $p_t = \frac{p}{2}\sin 2\theta$

From which $p = \dfrac{2p_t}{\sin 2\theta} = \dfrac{2 \times 1}{\sin 60^\circ} = 2.309\ \text{N/mm}^2$ ...(*ii*)

The safe stress p will be the lesser of the two values.

$\therefore$ $p = 2.309\ \text{N/mm}^2$

$\therefore$ $P = p\,.A = 2.309 \times 20000 \times 10^{-3} = \mathbf{46.18\ kN}$

Example 4.3. *A prismatic bar in compression has a cross-sectional area $A = 900\,mm^2$ and carries a load of 60 kN (Fig. 4.13 a). Determine the stresses acting on a plane cut through the bar at $\theta = 30°$. Then show the complete state of stress for $\theta = 30°$ by determining the stresses on all faces of a stress element.*

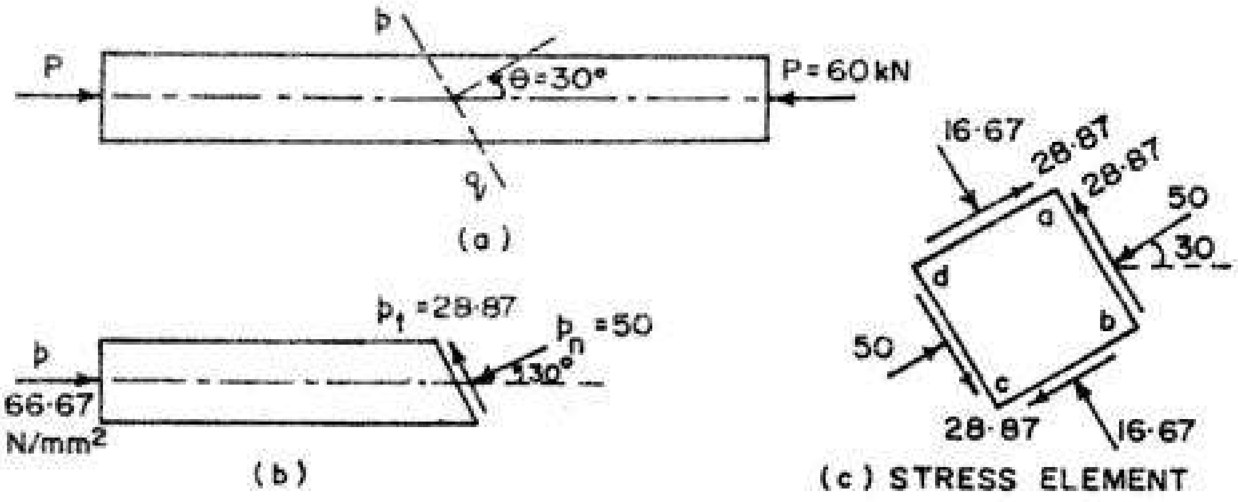


FIG. 4.13

Solution

$$p = \frac{P}{A} = \frac{60 \times 10^3}{900} = 66.67\ \text{N/mm}^2\ \text{(compression)}$$

Now $\quad p_n = p\cos^2\theta = (-66.67)\cos^2 30° = -50\ \text{N/mm}^2$ (i.e. comp.)

and $\quad p_t = -\frac{p}{2}\sin 2\theta = \frac{66.67}{2}\sin 60° = 28.87\ \text{N/mm}^2$

Fig. 4.13 (*a*) shows the magnitude and direction of both p_n and p_t on the inclined plane. Fig. 4.13 (*c*) shows the stresses on all the four faces of the stress element. Since face *ab* has the same orientation as the inclined plane *pq*, the stresses are the same.

For stresses of face *cd*, substitute $\theta = 30° + 180° = 210°$ in Eqs. 4.2*a* and 4.2*b*,

Thus, $\quad p_n = -66.67\cos^2 210° = -50\ \text{N/mm}^2$

and $\quad p_t = \frac{66.67}{2}\sin(2 \times 210°) = 28.87\ \text{N/mm}^2$

Similarly, for face *ad*, we have $\theta = 30° + 90° = 120°$

Hence $\quad p_n = -66.67\cos^2 120° = 16.67\ \text{N/mm}^2$

and $\quad p_t = \frac{66.67}{2}\sin(2 \times 120°) = -28.87\ \text{N/mm}^2$

Lastly, the values of p_n and p_t for face *cd* can be obtained by substituting $\theta = -60°$. The stresses on all the four faces of the stress element are shown in Fig. 4.13(c).

Example 4.4. *A piece of material is subjected to tensile stresses of 70 N/mm² and 50 N/mm² at right angles to each other. Find fully the stresses on a plane the normal of which makes an angle of 35° with the 70 N/mm² stress.*

Solution

$$p_n = p_1 \cos^2\theta + p_2 \sin^2\theta \quad ...(4.7)$$

$$= 70 \cos^2 35° + 50 \sin^2 35° = \mathbf{63.42\ N/mm^2}$$

$$p_t = -\frac{p_1 - p_2}{2} \sin 2\theta \quad ...(4.8)$$

$$= -\frac{70 - 50}{2} \sin 70° = \mathbf{9.4\ N/mm^2} \text{ (numerically)}$$

$$p_r = \sqrt{p_n^2 + p_t^2} = \sqrt{(63.42)^2 + (9.4)^2} = \mathbf{64.11\ N/mm^2}$$

$$\tan\alpha = \frac{p_2}{p_1}\tan\theta = \frac{50}{70}\tan 35° = 0.5$$

From which $\alpha = \mathbf{26.57°}$ with the direction of p_1.

Example 4.5. *A piece of material is subjected to tensile stress of 70 N/mm² in one direction and a compressive stress of 50 N/mm² in a direction at right angles to the previous one.*

Find fully the stresses on a plane the normal of which makes an angle of 40° with the 70 N/mm² stress.

Solution

In this case, p_1 is tensile while p_2 is compressive.

Hence $$p_n = p_1 \cos^2\theta - p_2 \sin^2\theta = 70 \cos^2 40° - 50 \sin^2 40° = \mathbf{20.42\ N/mm^2}$$

$$p_t = -\frac{p_1 + p_2}{2} \sin 2\theta = -\frac{70 + 50}{2} \sin 80° = \mathbf{59.09\ N/mm^2} \text{ (numerically)}$$

$$p_r = \sqrt{p_n^2 + p_t^2} = \sqrt{(20.42)^2 + (59.09)^2} = \mathbf{62.52\ N/mm^2}$$

$$\tan\alpha = \frac{p_2}{p_1}\tan\theta = \frac{50}{70}\tan 40° = 0.5994$$

From which $\alpha = \mathbf{30.94°}$

Example 4.6. *The stresses on two perpendicular planes through a point are 60 N/mm² tension, 40 N/mm² compression and 30 N/mm² shear (Fig 4.14 a). Find the stress components and the resultant stress on a plane at 60° to that of the tensile stress.*

Solution :

Given $p_1 = 60\ \text{N/mm}^2$ (tension)

$p_2 = 40\ \text{N/mm}^2$ (comp.) $q = 30\ \text{N/mm}^2$

and $\theta = 60°$

$$p_n = p_1 \cos^2\theta + p_2 \sin^2\theta + q \sin 2\theta \quad \text{(Eq. 4.14)}$$

$$= 60 \cos^2 60° - 40 \sin^2 60° + 30 \sin(2 \times 60°) = \mathbf{10.98\ N/mm^2}$$

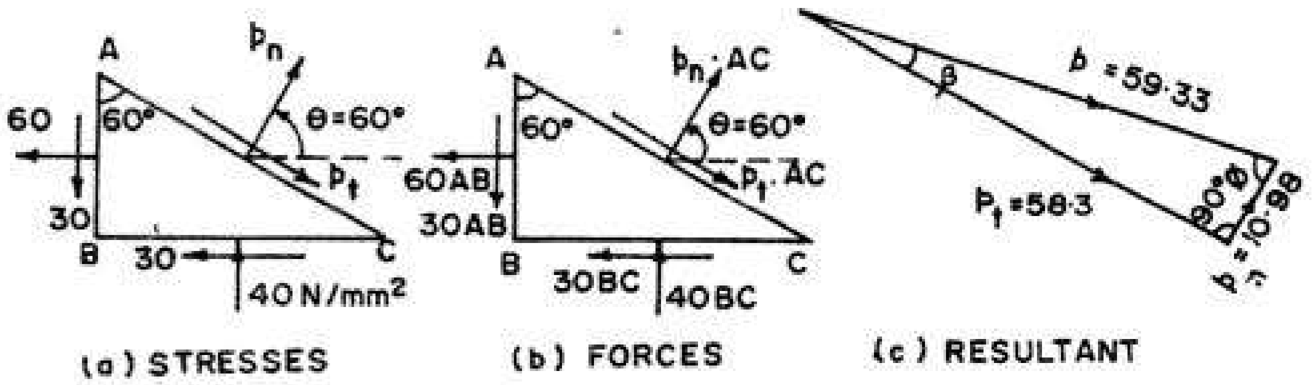


FIG. 4.14

$$p_t = -\frac{p_1 - p_2}{2}\sin 2\theta + q\cos 2\theta \qquad \text{...(4.16)}$$

$$= -\frac{60 + 40}{2}\sin 120^\circ + 30\cos 120^\circ = -\mathbf{58.3\ N/mm^2}$$

(*i.e.* in the direction as marked in Fig. 4.14)

$\therefore$ $$p_r = \sqrt{(10.98)^2 + (58.3)^2} = \mathbf{59.33\ N/mm^2} \text{ (tensile)}$$

$$\varphi = \tan^{-1}\frac{p_t}{p_n} = \tan^{-1}\left(\frac{58.3}{10.98}\right) = \mathbf{79.33^\circ}$$

(or $$\beta = \tan^{-1}\frac{p_n}{p_t} = \tan^{-1}\frac{10.98}{58.3} = 10.67^\circ)$$

Example 4.7. *A piece of material is subjected to tensile stresses of p_1 and p_2 at right angles to each other ($p_1 > p_2$).*

Find the plane across which the resultant stress is most inclined to the normal. Find the value of this inclination and the resultant stress when $p_1 = 60\ N/mm^2$ and $p_2 = 40\ N/mm^2$ (both tensile).

Solution :

Let φ be the inclination of the resultant with the normal (Fig. 4.9).

We have $$\tan\varphi = \frac{p_t}{p_n} = \frac{(p_1 - p_2)\sin\theta\cos\theta}{p_1\cos^2\theta + p_2\sin^2\theta} \qquad \text{...}(i)$$

For φ to be maximum, $\tan\varphi$ is also maximum, and $\dfrac{d(\tan\varphi)}{d\theta} = 0$

Hence, differentiating and dividing out common factors,

$$\left(p_1\cos^2\theta + p_2\sin^2\theta\right)\cos 2\theta + \left(p_1 - p_2\right)\sin\theta\cos\theta \times \sin 2\theta = 0$$

or $$p_n\cos 2\theta + p_t\sin 2\theta = 0$$

Hence $$\tan 2\theta = -\frac{p_n}{p_t} = -\cot\varphi = \tan\left(\frac{\pi}{2} + \varphi\right)$$

or $$2\theta = \frac{\pi}{2} + \varphi \qquad ...(ii)$$

Substituting the value of θ in (i), we get

$$\tan\varphi = \frac{(p_1 - p_2)\cos\varphi}{p_1(1 - \sin\varphi) + p_2\left(1 + \sin\varphi\right)}$$

or $$\frac{p_2}{p_1} = \frac{1 - \sin\varphi}{1 + \cos\varphi}$$

or $$\sin\varphi = \frac{p_1 - p_2}{p_1 + p_2} \qquad ...(iii)$$

Hence from (ii) $$\theta = \frac{1}{2}\left(\frac{\pi}{2} + \sin^{-1}\frac{p_1 - p_2}{p_1 + p_2}\right)$$

which gives the position of the required plane.

Substituting the values of p_1 and p_2, we get

$$\theta = \frac{1}{2}\left(90^\circ + \sin^{-1}\frac{60^\circ - 40^\circ}{60^\circ + 40^\circ}\right) = \frac{1}{2}\left(90^\circ + 11^\circ 32'\right)$$

$$= \mathbf{50^\circ\ 46'}$$

$$\varphi = \sin^{-1}\frac{p_1 - p_2}{p_1 + p_2} = \sin^{-1}\frac{60^\circ - 40^\circ}{60^\circ + 40^\circ} = \mathbf{11^\circ\ 32'}$$

$$p_r = \sqrt{(60\cos 50^\circ 46')^2 + (40\sin 50^\circ 46')^2} = \mathbf{48.9\ N/mm^2}$$

Example 4.8. *An element in plane stress is subjected to stresses $p_1 = 120\ N/mm^2$ and $p_2 = 45\ N/mm^2$ (both tensile) and shearing stress of $30\ N/mm^2$, as shown in Fig. 4.15(a). Determine the stresses acting as an element rotated through an angle $\theta = 45°$.*

Solution :

The required stresses p_n and p_t can be found by Eqs. 4.15 and 4.16

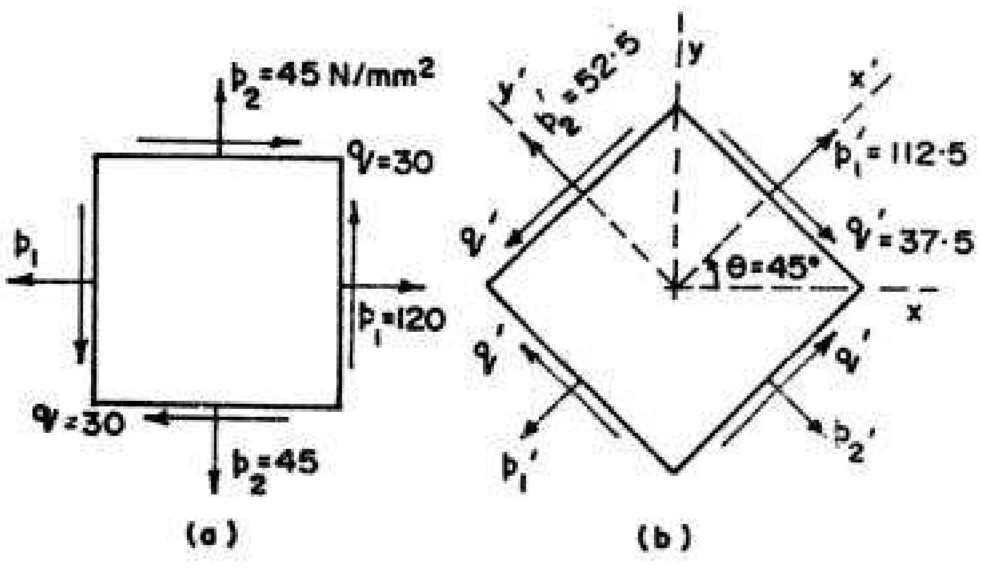


FIG. 4.15

Here

$$\frac{p_1 + p_2}{2} = \frac{120 + 45}{2} = 82.5\ \text{N/mm}^2$$

and

$$\frac{p_1 - p_2}{2} = \frac{120 - 45}{2} = 37.5\ \text{N/mm}^2$$

$$q = 30\ \text{N/mm}^2$$

$$\sin 2\theta = \sin(2 \times 45^\circ) = 1$$

and

$$\cos 2\theta = \cos(2 \times 45^\circ) = 0$$

Hence $$p_n = p_1' = \tfrac{1}{2}(p_1 + p_2) + \tfrac{1}{2}(p_1 - p_2)\cos 2\theta + q\sin 2\theta$$

$$= 82.5 + 37.5 \times 0 + 30 \times 1 = \mathbf{112.5\ N/mm^2}$$

and
$$p_t = q' = -\frac{p_1 - p_2}{2}\sin 2\theta + q\cos 2\theta$$

$$= -37.5 \times 1 + 30 \times 0 = -37.5\ \text{N/mm}^2$$

The normal stress on the other plane, at right angles to the previous one can be found by substituting $\theta = 90° + 45° = 135°$, so that $\sin 2\theta = \sin(2 \times 135°) = -1$ and $\cos 2\theta = \cos(2 \times 135°) = 0$

$$\therefore \quad p_2' = \frac{1}{2}(p_1 + p_2) + \frac{1}{2}(p_1 - p_2)\cos 2\theta + q\sin 2\theta$$

$$= 82.5 + 37.5 \times 0 + 30(-1) = \mathbf{52.5\ N/mm^2}$$

The stresses on the original stress element are shown in Fig. 4.15 *(a)* while the stresses on the stress element rotated through 45° are shown in Fig. 4.15 *(b)*. It is to be noted that $p_1 + p_2 = p_1' + p_2'$.

Example 4.9. *A plane stress condition exists at a point in a loaded structure. The stresses have the magnitude and directions shown on the stress element of Fig. 4.16 (a). Calculate the stresses acting on the planes obtained by rotating the element clockwise through an angle of 16°.*

Solution :

Here we have

$$p_1 = -100\ \text{N/mm}^2$$

$$p_2 = 25\ \text{N/mm}^2$$

and
$$q = -40\ \text{N/mm}^2$$

$$\frac{p_1 + p_2}{2} = \frac{-100+25}{2} = -37.5\ \text{N/mm}^2$$

$$\frac{p_1 - p_2}{2} = \frac{-100-25}{2} = -62.5\ \text{N/mm}^2$$

$$q = -40\ \text{N/mm}^2$$

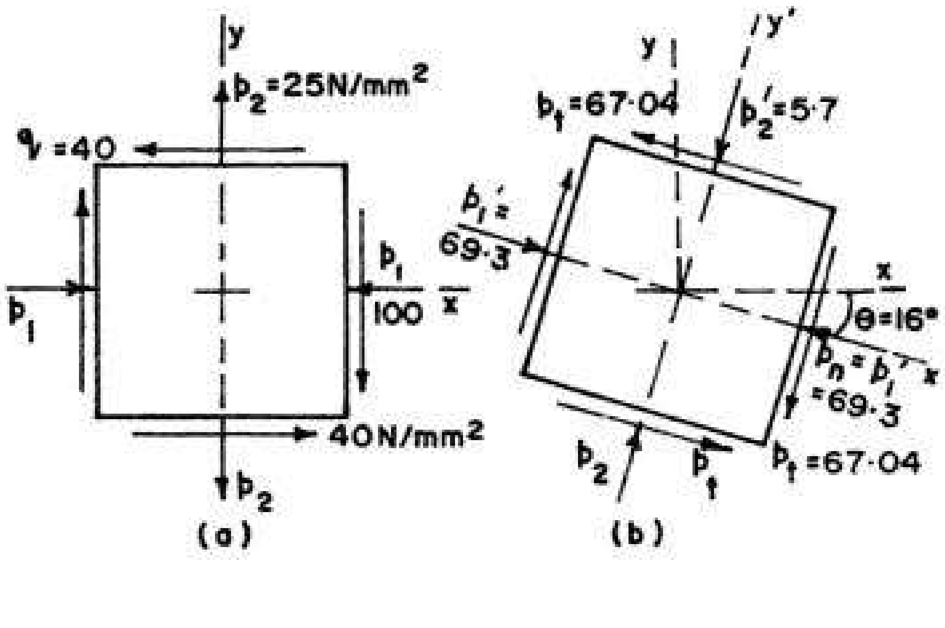


FIG. 4.16

For clockwise rotation, $\theta = -16°$

$$\therefore \quad \sin 2\theta = \sin(-32°) = -0.5299 \text{ and } \cos 2\theta = \cos(-32°) = 0.8480$$

$$\therefore \quad p_1' = \frac{1}{2}(p_1 + p_2) + \frac{1}{2}(p_1 - p_2)\cos 2\theta + q\sin 2\theta$$

$$= -37.5 - 62.5(0.8480) - 40(-0.5299) = \mathbf{-69.3\ N/mm^2}$$

(i.e. compressive)

$$p_t = -\frac{p_1 - p_2}{2}\sin 2\theta + q\cos 2\theta$$

$$= +62.5(-0.5299) + (-40)(0.8480) = -67.04\ \text{N/mm}^2$$

For finding stresses on the y' face, substitute $\theta = \theta + 90° = -16° + 90° = 74°$

Hence $\sin 2\theta = \sin(2 \times 74°) = 0.5299$ and $\cos 2\theta = \cos(2 \times 74°) = -0.8480$

$$\therefore \quad p_2' = \frac{1}{2}(p_1 + p_2) + \frac{1}{2}(p_1 - p_2)\cos 2\theta + q \sin 2\theta$$

$$= -37.5 - 62.5(-0.8480) - 40(0.5299) = \mathbf{-5.7\ N/mm^2}$$

(i.e. compression)

Here $p_1' + p_2' = -69.3 - 5.7 = -75$, and $p_1 + p_2 = -100 + 25 = -75$

Hence $p_1' + p_2' = p_1 + p_2$

The stresses acting on the stress element rotated through 16° in the clockwise direction are shown in Fig 4.16 (*b*) :

4.7. CIRCULAR DIAGRAM FOR STRESSES : MOHR CIRCLE

In the previous articles, we have discussed the analytical methods of computing the normal, shear and resultant stresses on any inclined plane. The so called *transformation equations*, used for such computations can be easily represented in a *graphical form* known as Mohr's circle. Such a graphical representation, commonly known as *circular diagram for stress*, is extremely useful in visualising the relationships between normal and shear stresses acting on various inclined planes at a point is a stressed body. We shall consider here different cases.

(a) Mohr's Circle for two like stresses p_1 and p_2

From Eq. 4.7,

$$p_n = p_1 \cos^2\theta + p_2 \sin^2\theta \qquad ...(4.7)$$

Using the identities : $\cos^2\theta = \frac{1}{2}(1 + \cos 2\theta)$ and $\sin^2\theta = \frac{1}{2}(1 - \cos 2\theta)$

Eq. 4.7 can be written as :

$$p_n = \frac{1}{2}(p_1 + p_2) + \frac{1}{2}(p_1 - p_2)\cos 2\theta \qquad ...(4.17)$$

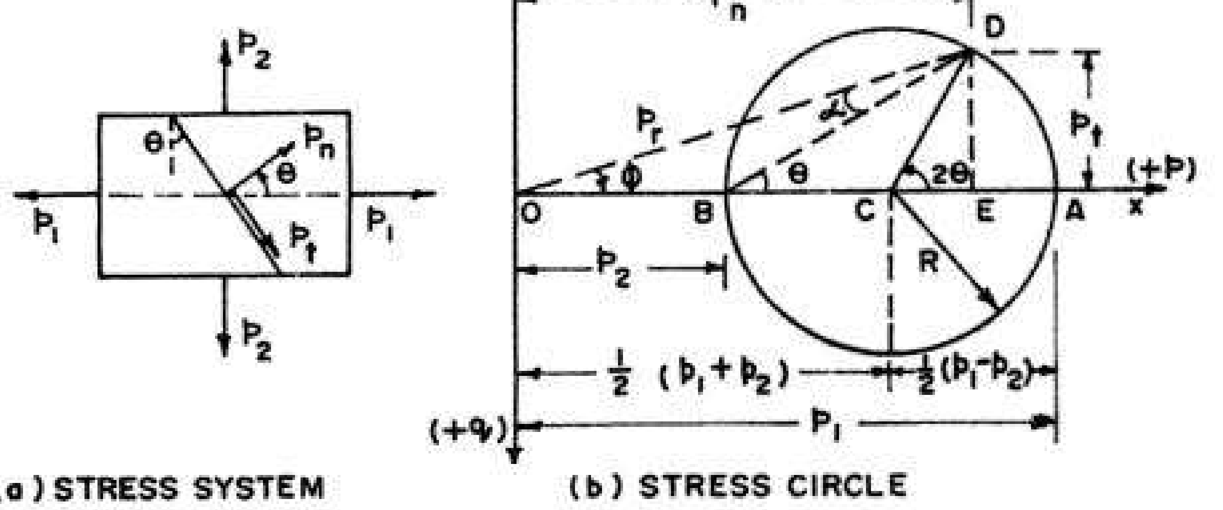


FIG. 4.17

Also, from Eq. 4.8,

$$p_t = -\frac{p_1 - p_2}{2} \sin 2\theta \qquad ...(4.18)$$

These two equations are the equations of a circle in parametric form, with the angle 2θ as the parameter. This parameter can be eliminated by squaring both sides of each equations and then adding. Thus :

$$\left(p_n - \frac{p_1 + p_2}{2}\right)^2 + p_t^2 = \left(\frac{p_1 - p_2}{2}\right)^2 \qquad ...(4.19\ a)$$

Introducing $p_{av} = \frac{p_1 + p_2}{2}$ and $R = \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2}$ ...(4.19 *b*)

Equation 4.19 (*a*) takes the following form :

$$(p_n - p_{av})^2 + p_t^2 = R^2 \qquad ...(4.19)$$

This is the equation of a circle with p_n and p_t as the coordinates. The radius of the circle is $\frac{1}{2}(p_1 - p_2)$ and its centre has coordinates $p_n = p_{av}$ and $p_t = 0$

Procedure for construction of Mohr's circle

To construct Mohr's circle, we take p_n as the abscissa and p_t as the ordinate. We take p_n *positive to the right* and p_t *positive downwards* ; then angle 2θ is positive when taken counter clockwise.

(*i*) Locate centre C of the circle at the point having coordinates $p_{av} = \frac{1}{2}(p_1 + p_2)$ and $p_t = 0$.

(*ii*) Measure OA and OB equal to p_1 and p_2 respectively to some scale to the same side of O.

(*iii*) With C (which is midway between B and A) as centre, and CA or CB as radius $= \frac{1}{2}(p_1 - p_2)$, draw a circle.

(*iv*) At the centre C, draw a line CD at an angle 2θ, in the same direction as the normal to the plane makes with the direction of p_1. In Fig. 4.17 (*a*), which represents the stress system, the normal to the plane makes an angle θ in the anticlockwise direction with the directions of p_1.

(*v*) From D, draw a perpendicular DE on the axis OX.

(*vi*) Then DE represents p_t and OE represents p_n.

From the stress diagram :

$$CD = CA = \frac{1}{2}(p_1 - p_2)$$

$$\therefore \quad DE = CD \sin 2\theta = \frac{1}{2}(p_1 - p_2) \sin 2\theta = p_t$$

Similarly, $\quad OE = OC + CE = \frac{1}{2}\left(p_1 + p_2\right) + \frac{1}{2}\left(p_1 - p_2\right)\cos 2\theta = p_n.$

Hence it is proved that the two rectangular projections of the radius vector give p_t and p_n.

From the stress circle, p_t is maximum when $2\theta = 90°$ (or $\theta = 45°$) and $p_{t,max} = \frac{1}{2}\left(p_1 - p_2\right)$.

In Fig 4.17 (*b*) p_t is above the axis and hence it is *negative*. Hence, in the stress system shown in Fig. 4.17 (*a*), p_t has been shown in the clockwise direction (*i.e.* negative). Hence we follow the following sign conventions for Mohr's stress circle.

Sign Conventions

(*i*) Tensile stresses are reckoned positive and are plotted to the right of origin *O*. Similarly, compressive stress is reckoned negative and is plotted to the left of origin *O*.

(*ii*) p_t is taken positive when it gives anticlockwise rotation of the element and is plotted below the axis. Similarly, p_t that gives clockwise rotation is reckoned *negative* and is plotted above the axis.

(*iii*) θ is reckoned positive when it is in the anticlockwise direction,

Thus, as per the above sign conventions, both p_1 and p_2 marked in the stress system of Fig 4.17 (*a*) are positive and have been plotted to the R.H. side of the origin in Fig. 4.17 (*b*). The angle θ is positive since it is measured in the anticlockwise direction. The resulting normal stress p_n is thus positive (i.e. tensile) while the shear stress p_t (= *DE*), is *negative* (since it falls above the *x*-axis of the stress circle) and has been adequately marked in Fig 4.17 (*a*).

Mohr's circle construction for two unlike stresses p_1 and p_2

Let p_1 be tensile (*i.e.* +ve) and p_2 be compressive (*i.e.* –ve). It is required to find the stresses on a plane normal to which makes an angle θ in the anticlockwise direction, with the direction of p_1, as marked in the stress system shown in Fig ·4.18 (*a*).

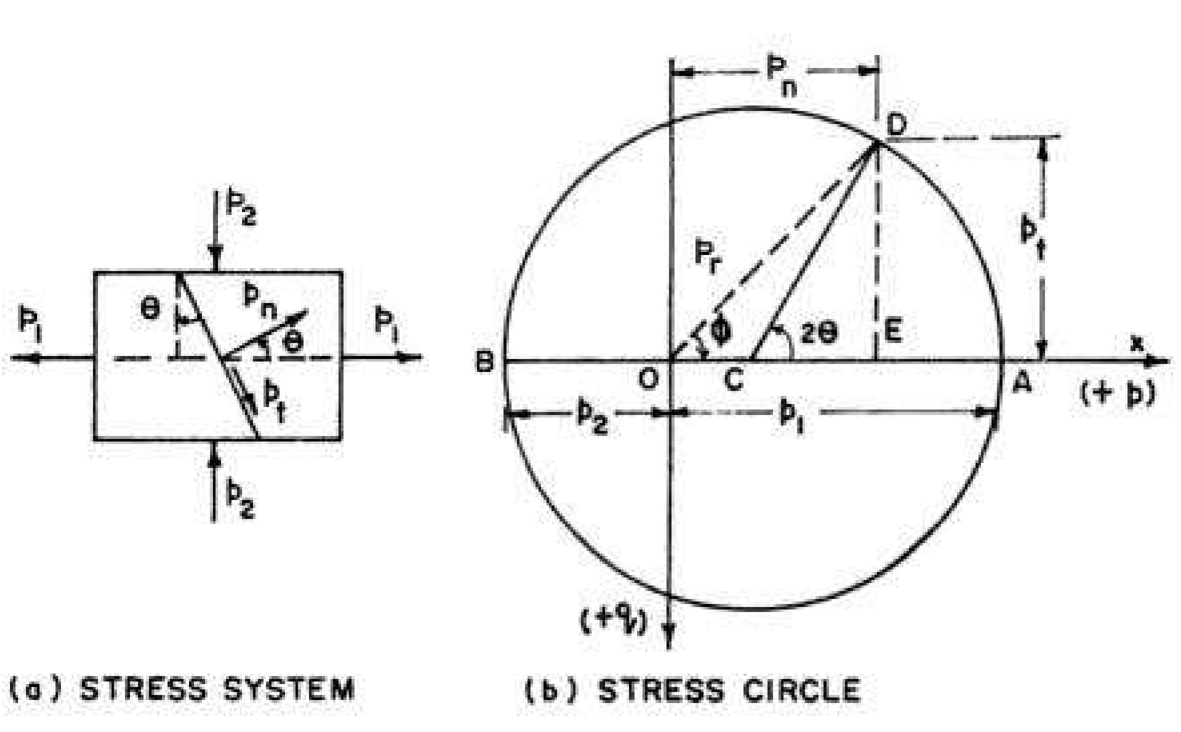
Steps.

(*i*) Mark $OA = p_1$ to the right of *O* and $OB = p_2$ to the left.

(*ii*) Find point *C*, midway between *A* and *B*. Thus, $OC = \left(\frac{p_1 - p_2}{2}\right)$.

If $p_1 > p_2$, *C* will be to the right *O*; otherwise it will be to the left.

(*iii*) At *C*, draw radius vector *CD*, at angle 2θ in the anticlockwise



(a) STRESS SYSTEM (b) STRESS CIRCLE

FIG. 4.18

direction. We have radius $CD = \frac{p_1 + p_2}{2} = CA$.

(*iv*) Draw DE perpendicular to the x-axis.

Then $$OE = p_n \text{ and } DE = p_t$$

It is clear that the direction of p_n will depend upon the position of point E. If it is to the right of O, the direction of p_n will be the same as that of p_1. For the stress system marked and shown in Fig. 4.18, p_n is positive (*i.e.* tensile), while $p_t\,(=DE)$ is negative.

The *resultant stress* $p_r = OD$ and it makes an angle $\varphi\,(=DOC)$ with the direction of p_n or an angle $\alpha\,(=ODC)$ with the direction of p_1.

(c) Mohr's circle for the general case of plane stress

Let is now take the general case of plane stress in which the material is subjected to direct stresses p_1 and p_2 accompanied with the state of simple shear as marked in Fig 4.19 (*a*). All the stresses (*i.e.* p_1, p_2 and q) marked in the stress system are positive according to our sign convention discussed earlier.

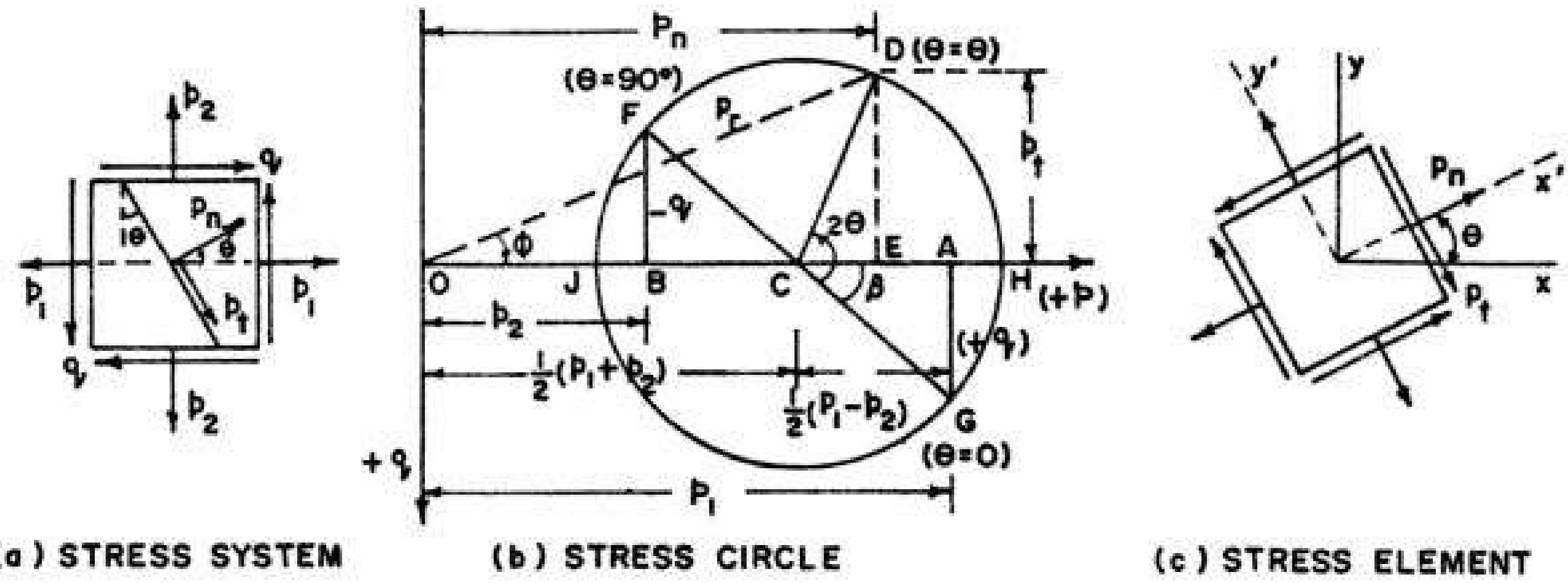


FIG. 4.19

From Eq. 4.15

$$p_n = \frac{1}{2}\left(p_1 + p_2\right) + \frac{1}{2}\left(p_1 - p_2\right)\cos 2\theta + q\sin 2\theta \qquad ...(4.15)$$

Also, from Eq. 4.16, $$p_t = -\frac{p_1 - p_2}{2}\sin 2\theta + q\cos 2\theta \qquad ...(4.16)$$

These two equations are the equations of a circle in parametric form, with angle 2θ as the parameter. This parameter can be eliminated by squaring both sides of each equation and then adding. Thus.

$$\left(p_n - \frac{p_1 + p_2}{2}\right)^2 + p_t^2 = \left(\frac{p_1 - p_2}{2}\right)^2 + q^2 \qquad ...(4.20\ a)$$

Introducing $$p_{av} = \frac{1}{2}\left(p_1 + p_2\right) \text{ and } R = \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2} \qquad ...(4.20\ b)$$

Eq. 4.20 (*a*) takes the following form.

$$(p_n - p_{av})^2 + p_t^2 = R^2 \qquad \text{...(4.20)}$$

This is the equation of a circle with p_n and p_t as coordinates. The radius R of the circle is $\sqrt{\left[\frac{1}{2}(p_1 - p_2)\right]^2 + q^2}$ and its centre has the co-ordinates $p_n = p_{av}$ and $p_t = 0$. The corresponding Mohr's circle is shown in Fig. 4.19 (*b*), with the procedure for its construction described below.

Procedure for construction of Mohr's circle

To construct Mohr's circle, we take p_n as the abscissa and p_t as ordinate. We take p_n *positive to the right* and p_t *positive downwards ; then angle* 2θ *is positive when taken counter clockwise.*

(*i*) Make $OA = p_1$ and $OB = p_2$ to the same side of O, since both p_1 and p_2 are alike, and are positive.

(*ii*) The shear stress q marked on x-face (*i.e.* face *ab*) in Fig 4.19 (*a*) is positive as per our sign conventions. Hence draw AG perpendicular to OX, and equal to q, in the downward direction, since positive shear is measured downwards. Similarly make BF perpendicular to OX in *upward direction* and make in equal to q; it is to be noted that shear on face *da* (on which stress p_2 is acting) is negative since it tends to rotate the element in clockwise direction, and negative shear is plotted upwards.

(*iii*) Join F and G, cutting the x-axis in C, which is the centre of the stress circle (i.e. Mohr's circle). The stress corresponding to point C is evidently equal to $\frac{1}{2}(p_1 + p_2)$ since point C is midway between points A and B.

(*iv*) With C as centre and radius equal to CG (or CF) draw a circle.

(*v*) At C, make CD at an angle 2θ with CG in the anticlockwise direction. Line CD is the required radius vector, and the coordinates of point D give the required stresses p_n and p_t. To find their values, draw DE perpendicular to x-axis.

Then $DE = p_t$, $OE = p_n$ and $OD = p_r$.

Here p_n is positive and p_t negative as per over sign conventions. It is interesting to note that point G represents the stress conditions on x-face (i.e. face *ab*) for which $\theta = 0$ where $p_n = OA = p_1$ and $p_t = AG = +q$. Similarly, point F represents the stress conditions on y-face (*i.e.* face *da*) for which $2\theta = 180°$ (or $\theta = 90°$), where stress $p_n = OB = p_2$ and $p_t = BF = -q$

Proof : Let R = radius of the stress circle

$$= \sqrt{CA^2 + AG^2} = \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2}$$

$$R\cos\beta = CA = \frac{p_1 - p_2}{2}$$

$$R\sin\beta = AG = q$$

Now $$OE = OC + CE = OC + R\cos(2\theta - \beta)$$

$$= OC + R\cos\beta\cos 2\theta + R\sin\beta\sin 2\theta$$

$$= \frac{1}{2}\left(p_1 + p_2\right) + \frac{1}{2}\left(p_1 - p_2\right)\cos 2\theta + q\sin 2\theta$$

$$= p_n \text{ (as per Eq. 4.15)}$$

Similarly, $DE = R\sin(2\theta - \beta) = R\cos\beta\sin 2\theta - R\sin\beta\cos 2\theta$

$$= \frac{1}{2}\left(p_1 - p_2\right)\sin 2\theta - q\cos 2\theta$$

$$= -p_t \text{ (as per Eq. 4.16)}$$

The stress conditions are marked on the stress element shown in Fig 4.19 (*c*).

Conclusions

From the stress circle, we draw the following conclusions :

(*i*) When D coincides with H (*i.e.* $2\theta = \beta$), p_n attains its *maximum* value :

$$p_{n,max} = OC + CH$$

$$= \frac{1}{2}\left(p_1 + p_2\right) + \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2}$$

where $p_{n,max}$ is known as the *major principal stress* σ_1 (see § 4.9).

Also, $p_t = 0$ and $p_{r,max} = p_{t,max}$

$$\tan 2\theta = \tan\beta = \frac{q}{\frac{1}{2}(p_1 - p_2)} = \frac{2q}{p_1 - p_2}$$

(see § 4.9)

(*ii*) When D coincides with J, p_n attains the minimum value :

$$p_{n,min} = OJ = OC - CJ$$

$$= \frac{1}{2}\left(p_1 + p_2\right) - \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2}$$

$p_{n,min}$ is known as the minor principal stress σ_2 (see § 4.9)

$p_t = 0$; $p_{r,min} = p_{n,min}$; $2\theta = 180° + \beta$ or $\theta = 90° + \dfrac{\beta}{2}$

(*iii*) When $2\theta = \beta + 90°$, p_t attains its maximum value

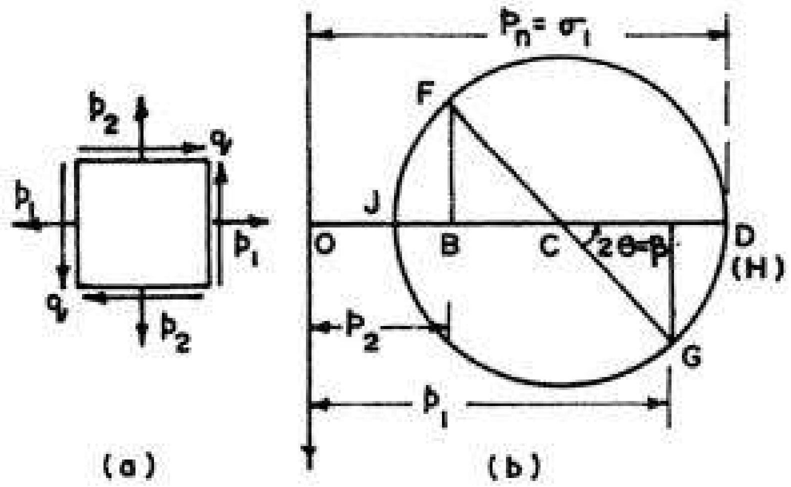


FIG. 4.20

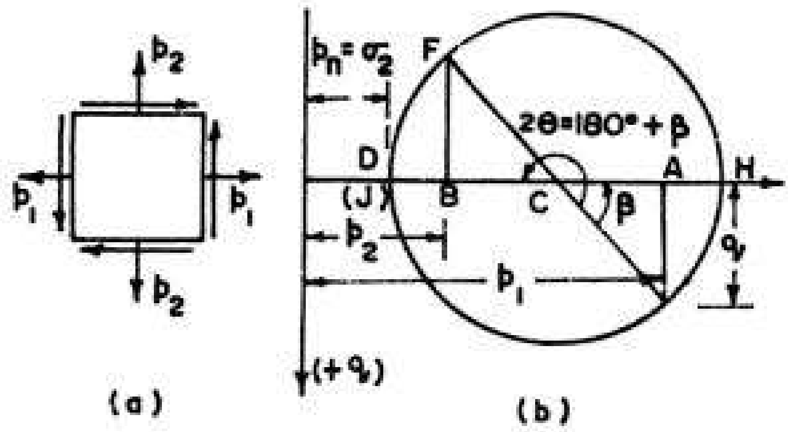


FIG. 4.21

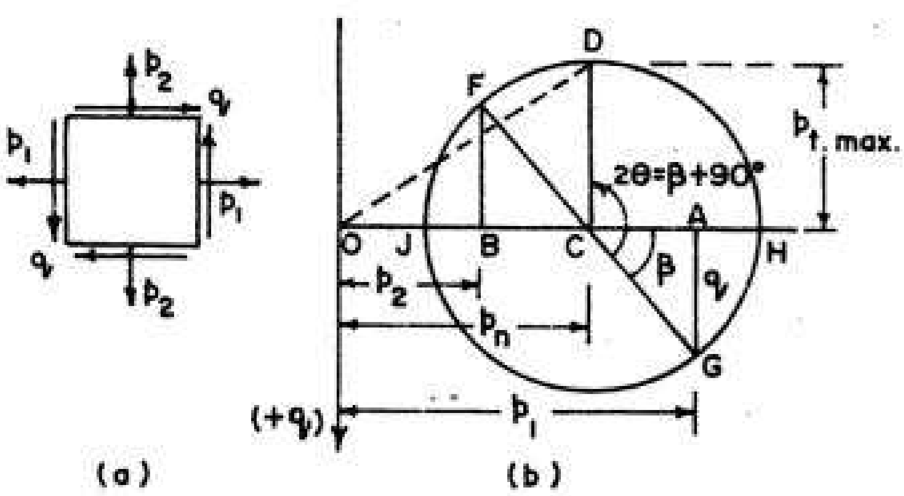


FIG. 4.22

$$p_{t,max} = R = -\sqrt{\frac{1}{4}(p_1 - p_2)^2 + q^2} = -\frac{\sigma_1 - \sigma_2}{2}$$

(*iv*) Similarly, when $2\theta = \beta + 270°$

$$p_{t,max} = R = \sqrt{\frac{1}{4}(p_1 - p_2)^2 + q^2} = \frac{1}{2}(\sigma_1 - \sigma_2)$$

Example 4.10. *A piece of material is subjected to tensile stresses of 70 N/mm² and 30 N/mm² at right angles to each other. Find fully the stresses on a plane the normal of which makes an angle of 35° with the 70 N/mm² stress.*

Solution :

To construct Mohr circle, mark $OA = p_1 = 70$ N/mm² and $OB = p_2 = 30$ N/mm² to the same side of O, since both of them are positive. There are no shearing stresses present in the stress system. Bisect AB to get C, the centre of the circle. Draw the stress circle with C as centre and CA (or CB) as radius. Draw CD at $2\theta = 70°$ to x-axis, in the counter clockwise direction. Drop perpendicular DE. The stress circle is shown in Fig. 4.23 (b) from which we obtain

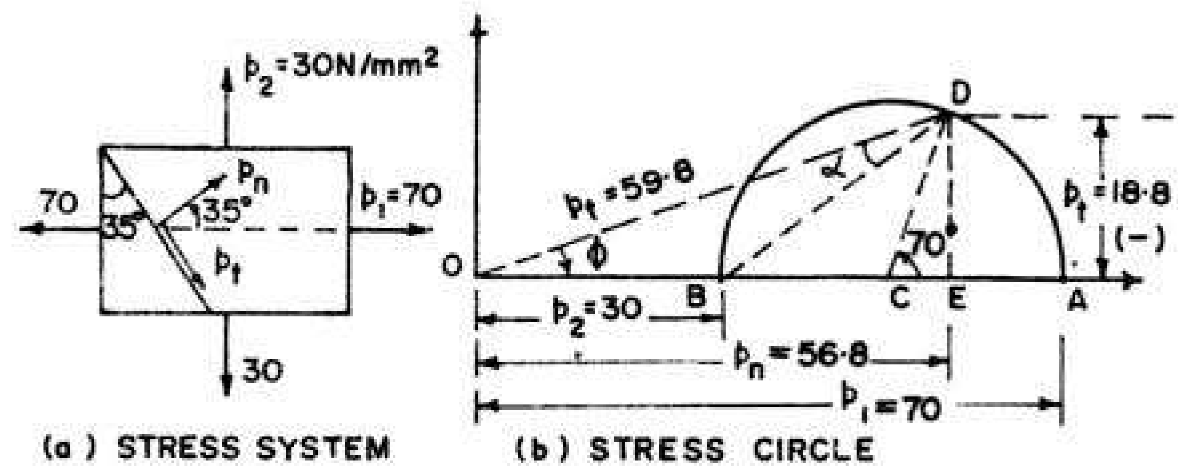


FIG. 4.23

$p_n = OE \approx$ **56.8 N/mm²**;

$p_t = DE \approx$ **− 18.8 N/mm²** (*i.e.* anticlockwise rotation) $p_r = OD \approx$ **59.8 N/mm²**

Also $\varphi \approx$ **− 18.5°** and $\alpha \approx$ **16.5°**.

Example 4.11. *At a point in a material, the stresses on two mutually perpendicular planes are 50 N/mm² (tensile) and 30 N/mm² (tensile). The shear stress across these planes is 12 N/mm². Using Mohr circle, find magnitude and direction of the resultant stress on a plane making an angle of 35° with the plane of the first stress. Find also, the normal and tangential stresses on this plane.*

Solution :

To draw Mohr's circle (Fig 4.24 b), make $OA = 50$ N/mm² and $OB = 30$ N/mm², to the same side of O. Erect perpendicular $AG = 12$ N/mm² in the downward direction (since q is positive on x-face) and erect $BF = 12$ N/mm² (since q is negative on y-face). Join F and G to get point C, the centre of the stress circle. With C as the centre and CG (or CF) as radius, draw Mohr's circle. Draw CD at angle $2\theta = 70°$ in the anti-clockwise direction with CG. The coordinates of point D give the values of p_n and p_t. Thus, we obtain :

$p_n = OE \doteq$ **54.5 N/mm²**; $p_t = ED =$ **− 5.5 N/mm²**; $p_r = OD =$ **54.5 N/mm²**

$\varphi =$ **− 5.5°**

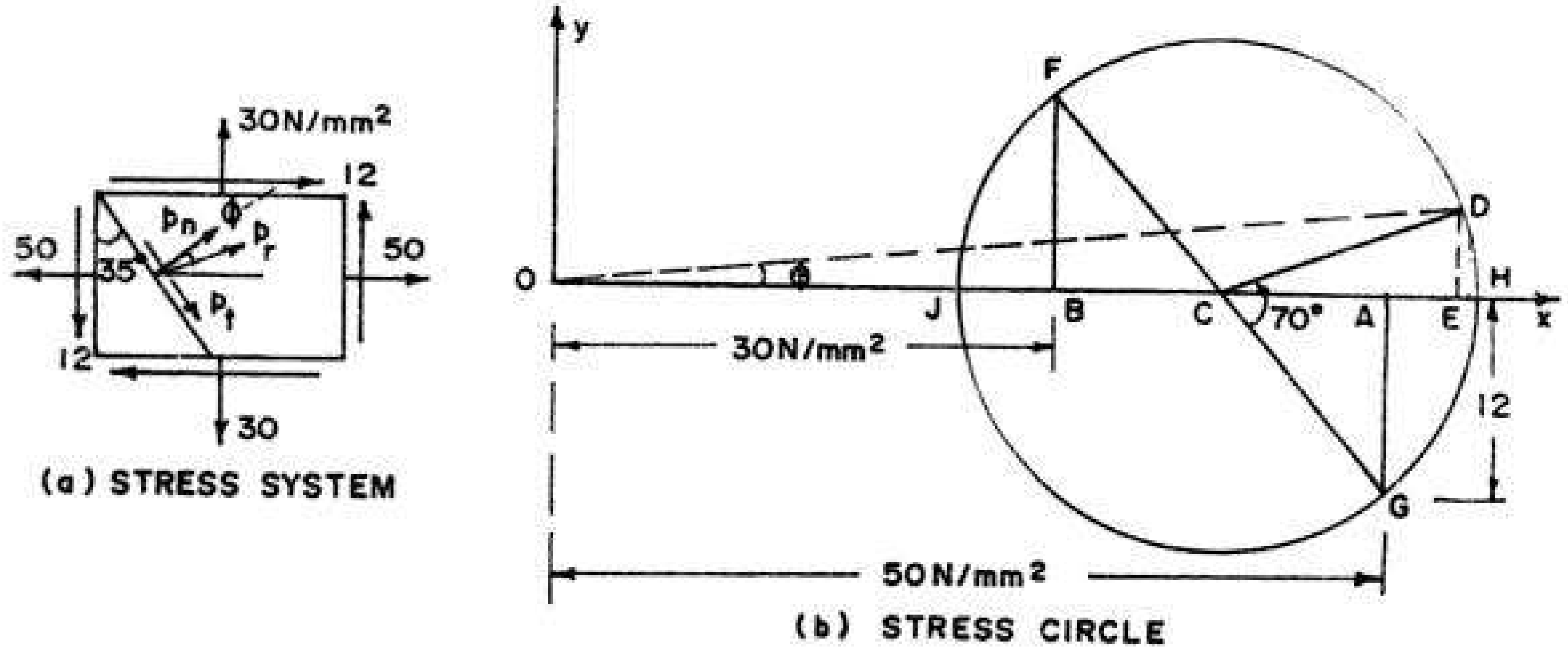


FIG. 4.24

From the theoretical solution, the corresponding values will be as under :

$$p_n = \frac{1}{2}\left(p_1 + p_2\right) + \frac{1}{2}\left(p_1 - p_2\right)\cos 2\theta + q\sin 2\theta$$

$$= \frac{1}{2}\left(50 + 30\right) + \frac{1}{2}\left(50 - 30\right)\cos 70° + 12\sin 70° = \mathbf{54.7\ N/mm^2}$$

$$p_t = -\frac{p_1 - p_2}{2}\sin 2\theta + q\cos 2\theta$$

$$= -\frac{50-30}{2}\sin 70° + 12\cos 70° = -\mathbf{5.29\ N/mm^2}$$

$$p_r = \sqrt{(54.7)^2 + (-5.29)^2} = \mathbf{54.96\ N/mm^2}$$

$$\varphi = \tan^{-1}\left(\frac{p_t}{p_n}\right) = \tan^{-1}\left(\frac{-5.29}{54.7}\right) = -\mathbf{5.52°}$$

(*i.e.* in the clockwise direction as marked in Fig 4.24 *a*)

Example 4.12. *For the data of Example 4.11, find the stresses on a plane making an angle of 15° with the plane of the first stress.*

Solution :

The stress circle is shown in Fig 4.25 (*b*). Here the point *D* falls below the *x*-axis. Hence the shear stress p_t (= *DE*) comes out to be positive, as marked in Fig. 4.25 (*a*). From the stress circle we get

$$p_n = OE = \mathbf{54.5\ N/mm^2};\ p_t = ED = \mathbf{5.5\ N/mm^2};\ p_r = OD = \mathbf{54.5\ N/mm^2}$$

$$\varphi \approx +\mathbf{5.5°}$$

Analytical check :

$$p_n = \frac{1}{2}(50 + 30) + \frac{1}{2}(50 - 30)\cos 30° + 12\sin 30° = \mathbf{54.67\ N/mm^2}$$

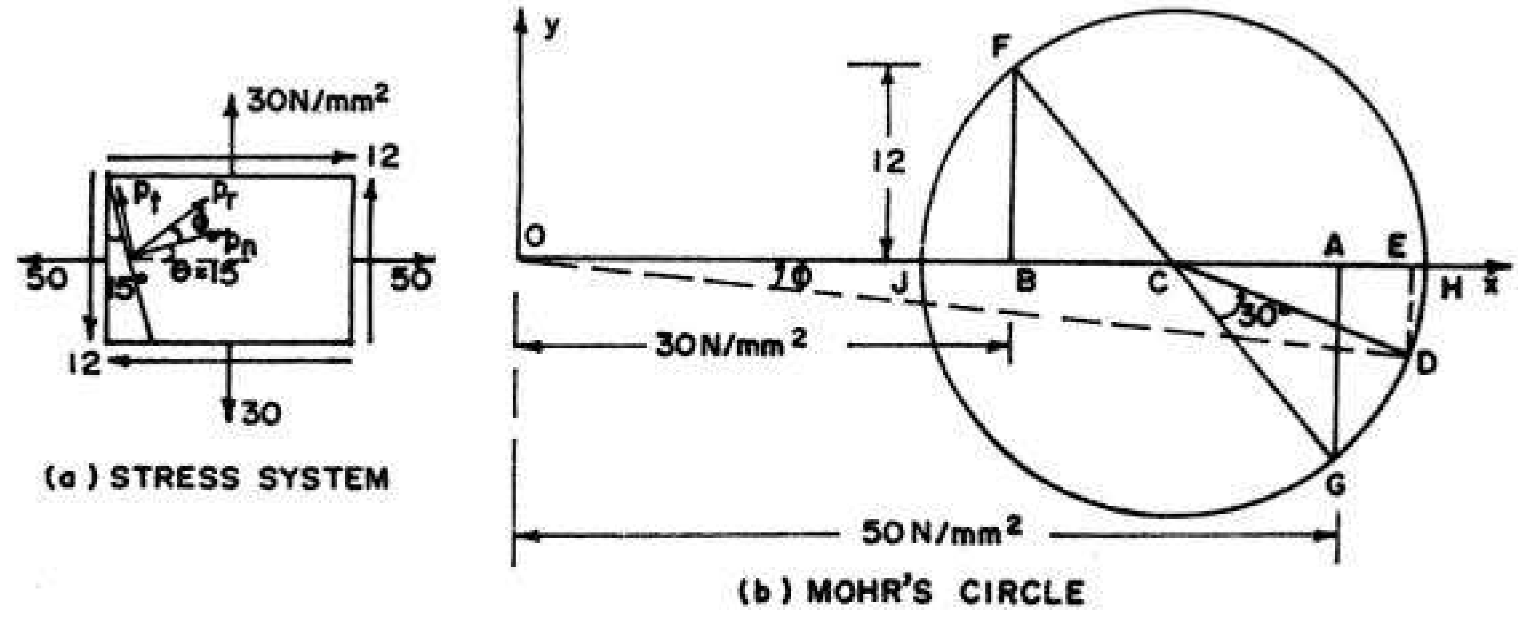


FIG. 4.25

$$p_t = -\frac{50-30}{2}\sin 30° + 12\cos 30° = \mathbf{+\,5.39\ N/mm^2}$$

$$p_r = \sqrt{(54.67)^2 + (5.39)^2} = \mathbf{54.94\ N/mm^2}\ ;\ \varphi = \tan^{-1}\left(\frac{p_t}{p_n}\right) = \tan^{-1}\left(\frac{5.39}{54.67}\right) = \mathbf{+\,5.63°}$$

Example 4.13. *Draw Mohr's stress circle for direct stresses of 45 N/mm² (tensile) and 25 N/mm² (comp.) and find the magnitude and direction of resultant stresses on planes making angles of 30° and 60° with the plane of first principal stress. Also, find the normal and tangential stresses on these planes.*

Solution : To draw Mohr's circle, plot $OA = 45\ \text{N/mm}^2$ to the right and $OB = 25\ \text{N/mm}^2$ to the left of origin O. Bisect AB to get C. With C as centre and CA (or CB) as radius, draw the stress circle.

(*a*) Draw CD_1, at an angle $2\theta_1 = 2 \times 30° = 60°$ with CA. Draw D_1E_1 perpendicular to x-axis. Then for the first plane (Fig 4.26 *a*), we get

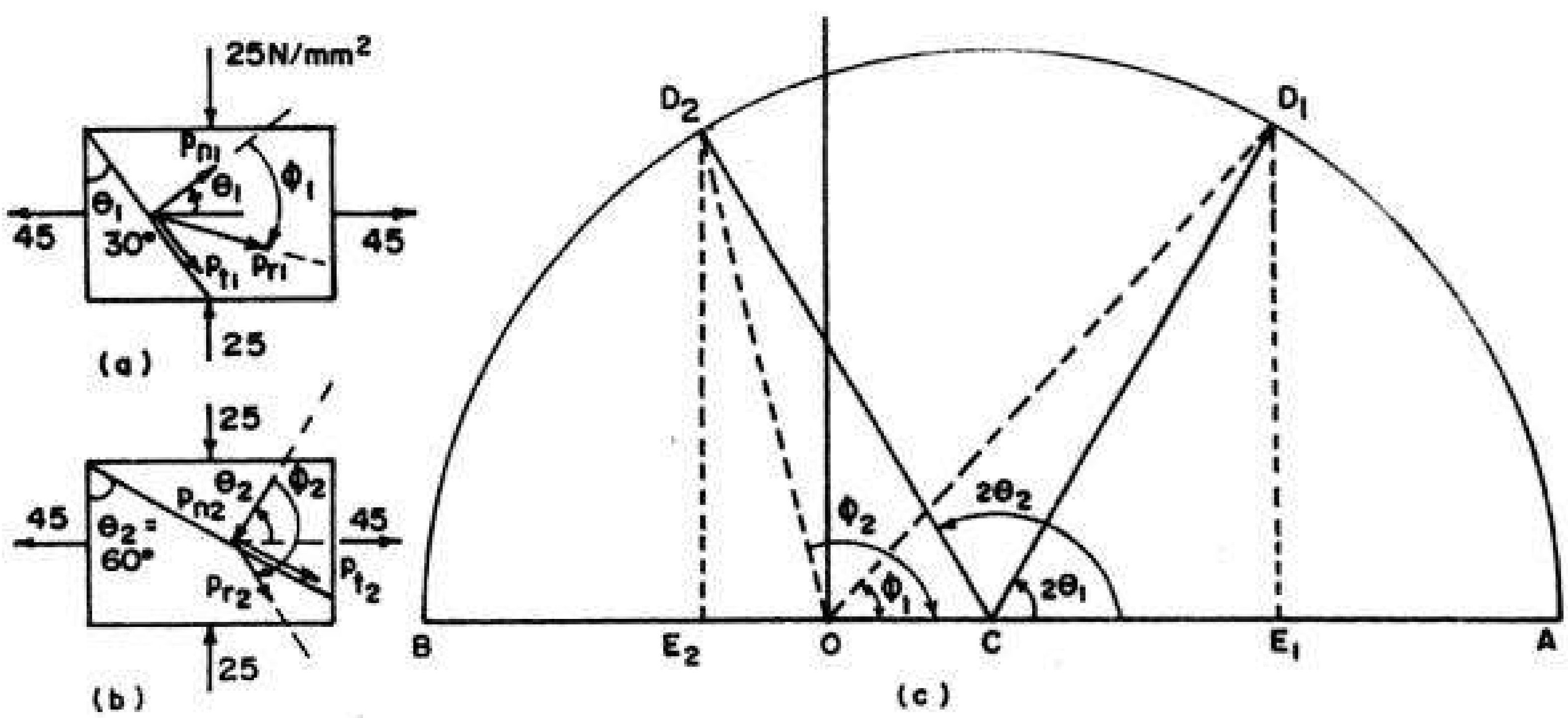


FIG. 4.26

$p_{n1} = OE_1 = \mathbf{27.5\ N/mm^2}$ (tensile)

$p_{t1} = E_1 D_1 = -\mathbf{30\ N/mm^2}$ (*i.e.* clockwise rotation as marked in Fig. 4.26 *a*)

$p_{r1} = OD_1 = \mathbf{41\ N/mm^2}$

and $\varphi_1 = -\mathbf{48^\circ}$

(*b*) Draw CD_2 at an angle $2\theta_2 = 2 \times 60^\circ = 120^\circ$ with CA. Draw D_2E_2 perpendicular to x-axis. Then, for the second plane (Fig. 4.26 *b*) we get

$p_{n2} = OE_2 = -\mathbf{7.5\ N/mm^2}$ (i.e. comp.)

$p_{t2} = E_2 D_2 = -\mathbf{30\ N/mm^2}$ (i.e. clockwise rotation)

$p_{r2} = OD_2 = \mathbf{31\ N/mm^2}$

and $\varphi_2 = -\mathbf{104^\circ}$

Example 4.14. *Solve example 4.7, using Mohr's stress circle solution.*

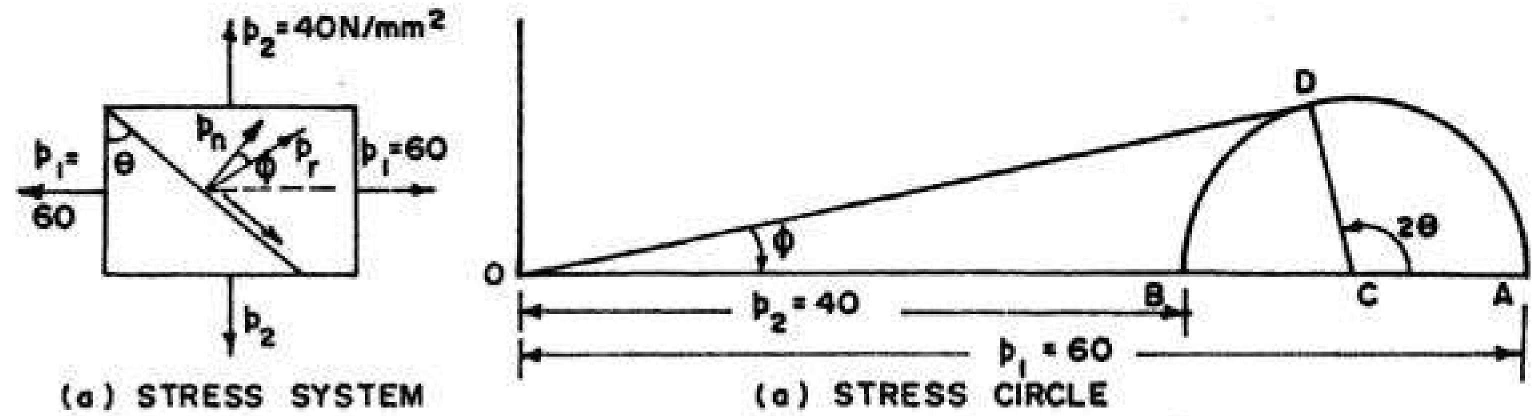


FIG. 4.27

Solution :

To draw Mohr's circle (Fig 4.27 *b*), make $OA = 60\ \text{N/mm}^2$ and $OB = 40\ \text{N/mm}^2$. Obtain centre C midway between A and B and draw the circle with radius $= CA = CB$. To obtain maximum obliquity of the resultant, draw OD tangential to the circle.

$$\text{From Mohr's circle, } \sin\varphi = \frac{DC}{OC} = \frac{\frac{1}{2}(p_1 - p_2)}{\frac{1}{2}(p_1 + p_2)} = \frac{p_1 - p_2}{p_1 + p_2}$$

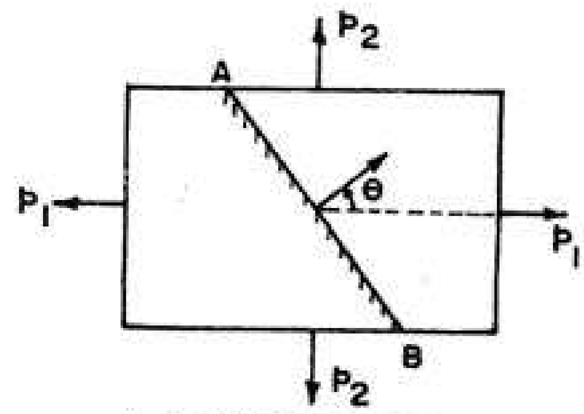
and
$$\theta = \frac{1}{2}(90 + \varphi)$$

From the measurements, we obtain

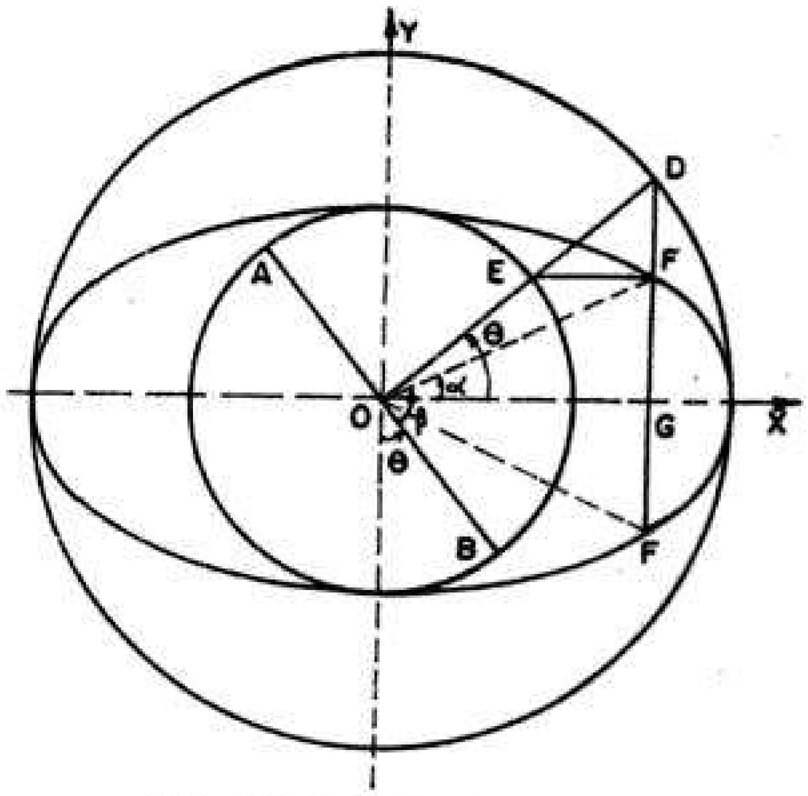
$$\varphi \approx \mathbf{11.5^\circ}\ ;\ \theta \approx \mathbf{51^\circ} \text{ and } p_r = \mathbf{49\ N/mm^2}$$

4.8. ELLIPSE OF STRESS

There are two graphical methods available for finding stresses on an inclined plane in a material : (*i*) method of *stress circle* (Mohr's circle) and (*ii*) method of *ellipse of stress*. Though the first method (*i.e.* stress circle method) is applicable for a two dimensional generalised stress system, the method of ellipse of stress is suitable when the material is subjected to direct stresses p_1 and p_2 only (Fig 4.28 *a*).



(a) STRESS SYSTEM



(b) ELLIPSE OF STRESS

FIG. 4.28

Steps for constructing ellipse of stress

1. Let O be the origin through which x and y axes pass. Through O, draw two concentric circles with radius $R_1 = p_1$ and $R_2 = p_2$, to some suitable scale.

2. Draw the plane AB at an angle θ with the y-axis. Draw its normal OD cutting the inner and outer circles at E and D respectively. Obviously, the normal OD makes an angle θ with the x-axis which represents the direction of p_1. It is to be noted that the y-axis represents the direction of p_2.

3. Through E, draw a line EF parallel to x-axis and through D, draw a line DG parallel to the y-axis, so that both the lines meet at point F. Join OF.

4. Point F is one of the points on the ellipse. For different values of angle θ, the point F may be established. *The locus of F will evidently be an ellipse*. The diagram is, therefore, known as ellipse of stress (Fig. 4.28 *b*). From the diagram,

$$OG = OD\cos\theta = p_1\cos\theta;\quad FG = p_2\sin\theta.$$

$$\therefore \qquad OF = \sqrt{p_1^2\cos^2\theta + p_2^2\sin^2\theta} = p_r$$

Hence OF gives the resultant stress p_r of this plane.

Also, $\tan\alpha = \dfrac{p_2 \sin\theta}{p_1 \cos\theta} = \dfrac{p_2}{p_1}\tan\theta$, as earlier.

4.9. PRINCIPAL STRESSES AND PRINCIPAL PLANES

We have seen that when a material is subjected to a plane stress system (consisting of p_1, p_2 and q) the resulting normal and shear stress on a plane through the point varies with the angle θ. This variation is continuous as the stress element is rotated. The magnitudes of p_n and p_t depend upon the value of θ. For design purposes, the *largest positive and negative* values of p_n are needed. These largest values of normal stresses are known as *principal stresses* and planes on which these stresses act are known as *principal planes*. From Figs 4.20 and 4.21, we observe that for such planes the shear stresses are zero. *Hence principal plane through a point within a material under stress is that plane the stress across which is wholly normal*; evidently, no shear stress exists along this plane. *The normal stress across the principal plane is known as principal stress*. It may be stated in general, that at any point in a strained material under three dimensional stress system, there are *three such planes*, mutually orthogonal to each other, carrying direct stresses and no tangential stress. Out of these, the plane carrying the *maximum normal stress is called the major principal plane* and the normal stress is called the *major principal stress*. The plane carrying the *minimum* normal stress is known as *minor principal plane* and the corresponding normal stress is called the *minor principal stress*. Here, however, we shall be considering only the case of plane stress or the two dimensional stress system where the third principal stress is zero. We shall be denoting the major principal stress by σ_1 and minor principal stress by σ_2.

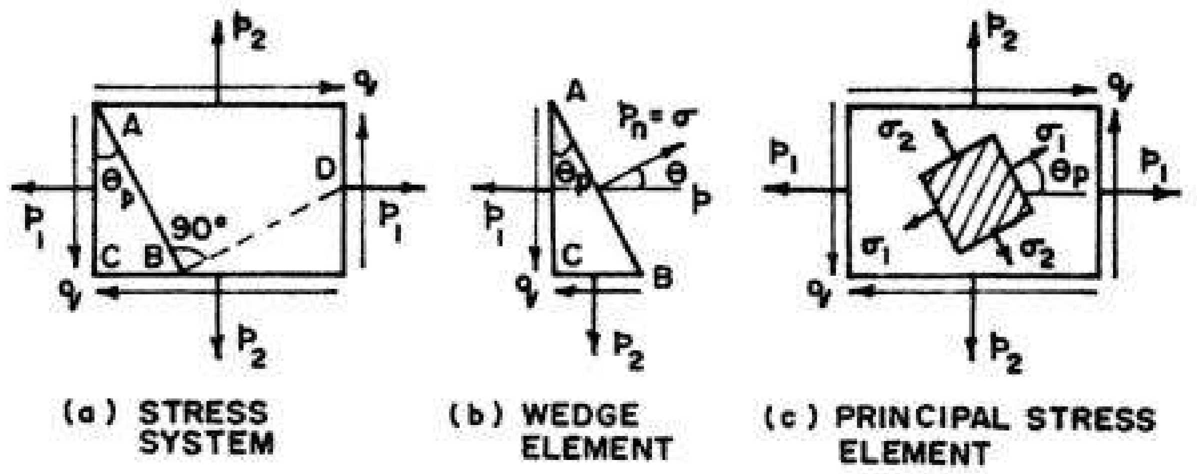


FIG. 4.29

Fig 4.29 (*a*) shows the plane stress case where the element is subjected to p_1, p_2 and q. Fig. 4.29 (*b*) shows wedge shaped element in equilibrium, where the resultant stress $p_n = \sigma$ is wholly normal to the inclined plane AB called the *principal plane*.

Resolving the *forces* parallel to BC, we have

$$(\sigma . AB)\cos\theta_p = p_1 . AC + q . BC$$

or $$\sigma\cos\theta_p = p_1\frac{AC}{AB} + q . \frac{BC}{AB} = p_1\cos\theta_p + q\sin\theta_p.$$

$$\therefore\quad (\sigma - p_1)\cos\theta_p = q\sin\theta_p$$

or $$(\sigma - p_1) = q\tan\theta_p \qquad ...(1)$$

Similarly, resolving the forces along AC, we have

$$(\sigma . AB) \sin\theta_p = p_2 . BC + q . AC$$

or $$\sigma \sin\theta_p = p_2 . \frac{BC}{AB} + q . \frac{AC}{AB} = p_2 \sin\theta_p + q \cos\theta_p$$

or $$(\sigma - p_2) = q \cot\theta_p \qquad ...(2)$$

Subtracting (1) from (2), we get

$$(p_1 - p_2) = q\left(\cot\theta_p - \tan\theta_p\right) = \frac{2q}{\tan 2\theta_p}$$

$$\therefore \qquad \tan 2\theta_p = \frac{2q}{p_1 - p_2} \qquad ...(4.21)$$

where θ_p defines the position of the principal planes,

Again, multiplying (1) and (2), we get

$$(\sigma - p_1)(\sigma - p_2) = q^2$$

or $$\sigma^2 - p_1\sigma - p_2\sigma + p_1p_2 - q^2 = 0$$

or $$\sigma^2 - \sigma(p_1 + p_2) + (p_1p_2 - q^2) = 0$$

which is the quadratic equation for the principal stress σ, from which

$$\sigma = \frac{1}{2}(p_1 + p_2) \pm \sqrt{\frac{1}{4}(p_1 + p_2)^2 + (q^2 - p_1p_2)}$$

or $$\sigma = \frac{1}{2}(p_1 + p_2) \pm \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2}$$

Hence $$\sigma_1 = \frac{p_1 + p_2}{2} + \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2} \qquad ...(4.22\ a)$$

and $$\sigma_2 = \frac{p_1 + p_2}{2} - \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2} \qquad ...(4.22\ a)$$

where σ_1 is the *major principal stress* and σ_2 is the *minor principal stress.*

The orientation of principal planes are shown in Fig. 4.29 (*c*)

Alternative method for finding θ_p, σ_1 and σ_2

We have $$p_n = \frac{p_1 + p_2}{2} + \frac{p_1 - p_2}{2}\cos 2\theta + q \sin 2\theta \qquad ...(\text{Eq. } 4.15)$$

For getting maximum (or minimum) value of p_n, differentiate the above expression with θ and equate it to zero.

$$\therefore \qquad \frac{dp_n}{d\theta} = -(p_1 - p_2)\sin 2\theta + 2q\cos 2\theta$$

From which, we get $$\tan 2\theta_p = \frac{2q}{p_1 - p_2} \qquad ...(4.21)$$

Eq. 4.21 can be represented *diagrammatically* by Fig. 4.30, in which

$$\tan 2\theta_p = \frac{q}{\frac{p_1 - p_2}{2}} \left(= \frac{2q}{p_1 - p_2} \right)$$

Hence diagonal $R = \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2}$

(Taking only the positive root)

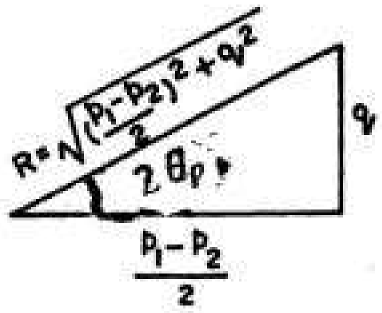


FIG. 4.30. DIAGRAMMATIC REPRESENTATION OF EQ. 4.21

Thus, $\cos 2\theta_p = \frac{p_1 - p_2}{2R}$ and $\sin 2\theta_p = \frac{q}{R}$ ...(4.21 *a*, *b*)

Substituting the values of $\cos 2\theta_p$ and $\sin 2\theta_p$ in Eq. 4.15 and setting $p_n = \sigma_1$, we get

$$\sigma_1 = \frac{p_1 + p_2}{2} + \sqrt{\left(\frac{p_1 - p_2}{2}\right) + q^2} \quad ...(4.22\ a)$$

The minor principal stress σ_2 is obtained from the relation :

$$\sigma_1 + \sigma_2 = p_1 + p_2$$

$$\therefore \qquad \sigma_2 = \left(p_1 + p_2\right) - \left[\frac{p_1 + p_2}{2} + \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2}\right]$$

From which $\sigma_2 = \frac{p_1 + p_2}{2} - \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2}$ ...(4.22 *b*)

Hence $\sigma_{1,2} = \frac{1}{2}\left(p_1 + p_2\right) \pm \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2}$ ...(4.22)

Note : An important characteristic concerning the principal planes can be observed from Eq. 4.16 for shear stress :

$$p_t = -\frac{p_1 - p_2}{2}\sin 2\theta + q\cos 2\theta$$

Setting the above to zero and solving for θ, we get

$$\tan 2\theta = \frac{2q}{p_1 - p_2}$$

Comparing this with Eq 4.21, we observe that *shear stresses are zero on the principal planes*.

Orientation of principal planes

The orientation of principal planes for the cases of uniaxial, and biaxial stress systems are shown in Fig. 4.31. The principal planes for such stresses are the x and y planes them selves because than $2\theta_p = 0$ and hence the

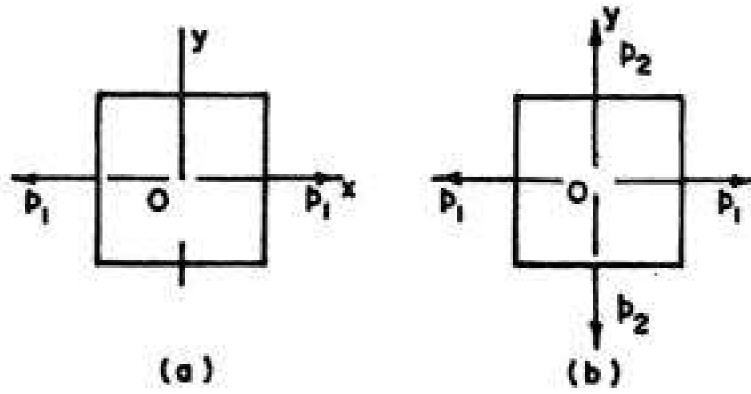


FIG. 4.31. ELEMENTS IN UNIAXIAL AND BIAXIAL STRESS

two values of θ_p are 0° and 90°, Here, $\sigma_1 = p_1$ and $\sigma_2 = p_2$ (if p_2 exists).

For an element in pure shear (Fig. 4.32 *a*), $\tan 2\theta_p \left(= \frac{2q}{p_1 - p_2} \right)$ is *infinite* and hence the two values of θ_p are 45° and 135° (Fig 4.32 *b*). For positive values of q (Fig 4.32 *a*), we get (from Eq 4.22), $\sigma_1 = q$ and $\sigma_2 = -q$.

For a general case of two dimensional stress system (Fig 4.33 *a,c*) or the plane stress case, the orientation of principal plane are shown in Fig 4.33 (*b*) and 4.33 (*c*).

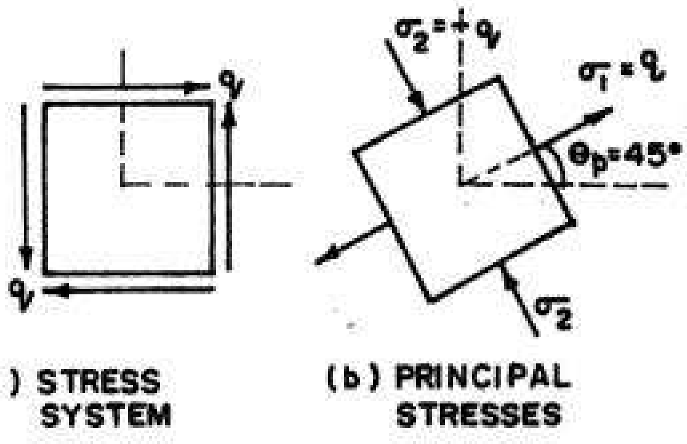


FIG. 4.32. ELEMENT IN PURE SHEAR

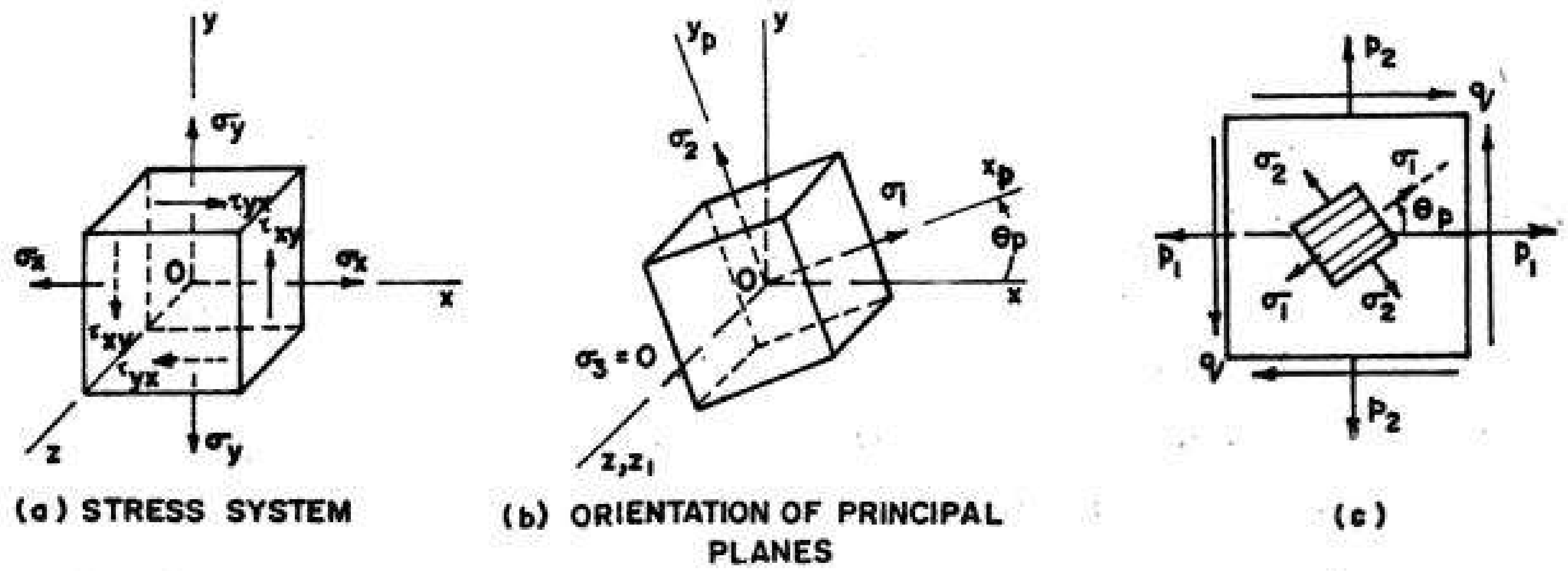


FIG. 4.33. PRINCIPAL PLANES FOR AN ELEMENT IN PLANE STRESS

Mohr's stress circle method

Principal stresses and principal planes can also be found by Mohr's circle method, illustrated in Fig. 4.34(*b*). To locate the principal plane, mark *OA* and *OB* equal to p_1 and p_2 respectively to some scale. At *A*, perpendicular $AG=q$ in downward direction (since q is positive). Similarly, erect perpendicular $BF = q$ in upward direction. Joint *F* and *G* intersecting the axis in *C*. Since, by definition, principal plane is the one at which shear stress is zero, *CH* represents major principal plane (point *D* coinciding with *H*) and angle $2\theta_p = \beta$. Similarly, *CJ* represents the minor principal plane (point *D'* coinciding with *J*), and angle $2\theta_p = 180° + \beta$.

Now $\quad OH = OC + CH = \frac{1}{2}\left(p_1 + p_2\right) + R$, where R=radius of circle

$$= \frac{1}{2}\left(p_1 + p_2\right) + \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2} = \sigma_1$$

Similarly, $\quad OJ = OC - CJ = \frac{1}{2}(p_1 + p_2) - R$, where R= radius of circle

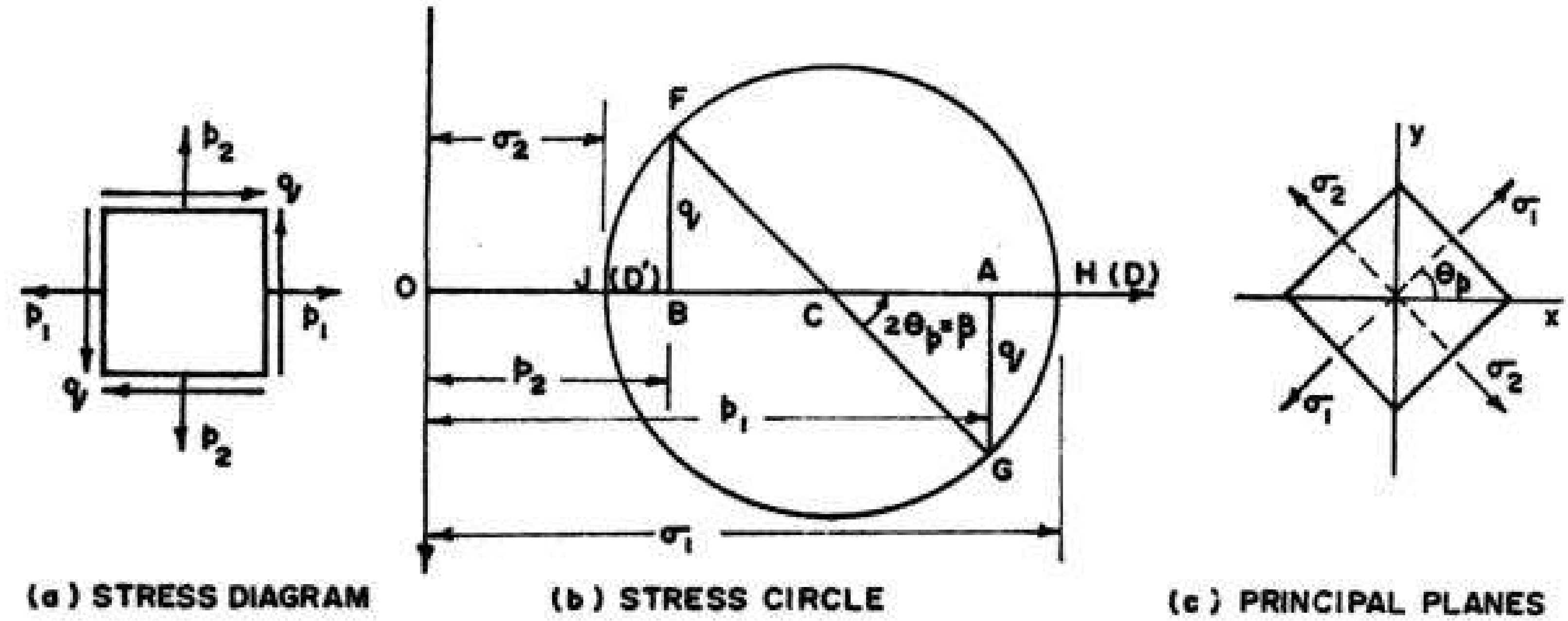


FIG. 4.34. MOHR'S CIRCLE METHOD

$$= \frac{1}{2}(p_1 + p_2) - \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2} = \sigma_2$$

Also, $$\tan\beta = \frac{AG}{AC} = \frac{q}{\frac{p_1 - p_2}{2}} = \frac{2q}{p_1 - p_2} = \tan 2\theta_p$$

where $$2\theta_p = \beta$$

The stress element indicating principal planes is shown in Fig. 4.34 (*c*).

4.10. MAXIMUM SHEAR STRESSES

In the previous article, we have found the maximum normal stresses acting on an element in plane stress and have seen that the shearing stresses are zero on the planes (called principal planes) on which these maximum normal stresses (called principal stresses) act. Let us now determine the maximum shear stresses and the planes on which they act.

In general, the shear stress varies with angle θ, and its magnitude is given by :

$$p_t = -\left(\frac{p_1 - p_2}{2}\right)\sin 2\theta + q\cos 2\theta \qquad ...(4.16)$$

For maximum value of p_t, differentiate it with respect to θ and equate it to zero.

$$\therefore \qquad \frac{dp_t}{d\theta} = -\left(p_1 - p_2\right)\cos 2\theta - 2q\sin 2\theta = 0$$

from which $$\tan 2\theta = \tan 2\theta_s = -\frac{p_1 - p_2}{2q} \qquad ...(4.23)$$

Eq. 4.23 gives one value of θ_s between 0° and 90° and another value of θ_s between 90° and 180°, both values differing by 90°. This means that the maximum and minimum values of p_t occur on perpendicular planes. However, since shear stresses on perpendicular planes are equal in absolute value, the *maximum and minimum shear stress differ only in sign.*

Also, by comparing Eqs. 4.23 with Eq. 4.21, we observe that

$$\tan 2\theta_s = -\frac{1}{\tan 2\theta_p} = -\cot 2\theta_p \qquad ...(4.24)$$

Since $\tan(\alpha \pm 90°) = -\cot\alpha$, we have :

$$2\theta_s = 2\theta_p \pm 90° \text{ or } \theta_s = \theta_p \pm 45° \qquad ...(4.25)$$

Hence *planes of maximum shear stress occur at 45° to the principal planes.*

Eq. 4.23 is diagrammatically represented by Fig. 4.35, where

$$\tan 2\theta_s = \frac{-\frac{p_1 - p_2}{2}}{q} = -\frac{p_1 - p_2}{2q}$$

$$\therefore \quad \cos 2\theta_{s1} = \frac{q}{R} \text{ and } \sin 2\theta_{s1} = -\frac{p_1 - p_2}{2R} \qquad ...(4.26)$$

where

$$R = \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2}$$

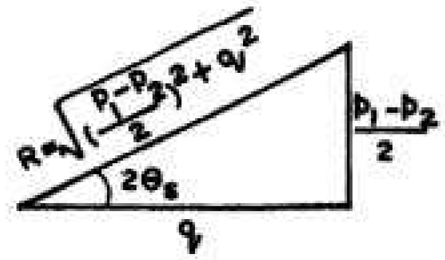


FIG. 4.35

Also, θ_{s1} is related to θ_{p1} by the relation

$$\theta_{s1} = \theta_{p1} - 45° \qquad ...(4.27)$$

Substituting the values of $\cos 2\theta_{s1}$ and $\sin 2\theta_{s1}$ in Eq. 4.16, we get

$$p_{t,\max} = -\left(\frac{p_1 - p_2}{2}\right) \times \left(-\frac{p_1 - p_2}{2R}\right) + q \cdot \frac{q}{R}$$

or

$$p_{t,\max} = \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2} \qquad ...(4.28)$$

Note that, algebraically, minimum shear stress $p_{t,\min}$ has the same magnitude but has opposite sign. Again, subtracting σ_2 from σ_1 (Eq. 4.22 *b* and 4.22 *a*), we get

$$\sigma_1 - \sigma_2 = 2\sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2} \qquad ...(4.22\ c)$$

Comparing this with Eq. 4.28, we get

$$p_{t,\max} = \frac{\sigma_1 - \sigma_2}{2} \qquad ...(4.29)$$

Hence the maximum shear stress is equal to one-half the difference of principal stresses.

It should be very clearly noted that *normal stresses also act on the planes of maximum shear stress*. The values of these normal stresses (p_{ns}, say) can be found by substituting the value of θ_{s1} in Eq. 4.15. Thus, we have

$$p_{ns} = \frac{p_1 + p_2}{2} + \frac{p_1 - p_2}{2} \times \frac{q}{R} + q\left(-\frac{p_1 - p_2}{2R}\right)$$

or

$$p_{ns} = \frac{p_1 + p_2}{2} = p_{av} \qquad ...(4.30)$$

Thus stress p_{av} acts on the plane of maximum shear stress and the plane of minimum shear stress.

Mohr's circle method : Mohr's stress circle method can also be used for locating the planes of maximum shear stress. Fig. 4.36 (*b*), shows Mohr's stress circle, for the stress system of Fig. 4.36 (*a*). Maximum shear stress corresponds to point D, when $2\theta_s = \beta + 90°$.

Evidently,

$$p_{t,\,max} = R = \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2}$$

$$= \frac{\sigma_1 - \sigma_2}{2} \text{ (numerically).}$$

The corresponding normal stress is evidently equal to OC.

$$\therefore\ p_{ns} = OC = \frac{1}{2}\left(p_1 + p_2\right)$$

Similarly, when $2\theta_s = \beta + 270°$ (corresponding to point D'),

$$p_{t,\,max} = R$$

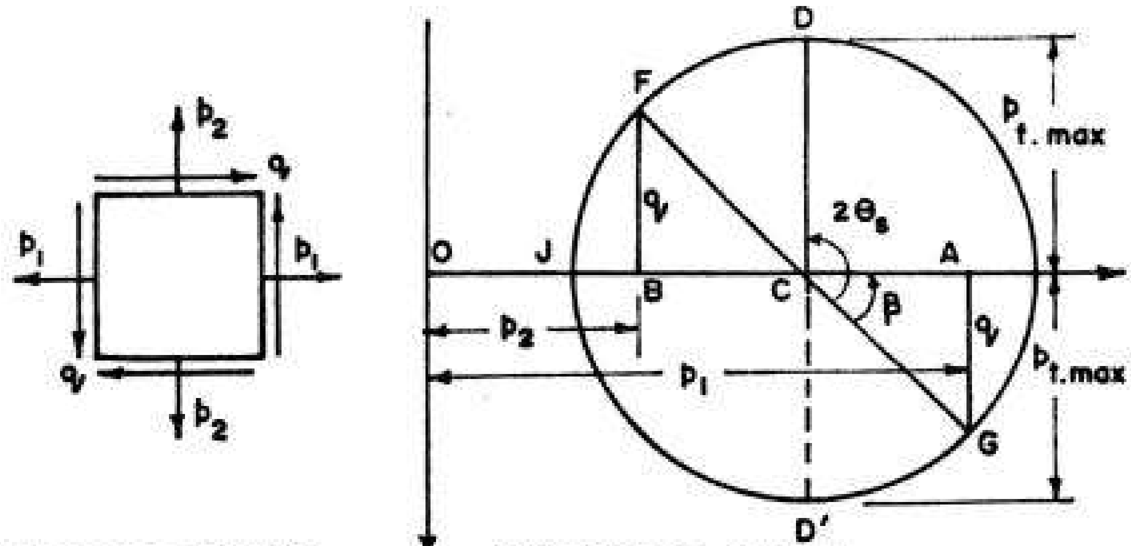
$$= \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2}$$

$$= \frac{\sigma_1 - \sigma_2}{2} \qquad \text{...(4.28, 4.29)}$$

Also, $2\theta_s = \beta + 90°$

$$\therefore\ \tan 2\theta_s = \tan\left(\beta + 90°\right)$$

$$= -\cot\beta = -\frac{p_1 - p_2}{2q}$$



(a) STRESS SYSTEM (b) STRESS CIRCLE

FIG. 4.36

Example 4.15. *At a point in an elastic material under strain, there are normal stresses of 60 N/mm² and 40 N/mm² (both tensile) respectively at right angles to each other, with positive shearing stress of 20 N/mm². Find (i) principal stresses and the position of principal planes, and (ii) maximum shear stress and its plane.*

(a) Analytical Solution

$$\sigma = \frac{p_1 + p_2}{2} \pm \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2} = \frac{60 + 40}{2} \pm \sqrt{\left(\frac{60 - 40}{2}\right)^2 + (20)^2}$$

$$= 50 \pm 22.36$$

$$\therefore \qquad \sigma_1 = \mathbf{72.36\ N/mm^2} \text{ (tensile) and } \sigma_2 = \mathbf{27.64\ N/mm^2} \text{ (tensile)}$$

$$\tan 2\theta_p = \frac{2q}{p_1 - p_2} = \frac{2 \times 20}{60 - 40} = 2 \text{ or } 2\theta_p = 63.43 = 63°26'$$

$$\therefore \qquad \theta_{p1} = \mathbf{31°43'} \text{ and } \theta_{p2} = \mathbf{121°43'}$$

$$p_{t.\,max} = \frac{\sigma_1 - \sigma_2}{2} = \frac{72.36 - 27.64}{2} = \mathbf{22.36\ N/mm^2}$$

$$\theta_{s1} = 45° + \theta_{p1} = 45° + 31°\,31' = \mathbf{76°\,43'}$$

(b) Graphical Solution

Fig. 4.37 shows the graphical solution in the form of Mohr's circle. The method is self-explanatory. From the Mohr's stress circle, we obtain the following results :

$\sigma_1 = OH \approx 72.5\ \text{N/mm}^2$ (tension)

$\sigma_2 = OJ \approx 28\ \text{N/mm}^2$ (tension)

$\theta_{p1} \approx 31°\ 30'$; $\theta_{p2} \approx 121°30'$

$p_{t\,.\,\max} = CD' = 22.3\ \text{N/mm}^2$;

$\theta_{s1} = 76°\ 30'$

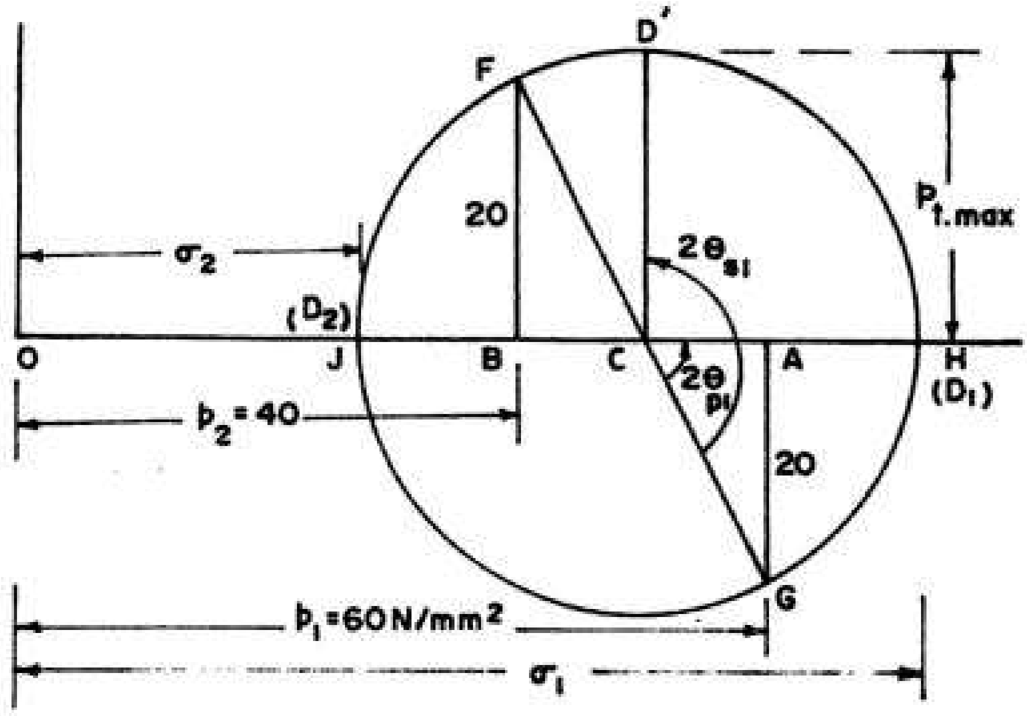


FIG. 4.37

Example 4.16. *Solve example 4.15 if the 60 N/mm² stress is tensile while the 40 N/mm² stress is compressive.*

(a) Analytical solution : Here $p_1 = +60$ and $p_2 = -40\ \text{N/mm}^2$.

$$\sigma = \frac{p_1 + p_2}{2} \pm \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2} = \frac{60 - 40}{2} \pm \sqrt{\left(\frac{60 + 40}{2}\right)^2 + (20)^2} = 10 \pm 53.85$$

$\therefore \quad \sigma_1 = \mathbf{63.85\ N/mm^2}$ (tensile)

and $\sigma_2 = -\mathbf{43.85\ N/mm^2}$ (i.e. comp.)

$$\tan 2\theta_p = \frac{2q}{p_1 - p_2} = \frac{2 \times 20}{60 + 40} = 0.4$$

$\therefore\ 2\theta_p = 21.8°$ or $\theta_{p1} = \mathbf{10°\ 34'}$
and $\theta_{p2} = \mathbf{100°\ 34'}$

$$p_{t\,,\,\max} = \frac{\sigma_1 - \sigma_2}{2} = \frac{63.85 - (-43.85)}{2}$$

$= \mathbf{53.85\ N/mm^2}$

$\theta_{s1} = 45° + \theta_{p1} = 45° + 10°34' = \mathbf{55°34'}$

Graphical Solution : Graphical solution is shown in Fig 4.38. From Mohr's circle, we obtain the following results;

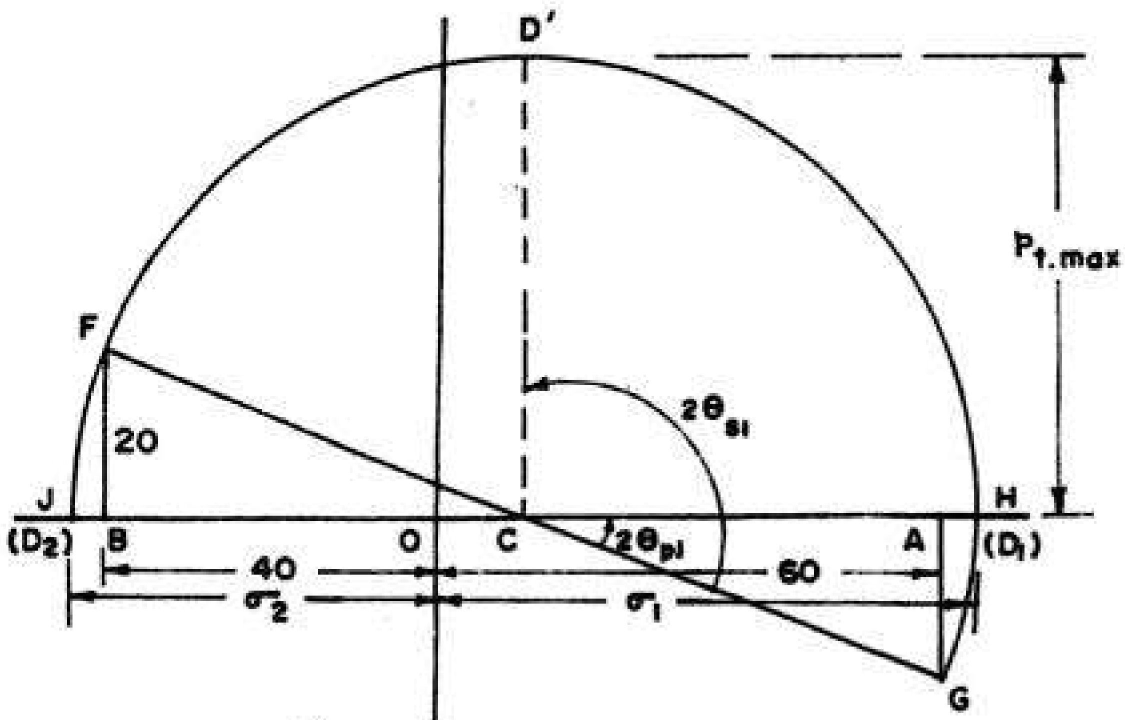


FIG. 4.38

$\sigma_1 \approx 64\ \text{N/mm}^2$ (tension) ; $\sigma_2 \approx 44\ \text{N/mm}^2$ (comp.)

$\theta_{p1} = 10°30'$; $\theta_{p2} = 100°30'$; $p_{t\,,\,\max} \approx 54\ \text{N/mm}^2$; $\theta_{s1} = 55°30'$

Example 4.17. *At a certain point in a piece of elastic material, there are normal stresses of 40 N/mm² (tension) and 30 N/mm² (compression) on two planes at right angles to one another,* *together with shearing stress of 20 N/mm² on the same planes. If the loading on the material is increased so that stresses reach values of k times those given, find the maximum value of k if the maximum direct stress in the material is not to exceed 100 N/mm², and maximum shear stress is not to exceed 70 N/mm².*

Solution :

$$\sigma = \frac{p_1 + p_2}{2} \pm \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2} = \frac{40 - 30}{2} \pm \sqrt{\left(\frac{40 + 30}{2}\right)^2 + (20)^2} = 5 \pm 40.31$$

Hence $\sigma_1 = 45.31\ \text{N/mm}^2$ (tension) and $\sigma_2 = 35.31\ \text{N/mm}^2$ (comp.)

$$p_{t,\max} = \frac{\sigma_1 - \sigma_2}{2} = \frac{45.31 - (-35.31)}{2} = 37.81\ \text{N/mm}^2$$

Now if the stresses use increased by k times, we have

(*a*) Maximum principal stress (or direct stress) becomes 45.31 k, which is not to exceed 100 N/mm².

$\therefore$ $45.31\,k = 100$

From which $k = 100/45.31 = 2.207$

(*b*) Maximum shear stress becomes 37.81 k, which is not to exceed 70 N/mm².

$\therefore$ $37.81\,k = 70$

From which $k = 70/37.81 = 1.851$

Hence permissible value of k = **1.851**

Example 4.18. *At a point in a material under stress, the intensity of resultant stress on a certain plane is 60 N/mm² (tensile) inclined at 30° to the normal of that plane. The stress on a plane at right angles to this has a normal tensile component of intensity 40 N/mm². Find fully (i) the resultant stress on the second plane (ii) the principal planes and stresses, and (iii) the plane of maximum shear and its location (Based on U.L.).*

Analytical Solution : The problem has been illustrated in Fig. 4.39 (*a*).

On the first plane AB, the tangential stress is

$$q = 60 \sin 30° = 30\ \text{N/mm}^2$$

Hence on the second plane CA, the complimentary shear stress is 30 N/mm².

$\therefore$ Resultant stress on the second plane is

$$p_r = \sqrt{(40)^2 + (30)^2}$$
$$= \mathbf{50\ N/mm^2}$$

The intensity of stress normal to the first plane (AB) is

$$p_n = 60 \cos 30° \approx 52\ \text{N/mm}^2$$

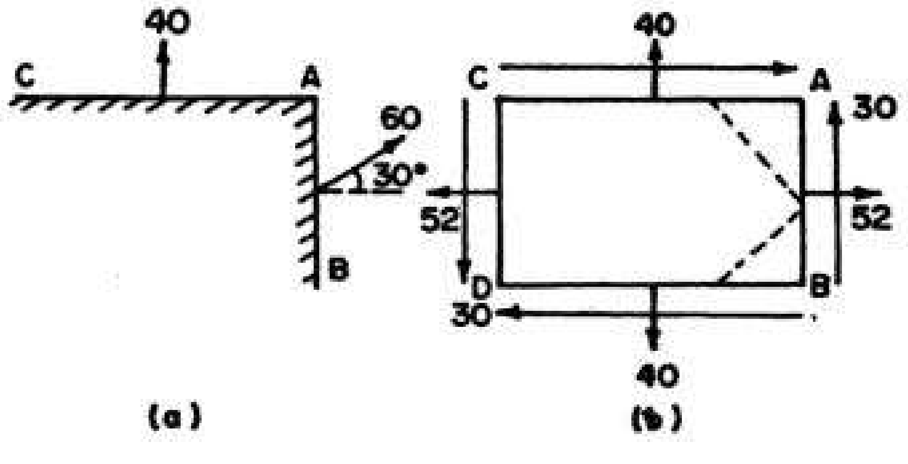


FIG. 4.39

The final stress system has been represented in Fig. 4.39 (*b*) from which the principal stresses are

$$\sigma = \frac{p_1 + p_2}{2} \pm \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2}$$

$$= \frac{52 + 40}{2} \pm \sqrt{\left(\frac{52 - 40}{2}\right)^2 + (30)^2} = 46 \pm 30.6$$

$\therefore$ $\sigma_1 = \mathbf{76.6\ N/mm^2}$ (tension) and $\sigma_2 = \mathbf{15.4\ N/mm^2}$ (tension)

$$\tan 2\theta_p = \frac{2q}{p_1 - p_2} = \frac{2 \times 30}{52 - 40} = \frac{60}{12} = 5, \text{ from which } 2\theta_p = 78°42'$$

$\therefore$ $\theta_{p1} = \mathbf{39°21'}$ and $\theta_{p2} = \mathbf{129°21'}$

$$p_{t,\max} = \frac{\sigma_1 - \sigma_2}{2} = \frac{76.6 - 15.4}{2} = \mathbf{30.6\ N/mm^2}$$

$$\theta_{s1} = 45° + \theta_{p1} = 45° + 39°21' = \mathbf{84°\ 21'}$$

Graphical Solution (Refer Fig. 4.40)

On the plane AB, p_n is positive (being tensile) and p_t is positive being in the anticlockwise direction; the resultant stress is 60 N/mm². Hence draw OD at an inclination of 30° to OX, in the clockwise direction and make OD = 60 N/m m². Draw DE perpendicular to OX. Since stress of CA is tensile (or is positive), set off OE' = 40 N/mm² to the right of O. Bisect EE' to get the centre C. With C as centre and CD as radius, draw the Mohr's circle. The circle will cut the x-axis in H and J. From

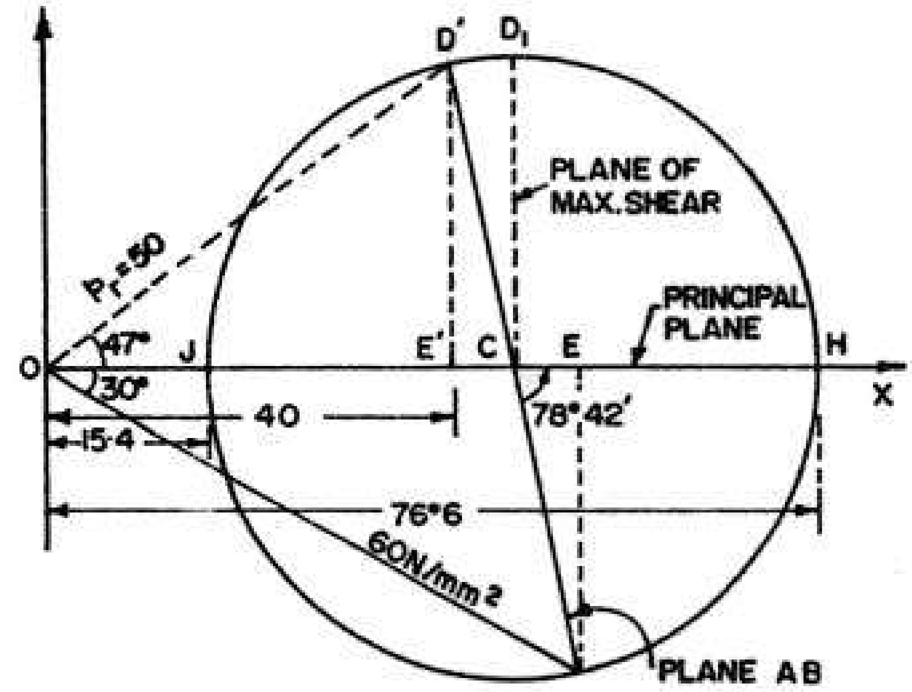


FIG. 4.40

the circle, we have $OH = \sigma_1$ = 76.5 N/mm²; $OJ = \sigma_2$ = 15.4 N/mm²; $\angle HCD = 2\theta_p = 78°42'$ or $\theta_p = 39°21'$. Thus, normal to the principal plane is inclined at 39°21', in the anticlockwise direction, to the normal to AB. In other words, the principal plane is inclined at 39°21' to AB, in the anticlockwise direction.

To find the resultant stress on the second plane, prolong DC to cut the circle in D'. Join O and D'. Thus OD' = 50 N/mm² is the resultant on the second plane inclined at 47°.

To locate the plane of maximum shear stress, draw CD_1 perpendicular to the axis OC. Thus $p_{t,\max} = CD_1$ = 30.6 N/mm². The angle of inclination of the plane of maximum shear stress $= \frac{1}{2} DCD_1 \approx 84°15'$, anticlockwise with line CD (i.e. with plane AB).

Example 4.19. *Fig. 4.41 shows the normal and tangential stresses on two planes. Determine the principal stresses. Also determine shear stress along CB.*

Solution :

Let the plane AB make an angle θ, in the anticlockwise direction, with the principal plane. Hence the angle made by plane BC with the principal plane will be $\theta + 40°$.

Let σ_1 and σ_2 be the principal stresses.

The normal and tangential stresses on plane AB, due to stresses σ_1 and σ_2 on the principal planes are given by

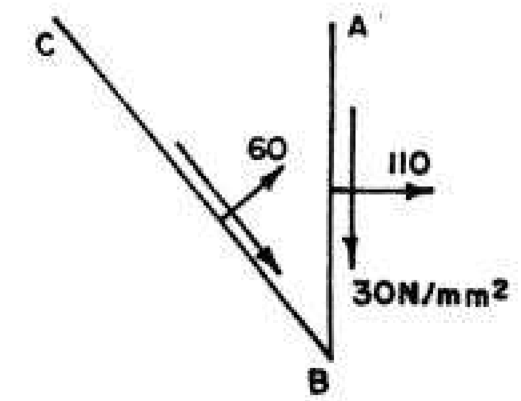


FIG. 4.41

$$(p_n)_{AB} = 110 = \frac{\sigma_1 + \sigma_2}{2} + \frac{\sigma_1 - \sigma_2}{2} \cos 2\theta \qquad ...(4.15)$$

and $$(p_t)_{AB} = -30 = -\frac{\sigma_1 - \sigma_2}{2} \sin 2\theta \qquad ...(4.16)..(1)$$

or $$30 = \frac{\sigma_1 - \sigma_2}{2} \sin 2\theta \qquad ...(2)$$

Similarly, the normal stress on plane BC, due to stresses σ_1 and σ_2 on the principal planes is given by

$$(p_n)_{BC} = 60 = \frac{\sigma_1 + \sigma_2}{2} + \frac{\sigma_1 - \sigma_2}{2} \cos\left(2\theta + 80°\right) \qquad ...(4.15)...(3)$$

Subtracting Eq. (3) from Eq. (1),

$$\frac{\sigma_1 - \sigma_2}{2}\left[\cos 2\theta - \cos\left(2\theta + 80°\right)\right] = 50 \qquad ...(4)$$

Dividing Eq. (4) by Eq. (2),

$$\frac{\cos 2\theta - \cos(2\theta + 80°)}{\sin 2\theta} = \frac{50}{30} = 1.6667$$

or $\cos 2\theta - \cos 2\theta \cos 80° + \sin 2\theta \sin 80° = 1.6667 \sin 2\theta$

or $\cos 2\theta - 0.1736 \cos 2\theta = 1.6667 \sin 2\theta - 0.9848 \sin 2\theta$

or $0.6819 \sin 2\theta = 0.8264 \cos 2\theta$

or $\tan 2\theta = 1.2119$

From which $2\theta = 50.48°$

or $\theta = 25.24° = 25°14'$

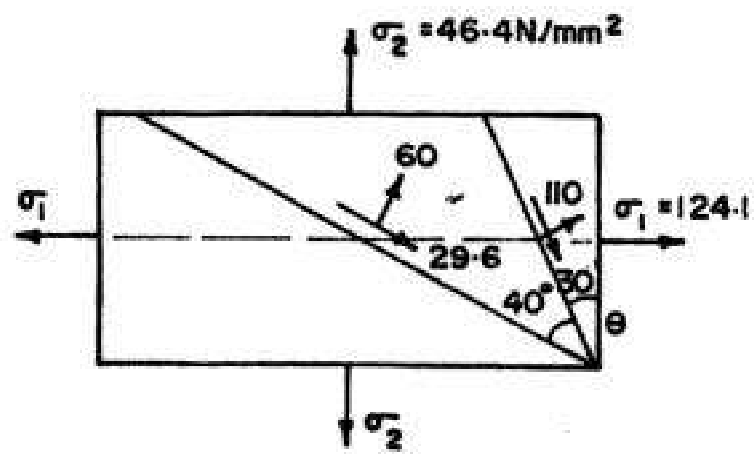


FIG 4.42

Now from (2) $$\frac{\sigma_1 - \sigma_2}{2} = \frac{30}{\sin 2\theta} = 38.89$$

or $$\sigma_1 - \sigma_2 = 77.79 \qquad ...(i)$$

From (i), $$\frac{\sigma_1 + \sigma_2}{2} + \frac{\sigma_1 - \sigma_2}{2} \cos 2\theta = 110$$

or $$\frac{\sigma_1 + \sigma_2}{2} + \frac{77.79}{2} \cos 50.48° = 110$$

From which $$\sigma_1 + \sigma_2 \approx 170.5 \qquad ...(ii)$$

From (*i*) and (*ii*) we get $\sigma_1 = \mathbf{124.1\ N/mm^2}$ and $\sigma_2 = \mathbf{46.4\ N/mm^2}$

Also, $$(p_t)_{BC} = -\frac{\sigma_1 - \sigma_2}{2}\sin\left(2\theta + 80\right)^\circ = -\frac{77.79}{2}\sin\left(50.48^\circ + 80^\circ\right)$$
$$\approx -29.6\ \text{N/mm}^2 \quad (\textit{i.e}\ 29.6 \text{ in the direction marked in Fig 4.41})$$

The position of various planes are marked in Fig. 4.42.

Example 4.20. *At a point in a material subjected to two dimensional stresses, the stresses on a certain plane are 75 N/mm² tension and 50 N/mm² shear and on another plane the stresses are 45 N/mm² tension and 40 N/mm² shear as shown in Fig. 4.43. Find the principal stresses and their directions relative to the given planes.*

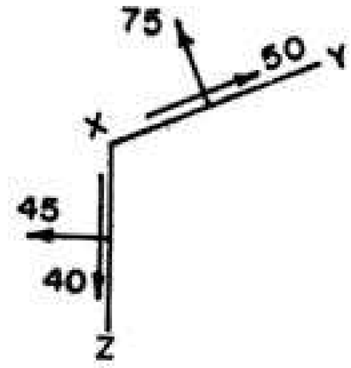


FIG. 4.43

Solution :

For Plane AB

$p_n = +75$ (being tensile) and $p_t = -50$ (being clockwise)

Hence make $OE = +75\ \text{N/mm}^2$ and $ED = -50\ \text{N/mm}^2$ (*i.e.* upwards)

For plane AC : $p_n = +45$ (being tensile) and $p_t = +40$ (being anticlockwise)

Hence make $OE' = +45\ \text{N/mm}^2$ and $E'D' = +40\ \text{N/mm}^2$ (i.e downwards), as shown in Fig. 4.44 (*a*).

The Mohr's circle should pass through D and D'. Hence join D and D' and draw its perpendicular bisector to cut the x-axis is C, giving the centre of the circle. In this special case, C coincides with E. Join E and D'. Angle $DCH = 90°$. The plane AB is therefore inclined

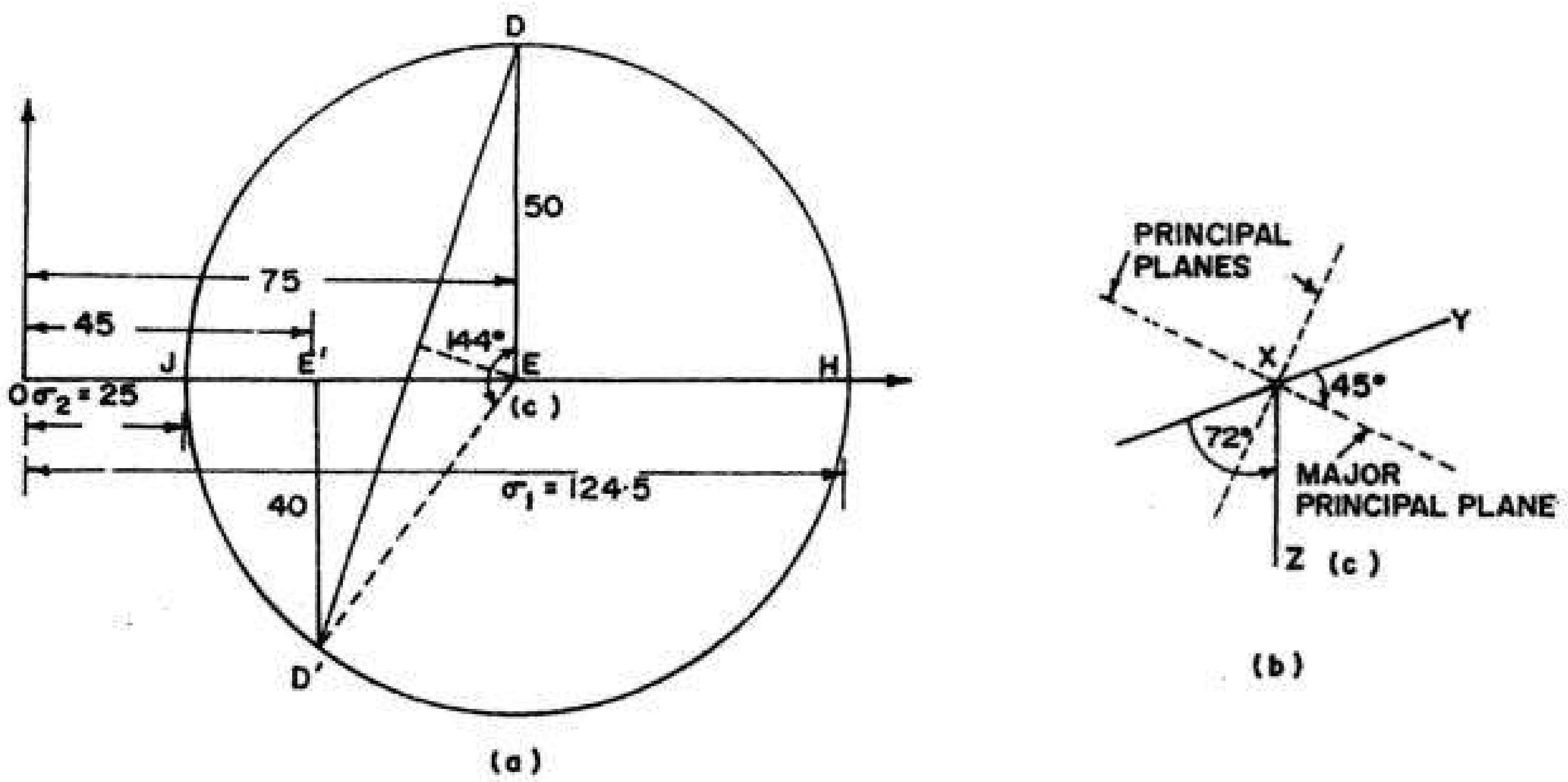


FIG. 4.44

at 45° to the principal planes in anticlockwise direction. Thus the principal planes are inclined at 45° to the AB plane in clockwise direction. The position of principal planes have been marked in Fig 4.44 (b). From Mohr's circle, we get

$$\sigma_1 = OH = +\mathbf{124.5\ N/mm^2} \text{ and } \sigma_2 = OJ = +\mathbf{25\ N/mm^2}.$$

Also, $\angle DED' = 144°$. Hence angle between AB and $AC = 72°$ in the anticlockwise direction, as marked in Fig. 4.44 (b).

Example 4.21. *At a point in an elastic material a direct tensile stress of 70 N/mm² and a direct compressive stress of 50 N/mm² are applied on planes at right angles to each other. If the maximum principal stress in the material is limited to 75 N/mm², find out the shear stress that may be allowed on the planes. Also, determine the magnitude and direction of the minimum principal stress and the maximum shear stress.*

Solution : Refer Fig 4.29.

Given $p_1 = +70\ \text{N/mm}^2$; $p_2 = -50\ \text{N/mm}^2$ and $\sigma_1 = 75\ \text{N/mm}^2$

Now
$$\sigma_1 = \frac{p_1 + p_2}{2} + \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2}$$

or
$$75 = \frac{70 - 50}{2} + \sqrt{\left(\frac{70 + 50}{2}\right)^2 + q^2}$$

From which $q = \mathbf{25\ N/mm^2}$

Also,
$$\sigma_2 = \frac{p_1 + p_2}{2} - \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2} = \frac{70 - 50}{2} - \sqrt{\left(\frac{70 + 50}{2}\right)^2 + (25)^2}$$
$$= 10 - 65 = -\mathbf{55\ N/mm^2} \text{ (i.e. compressive).}$$

$$p_{t,\max} = \frac{\sigma_1 - \sigma_2}{2} = \frac{75 - (-55)}{2} = \mathbf{65\ N/mm^2}$$

$$\tan 2\theta_p = \frac{2q}{p_1 - p_2} = \frac{2 \times 25}{70 - (-50)} = 0.4167$$

From which $2\theta_p = 22.62°$ or $\theta_{p1} \approx \mathbf{11°37'}$

Thus, the major principal plane is oriented at $11°37'$ in the anticlockwise direction with the plane on which p is acting : The minor principal plane is inclined at $\theta_{p2} = 90° + 11°37' = \mathbf{101°37'}$ with the plane of p_1, in the anticlockwise direction.

Also, $\theta_{s1} = \theta_{p1} + 45° = 11°37' + 45° = \mathbf{56°37'}$

Thus the plane of maximum shear is inclined at $56°37'$ with that of p_1, in the anticlockwise direction.

Example 4.27. *A piece of material is subjected to three perpendicular tensile stresses. The strains in three directions are in the ratio of 3:4:5. If Poisson's ratio is 0.286, find the ratio of stresses and their values if the greatest is 60 N/mm².*

Solution : Let the stresses be σ_1, σ_2 and σ_3 and the corresponding strains be 3ε, 4ε and 5ε.

$\therefore$
$$3\varepsilon E = \sigma_1 - 0.286\left(\sigma_2 + \sigma_3\right) \quad ...(i)$$

$$4\,\varepsilon E = \sigma_2 - 0.286\,(\sigma_3 + \sigma_1) \qquad ...(ii)$$
$$5\,\varepsilon E = \sigma_3 - 0.286\,(\sigma_1 + \sigma_2) \qquad ...(iii)$$

$\therefore$ Subtracting (*i*) from (*iii*), we get $2\,\varepsilon E = (\sigma_3 - \sigma_1) - 0.286\,(\sigma_1 - \sigma_3)$

from which $\sigma_3 - \sigma_1 = \dfrac{2\,\varepsilon E}{1.286}$ ...(*iv*)

or $1.286\,\sigma_3 - 1.286\,\sigma_1 = 2\,\varepsilon E$ ...(*a*)

Again, from (*iii*), $\dfrac{\sigma_3}{0.286} - \sigma_1 - \sigma_2 = \dfrac{5\,\varepsilon E}{0.286}$ ...(*v*)

and from (*ii*), $\sigma_2 - 0.286\,\sigma_3 - 0.286\,\sigma_1 = 4\,\varepsilon E$ ...(*vi*)

Adding (*v*) and (*vi*), $3.21\,\sigma_3 - 1.286\,\sigma_1 = 21.5\,\varepsilon E$

Subtracting (*a*) and (*b*), we get $1.924\,\sigma_3 = 19.5\,\varepsilon E$

$\therefore \quad \sigma_3 = 10.14\,\varepsilon E$

Hence $\sigma_1 = 8.58\,\varepsilon E$ (from *iv*) and $\sigma_2 = 9.35\,\varepsilon E$ (from *ii*)

$\therefore \quad \sigma_1 : \sigma_2 : \sigma_3 = 8.58 : 9.35 : 10.14$

$\therefore \quad \sigma_1 : \sigma_2 : \sigma_3 = 0.847 : 0.923 : 1$

Taking $\sigma_3 = 60\text{ N/mm}^2$, we get

$\sigma_1 = 0.847 \times 60 = \mathbf{50.8\ N/mm^2}$ and $\sigma_2 = 0.923 \times 60 = \mathbf{55.4\ N/mm^2}$

4.11. ADDITIONAL ILLUSTRATIVE EXAMPLES

Example 4.22 *At a point in a piece of elastic material, there are three mutually perpendicular planes on which the stresses are as follows: tensile stress 50 N/mm² and shear stress 40 N/mm² on one plane, compressive stress 35 N/mm² and complimentary shear stress 40 N/mm² on the second plane, and no stress on the third plane. Find (a) the principal stresses and the positions of the planes on which they act (b) the position of planes on which there is no normal stress.*

Solution :

(a) Graphical solution (Fig 4.45 *a*)

(*i*) Mark off $OA = 50\text{ N/mm}^2$

$AG = -40\text{ N/mm}^2$ (*i.e.* upwards) $OB = -35\text{ N/mm}^2$ and $BF = +40\text{ N/mm}^2$ (*i.e* . downwards)

(*ii*) Join *F* and *G*, cutting *AB* in *C*. Draw the circle with *C* as the centre and *CG* (or *CF*) as radius, cutting the *x*-axis in *H* and *J*.

From Mohr circle, we get

$$\sigma_1 = OH = \mathbf{65.9\ N/mm^2}\ \text{(tensile)}$$

$$\sigma_2 = OJ = \mathbf{50.9\ N/mm^2}\ \text{(comp.)}$$

$$2\,\theta_p = 43^\circ \text{ or } \theta_{p1} = 21^\circ\,30'$$

The relative positions of the planes are shown in Fig. 4.45 (*a*).

If there is no normal stress on a plane, then for that plane *O* and *A* will coincide. The radius vector for that plane will be *CG'* in Fig. 4.45 (*a*) for which $2\theta' = 97^\circ$ or $\theta' = 48^\circ\,30'$

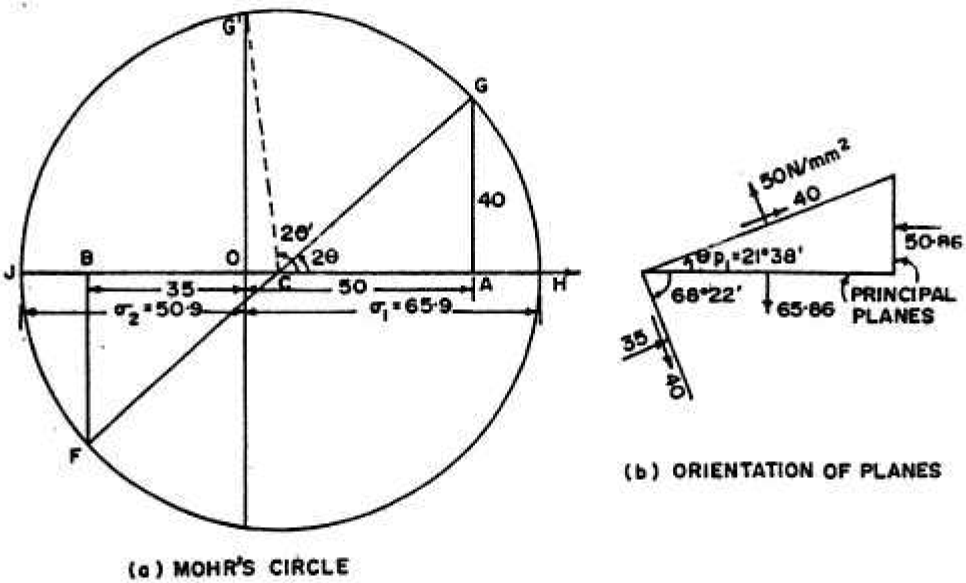


FIG. 4.45

(b) Analytical Solution

$$\sigma = \frac{50-35}{2} \pm \sqrt{\left(\frac{50+35}{2}\right)^2 + (40)^2} = 7.5 \pm 58.36$$

$\therefore$ $\sigma_1 = \mathbf{65.86\ N/mm^2}$ (tensile) and $\sigma_2 = -\mathbf{50.86\ N/mm^2}$ (*i.e.* compressive)

$$2\theta_p = \tan^{-1}\frac{2\times 40}{50+35} = 43°\,16'$$

$\therefore$ $$\theta_{p1} = \mathbf{21°\,38'}$$

If there is no normal stress on a plane, then for that plane θ and A will coincide and therefore the radius vector for that plane will be CG' (Fig. 4.45 *a*).

Then $$2\theta' = 180° - \cos^{-1}\frac{OC}{CG'} \text{ where } OC = OA - CA = 50 - \tfrac{1}{2}(50+35) = 7.5$$

and $CG' = R = CG = \sqrt{(40)^2 + (42.5)^2} = 58.36$

$\therefore$ $$2\theta' = 180° - \cos^{-1}\frac{7.5}{58.36} = 180° - 82°\,37' = 97°23'$$

$\therefore$ $\theta' = 48°\,41.5'$ to the principal plane.

Example 4.24. *An element in plane stress is subjected to stresses $p_1 = 120\ N/mm^2$, $p_2 = -40\ N/mm^2$ and $q = -45\ N/mm^2$, as shown in Fig 4.46 (a). Determine (a) the principal stresses and show them on a sketch of a properly oriented element. (b) the maximum shear stresses and show them on a properly oriented element. Consider only the in-plane stresses.*

Solution : Here $p_1 = +120\ N/mm^2$; $p_2 = -40\ N/mm^2$ and $q = -45\ N/mm^2$

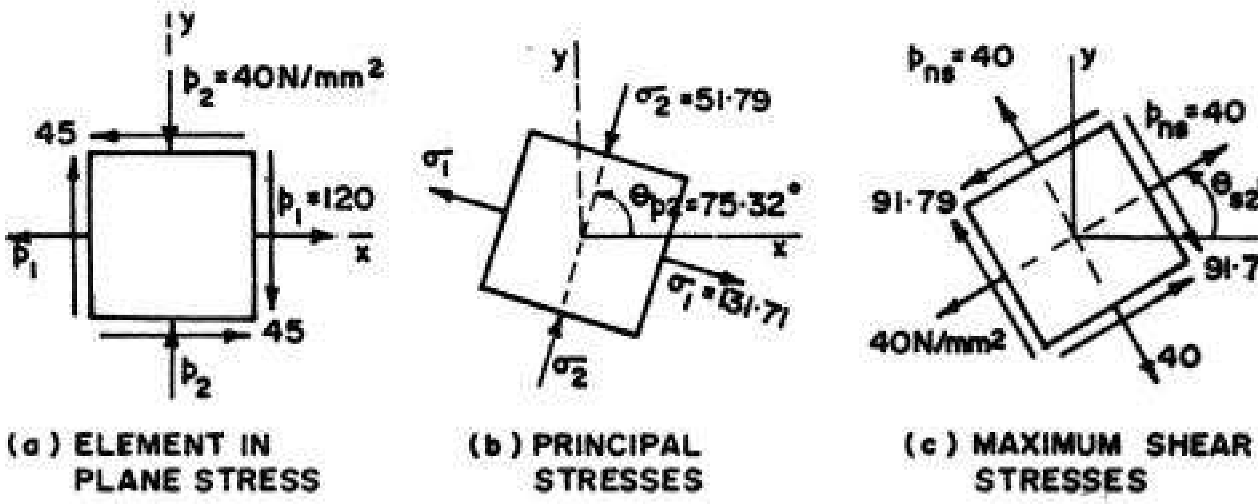


FIG. 4.46

(a) ***Principal Stresses***

$$\sigma = \frac{p_1 + p_2}{2} \pm \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2} = \frac{120 + (-40)}{2} \pm \sqrt{\left(\frac{120 + 40}{2}\right)^2 + (-45)^2}$$
$$= 40 \pm 91.79$$

From which $\sigma_1 = 131.79\ \text{N/mm}^2$ (tensile) and $\sigma_2 = -51.79\ \text{N/mm}^2$ (i.e. compressive)

$$\tan 2\theta_p = \frac{2q}{p_1 - p_2} = \frac{2 \times (-45)}{120 + 40} = -0.5625, \text{ from which } 2\theta_p = -29.36°$$

Since $2\theta_p$ varies in the range from 0° to 360°, we have

$$2\theta_p = 360° - 29.36° = 330.64° \text{ and } \theta_p = 165.32°$$

or
$$2\theta_p = 180° - 29.36° = 150.64° \text{ and } \theta_p = 75.32°$$

One of this angle is associated with the major principal stress while the other is associated with minor principal stress. Both these angles differ by 90°. However, in order to remove confusion, we study Eqs. 4.21 (*a,b*) and Fig 4.30, correlating principal angles and stresses. The only angle that satisfies both the equations is θ_{p1}. Hence we can rewrite these equations as follows :

$$\cos 2\theta_{p1} = \frac{p_1 - p_2}{2R} \text{ and } \sin 2\theta_{p1} = \frac{q}{R}$$

where $R = \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2} = 91.79$ (found earlier)

$$\therefore \cos 2\theta_{p1} = \frac{120 + 40}{2 \times 91.79} = 0.8716 \text{ from which } 2\theta_{p1} = 29.36° \quad (\text{or } 360° - 29.36° = 330.64°)$$

$$\text{and } \sin 2\theta_{p1} = -\frac{45}{91.79} = -0.4905, \text{ from which } 2\theta_{p1} = -29.30° \ (\text{or } 360° - 29.36°) = 330.64°$$

The only angle between 0° and 360°, satisfying both of these conditions is $2\theta_{p1} = 330.64°$. Hence $\theta_{p1} = \mathbf{165.32°}$

The other angle θ_{p2} is 90° larger or smaller than θ_{p1}

Hence $\theta_{p2} = 165.32° - 90° = \mathbf{75°32'}$

Fig. 4.46 (*b*) shows the rotated stress element showing the principal planes and principal stresses.

(b) Maximum shear stresses

$$q_{max} = \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2} = \sqrt{\left(\frac{120 + 40}{2}\right)^2 + (-45)^2} = 91.79$$

Also $\theta_{s1} = \theta_{p1} - 45° = 165.32° - 45° = 120.32°$

Negative shear stress acts as the plane for which $\theta_{s2} = 120.32° - 90°$
$= 30.32°$ (or $\theta_{s2} = \theta_{p2} - 45° = 75.32° - 45° = 30.32°$)

The normal stresses acting on the plane of maximum shear stresses are given by Eqs. 4.30 :

$$p_{ns} = p_{av} = \frac{p_1 + p_2}{2} = \frac{120 - 40}{2} = 40 \text{ N/mm}^2$$

Fig. 4.46(*c*) shows the rotated stress element showing the maximum shear stresses and the planes on which they act, along with the corresponding normal stresses.

Example 4.25. *An element in plane stress is subjected to stresses $p_1 = 150$ N/mm², $p_2 = 50$ N/mm² and $q = 40$ N/mm², as shown in Fig. 4.47 (a). Using Mohr's circle, determine (a) the stresses acting on an element rotated through an angle $\theta = 41°$ (b) the principal stresses, and (c) the maximum shear stresses. Show all results on sketches of properly oriented elements.*

Solution :

The Mohr's circle is shown in Fig. 4.47 (*b*), the construction of which is self explanatory.

From the stress circle of Fig. 4.47 (*b*), we get the following values :

$(p_n)_D = OE = +147$ N/mm²; $(p_t)_D = ED = -44$ N/mm²; $(p_n)_{D'} = OE' = 54$ N/mm²

$\sigma_1 = OH = +164$ N/mm²; $\sigma_2 = OJ = 36$ N/mm²

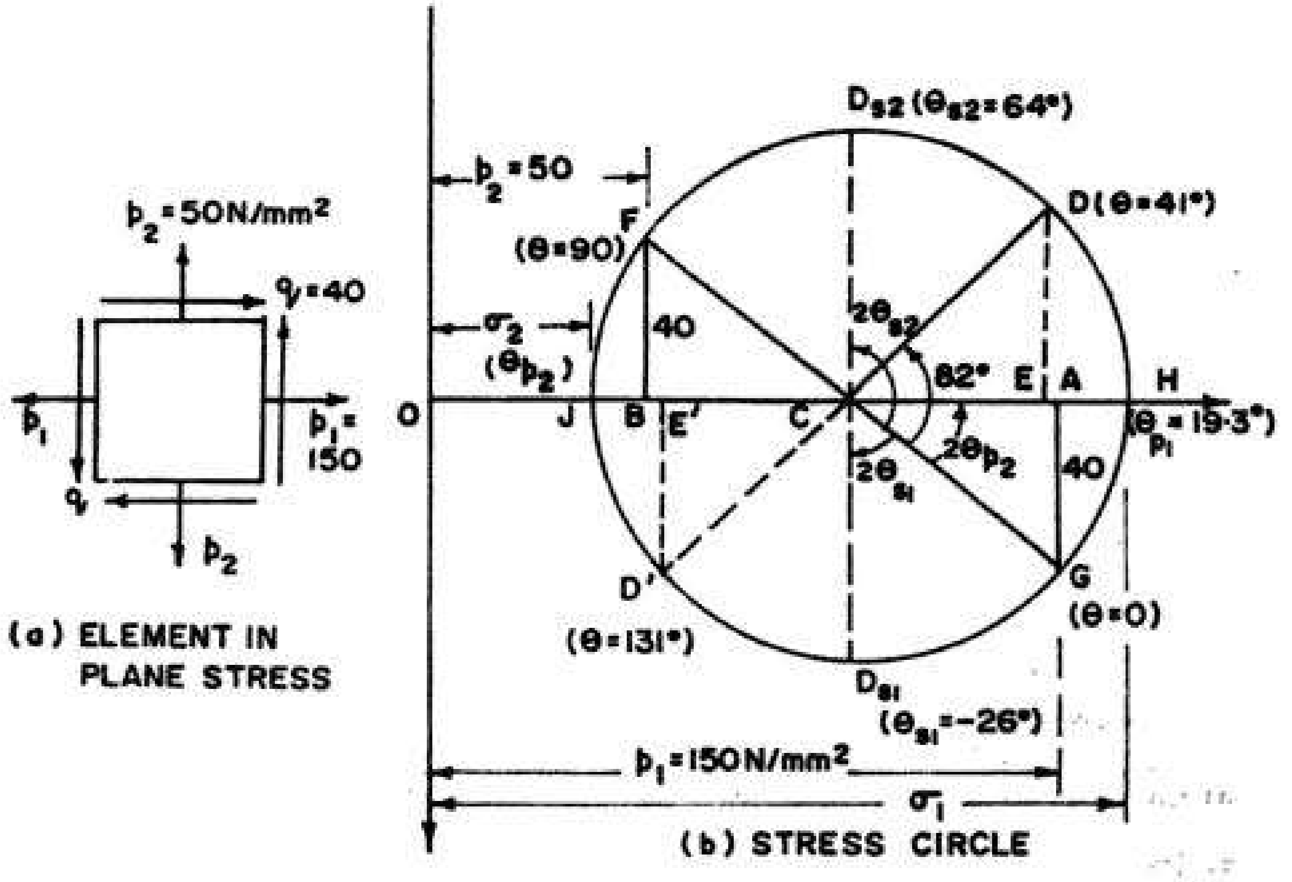


FIG. 4.47

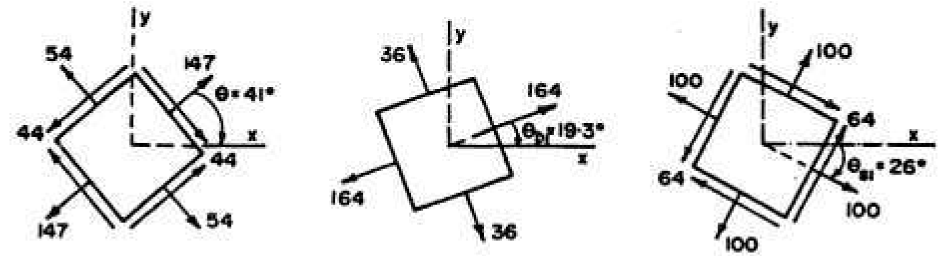
$$2\theta_{p1} = +38.6^\circ \ ; \ \theta_{p1} = 19.3^\circ$$

$$(p_t)_{max} = CD_1 = +64 \text{ N/mm}^2; \ (p_t)_{min} = CD_2 = -64 \text{ N/mm}^2$$

$$2\theta_{s1} = -52^\circ; \ \theta_{s1} = -26^\circ \ ; \ 2\theta_{s2} = +128^\circ; \ \theta_{s2} = +64^\circ$$

$$(p_n)_{D1} = p_{av} = 100 \text{ N/mm}^2 = (p_n)_{D2}$$

Fig 4.48(*a*) shows the stresses on stress element at $\theta = +41^\circ$



(a) STRESSES ON ELEMENT AT θ=41° (b) PRINCIPAL STRESSES (c) MAXIMUM SHEAR STRESSES

FIG. 4.48

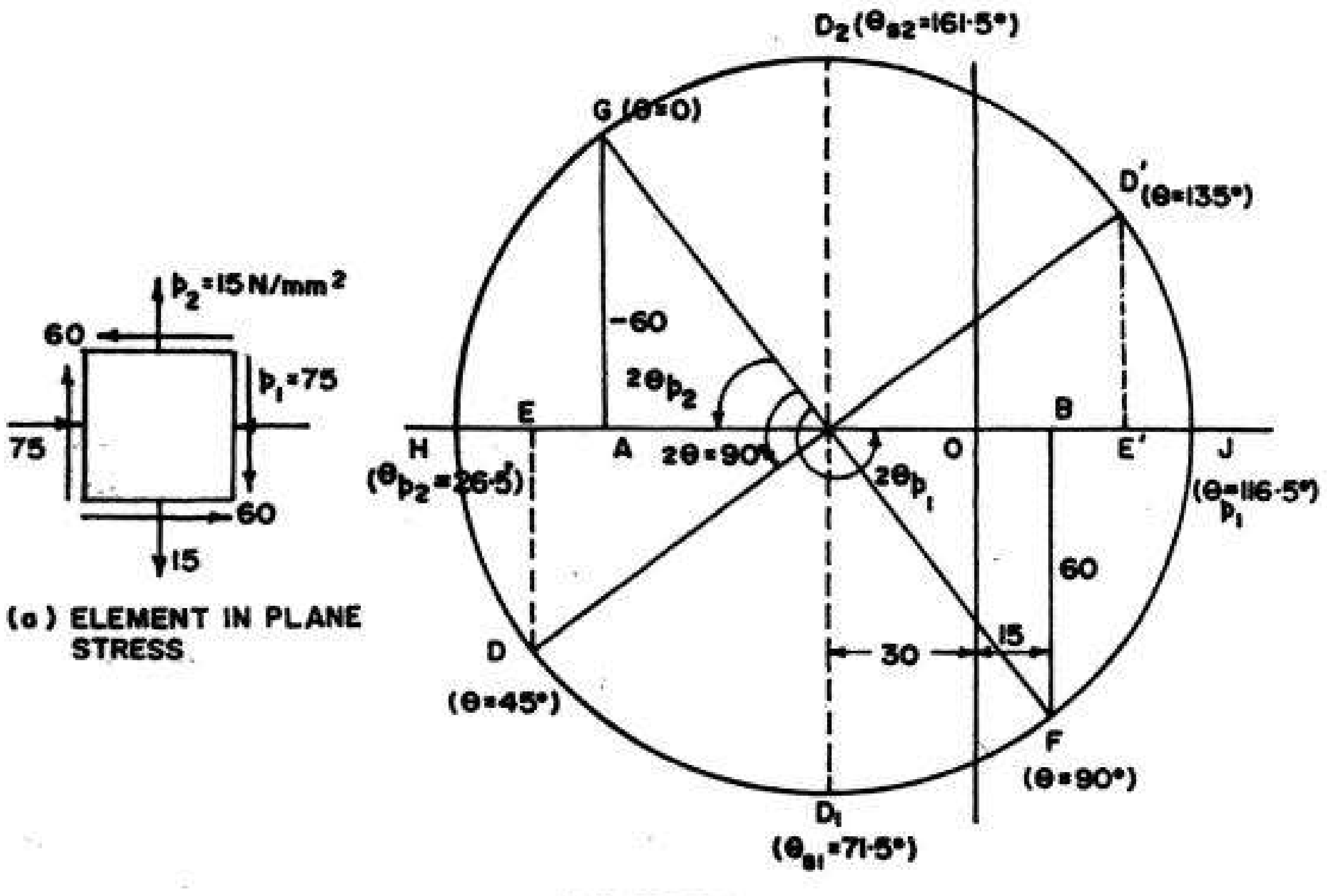
Fig 4.48(*b*) shows the stress element containing principal planes on which only the principal stresses act and there are no shearing stresses. Fig. 4.48(*c*) shows the orientation of stress element containing the planes on which maximum shear stresses act. The corresponding normal stresses on these planes are evidently equal to the average stress.

Example 4.26. *An element in plane stress is subjected to stresses $p_1 = -75$ N/mm², $p_2 = 15$ N/mm² and $q = -60$ N/mm², as shown in Fig. 4.49 (a). Using Mohr's circle, determine (a) the stresses acting on an element rotated through an angle = 45° in the anticlockwise direction, (b) the principal stresses and (c) the maximum shear stresses. Show all results on sketches of properly oriented elements.*

Solution : Given : $p_1 = -75$ N/mm²; $p_2 = 15$ N/mm² and $q = -60$ N/mm². The Mohr's circle is shown in Fig 4.49 (*b*). The centre *C* of the circle has *x*-coordinate $= p_{av} = \dfrac{p_1 + p_2}{2} = \dfrac{-75 + 15}{2} = -30$ N/mm². The coordinates of point G ($\theta = 0$) are given by the stresses on the *x*-face of the element (Fig. 4.49 *a*). Hence mark $OA = -75$ and $AG = -60$ (*i.e.* upwards). Similarly, the co-ordinates of point F ($\theta = 90^\circ$) are given by the stresses on the *y*-face of the element of Fig. 4.49 (*a*). Hence make $OB = +15$ and $BF = +60$ (*i.e.* downwards). Join *G* and *F*, to intersect x-axis on *C*, giving the centre of the circle. With *C* as centre, and *CG* (or *CF*) as radius $R \left(= \sqrt{(45)^2 + (60)^2} = 75 \text{ N/mm}^2 \right)$. The circle cuts the *x*-axis in points *H* and *J*.

(a) Stresses on element rotated through $\theta = 45^\circ$

Draw *CD* at $2\theta = 2 \times 45^\circ = 90^\circ$ with *CG* (i.e. *x* face) in the anticlockwise direction. Then, stresses acting on a plane at $\theta = 45^\circ$ is represented by *D*. Similarly, the coordinates of point *D′* represent the stresses on the other face, (*i.e.* at $\theta = 90^\circ + 45^\circ = 135^\circ$ with *x*-face). From Mohr's circle, we get the following values :



(a) ELEMENT IN PLANE STRESS

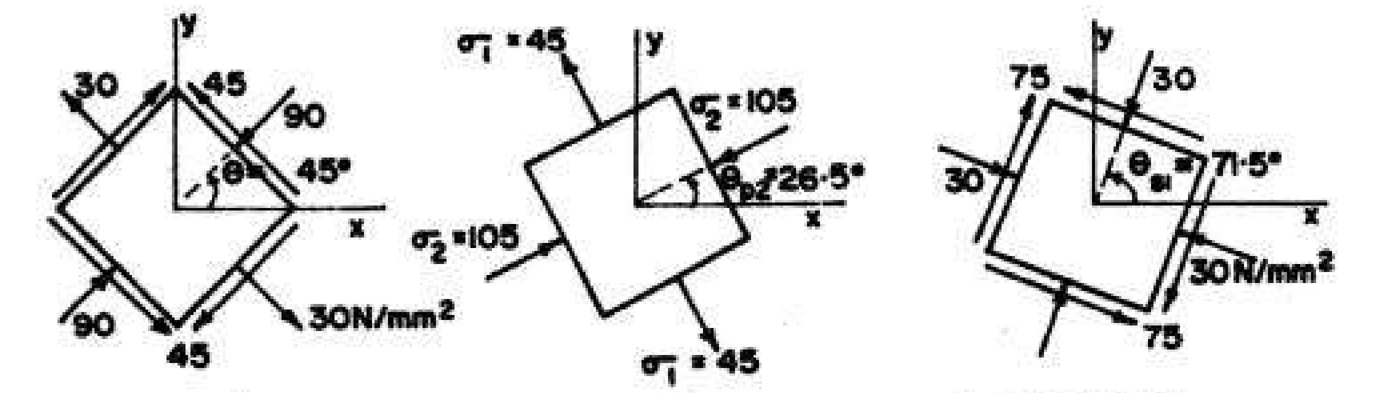
(b) STRESS CIRCLE

FIG. 4.49

$$(p_n)_D = -90\ \text{N/mm}^2;\ (p_t)_D = +45\ \text{N/mm}^2$$

$$(p_n)_{D'} = +30\ \text{N/mm}^2;\ (p_t)_{D'} = -45\ \text{N/mm}^2.$$

The corresponding stress element is shown in Fig. 4.50 (*a*)



(a) STRESS ON ELEMENT AT θ = 45°
(b) ELEMENT CORRESPONDING TO PRINCIPAL STRESSES
(c) ELEMENT CORRESPONDING TO MAX. SHEAR STRESS

FIG. 4.50

(b) Principal planes and principal stresses : The principal stresses are represented by point H and J on stress circle. The angle GCH is the angle $2\theta_{p2}$, from point $G\ (\theta = 0)$ to point H, representing the principal plane having the algebraically smaller principal stress. Hence from the stress circle, we obtain.

$$\sigma_1 = OJ = +45\ \text{N/mm}^2;\ \sigma_2 = OH = -105\ \text{N/mm}^2$$
$$2\theta_{p1} = 233°\ ;\ \ \theta_{p1} = 116.5°\ ;\ 2\theta_{p2} = 53°;\ \theta_{p2} = 26.5°$$

The stress element representing principal planes and corresponding principal stresses is shown in Fig. 4.50(*b*).

(c) Maximum and Minimum shear stresses : Maximum shear stress is represented by point D, (measured vertically downwards) and the corresponding values are

$$(p_t)_{max} = CD_1 = +75\ \text{N/mm}^2;\ (p_n)_{D1} = -30\ \text{N/mm}^2\ (= p_{av})$$
$$2\theta_{s1} = \angle GCD_1 = 90° + 2\theta_{p2} = 90° + 53° = 143°\ ;\ \theta_{s1} = 71.5°$$

The minimum shear stress is presented by point D_2.

$$(p_t)_{min} = CD_2 = -75\ \text{N/mm}^2;\ (p_n)_{D2} = -30\ \text{N/mm}^2\ (= p_{av})$$
$$2\theta_{s2} = 2\theta_{s1} + 180° = 143° + 180° = 323°\ ;\ \theta_{s2} = 161.5$$

The stress element representing maximum shear stresses (and corresponding planes) is shown in Fig. 4.50(*c*).

Example 4.27. *At a point in a material under stress, the resultant intensity of stress across a certain plane is 100 N/mm² (compressive) inclined at 30° to its normal. On another plane, the intensity of resultant stress is 75 N/mm² (tensile) inclined at 45° to its normal, as shown in Fig. 4.51. Find the principal stresses and their directions to the given plane. Find also the angle between the two planes.*

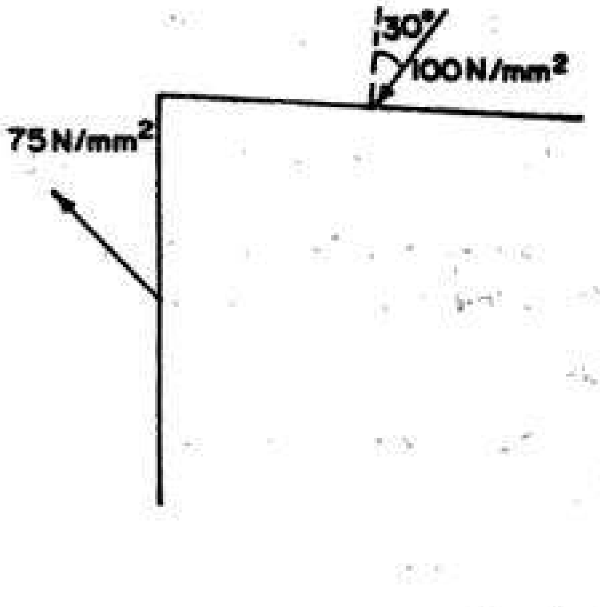


FIG. 4.51

Solution :

For the Plane AB : $p_r = 100\ \text{N/mm}^2$, p_n is negative and p_t is positive. Hence draw $OD' = 100\ \text{N/mm}^2$ at $\varphi = 30°$ in the anticlockwise direction, as shown in Fig. 4.52, so that p_n is negative and p_t is positive.

For Plane AC

$p_r = 75\ \text{N/mm}^2$. p_n is positive and p_t is negative.

Hence draw $OD\ (= 75\ \text{N/mm}^2)$ at 45° with OX (in the counter clockwise direction) so that p_n is positive but p_t is negative. The circle has to pass through point D and D'. Hence join D and D', and draw the perpendicular bisector of DD' to cut x-axis in C. With C as centre and CD' (or CD) as radius, draw Mohr circle to cut the x-axis in H and J.

From Mohr's circle, we obtain the following values :

$\sigma_1 = OJ = \mathbf{-82\ N/mm^2}$ (*i.e.* compressive)

$\sigma_2 = OH = \mathbf{+58\ N/mm^2}$ (*i.e.* tensile)

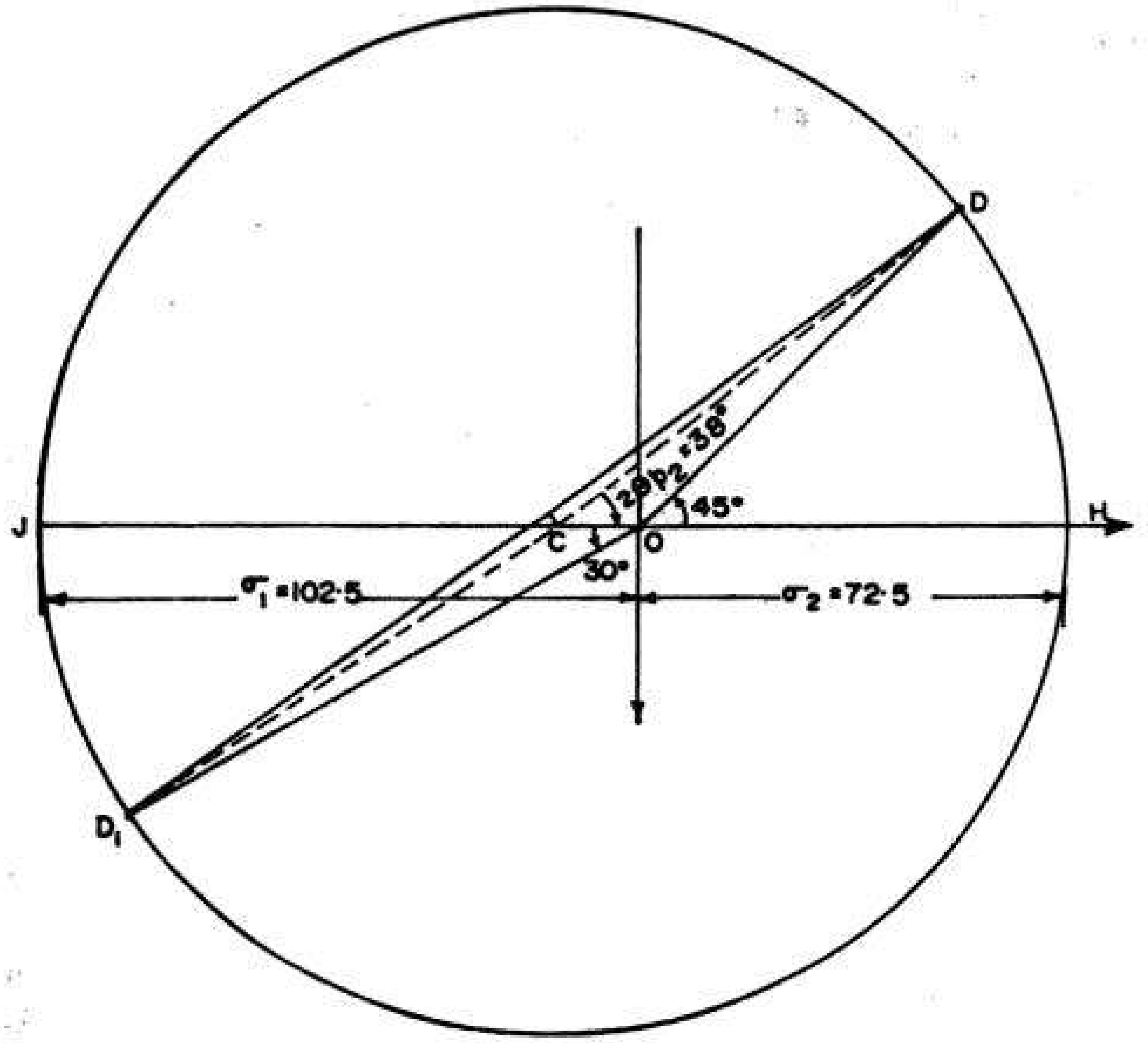


FIG. 4.52

$2\theta_{p2} = \angle HCD = 38°$; $\theta_{p2} = 19°$ (clockwise with plane AC)

Inclination of plane of $\sigma_1 = \theta_{p1} = 90 + 19° = 109°$, (clockwise with plane AC)

Angle between the two planes $= \frac{1}{2} \angle D'CD = \frac{1}{2} \times 178° = 89°$.

Example 4.28. *Fig. 4.53(a) shows the direct stresses in three co-planer directions differing by 60°. Find the magnitude and directions of principal stresses.*

Solution :

The problem can be solved more conveniently, using Mohr's circle which can be drawn in the following steps (Fig. 4.53 *b*) :

1. Draw vertical line representing *y*-axis. Set off three vertical lines representing $+ 80\ \text{N/mm}^2$, $+ 30\ \text{N/mm}^2$ and $- 50\ \text{N/mm}^2$ normal stresses.

2. On the intermediate vertical line corresponding to a normal stress of $+ 30\ \text{N/mm}^2$, select any arbitrary point *P*.

3. Through point *P*, draw a line *PN* at $+ 60°$ (*i.e.* in anticlockwise direction) with vertical, intersecting vertical line of $80\ \text{N/mm}^2$ in point *N*. Similarly draw another line at $+ 120°$ with vertical, intersecting the vertical line of $- 50\ \text{N/mm}^2$ in point *M*, when produced back.

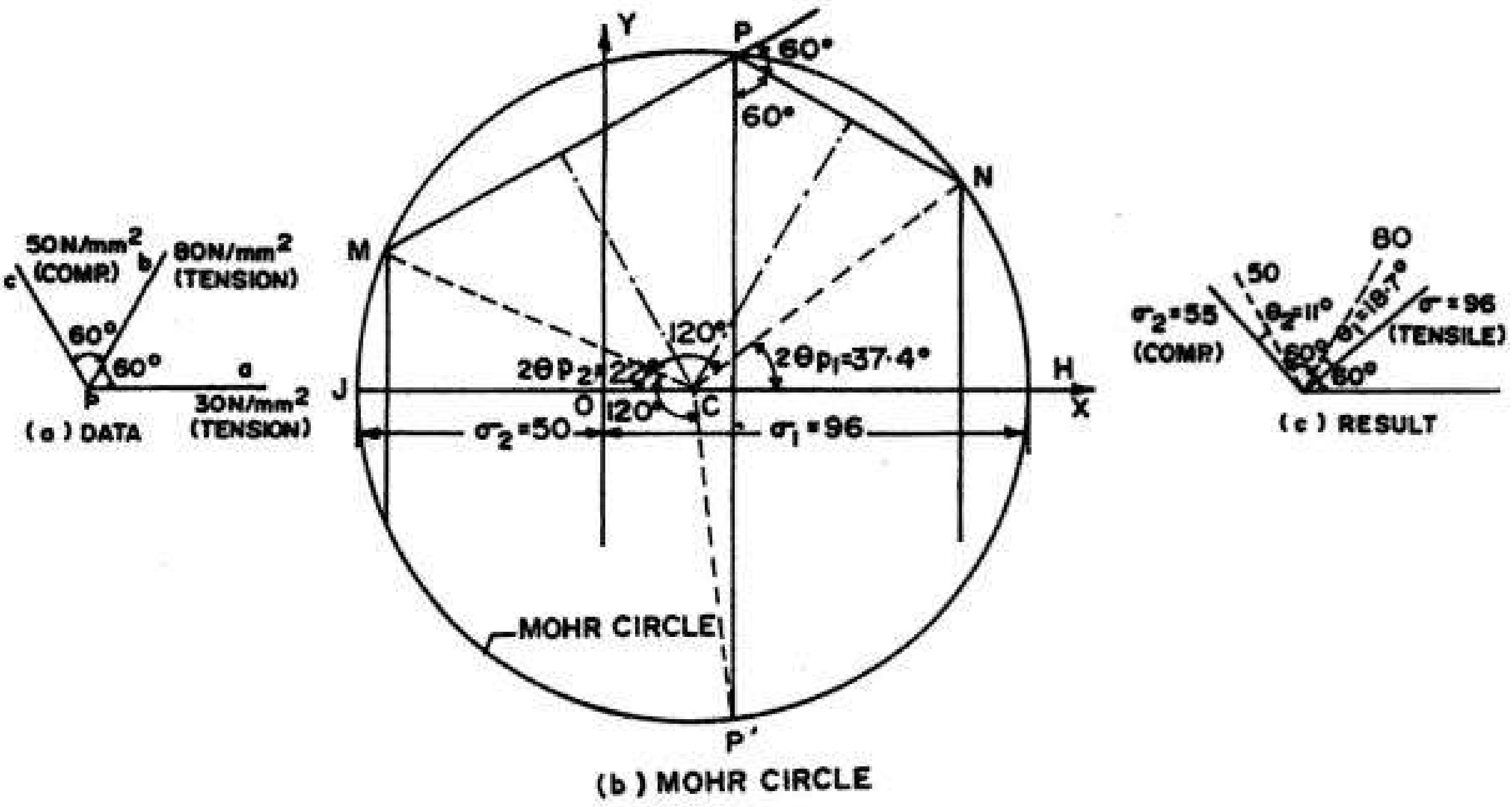


FIG. 4.53

4. Thus, we have three points P, N and M on three vertical lines corresponding to three normal stresses. The Mohr's circle will pass through P, N and M. To locate its centre, draw perpendicular bisectors of MP and PN, intersecting at C.

5. Through C, draw horizontal lines, representing x-axis, intersecting y-axis in O. Thus, we get the origin O and centre C. Draw the Mohr's circle with radius CM ($= CN$). The circle will pass through point M, P and N.

6. The stress conditions on the three given planes are represented by points M, N and P, where P' is the point on the circle, vertically below P. The justification of the construction lies from the fact that the angles between the three radius vectors CM, CN and CP' are 120°, which is twice the angle between each pair of given direct stresses.

Result : From Fig. 4.53(b), we obtain the following results :

$\sigma_1 = OH = \mathbf{+96\ N/mm^2}$ ($i.e.$ tension); $\sigma_2 = OJ = \mathbf{-55\ N/mm^2}$ ($i.e.$ compressive)

$\theta_{p1} = \frac{1}{2} \angle NCH = \mathbf{18.7°}$ to the $80\ N/mm^2$ stress.

$\theta_{p2} = \frac{1}{2} \angle CJM = \mathbf{11°}$ to the $-55\ N/mm^2$ stress.

The directions of σ_1 and σ_2 relative to the given normal stresses are shown in Fig 4.53 (c).

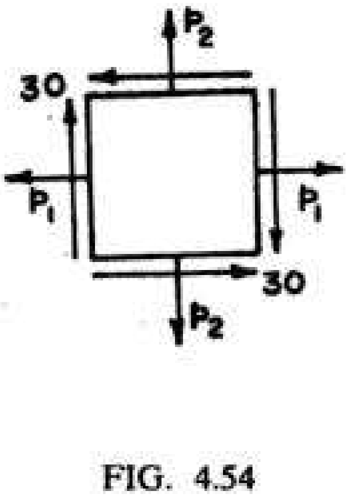


FIG. 4.54

Example 4.29. *For the element shown in Fig. 4.54 determine the values of p_1 and p_2 if the principal stresses are known to be $20\ N/mm^2$ and $-80\ N/mm^2$.*

Solution :

Given : $\sigma_1 = 20\ N/mm^2$, $\sigma_2 = -80\ N/mm^2$

and $q = -30\ N/mm^2$.

$$\sigma = \frac{p_1 + p_2}{2} \pm \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + q^2}$$

$$\therefore \qquad \sigma_1 = 20 = \frac{p_1 + p_2}{2} \pm \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + (-30)^2} \qquad ...(1)$$

and $$\sigma_2 = -80 = \frac{p_1 + p_2}{2} - \sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + (-30)^2} \qquad ...(2)$$

Subtracting (2) from (1), we get

$$2\sqrt{\left(\frac{p_1 - p_2}{2}\right)^2 + 900} = 100$$

or $$\left(\frac{p_1 - p_2}{2}\right)^2 + 900 = 2500, \text{ from which } p_1 - p_2 = 80 \qquad ...(a)$$

Also, adding (1) and (2), we get $p_1 + p_2 = -60$ ...(*b*)

From (*a*) and (*b*), we get p_1 = **10 N/mm²** and p_2 = **− 70 N/mm²**.

Example 4.30. *The state of stress at a point is the result of the three separate actions that produce the three states of stresses shown in Fig. 4.55. Determine the principal stresses and principal planes caused by the superposition of these three stress states.*

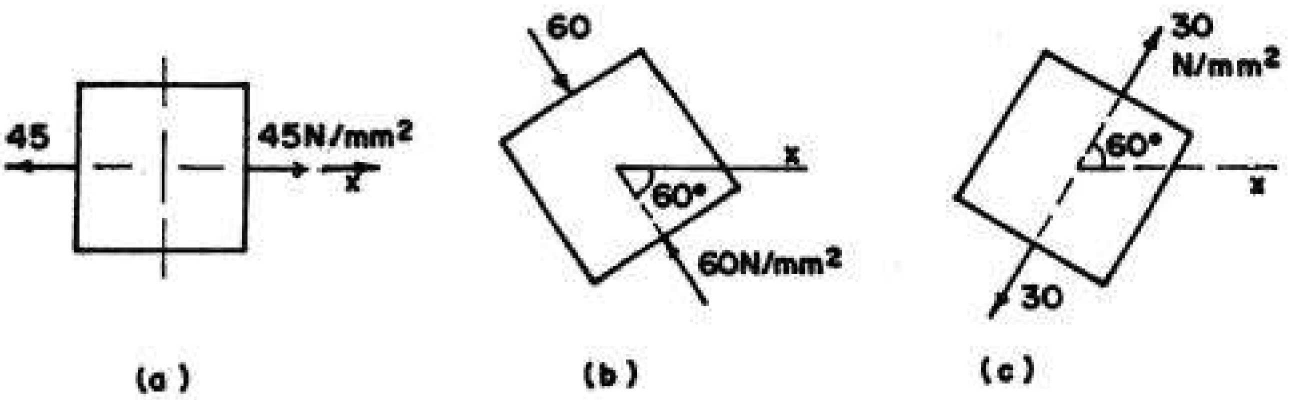


FIG 4.55

Solution : Let the principal stresses be σ and its inclination be θ_p with x-axis as shown in Fig. 4.56.

Hence stress on the element shown in Fig. 4.55 (*a*) is given by

$$p_{na} = \frac{1}{2}\left(\sigma_1 + \sigma_2\right) + \frac{1}{2}\left(\sigma_1 - \sigma_2\right)\cos 2\theta \qquad ...(4.15)$$

Here, $p_{na} = +45$, and $\theta = -\theta_p$ (since x axis is rotated by θ_p in the clockwise direction with respect to x_p axis.

$$\therefore \qquad 45 = \frac{1}{2}\left(\sigma_1 + \sigma_2\right) + \frac{1}{2}\left(\sigma_1 - \sigma_2\right)\cos\left(-2\theta_p\right)$$

or $$(\sigma_1 + \sigma_2) + (\sigma_1 - \sigma_2)\cos 2\theta_p = 90 \quad ...(1)$$

Also, stress on the element shown in Fig. 4.55 (*b*) is given by Eq. 4.15 where

$$p_{na} = -60 \text{ and } \theta = -\left(60° + \theta_p\right)$$

$$\therefore \qquad -60 = \frac{1}{2}\left(\sigma_1 + \sigma_2\right) + \frac{1}{2}\left(\sigma_1 - \sigma_2\right)\cos\left\{-2\left(60 + \theta_p\right)\right\}$$

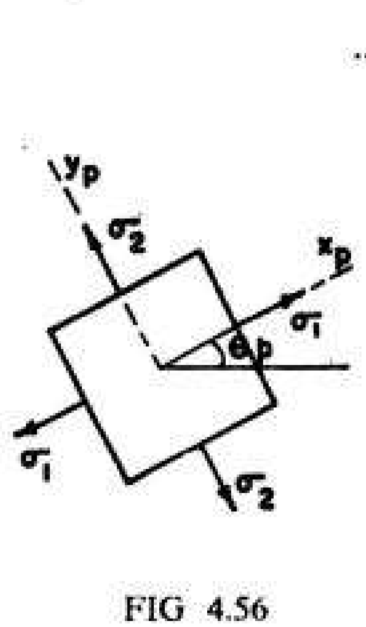


FIG 4.56

or $\quad (\sigma_1 + \sigma_2) + (\sigma_1 - \sigma_2) \cos\left(120^\circ + 2\,\theta_p\right) = -120$

or $\quad (\sigma_1 + \sigma_2) + (\sigma_1 - \sigma_2) \left\{\cos 120^\circ \cos 2\,\theta_p - \sin 120^\circ \sin 2\,\theta_p\right\} = -120$

or $\quad (\sigma_1 + \sigma_2) + (\sigma_1 - \sigma_2) \left\{0.5 \cos 2\,\theta_p + 0.866 \sin 2\,\theta_p\right\} = -120$

Lastly, the stress on element shown in Fig. 4.55 (*c*) is given by Eq. 4.15, where $p_{na} = +30$ and $\theta = (60^\circ - \theta_p)$

$$\therefore \quad 30 = \tfrac{1}{2}(\sigma_1 + \sigma_2) + \tfrac{1}{2}(\sigma_1 - \sigma_2) \cos 2\,(60^\circ - \theta_p)$$

or $\quad (\sigma_1 + \sigma_2) + (\sigma_1 - \sigma_2) \cos(120^\circ - 2\theta_p) = 60$

or $\quad (\sigma_1 + \sigma_2) + (\sigma_1 - \sigma_2) \left\{\cos 120^\circ \cos 2\,\theta_p + \sin 120^\circ \sin 2\,\theta_p\right\} = 60$

or $\quad (\sigma_1 + \sigma_2) + (\sigma_1 - \sigma_2) \left\{-0.5 \cos 2\,\theta_p + 0.866 \sin 2\,\theta_p\right\} = 60 \qquad ...(3)$

Thus we have three equations (*i.e.* Eqs 1, 2 and 3) corresponding to the three unknowns σ_1, σ_2 and θ_p. Adding Eqs (2) and (3) we get

$$(\sigma_1 - \sigma_2)\left\{1.732 \sin 2\,\theta_p\right\} = 180 \qquad ...(I)$$

Also, subtracting (3) from (1), we get

$$(\sigma_1 - \sigma_2)\left(1.5 \cos 2\,\theta_p - 0.866 \sin 2\,\theta_p\right) = 30 \qquad ...(II)$$

Dividing (II) by (I), we get $\quad \dfrac{1.5 \cos 2\,\theta_p - 0.866 \sin 2\,\theta_p}{1.732 \sin 2\,\theta_p} = \dfrac{30}{180} = \dfrac{1}{6}$

or $\quad 0.866 \cot 2\,\theta_p - 0.5 = 0.1667$

From which $\tan 2\,\theta_p = 1.2989$; $2\,\theta_p = 52.4^\circ$; $\theta_p = \mathbf{26.2^\circ}$

Also, from I, $(\sigma_1 - \sigma_2) = \dfrac{180}{1.732 \sin 52.4^\circ} = 131.16 \qquad ...(a)$

and from (1), $(\sigma_1 + \sigma_2) = 90 - 131.16 \cos 52.4^\circ \approx 10 \qquad ...(b)$

From (*a*) and (*b*), we get $\sigma_1 =$ **70.58 N/mm²** (tensile) and $\sigma_2 =$ **− 60.58** (*i.e.* comp.)

Example 4.31. *At a point in a piece of material, the principal stress on a plane P is 35 N/mm² tensile and the component stresses on a second plane Q at the same point are a normal stress of 126 N/mm² tensile and a shear stress of 70 N/mm². The principal stress on a third plane perpendicular to plane P and Q is zero. Determine by Mohr circle method (a) the angle between planes P and Q (b) the other principal stress and location of its plane (c) maximum shear stress, and (d) the maximum obliquity of the resultant stress with the normal stress on any plane.*

Solution :

Fig 4.57 shows the position of planes and the corresponding stresses. Fig. 4.58 shows the Mohr's circle of stress which is drawn in the following steps.

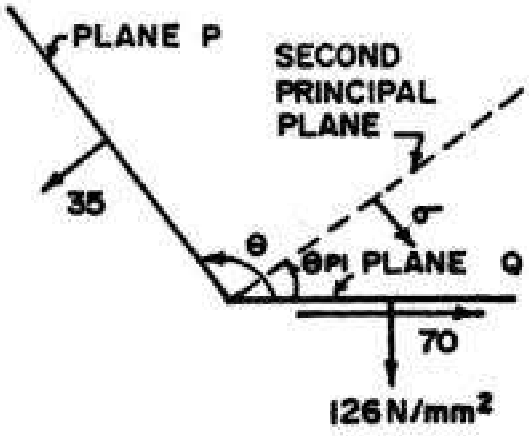


FIG 4.57

1. Mark $OA = 126\ N/mm^2$ in the positive direction.

2. Draw perpendicular AG in the downward direction (since q is positive on plane Q) and make it equal to $70\ N/mm^2$.

3. Mark $OB = 35\ N/mm^2$ in the positive direction.

4. Since plane P is the principal plane (q being zero on it), the circle must pass through point B. Circle must also pass through point G. Hence join points B and G and draw its perpendicular bisector to cut the x-axis in point C, the centre of the circle.

5. With C as the centre and CB or CG as radius, draw the circle cutting the x-axis in point $B(J)$ and H.

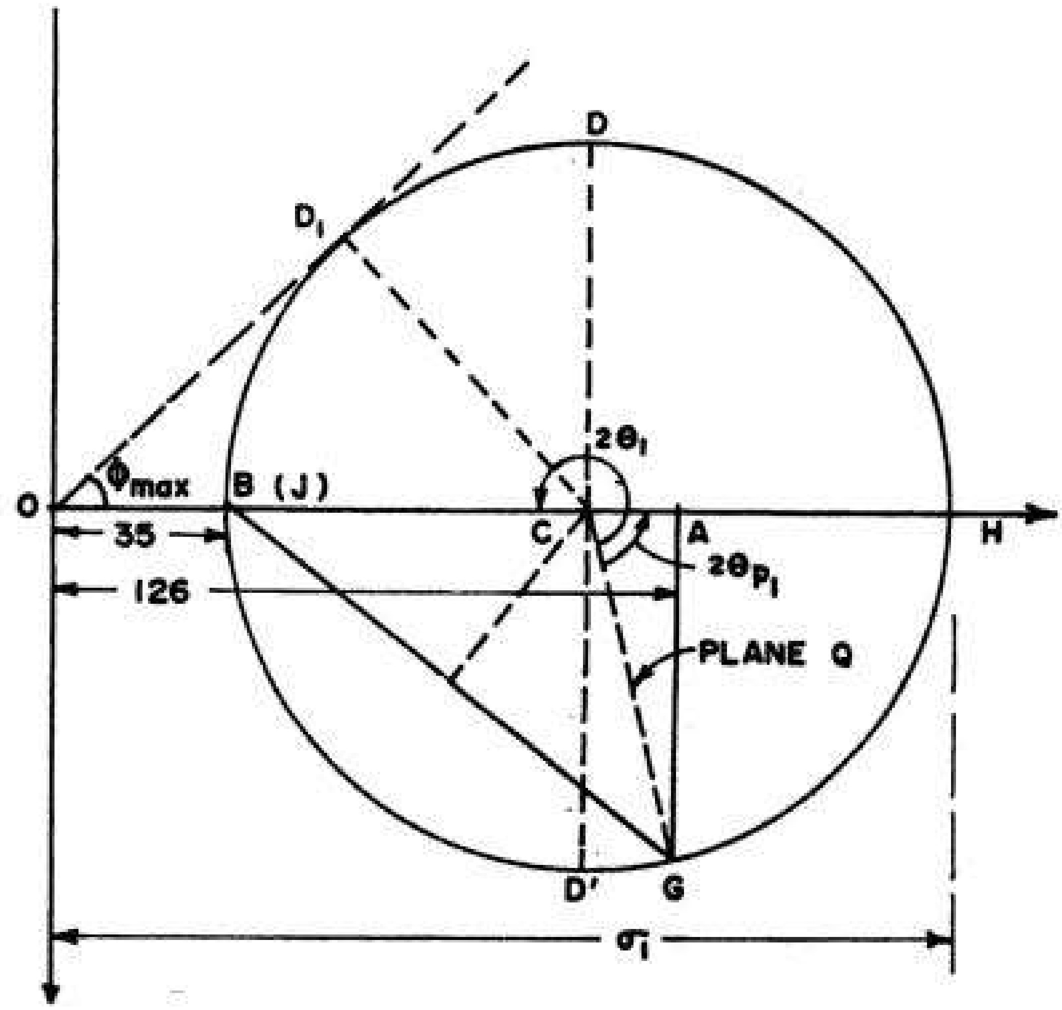


FIG 4.58

6. Through C, draw CD perpendicular to x-axis. Point D corresponds to maximum shear stress.

7. To find the maximum obliquity of the resultant stress, draw line OD_1 tangential to the circle.

From the above construction, we obtain the following :

(i) Angle between planes P and $Q = \theta$ (or θ_{P2}) $= \frac{1}{2} \angle GCO = \frac{1}{2}(256°)$ $= \mathbf{128°}$

(ii) Other principal stress $= \sigma_1 = OH = \mathbf{180\ N/mm^2}$. Inclination of other principal plane with plane $Q = \theta_{p1} = \frac{1}{2} \angle GCH = \frac{1}{2} \times 76° = \mathbf{38°}$

(iii) Maximum shear stress $= CD$ (or CD')$= \mathbf{71.7\ N/mm^2}$

(iv) Maximum obliquity of the resultant stress $\varphi_{max} = \angle D_1OC = \mathbf{41.5°}$

Example 4.32. *An element of plane stress is rotated through a known angle $\theta = 30°$ and the stresses on this rotated element are as shown in Fig. 4.59 (a). Determine the normal and shear stresses (i.e. p_1, p_2 and q) on the original element (Fig. 4.59 b) whose sides are parallel to the x-y axes.*

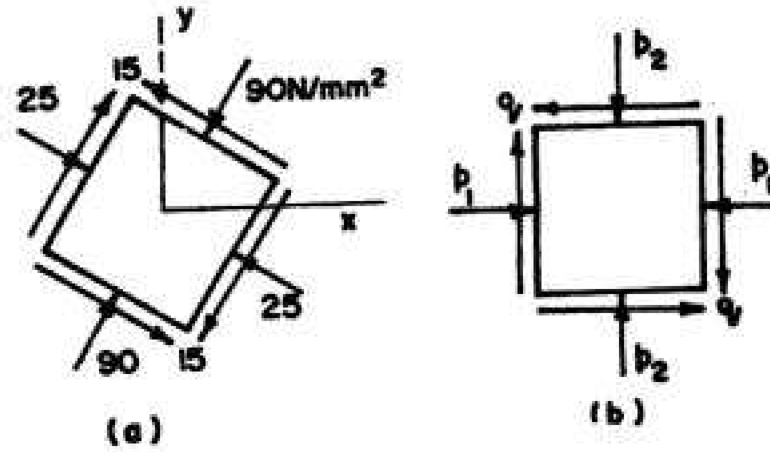


FIG 4.59

Solution :

Let the stresses be p_1, p_2 and q.

Here $\theta = +30°$.

Hence $$p_{n1} = -90 = \frac{p_1 + p_2}{2} + \frac{p_1 - p_2}{2}\cos 60° + q \sin 60°$$

or $$(p_1+p_2)+0.5\,(p_1-p_2)+1.732\,q = -180 \quad ...(1)$$

Also $$p_t = 15 = -\frac{p_1 - p_2}{2}\sin 60° + q\cos 60°$$

or $$-(p_1 - p_2)\,0.866 + q = 30 \quad ...(2)$$

At $\theta = 30° + 90° = 120°$, we have

$$p_{n2} = -25 = \frac{p_1 + p_2}{2} + \frac{p_1 - p_2}{2}\cos 240° + q\sin 240°$$

or $$(p_1 + p_2) - 0.5\,(p_1 - p_2) - 1.732\,q = -50 \quad ...(3)$$

Thus, we have three equations corresponding to the three unknowns.

Adding (1) from (3), we have

$$(p_1 + p_2) = -115 \quad ...(a)$$

From (2), $$p_1 - p_2 = \frac{q - 30}{0.866} = 1.1547\,(q - 30) \quad ...(b)$$

Substituting the values of $(p_1 + p_2)$ and $(p_1 - p_2)$ in (1), we get

$$-115 + 0.5 \times 1.1547\,(q - 30) + 1.732\,q = -180$$

From which $q = -20.65$ N/mm^2. Substituting the values of q in (b), we get

or $$p_1 - p_2 = -58.48 \quad ...(c)$$

From (a) and (c), we get $p_1 = \mathbf{-86.74}$ N/mm^2 and $p_2 = \mathbf{-28.26}$ N/mm^2.

Example 4.33. *An element in a structure subjected to the plane stress; the stresses have the magnitudes and directions shown acting on a element A (Fig 4.60 a). Element B located at the same point in the structure, is rotated through an angle θ_b of such magnitude that the stresses have the values shown in Fig 4.60 (b). Calculate the normal stress p_b and the angle θ_b.*

Solution :

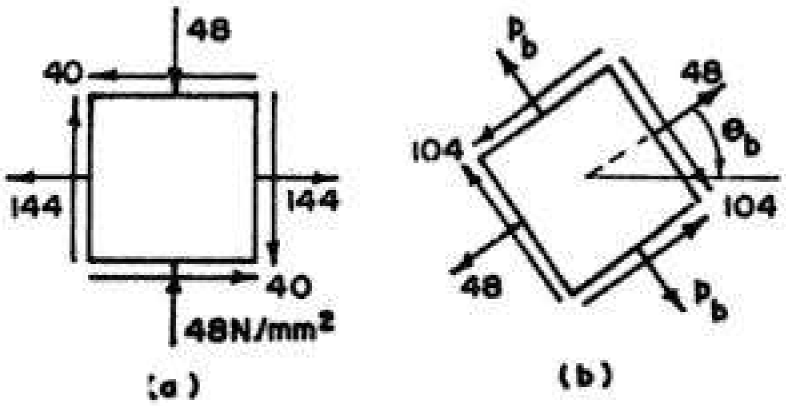


FIG. 4.60

Here $p_1 = +144$ N/mm^2,

$p_2 = -48$ N/mm^2 and $q = -40$

$$p_{n1} = 48 = \frac{144 - 48}{2} + \frac{144 + 48}{2}\cos 2\theta_b - 40\sin 2\theta_b$$

or $$96\cos 2\theta_b - 40\sin 2\theta_b = 0$$

From which $\tan 2\theta_b = 2.4$

or $2\theta_b = 67.38°$ or $\theta_b = 33.69°$.

Also, at $\theta = 90° + \theta_b = 90° + 33.69° = 123.69°$,

we have

$$p_{n2} = p_b = \frac{144 - 48}{2} + \frac{144 + 48}{2}\cos(2 \times 123.69°) - 40\sin(2 \times 123.69)$$

or $$p_b = 48 + (-36.92) - 40\,(-0.923) = \mathbf{48}\ \text{N/mm}^2$$

PROBLEMS

1. At a point in a bracket, the stresses on two mutually perpendicular planes are 100 N/mm^2 (tensile) and 50 N/mm^2 (tensile). The shear stress across these planes is 25 N/mm^2. Find, using Mohr's circle, the magnitude and direction of the resultant stress on a plane making an angle of 30° with the plane of the first stress. Find also, the normal and tangential stresses on this plane.

2. At a point in a loaded specimen, the principal stresses acting on two mutually perpendicular planes are 90 N/mm^2 and 60 N/mm^2, both being compressive. Determine the resultant stress acting on a plane inclined at 60° measured clockwise to the plane on which the larger normal stress is acting.

3. In a piece of material, a tensile stress p_1 and shearing stress q act on a given plane. Show that principal stresses are always of opposite sign.

If an additional tensile stress p_2 acts on a plane perpendicular to that on which p_1 acts and all the stresses are coplanar, find the condition that both principal stresses may be of the same sign.

4. The principal stresses at a point subjected to two dimensional stresses are 56 N/mm^2 and 40 N/mm^2 tensile. Find the plane on which the resultant stress has maximum obliquity and the magnitude of the obliquity. Find also the resultant stress.

5. In each of the following cases, p_1 and p_2 are perpendicular direct stresses. Calculate the normal, tangential and resultant stress on planes making angle of (*i*) 20°, (*ii*) 40° and (*iii*) 80° with the plane of p_1 and also for the plane having maximum shear stress. Find also the obliquity of the resultant (i.e. its inclination to the normal).

(*a*) $p_1 = +64$ N/mm^2; $p_2 = +24$ N/mm^2

(*b*) $p_1 = +64$ N/mm^2; $p_2 = -24$ N/mm^2

(*c*) $p_1 = +32$ N/mm^2; $p_2 = -64$ N/mm^2

Use Mohr's circle method.

6. At a certain point in stressed body, the principal stress are $\sigma_x = 120$ N/mm^2 and $\sigma_y = -60$ N/mm^2. Determine p_n and p_t on planes whose normals are at $+30°$ and $+120°$ with the x-axis. Show your results on a sketch of a differential element.

7. Calculate the principal stresses and maximum shear stress for the following cases of two perpendicular direct stresses together with complimentary shear stresses. Draw diagram showing the positions of principal planes and planes of maximum shear stress relative to the planes the applied stresses.

(*i*) $p_1 = 90$ N/mm^2 (tensile) ; $p_2 = 30$ N/mm^2 (tensile) and $q = +15$ N/mm^2

(*ii*) $p_1 = 30$ N/mm^2 (tensile) ; $p_2 = +75$ N/mm^2 (comp.) and $q = +15$ N/mm^2

(*iii*) $p_1 = 0$; $p_2 = 30$ N/mm^2(comp.) and $q = +60$ N/mm^2

8. The stresses at a particular point in a piece of material act upon three planes whose relative angular positions are given by triangle ABC, in which B is a right angle and C is 30°. The normal stresses on these planes are 105 N/mm^2 (tension) on AB, 45 N/mm^2 (compression) on BC and 60 N/mm^2 (tension) on AC. Determine the magnitude and direction of the shearing stresses on the given planes and the magnitude of the greatest direct stress and the greatest shearing stress at the point.

9. A right angled triangle ABC with the right angle at C represents planes in an elastic material. There are shearing stresses of 50 N/mm^2 acting along planes AC and CB towards C, and normal tensile stresses on AC and CB of 80 N/mm^2 and 60 N/mm^2 respectively. There is no stress on the plane perpendicular to planes AC and CB.

Determine the positions of the plane AB when the resultant stress on AB is

(a) the greatest magnitude

(b) the least magnitude

(c) the greatest component normal to AB

(d) the greatest tangential component along AB

(e) the least inclination to AB.

Analytical or graphical methods may be used. State for each plane found, its angular position relative to AC and the magnitude of the stress referred to.

10. At a point in two dimensional stress system, the normal stress on two mutually perpendicular planes are p and p' (both alike) and the shear stress q. Show that one of the principal stresses is zero if $q^2 = pp'$.

11. An element in plane stress is rotated through a known angle $\theta = 60°$ (Fig. 4.61). On the rotated element, the normal and shear stresses have the magnitude and direction as shown. Determine the normal and shear stresses on an element whose sides are parallel to x-y axes.

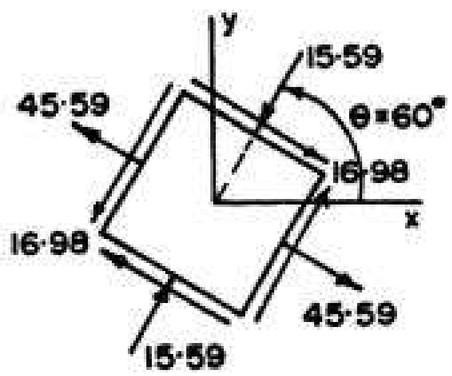


FIG. 4.61

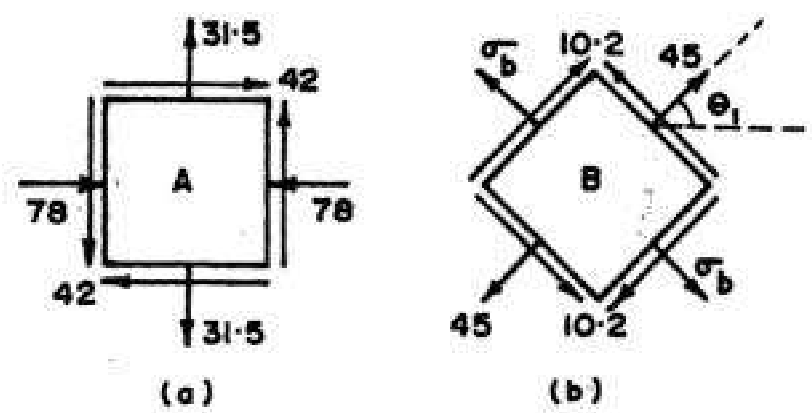


FIG. 4.62

12. At a point in a structure subjected to plane stress, the stresses have the magnitudes and directions shown acting an element A in Fig. 4.62. Element B, located at the same point in the structure is rotated through an angle θ_1 of such magnitude that the stresses have the values shown in Fig. 4.62 (b). Determine σ_b and θ_1.

13. Construct Mohr's circle for an element in uniaxial stress p_1.

(a) From the circle, derive the following stress transformation equations

$$p_n = \frac{p_1}{2}(1 + \cos 2\theta)\,;\, p_t = -\frac{p_1}{2}\sin 2\theta.$$

(b) Show from the circle that the principal stresses are $\sigma_1 = p_1$ and $\sigma_2 = 0$

(c) Obtain from the circle the maximum shear stresses and show them on a sketch of a properly oriented element.

14. Construct Mohr's circle for an element in pure shear q.

(a) From the circle, derive the following stress transformation equations :

$$p_n = q \sin 2\theta \;;\; p_t = q \cos 2\theta.$$

(*b*) Obtain from the circle the principal stresses and show them on a sketch of properly oriented element.

(*c*) Show from the circle that the maximum and minimum shear stresses are $\pm q$.

ANSWERS

1. $p_n = 109.15\ \text{N/mm}^2$; $p_t = -9.15\ \text{N/mm}^2$; $p_r = 109.53\ \text{N/mm}^2$; $\theta = -4.8°$

2. $p_n = -67.5\ \text{N/mm}^2$; $p_t = 12.99\ \text{N/mm}^2$; $p_r = 68.74\ \text{N/mm}^2$

3. $p_1 p_2 > q^2$

4. $\theta = 49.8°$; $\varphi = 9.59°$; $p_r = 47.33\ \text{N/mm}^2$

5. Table below :

		p_n (N/mm^2)	p_t (N/mm^2)	p_r (N/mm^2)	φ	$p_{t,\max}$ (N/mm^2)
(*a*)	(i)	59.32	–12.86	60.7	–12.23°	20
	(ii)	47.47	–19.70	51.4	–22.54°	20
	(iii)	25.21	–6.84	26.12	–15.18°	20
(*b*)	(i)	53.70	–28.28	60.69	–27.77°	44
	(ii)	27.64	–43.33	51.40	–57.47°	44
	(iii)	–21.35	–15.05	26.12	+35.18°	44
(*c*)	(i)	20.77	–30.85	37.19	– 56°	48
	(ii)	–7.66	–47.27	47.89	–80.8°	48
	(iii)	–61.11	–16.40	63.27	–15.02°	48

6. (*a*) $p_n = 75\ \text{N/mm}^2$; $p_t = -77.9\ \text{N/mm}^2$

(*b*) $p_n = -15\ \text{N/mm}^2$; $p_t = 77.9\ \text{N/mm}^2$

7. Table below :

	σ_1 (N/mm^2)	σ_2 (N/mm^2)	θ_{p1}	θ_{p2}	$p_{t\max}$ (N/mm^2)
(i)	+ 93.54	+26.46	13.3°	103.3°	33.54
(ii)	+ 32.1	– 77.1	8°	98°	54.6
(iii)	+ 46.84	– 76.84	38°	128°	61.84

8. Shear on $AB = +77.94\ \text{N/mm}^2$; shear on $AC = -105\ \text{N/mm}^2$

$\sigma_1 = 138.16\ \text{N/mm}^2$ (tension); $p_{t,\max} = +108.16\ \text{N/mm}^2$.

9. (*a*) $121\ \text{N/mm}^2$ (tensile) at $39°21'$ to AC

(*b*) $19\ \text{N/mm}^2$ (tensile) at $129°21'$ to AC

(*c*) Same as (*a*)

(*d*) $51\ \text{N/mm}^2$ at $84°21'$ to AC

(*e*) $107°45'$

11. $p_1 = 45\ \text{N/mm}^2$; $p_2 = -15\ \text{N/mm}^2$; $q = -18\ \text{N/mm}^2$

12. $\sigma_b = -91.5\ \text{N/mm}^2$; $\theta_1 = 67°$

5

Analysis of Strain : Principal Strains

5.1. LONGITUDINAL AND LATERAL STRAINS

We have seen in chapter 3 that within the elastic limit, strain is proportional to stress. A direct stress produces a change in length in the direction of stress. *Strain* is a measure of the *deformation* produced in the member by the load. If a rod of length L is in tension and the stretch or elongation is ΔL, then the *direct strain* (ε) or the longitudinal strain is defined as the ratio.

$$\varepsilon = \frac{\Delta L}{L} = \frac{\text{Change in length}}{\text{Original length}} \qquad \text{...(5.1)}$$

It is to be noted that strain is a *ratio*, and hence is dimensionless. As per Hooke's law, *strain is proportional to stress producing it.*

In addition to the *longitudinal deformation* of the bar subjected to a direct load along its longitudinal axis, there also occurs deformations in its lateral dimensions. If an axial load is applied to a bar, its length will increase, and at the same time its lateral dimensions (*i.e.* either the diameter or the width and breadth) will decrease. *Thus, any direct stress produces a strain in its own direction and an opposite kind of strain in every direction at right angles to it.* The ratio of the lateral strain to the longitudinal strain is constant for a given material and is known as Poisson's ratio :

$$\text{Thus} \quad \frac{\text{Lateral Strain}}{\text{Longitudinal Strain}} = \text{Constant} = \frac{1}{m} = \mu$$

or

$$\text{Lateral Strain} = \frac{1}{m} \times \text{longitudinal strain} \qquad \text{...(5.2)}$$

where $\frac{1}{m}$ (or μ) is known as the Poisson's ratio, the value of which lies between 0.25 to 0.42 for most of the metals.

5.2. PRINCIPAL STRAINS IN THREE DIMENSIONS

Let us take a case of small element subjected to three dimensional principal stresses σ_1, σ_2 and σ_3 (Fig. 5.1), acting on three mutually perpendicular planes at a point in an isotropic material. Taking all the three stresses alike, the principal strains ε_1, ε_2 and ε_3 can be obtained from the generalised Hooke's law:

$$\varepsilon_1 = \frac{\sigma_1}{E} - \frac{\sigma_2 + \sigma_3}{m\,E}$$

$$\varepsilon_2 = \frac{\sigma_2}{E} - \frac{\sigma_3 + \sigma_1}{m\,E}$$

and $$\varepsilon_3 = \frac{\sigma_3}{E} - \frac{\sigma_1 + \sigma_2}{m\,E}$$

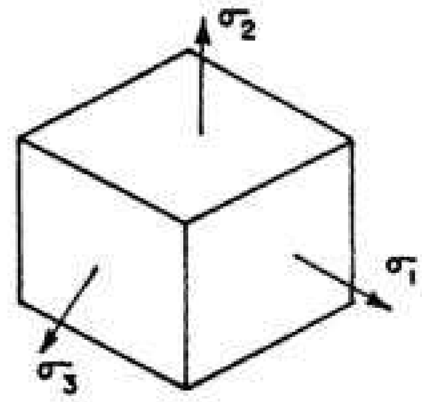


FIG. 5.1

Let σ_1 be the greatest (or major) principal stress, σ_2 be the least (or minor) principal stress and σ_3 be the *intermediate* principal stress. Then the greatest difference of principal strains is

$$\varepsilon_1 - \varepsilon_2 = \frac{\sigma_1 - \sigma_2}{E} + \frac{\sigma_1 - \sigma_2}{m\,E} = \frac{\sigma_1 - \sigma_2}{E}\left(1 + \frac{1}{m}\right) = \left(\frac{m+1}{m}\right)\left(\frac{\sigma_1 - \sigma_2}{E}\right) \quad ...(4)$$

As found earlier, $q_{max} = \dfrac{\sigma_1 - \sigma_2}{2}$ and $\dfrac{m\,E}{m+1} = 2\,N$

We have $$\varepsilon_1 - \varepsilon_2 = \frac{1}{2N}(\sigma_1 - \sigma_2) = \frac{q_{max}}{N} \quad ...(5.2\ a)$$

$$\therefore \quad \varphi_{max} = \frac{q_{max}}{N} = \varepsilon_1 - \varepsilon_2 \quad ...(5.2)$$

Thus, the greatest shear strain (φ_{max}) *is equal to greatest difference* $(\varepsilon_1 - \varepsilon_2)$ *of principal strains.*

It should be carefully noted that *stress and strain in any given direction are not proportional where stress exists in more than one direction.* In fact, strain can exist without a stress in the same direction (*i.e.* even if $\sigma_3 = 0$, the $\varepsilon_3 = -\dfrac{\sigma_1 + \sigma_2}{m\,E}$).

Example 5.1. *A piece of material is subjected to three perpendicular tensile stresses and the strains in the three directions are in the ratio of 2 : 3 : 4. If the Poisson's ratio is 0.286, find the ratio of the stresses and their values if the greatest is 100 N/mm².*

Solution :

Let the stresses be σ_1, σ_2 and σ_3 and the corresponding strains $2\,k, 3\,k$ and $4\,k$.

Then
$$2\,kE = \sigma_1 - 0.286\,(\sigma_2 + \sigma_3) \quad ...(1)$$
$$3\,kE = \sigma_2 - 0.286\,(\sigma_3 + \sigma_1) \quad ...(2)$$
$$4\,kE = \sigma_3 - 0.286\,(\sigma_1 + \sigma_2) \quad ...(3)$$

Subtracting (1) from (3), we get $(\sigma_3 - \sigma_1) - 0.286\,(\sigma_1 - \sigma_3) = 2\,k\,E$

or $$(\sigma_3 - \sigma_1) = \frac{2\,k\,E}{1.286} \quad ...(4)$$

Also, from (3) $$\frac{\sigma_3}{0.286} - \sigma_1 - \sigma_2 = \frac{4\,k\,E}{0.286} \quad ...(5)$$

and from (2), $\sigma_2 - 0.286\,\sigma_3 - 0.286\,\sigma_1 = 3\,k\,E$ ...(6)

Adding (5) and (6), $3.2105\,\sigma_3 - 1.286\,\sigma_1 = 16.986\,kE$ ...(7)

Again from (4) $1.286\,\sigma_3 - 1.286\,\sigma_1 = 2\,k\,E$ ...(8)

Subtracting (8) from (7), $1.9245\,\sigma_3 = 14.486\,k\,E$

From which $\sigma_3 = 7.527\,k\,E$

Hence from (4) $\sigma_1 = \sigma_3 - \dfrac{2\,k\,E}{1.286} = 7.527\,k\,E - 1.555\,k\,E = 5.972\,k\,E$

and from (2), $\sigma_2 = 3\,k\,E + 0.286\,(5.972\,k\,E + 7.527\,k\,E) = 6.861\,k\,E$

$\therefore$ $\sigma_1 : \sigma_2 : \sigma_3 = 5.972 : 6.861 : 7.527 = 0.7934 : 0.9115 : 1$

If the greatest $\sigma_3 = 100\text{ N/mm}^2$,

we have $\boldsymbol{\sigma_1 = 79.34\text{ N/mm}^2}$ **and** $\boldsymbol{\sigma_2 = 91.15\text{ N/mm}^2}$

Example 5.2. *A circle of 400 mm diameter is scribed on a mild steel plate before it is subjected to stresses as shown in Fig. 5.2. In stressing, the circle deforms to an ellipse. Calculate the lengths of the major and minor axes of the ellipse and also find their directions. Take* $\dfrac{1}{m} = 0.286$ *and* $E = 205\text{ kN/mm}^2$.

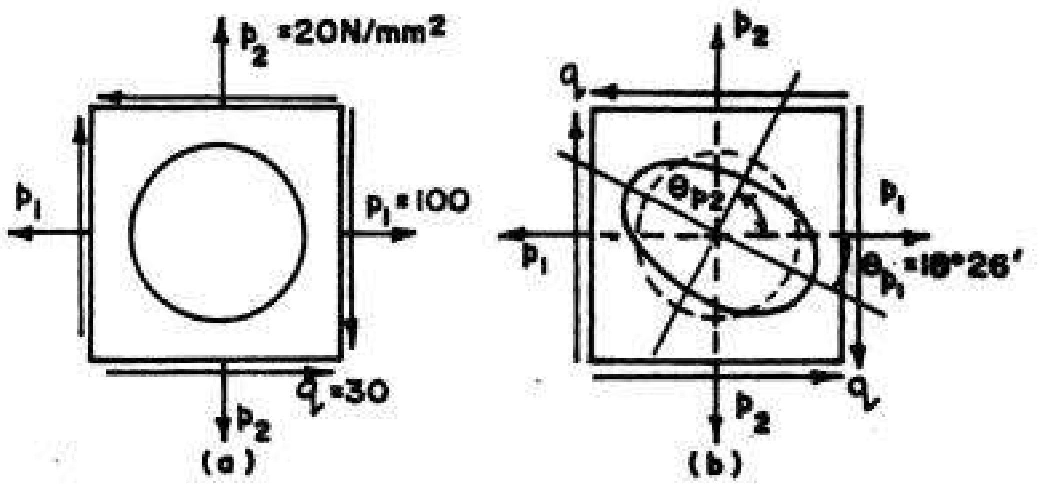


FIG. 5.2

Solution :

$$\sigma = \frac{p_1+p_2}{2} \pm \sqrt{\left(\frac{p_1-p_2}{2}\right)^2 + q^2}$$

$$= \frac{100+20}{2} \pm \sqrt{\left(\frac{100-20}{2}\right)^2 + (30)^2} = 60 \pm 50$$

$\therefore$ $\sigma_1 = +\,110\text{ N/mm}^2$ and $\sigma_2 = +\,10\text{ N/mm}^2$

$$\tan 2\theta = \frac{2q}{p_1 - p_2} = -\frac{2\times 30}{100-20} = -\,0.75\,;\ 2\,\theta_p = -\,36.87° = -\,36°52'$$

$\theta_{p1} = -\,18°26'$ (*i.e.* in the clockwise direction) and $\theta_{p2} = -\,18°26' + 90° = 71°34'$

Major principal strain, $\varepsilon_1 = \dfrac{\sigma_1}{E} - \dfrac{\sigma_2}{mE} = (110 - 0.286\times 10)\dfrac{1}{205\times 1000}$

$= +\,5.226\times 10^{-4}$ (increase)

Minor principal strain, $\varepsilon_2 = \dfrac{\sigma_2}{E} - \dfrac{\sigma_1}{mE} = (10 - 110\times 0.286)\dfrac{1}{205\times 1000}$

$= -\,1.047\times 10^{-4}$ (increase)$= 1.047\times 10^{-4}$ (decrease)

$\therefore$ Length of diameter along $\sigma_1 = d + \varepsilon_1 d = d\,(1+\varepsilon_1)$

$= 400\,(1 + 5.226\times 10^{-4}) =$ **400.209** mm

Length of the diameter along σ_2

$= d - \varepsilon_2 d = d\,(1 - \varepsilon_2) = 400\,(1 - 1.047 \times 10^{-4})$ = **399.958 mm**

Hence the circle will be converted into ellipse having major axis = 400.209 mm at 18°26′ with the x-axis (measured in clock wise direction) and minor axis = 399.958 mm at 71°34′ with x-axis (measured in anticlockwise direction).

5.3. COMPUTATION OF PRINCIPAL STRESSES FROM PRINCIPAL STRAINS

In most of the experimental investigations, strains are measured with the help of strain gauges or 'Strain Rosettes'. Hence principal strains are known from these measurements. Knowing the principal strains, principal stresses can be computed as follows.

Let $\varepsilon_1, \varepsilon_2$ and ε_3 be the principal strains and σ_1, σ_2 and σ_3 be the corresponding principal stresses. If $\mu\ (= 1/m)$ is the Poisson's ratio, we have :

$$E\,\varepsilon_1 = \sigma_1 - \mu\,\sigma_2 - \mu\,\sigma_3 \qquad ...(i)$$

$$E\,\varepsilon_2 = \sigma_2 - \mu\,\sigma_3 - \mu\,\sigma_1 \qquad ...(ii)$$

and

$$E\,\varepsilon_3 = \sigma_3 - \mu\,\sigma_1 - \mu\,\sigma_2 \qquad ...(iii)$$

Subtracting (*ii*) from (*i*), $E\,(\varepsilon_1 - \varepsilon_2) = (\sigma_1 - \sigma_2)\,(1 + \mu)$ ...(*iv*)

Eliminating σ_3 from (*i*) and (*iii*), $E\,(\varepsilon_1 + \mu\,\varepsilon_3) = \sigma_1\,(1 - \mu^2) - \sigma_2\,(1 + \mu)\,\mu$ ...(*v*)

Multiplying (*iv*) by μ and subtracting from (*v*) we get

$$E\left[(1-\mu)\,\varepsilon_1 + \mu\,(\varepsilon_2 + \varepsilon_3)\right] = \sigma_1\,(1 - \mu - 2\mu^2) = \sigma_1\,(1 + \mu)\,(1 - 2\mu)$$

From which,
$$\sigma_1 = \frac{E\,[\,(1-\mu)\,\varepsilon_1 + \mu\,(\varepsilon_2 + \varepsilon_3)\,]}{(1 + \mu)\,(1 - 2\mu)} \qquad ...(5.3\ a)$$

Similarly,
$$\sigma_2 = \frac{E\,[\,(1-\mu)\,\varepsilon_2 + \mu\,(\varepsilon_3 + \varepsilon_1)\,]}{(1 + \mu)\,(1 - 2\mu)} \qquad ...(5.3\ b)$$

and
$$\sigma_3 = \frac{E\,[\,(1-\mu)\,\varepsilon_3 + \mu\,(\varepsilon_1 + \varepsilon_2)\,]}{(1 + \mu)\,(1 - 2\mu)} \qquad ...(5.3\ c)$$

For *two dimensional stress system*, $\sigma_3 = 0$

$\therefore$ $E\,\varepsilon_1 = \sigma_1 - \mu\,\sigma_2$ and $E\,\varepsilon_2 = \sigma_2 - \mu\,\sigma_1$

Solving these we get :
$$\sigma_1 = \frac{E\,(\varepsilon_1 + \mu\,\varepsilon_2)}{1 - \mu^2} \qquad ...(5.4\ a)$$

and
$$\sigma_3 = \frac{E\,(\mu\,\varepsilon_1 + \varepsilon_2)}{1 - \mu^2} \qquad ...(5.4\ b)$$

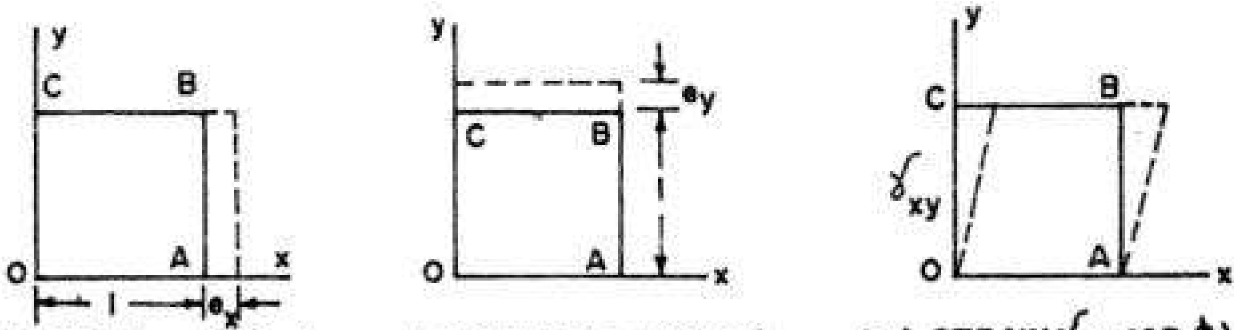
5.4. PLANE STRAIN

The strain caused by a normal stress is called the *normal strain or direct strain* denoted by e, while the strain caused by shear stress is called *shear strain* and is denoted by φ or γ. *The normal and shear strains at a point in a body vary with the direction*, in a manner *analogous to that for stresses*. If we consider any plane, say xy plane, three strain components may exist (Fig. 5.3) :

(*i*) Normal strain $e_x = e_1$, in x direction (*ii*) Normal strain $e_y = e_2$, in y direction

and (*iii*) Shear strain $\gamma_{xy} = \varphi_{xy} = \varphi$ (say)

An element of material subjected to only these strains is said to be in the *state of plane strain*. When an element is in the state of plane strain, the element does not have normal strain e_z (or e_3) in z-direction, and also does not have shear strains γ_{yz} and γ_{zx} in y-z and z-x planes respectively. Hence the *conditions of plane strain* are :



(a) STRAIN ex (OR e_1) (b) STRAIN e_y (OR e_2) (c) STRAIN γ_{xy} (OR ϕ)

FIG. 5.3. STRAIN COMPONENTS IN X-Y PLANE

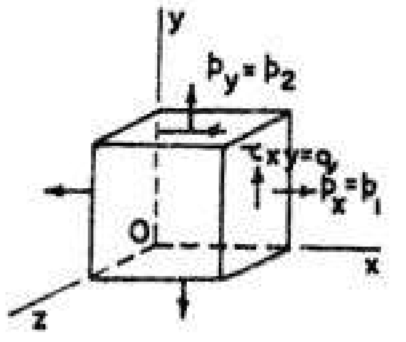
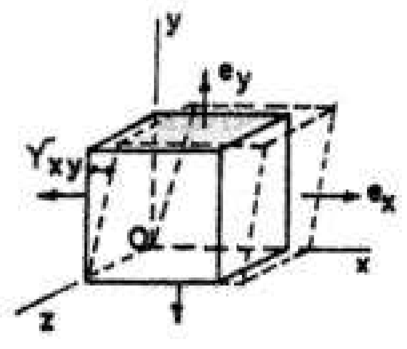
$$e_z \text{ (or } e_3) = 0 \;,\; \gamma_{yz} = 0 \;;\; \gamma_{zx} = 0 \qquad ...(5.5)$$

It should be remembered that the conditions of plane strain are analogous to the *conditions of plane stress* where

$$\sigma_3 \text{ (or } p_3) = 0 \;,\; \tau_{yz} \text{ (or } q_{yz}) = 0 \text{ and } \tau_{zx} = 0 \qquad ...(5.6)$$

This does not mean that the conditions of plane stress and plane strain occur simultaneously. In fact, an element in plane stress undergoes e_z is z-direction. The conditions of plane stress and those of plane strain are depicted in Table 5.1

TABLE 5.1
CONDITIONS OF PLANE STRESS AND PLANE STRAIN CASES

	Plane stress condition	***Plane strain condition***
Element	 FIG. 5.4(*a*)	 FIG. 5.4(*b*)
Stresses	σ_z (*or* p_3) $= 0$; $\tau_{yz} = 0$; $\tau_{zx} = 0$ σ_x (*or* p_1) $= 0$; σ_y (*or* p_2) and τ_{xy} (*or* q) may have non-zero values	$\tau_{yz} = 0$ $\tau_{zx} = 0$ σ_x (*or* p_1) ; σ_y (*or* p_2), σ_z ($= p_3$) and τ_{xy} ($= q$) may have non-zero values
Strains	$\gamma_{yz} = 0$; $\gamma_{zx} = 0$ e_x (*or* e_1), e_y (*or* e_2), e_z (*or* e_3) and γ_{xy} (*or* q) may have non-zero values	e_z ($= e_3$)$= 0$; $\gamma_{yz} = 0$; $\gamma_{zx} = 0$ and e_x ($= e_1$) , e_y ($= e_2$)and γ_{xy} ($= \varphi$) may have non-zero values

5.5. STRAIN COMPONENTS IN AN INCLINED DIRECTION

In the previous chapter, we have found stress components p_n and p_t on an inclined plane through a point when it is subjected to normal stresses p_1 and p_2 in two mutually perpendicular directions, along with shear stress q. In a similar manner, if direct and shearing strains are given in two mutually perpendicular directions, we can find direct and shearing strains in any other direction. In other words, we can also find the transformation equations for direct and shearing strains, as was done for direct and shearing stresses.

Fig. 5.5 shows a rectangular element $OABC$ in X-Y plane.

Let $e_1 =$ Strain in x-direction

$e_2 =$ Strain in y-direction

and $\varphi =$ Shearing strain relative to OX and OY

Let OB be any direction, inclined at θ with x-direction along which it is required to find *direct strain* (or normal strain) e_θ and *shearing strain* φ_θ.

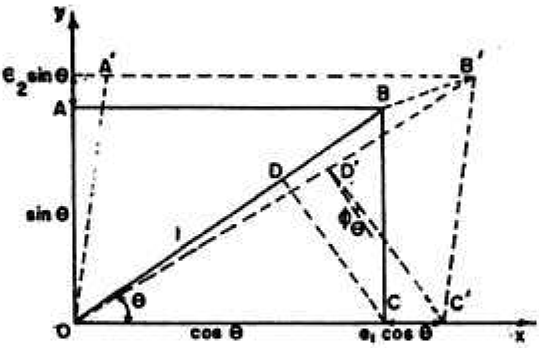


FIG. 5.5

Let us consider all the strains to be very small. Taking the length of diagonal OB as *unity*, side OA will be of length $\sin\theta$ while side OC will be of length $\cos\theta$. When the material is strained, OA extends by small amount $e_2 \sin\theta$ while OC extends by small amount $e_1 \cos\theta$. Also, due to shearing strain, OA rotates to OA' by small shearing angle φ. The distorted shape of the rectangle $OABC$ is represented by $OA'B'C'$, in which point A moves to A', B moves to B', and C moves to C', relative to point O.

Movement of B parallel to x-axis $= e_1 \cos\theta + \varphi \sin\theta$ ...(*i*)

Movement of B parallel to y-axis $= e_2 \sin\theta$ ...(*ii*)

$\therefore$ Movement of B parallel to $OB = (e_1 \cos\theta + \varphi \sin\theta)\cos\theta + (e_2 \sin\theta)\sin\theta$ ...(*iii*)

This movement of B will be approximately equal to the extension of OB in the strained condition. However, since OB has been taken to be of unit length, the extension of OB is equal to the direct strain e_θ in the direction OB.

$$\therefore \quad e_\theta = (e_1 \cos\theta + \varphi \sin\theta)\cos\theta + (e_2 \sin\theta)\sin\theta$$

or
$$e_\theta = e_1 \cos^2\theta + e_2 \sin^2\theta + \varphi \sin\theta\cos\theta \qquad \text{...(5.7 } a)$$

or
$$e_\theta = e_1 \cos^2\theta + e_2 \sin^2\theta + \frac{\varphi}{2}\sin 2\theta \qquad \text{...(5.7)}$$

Note that this expression is *analogous* to Eq. 4.14 for p_n, where p_1 and p_2 have been replaced by e_1 and e_2 respectively and q has been replaced by $\frac{\varphi}{2}$.

Eq. 5.7 can also be re-written in the form

$$e_\theta = \frac{1}{2}(e_1 + e_2) + \frac{1}{2}(e_1 - e_2)\cos 2\theta + \frac{1}{2}\varphi \sin 2\theta \qquad \text{...(5.8)}$$

which is analogous to Eq. 4.15 for p_n

Again, in order to find shearing strain φ_θ in the required direction, draw CD perpendicular to OB (Fig. 5.5). Point D' is the strained position of D. When OB extends to e_θ, OD extends by amount $= e_\theta . OD = e_\theta . \cos^2\theta$ (since $OD = OC\cos\theta = \cos\theta . \cos\theta = \cos^2\theta$).

During straining, the perpendicular CD rotates in the anti clockwise direction through a small angle

$$= \frac{e_1 \cos^2\theta - e_\theta \cos^2\theta}{\cos\theta \sin\theta} = (e_1 - e_\theta)\cot\theta$$

At the same time, OB rotates (in the clockwise direction) through a small angle $= -(e_1 \cos\theta + \varphi \sin\theta)\sin\theta + (e_2 \sin\theta)\cos\theta$. Now, the *shearing strain* φ_θ *in the direction* OB *is the amount by which angle ODC diminishes during straining.* Hence

$$\varphi_\theta = \text{Rotation of } OB - \text{Rotation of } CD$$

$$= [-(e_1 \cos\theta + \varphi \sin\theta)\sin\theta + e_2 \sin\theta \cos\theta] - [(e_1 - e_\theta)\cot\theta]$$

or

$$\varphi_\theta = -(e_1 - e_\theta)\cot\theta - (e_1 \cos\theta + \varphi \sin\theta)\sin\theta + (e_2 \sin\theta)\cos\theta$$

Substituting the value of e_θ from Eq. 5.7 (a), we have

$$\varphi_\theta = -2\,(e_1 - e_2)\cos\theta \sin\theta + \varphi\,(\cos^2\theta - \sin^2\theta) \qquad ...(5.9)$$

This can also be written as

$$\frac{1}{2}\varphi_\theta = -2\,(e_1 - e_2)\sin 2\theta + \frac{1}{2}\varphi \cos 2\theta \qquad ...(5.10)$$

The negative sign with φ_θ indicates that the angle at the origin will *increase* while positive sign shows that the angle will decrease.

It is interesting to note that Eq. 5.10 is analogous to Eq. 4.16, where p_1 and p_2 have been replaced by e_1 and e_2 respectively, q has been replaced by $\frac{1}{2}\varphi$ and p_t has been replaced by $\frac{1}{2}\varphi_\theta$.

Eqs. 5.8 and 5.10 are also known as *transformation equations* for strains. The direct strain $(e_{\theta 2})$ in the other direction, perpendicular to the previous one, can be found by replacing θ by $(90° + \theta)$ in Eq. 5.8 :

Thus

$$e_{\theta 2} = \frac{1}{2}(e_1 + e_2) + \frac{1}{2}(e_1 - e_2)\cos 2\,(90° + \theta) + \frac{1}{2}\varphi \sin 2\,(90° + \theta)$$

or

$$e_{\theta 2} = \frac{1}{2}(e_1 + e_2) - \frac{1}{2}(e_1 - e_2)\cos 2\theta - \frac{\theta}{2}\sin 2\theta \qquad ..(5.8\ a)$$

Taking $e_\theta = e_{\theta 1}$, we observe that

$$e_{\theta 1} + e_{\theta 2} = e_1 + e_2 \qquad ...(5.11)$$

Thus, the sum of normal strains in perpendicular directions is a constant.

5.6. PRINCIPAL STRAINS

We have seen in the above article that the direct strain (e_θ) changes with the angle θ. *The maximum and minimum direct strains in a material, subjected to complex stresses, are called the principal strains.* These strains act in the direction of principal stresses. The maximum and minimum values of direct strain occur when the position of plane OB is such that shearing strain φ_θ is zero. This occurs at values of θ obtained by equating $\frac{d\varphi_\theta}{d\theta}$ to zero.

Thus, differentiating Eq. 5.10 with respect to θ and equating it to zero, we get

$$\tan 2\theta_{pe} = \frac{\varphi}{e_1 - e_2} \qquad ...(5.12)$$

This equation is analogous to Eq. 4.21.

From Eq. 5.12, $$\tan 2\theta_{p\varepsilon} = \frac{\frac{q}{N}}{\left(\frac{p_1}{E} - \frac{p_2}{mE}\right) - \left(\frac{p_2}{E} - \frac{p_1}{mE}\right)} = \frac{q}{p_1 - p_2} \cdot \frac{E}{N\left(1 + \frac{1}{m}\right)}$$

or $$\tan 2\theta_{p\varepsilon} = \frac{2q}{p_1 - p_2} \qquad ...(5.12\ a)$$

which is same as Eq. 4.21

Study of Eqs. 5.12 and 4.21 reveal that *principal strains and principal stresses act in the same direction.*

Hence, $\theta_{p\varepsilon} = \theta_p$

Eq. 5.12 can be represented diagrammatically by Fig. 5.6, in which

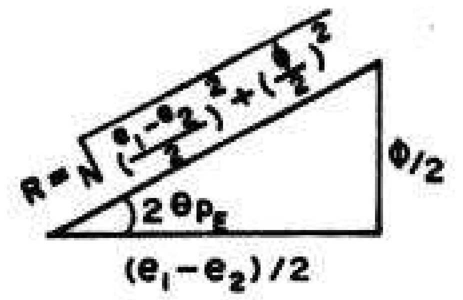


FIG. 5.6

$$\tan 2\theta_{p\varepsilon} = \frac{\varphi/2}{\frac{(e_1 - e_2)}{2}}$$

Hence diagonal $R = \sqrt{\left(\frac{e_1 - e_2}{2}\right)^2 + \left(\frac{\varphi}{2}\right)^2}$ (Taking the positive root)

Thus $$\cos 2\theta_{p\varepsilon} = \frac{e_1 - e_2}{2R} \quad \text{and} \quad \sin 2\theta_{p\varepsilon} = \frac{\varphi/2}{R} = \frac{\varphi}{2R} \qquad ...(5.13)$$

Substituting the values of $\cos 2\theta_{p\varepsilon}$ and $\sin 2\theta_{p\varepsilon}$ in Eq. 5.8 and setting

$e_\theta = \varepsilon_1$ we get

$$\varepsilon_1 = \frac{1}{2}(e_1 + e_2) + \sqrt{\left(\frac{e_1 - e_2}{2}\right)^2 + \left(\frac{\varphi}{2}\right)^2} \qquad ...(5.14\ a)$$

The minor principal strain ε_2 is obtained from the relation

$$\varepsilon_1 + \varepsilon_2 = e_1 + e_2$$

or $$\varepsilon_2 = (e_1 + e_2) - \left[\frac{e_1 + e_2}{2} + \sqrt{\left(\frac{e_1 - e_2}{2}\right)^2 + \left(\frac{\varphi}{2}\right)^2}\right]$$

or $$\varepsilon_2 = \frac{e_1 + e_2}{2} - \sqrt{\left(\frac{e_1 - e_2}{2}\right)^2 + \left(\frac{\varphi}{2}\right)^2} \qquad ...(5.14\ b)$$

Hence $$\varepsilon_{1,2} = \frac{e_1 + e_2}{2} \pm \sqrt{\left(\frac{e_1 - e_2}{2}\right)^2 + \left(\frac{\varphi}{2}\right)^2} \qquad ...(5.14)$$

5.7. MAXIMUM SHEAR STRAINS

In general, the shear strain varies with angle θ and its value is given by Eq. 5.10

$$\frac{1}{2}\varphi_\theta = -\frac{1}{2}(e_1 - e_2)\sin 2\theta + \frac{1}{2}\varphi\cos 2\theta \qquad ...(5.10)$$

For maximum value of φ_θ, differentiate it with respect to θ and equate it to zero.

$$\therefore \qquad \frac{d\left(\frac{\varphi_\theta}{2}\right)}{d\theta} = -(e_1 - e_2)\cos 2\theta - \varphi \sin 2\theta = 0$$

From which $$\tan 2\theta_{se} = -\frac{e_1 - e_2}{\varphi} \qquad \text{...(5.15)}$$

Comparing this with Eq. 5.12, we observe that

$$\tan 2\theta_{se} = -\frac{1}{\tan 2\theta_{pe}} = -\cot 2\theta_{pe} \qquad \text{...(5.16)}$$

Since $\tan(\alpha \pm 90°) = -\cot\alpha$, we have

$$2\theta_{se} = 2\varphi_{pe} \pm 90° \text{ or } \theta_{se} = \theta_{pe} \pm 45°$$

Thus, the maximum shear strains in the xy plane are associated with axes at 45° to the directions of principal planes. The algebraically maximum shear strain in the xy plane is given by

$$\frac{\varphi_{max}}{2} = \sqrt{\left(\frac{e_1 - e_2}{2}\right)^2 + \left(\frac{\varphi}{2}\right)^2} \qquad \text{...(5.17)}$$

The minimum shear strain has the same magnitude but is negative. In the directions of maximum shear strain, the normal strains are equal to $\frac{(e_1 + e_2)}{2}$.

5.8. MOHR'S CIRCLE FOR PLANE STRAIN

Since the expressions for linear strain (or direct strain) e_θ and shearing strain φ_θ on any plane are analogous to the corresponding expression for p_n and p_t, Mohr's circle method can be used for the graphical solution in a way similar to the one used for plane stress case.

Eq. 5.8 for e_θ can be rewritten as :

$$e_\theta - \frac{1}{2}(e_1 + e_2) = \frac{1}{2}(e_1 - e_2)\cos 2\theta + \frac{1}{2}\varphi \sin 2\theta \qquad \text{...}(i)$$

Similarly Eq. 5.10 for φ_θ can be rewritten as

$$\frac{1}{2}\varphi_\theta = -\frac{1}{2}(e_1 - e_2)\sin 2\theta + \frac{1}{2}\varphi \cos 2\theta \qquad \text{...}(ii)$$

Squaring (*i*) and (*ii*) and adding, we get

$$\left[e_\theta - \frac{1}{2}(e_1 + e_2)\right]^2 + \left[\frac{1}{2}\varphi_\theta\right]^2 = \left[\frac{1}{2}(e_1 - e_2)\right]^2 + \left[\frac{1}{2}\varphi\right]^2$$

Thus, all values of e_θ and $\frac{1}{2}\varphi_\theta$ lie on a circle whose radius is $\sqrt{\left[\frac{1}{2}(e_1 - e_2)^2\right] + \left[\frac{1}{2}\varphi\right]^2}$ and whose centre is at point $\left[\frac{1}{2}(e_1 + e_2), 0\right]$.

To draw the Mohr circle of strains, mark $OA = e_1$ and $OB = e_2$. Also, mark $AG = \frac{1}{2}\varphi = BF$. Join F and G, to get point C, the centre of the circle. Mohr circle is now drawn with C as the centre and FG or CF as radius, cutting the x-axis in points H and J. Draw the line CD, making an angle 2θ, in appropriate direction with CG. Here, the line CD corresponds to a positive angle while AG corresponds to a positive shear strain

$\left(\frac{1}{2}\varphi\right)$. Similarly, e_1 and e_2 are positive (being tensile). The coordinates of point D represents direct strain e_θ and shear strain $\frac{1}{2}\varphi_\theta$ (as marked) on a plane making an angle θ (in anti clockwise direction) with the direction of e_1. It should be noted that points H and J correspond to principal strains ε_1 and ε_2 respectively, since the shearing stain $\left(\frac{1}{2}\varphi_\theta\right)$ at these points are each zero.

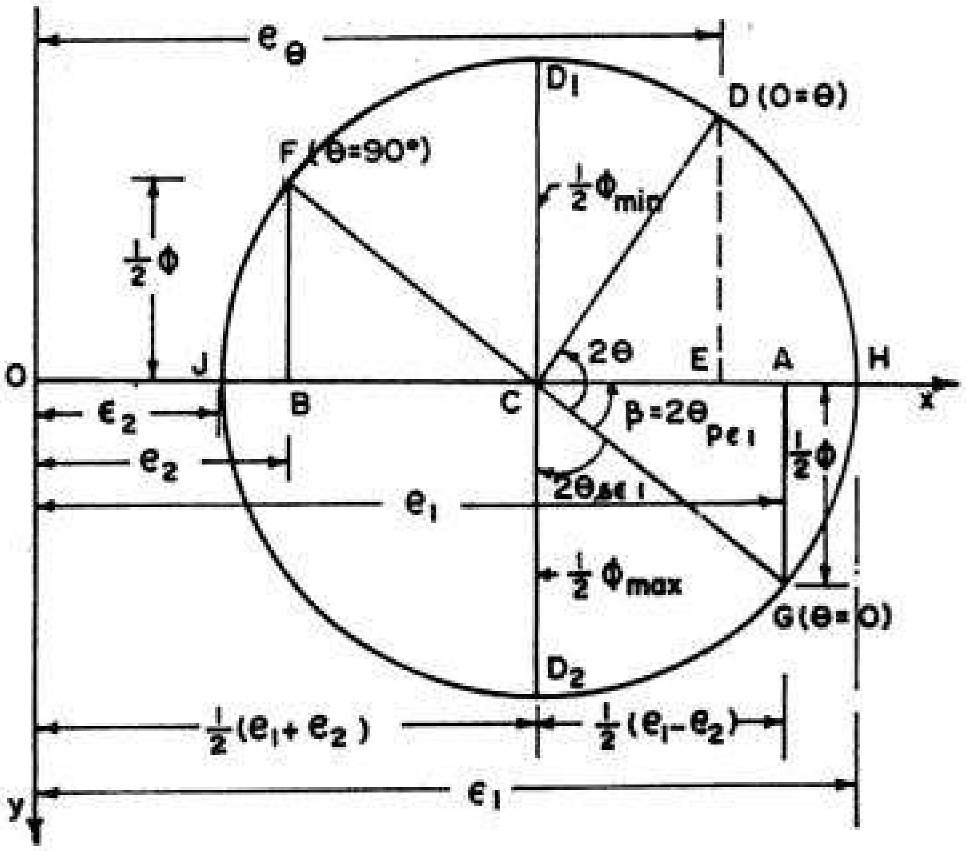


FIG. 5.7. MOHR CIRCLE OF STRAINS

Conclusions : From the strain circle, we draw the following conclusions :

(*i*) When D coincides with H (i.e. $2\theta = \beta$), e_θ attains its maximum value :

$$(e_\theta)_{\max} = \varepsilon_1 = OC + CH$$

$$= \frac{1}{2}(e_1 + e_2) + \sqrt{\left(\frac{e_1 - e_2}{2}\right)^2 + \left(\frac{1}{2}\varphi\right)^2}$$

where $(e_\theta)_{\max}$ is known as *major principal strain* (See Eq. 5.14 *a*)

$$\tan 2\theta_{p\varepsilon} = \tan\beta = \frac{\frac{1}{2}\varphi}{\frac{1}{2}(e_1 - e_2)} = \frac{\varphi}{e_1 - e_2}$$

(*ii*) when point D coincides with J, e_θ attains its minimum value :

$$(e_\theta)_{\min} = \varepsilon_2 = OC - CJ = \frac{1}{2}(e_1 + e_2) - \sqrt{\left(\frac{e_1 - e_2}{2}\right)^2 + \left(\frac{1}{2}\varphi\right)^2}$$

where $(e_\theta)_{\min} = \varepsilon_2$, is known as the *minor principal strain* (See Eq. 5.14 *b*)

(*iii*) When point D_1 occupies the highest position of the circle, $2\theta = \beta + 90°$ and $\frac{1}{2}\varphi_\theta$ attains its maximum value :

$$\frac{1}{2}\varphi_{\max} = \sqrt{\left(\frac{e_1 - e_2}{2}\right)^2 + \left(\frac{1}{2}\varphi\right)^2}$$

Direct strain (e_θ) corresponding to the point D_1, is obviously equal to $\frac{1}{2}(e_1 + e_2) = e_{av}$.

Example 5.3. *A flat bar 10 mm thick and 100 mm wide is subjected to a pull of 96 kN. One side of the bar is polished and lines are ruled on it to form a square of 50 mm side, one diagonal of the square being along the middle line of the polished side. Taking $E = 210\ kN/mm^2$ and $\mu = 0.28$, calculate the change in the sides and angles of the square.*

Solution :

Axial stress in bar,

$$p_1 = \frac{96 \times 1000}{100 \times 10} = 96 \text{ N/mm}^2 \text{ (tensile)}$$

Strain in one diagonal of square along the axis of the bar is

$$e_1 = \frac{p_1}{E} = \frac{96}{210 \times 10^3} = 4.571 \times 10^{-4}$$

(tensile)

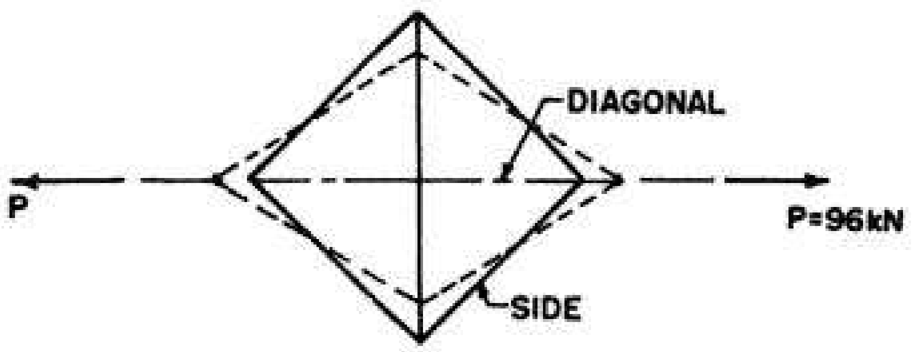


FIG. 5.8

$\therefore$ Strain in other diagonal $= e_2 = -\mu\, e_1 = -0.28 \times 4.571 \times 10^{-4}$

$$= -1.28 \times 10^{-4} \text{ (compressive)}$$

$\therefore$ Strain in the sides of square is given by

$$e_\theta = e_1 \cos^2 45^\circ + e_2 \sin^2 45^\circ \qquad ...(5.7)$$

$$= 4.571 \times 10^{-4} \times 0.5 - 1.28 \times 10^{-4} \times 0.5 = 1.645 \times 10^{-4} \text{ (tensile)}$$

Hence change in length of each side of square $= \left(1.645 \times 10^{-4}\right) \times 50$

$= \mathbf{0.008225\ mm}$ (increase)

Total shear strain or change in the angles of the square $= \varphi_\theta$

$$= -(e_1 - e_2) \sin 2\theta \qquad ...(5.10)$$

$$= -(4.571 + 1.280) \times 10^{-4} \sin 90^\circ = -\mathbf{5.851 \times 10^{-4}} \text{ radian}$$

Thus the two angles of the square which are along the line of pull will be decreased and the other two will be increased by the above amount.

Example 5.4. *A brass plate 3 mm thick, is stretched in the plane of the plate in two directions at right angles. An extensometer, arranged in the X-direction gave an extension of 0.0413 mm on a gauge length of 50 mm, while another, arranged in Y-direction gave an extension of 0.0187 mm on a 100 mm gauge length. If the normal stress on a plane making angle θ with the Y-axis is 66 N/mm², find angle θ. Find also the decrease in the thickness of the plate. Take E for brass = 80 N/mm² and μ = 0.3.*

Solution :

Let σ_x and σ_y be the principal stresses in two directions (say x and y directions respectively) and ε_1 and ε_2 be the corresponding principal strains.

Then $\qquad \varepsilon_x = \dfrac{0.0413}{50} = 8.26 \times 10^{-4}$ and $\varepsilon_y = \dfrac{0.0187}{100} = 1.87 \times 10^{-4}$

Also, $\qquad \varepsilon_x = \dfrac{\sigma_x}{E} - \dfrac{\mu\,\sigma_y}{E}$ and $\varepsilon_y = \dfrac{\sigma_y}{E} - \dfrac{\mu\,\sigma_x}{E}$

$$\therefore \quad 8.26 \times 10^{-4} = \frac{1}{80 \times 10^3}\left[\sigma_x - 0.3\,\sigma_y\right] \text{ or } \sigma_x - 0.3\,\sigma_y = 66.08 \qquad ...(i)$$